q.e.d.

A desktop app for your own website or podcast site, made without programming. This site is built with q.e.d. itself.

https://qed.informatom.com/

As of November 6, 2026

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q.e.d.



Your website, without programming

Write your pages, add pictures and episodes, choose colors and fonts: all in the app. q.e.d. turns it into a fast website and publishes it to your own webspace. No code, no server, no subscription.

→ Read the blog

q.e.d. is open source and in early development. It's built in the open with Claude Code[1], in Rust, Tauri, Svelte and TypeScript.

For podcasters

q.e.d. also makes podcast websites, and keeps everything on your own webspace: no hosting service, no tracking, nothing loaded from anywhere else.

- **Episodes** with season, number, chapters, people and transcripts, the shownotes written like any other page.
- **Feeds** for every audio format you offer (MP3, M4A, Ogg, Opus, FLAC), with Apple's and Podcasting 2.0 tags. If your old site was made with jekyll-octopod, your listeners' apps keep working.
- **The Podlove Web Player** with chapters, transcript, playlist and sharing, and the **Podlove Subscribe Button**, both self-hosted.
- **Contributors** entered once, with a page each listing their episodes.

This site is a q.e.d. site itself: a `site.toml` and a `pages/` folder, in the `docs/` folder of the q.e.d. repository[2].

[1] https://claude.com/claude-code
[2] https://github.com/informatom/qed
[3] https://github.com/informatom/qed/releases/latest

Download

The installers for the latest version are on the releases page[3]. They aren't signed yet, so your system asks once before the first start.

macOS

Download the `.dmg`, open it and drag qed into Applications. It runs on Apple Silicon and Intel Macs.

The first time, open qed. macOS says it can't check it: choose **Done**. Then open System Settings → **Privacy & Security**, scroll down to the note about qed, choose **Open Anyway** and confirm with your password. From then on qed opens as usual.

On macOS 14 and older, right-click qed in Applications and choose **Open** instead.

Windows

Download the `.exe` and run it.

When Windows shows "Windows protected your PC", choose **More info**, then **Run anyway**.

Linux

On Debian and Ubuntu, download the `.deb` and install it with

```
sudo apt install ./qed_*.deb
```

On any other distribution, download the `.AppImage`, make it executable (`chmod +x qed_*.AppImage`) and run it.

Learning along the way

The blog explains the Rust, Svelte and TypeScript behind q.e.d., step by step, for people who know other languages.

→ Start with the first post

Development

q.e.d. is a Rust and Tauri desktop app with a Svelte UI. The source is on GitHub[4] the plan lives in backlog.

Requirements

- Rust[5] (stable)
- Node.js[6] 24 or newer
- On Linux, the Tauri system libraries[7]

Running and checking

```
npm install
npm run tauri dev     # start the app
npm run check:all     # formatting, lints, Rust tests, Svelte type check
```

To build a site without the app:

```
cargo run -p qed-core --example build -- docs
```

The result lands in `docs/_site/`.

A test server to publish to

Publishing over SFTP needs a server. For development, the repository has a throwaway one in `dev/test-server/compose.yml`, run with Docker[8]:

```
docker compose -f dev/test-server/compose.yml up -d     # start
docker compose -f dev/test-server/compose.yml down -v   # stop and forget everything
```

It starts two containers that share one folder:

What	Where	Details
SFTP server	`127.0.0.1`, port 2222	user `qed`, password `qed`, folder `/www`
Web server	http://localhost:8080	shows whatever was published to `/www`

Both only listen on your own computer. The password is public, so never use this setup for anything real.

To try publishing from the app, add a target with exactly those details. On the first connection, q.e.d. shows the server's key fingerprint and asks whether to trust it. The server makes a new key each time it's recreated with `down -v`, so q.e.d. will warn that the key changed; that's expected here, and exactly the warning you'd want to take seriously with a real server. To continue, open the target with "Edit", click "Forget" next to the trusted key, save, and test the connection again.

Automated tests

The tests in `crates/qed-core/tests/test_server.rs` publish the demo site to this server and check the result over HTTP: first publish, unchanged re-publish, changed and deleted pages. They're skipped by a normal `cargo test`, so the test suite doesn't depend on Docker. To run them:

[4] https://github.com/informatom/qed
[5] https://www.rust-lang.org/tools/install
[6] https://nodejs.org/
[7] https://v2.tauri.app/start/prerequisites/
[8] https://docs.docker.com/get-docker/

```
docker compose -f dev/test-server/compose.yml up -d
cargo test -p qed-core --test test_server -- --ignored --test-threads=1
```

Testing the MCP server

AI assistants work on a site through `qed mcp <site folder>`: the app itself, started without a window, answering over stdin and stdout (see the manual's AI assistants for the user's side). The tools do their work through qed-core, which `cargo test` covers. Whether the program answers when an assistant starts it is checked by `dev/mcp-smoke.py`. It starts q.e.d. with pipes for stdin, stdout and stderr, as Claude Desktop and Claude Code do, makes the opening handshake, lists the tools, calls `site_info`, closes stdin and expects the program to end. It only reads, so any site will do:

```
python3 dev/mcp-smoke.py target/debug/qed docs
```

Instead of `target/debug/qed`, give it an installed copy: the `.AppImage` on Linux, `/Applications/qed.app/Contents/MacOS/qed` on macOS, `%LOCALAPPDATA%\qed\qed.exe` on Windows. The program path is the one App settings shows under the assistant settings.

Windows

Windows needs its own test. Release builds there are window programs (`windows_subsystem = "windows"` in `src-tauri/src/main.rs`), so no console window opens next to the app. Such a program has no console of its own, and it can still use pipes a caller hands it, which is how an assistant starts it. A debug build is a console program, so it doesn't show whether this works: test an installed release.

Without a Windows computer, use GitHub Actions: under Actions → *MCP on Windows* → *Run workflow* (`.github/workflows/mcp-windows.yml`). It installs a published release on a fresh Windows machine without questions (the latest, or the tag you give) and runs `dev/mcp-smoke.py` against it on the `docs` folder.

On a Windows computer, install q.e.d., then run the script against `%LOCALAPPDATA%\qed\qed.exe`. Don't judge by typing `qed mcp …` into cmd or PowerShell: they don't wait for window programs and don't show what they print, so it looks as if nothing happened, even when all is well.

Then try it with a real assistant, which also tests the settings App settings shows. In Claude Desktop, put them into the configuration file (Settings → Developer → Edit Config) and restart it; the q.e.d. server should appear with its tools. Ask for something small, like the site's title, then for a new page, and check in q.e.d. that the page and a version for it are there. If the server doesn't appear, Claude Desktop's log for it is in `%APPDATA%\Claude\logs\` (`mcp-server-<name>.log`), with whatever q.e.d. wrote to stderr. In Claude Code, run the `claude mcp add` command from App settings in PowerShell and cmd; paths in `C:\Program Files\…` or with spaces in the site folder are where the quoting can break.

Releasing

Installers are built by GitHub Actions (`.github/workflows/release.yml`) whenever a version tag is pushed to the GitHub repository:

System	File
macOS	`.dmg`, one for Apple Silicon and Intel Macs
Windows	`.exe` installer
Linux	`.deb` for Debian and Ubuntu, `.AppImage` for everything else

To make a release:

1. Set the new version in `Cargo.toml` (`[workspace.package]`) and in `package.json`. The app takes its version from `Cargo.toml` the build stops if the tag, `Cargo.toml` and `package.json` disagree.
2. Run `npm install` (so `package-lock.json` follows) and `npm run check:all`, then commit.
3. Tag and push the tag to GitHub:

```
git tag v0.2.0
git push github v0.2.0
```

4. After about 20 minutes, a draft release with the installers is on the repository's Releases page. Try them, write the notes, and publish the release. Check that it has a `latest.json` with entries for each system: `darwin-aarch64` and `darwin-x86_64` (both pointing to the universal `.app.tar.gz`), `windows-x86_64`, and on Linux one for the AppImage and one for the `.deb` (`linux-x86_64-deb`). Each installed copy looks for its own kind first (`linux-x86_64-deb`), then for the system (`linux-x86_64`).
5. Once it's published, installed copies of q.e.d. offer the update at their next start. Drafts stay invisible to them.

Updates

q.e.d. updates itself with Tauri's updater plugin[9]. At start (not in `npm run tauri dev`) it fetches `latest.json` from the latest published release, and if the version there is newer, a bar offers "Install and restart"; the App tab has "Check for updates". The download is checked against the public key in `src-tauri/tauri.conf.json` (`plugins.updater.pubkey`) before it's installed. On Windows the installer runs by itself, on macOS the `.app` is replaced, on Linux the `.AppImage`. A `.deb` install asks for the administrator password and installs the new `.deb` (with `pkexec dpkg -i`).

The update files and `latest.json` are only made in the release workflow, which adds `src-tauri/tauri.release.conf.json` to the build. Local builds don't need the signing key.

The signing key is made once, and its private half never goes into the repository:

1. `npx tauri signer generate -w ~/.tauri/qed.key` asks for a password and writes `~/.tauri/qed.key` (private) and `~/.tauri/qed.key.pub` (public). Keep both, and the password, somewhere safe: without them, installed copies can't take updates any more.
2. Put the contents of `qed.key.pub` into `plugins.updater.pubkey` in `src-tauri/tauri.conf.json` and commit it.
3. In the GitHub repository, under Settings → Secrets and variables → Actions, add two repository secrets: `TAURI_SIGNING_PRIVATE_KEY` with the contents of `qed.key`, and `TAURI_SIGNING_PRIVATE_KEY_PASSWORD` with the password.

The installers aren't signed yet (#070 in the backlog), so the systems warn before the first start. On macOS 15 and later, open the app once, then choose "Open Anyway" in System Settings → Privacy & Security (on macOS 14 and older, right-click the app and choose Open); or remove the download mark with `xattr -d com.apple.quarantine /Applications/qed.app`. On Windows, choose "More info" and "Run anyway" in the SmartScreen dialog.

To try the Linux packaging locally: `npm run tauri build -- --bundles deb,appimage`. The files land in `target/release/bundle/`.

[9] https://v2.tauri.app/plugin/updater/

Manual

This manual is for the people who make and look after a website with q.e.d.: no programming needed, and no need to know how websites work inside. The same pages are in the app: the 🕮 **Manual** button at the top right opens them in a window of their own, so you can read them offline, next to what you're doing. The manual is in English and German; **Language of the manual** in the **App** tab picks one (automatically your computer's).

- Before you start: what you need (and what you don't).
- Getting started: install q.e.d., make or open a site, find your way around, look at the result.
- Pages and blocks: write pages and posts, build them from blocks, columns and sections, link them.
- Icons: the icons for text and menus, and your own.
- Pictures and files: add photos and logos, files to download, and files the website needs as they are.
- Site settings: title, language, web address, logo, search and page width.
- Menus: the main and footer menus, made by themselves or by hand.
- Colors and fonts: a theme, where its colors go, fonts, and colors of your own.
- Versions: every save is kept; go back to an earlier state of a page or of the whole site.
- Backups: keep a copy of the site on another computer, and work on it from two computers.
- Publishing: put the website on your webspace.
- Podcasts: episodes, the people in them, audio and feeds.
- AI assistants: let an assistant such as Claude write and change pages for you, with q.e.d.'s rules.
- Moving a site over: from another computer or another system, keeping old addresses working.
- License and credits: q.e.d. is free software, your content stays yours, and the parts from others. Also where to report bugs.
- Libraries: every library q.e.d. is built from, with its version and license.

Screenshots in this manual show the app with its *catppuccin-mocha* theme (App → Theme of the q.e.d. app), so they look the same throughout.

Before you start

What you need

A computer

🏠 With macOS (Apple Silicon or Intel), Windows, or Linux (Ubuntu 22.04 or newer, or a distribution of the same age). q.e.d. runs there, and your site lives there as well: its pages, pictures and history.

Webspace

🌐 Space on a web server that you can reach over **SFTP**, as almost every hosting package offers: a server name, a user name and a password or key (Publishing). A website from q.e.d. needs no more than that.

A domain

🔗 An address such as `example.com`, usually from the same provider as the webspace. It goes into the **Web address** in the Site settings.

For a **podcast**, the webspace needs room for the audio: an hour of MP3 is about 50 to 80 MB per episode and format. Or the audio stays at a podcast host (Podcasts).

For **backups** on another computer, and for working on one site from two computers, a git server such as Codeberg or GitHub helps (Backups). It's optional.

What you don't need

- **Nothing on the server** but files: no PHP, no database, no WordPress, no updates to install there. A website that is only files can't be broken into through its software, and it's fast.
- **No programming** and no command line: everything is made in the app.
- **No account** with q.e.d. and no subscription. The app is free, and it works offline; only publishing needs the internet.
- **No other services** for fonts, the player, search or statistics: your website brings everything along and sends your visitors' data to no one.

Ready?

Install q.e.d. and make your first site.

→ Getting started

Getting started

Install

The installers are on the releases page[10]: a `.dmg` for macOS, an `.exe` for Windows, a `.deb` and an `.AppImage` for Linux. The q.e.d. website says how to get past the first warning of your system. q.e.d. looks for a newer version each time it starts and asks before installing one.

Your site is a folder

Everything a site consists of is in one folder on your computer: its pages, pictures, settings and history. There's no account and no server behind q.e.d.; the folder is the site. Keep it where your other documents are, and back it up like them.

→ Backups

Make a site, or open one

When no site is open, q.e.d. offers three ways in:

New site…

Asks for the site's title, its language and where to put its folder. q.e.d. makes the folder with a start page and starts its history.

Open site folder…

Opens a site that's already on your computer, made with q.e.d. here or elsewhere.

Get a site from git…

Brings a site over from a server where it was backed up; see Backups.

Later, the site's title at the top of the window is a menu: switch to a recently opened site, open another one, make a new one or close it.

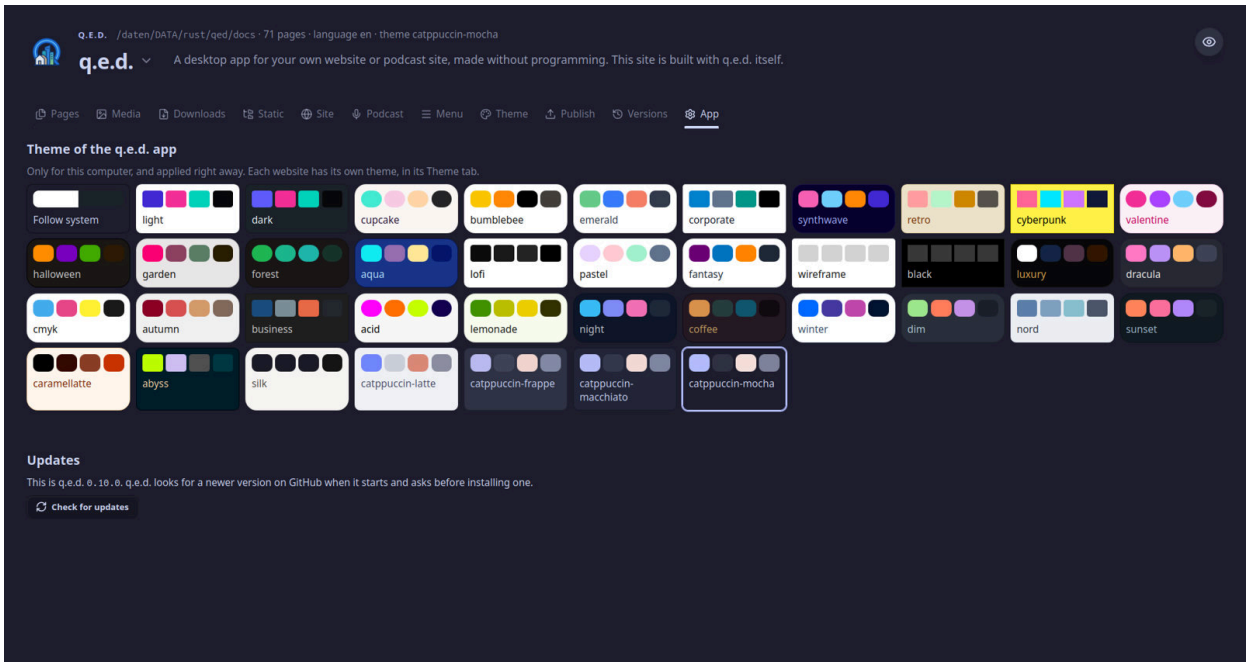
Finding your way around

The tabs below the title hold everything about the open site:

[10] https://github.com/informatom/qed/releases/latest

- **Pages**: the site's pages, posts and shared blocks, and the editor for the one you pick. See Pages and blocks.
- **Media**, **Downloads** and **Static**: pictures, files to download and files the website needs as they are. See Pictures and files.
- **Site**: title, description, language, the web address, logo, search and page width. See Site settings.
- **Menu**: the sidebar, and a menu made by hand if the automatic one isn't what you want. See Menus.
- **Theme**: colors and fonts. See Colors and fonts.
- **Publish**: build the website and put it online. See Publishing.
- **Versions**: the site's history. See Versions.
- **App**: settings for q.e.d. itself: its own colors (all screenshots here use *catppuccin-mocha*), the manual's language, updates, and the setup for AI assistants.

A podcast site has a few more: **Podcast**, **Audio**, **Transcripts** and **Contributors**. See Podcasts.



The **App** tab: q.e.d.'s own colors, and updates.

Look at the result

👁 The round button with the eye, in the top right corner, builds the site and opens it in your browser, at the page you're editing. Nothing is put online by that: the preview runs on your computer only. Change something, save, and press it again to see the change.

Saving

💾 Nothing is saved until you press **Save**. A dot on a tab and the words *Unsaved changes* say there's something to save, and **Discard** throws it away. Every save is kept in the site's history, so you can always go back; see Versions.

This manual

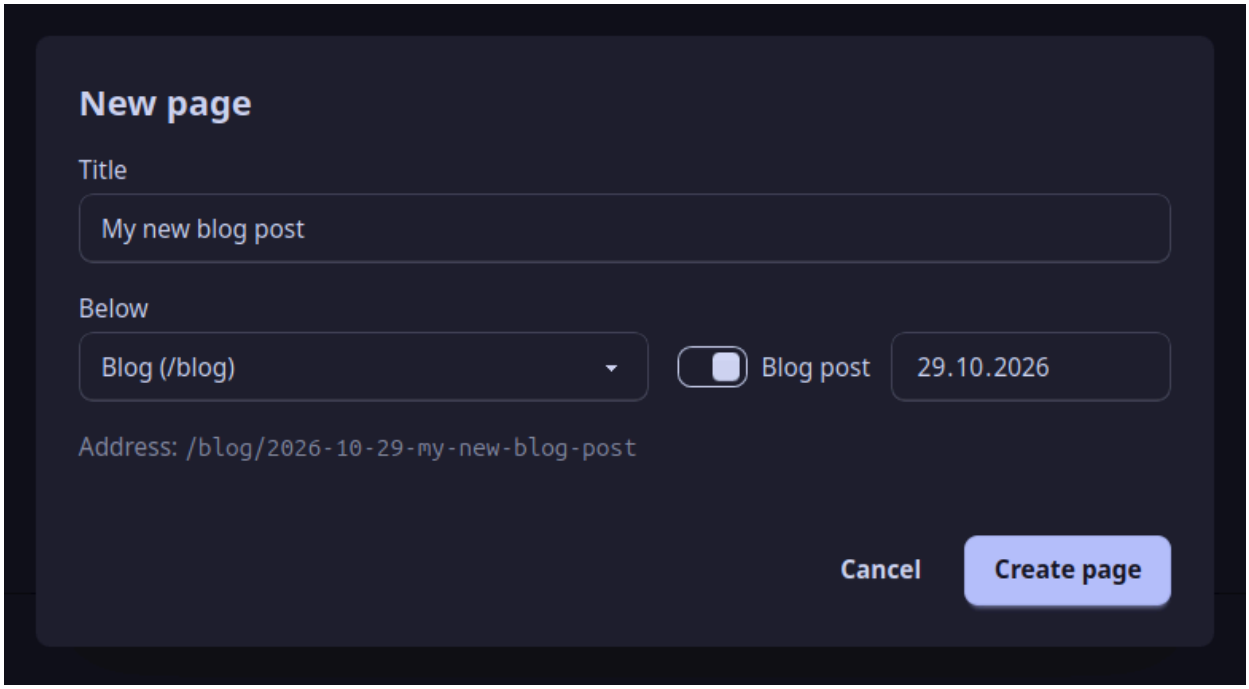
📖 **Manual**, next to the preview button, opens this manual in a window of its own, for the version of q.e.d. you have, without an internet connection. Some places in the app link to the page that explains them.

Pages and blocks

Pages and posts

The **Pages** tab lists the site's pages as a tree: a page below another one has its address below it, such as `/about/team` below `/about`. The start page is `/`. Drag a page in the list to put it elsewhere in the order, or below another page.

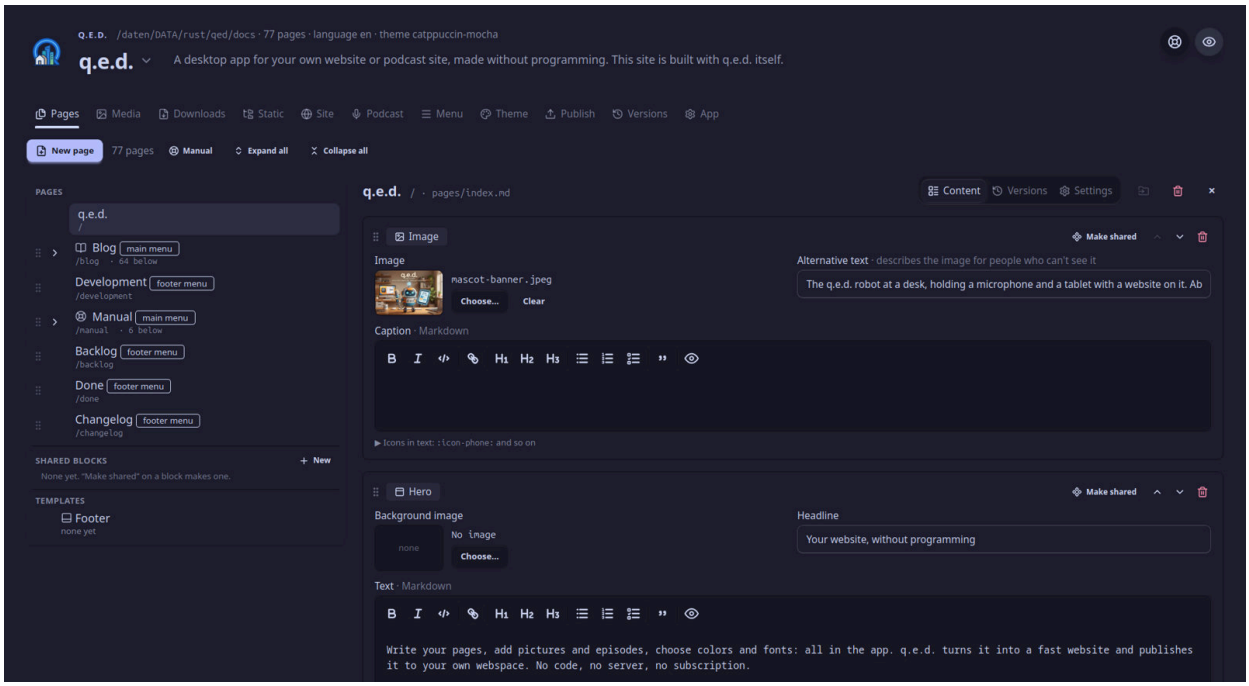
New page asks for a title and where the page goes (**Below**: the top level or another page). Switch on **Blog post** and pick a date to make a post; the time is filled in with the current one. Posts are listed by date and time, newest first, on the page they're below (usually the blog), and they get a feed. The time keeps posts of one day in order in feed readers; it's your computer's time zone, and you can change it or leave it empty in the page's Settings tab. A blog with many posts can group them with a page per day, month or year below it, named by the date (`/blog/2026-10-21`, `/blog/2026`) and with a Post list of its own: the posts below those still go into the blog's one feed. The dialog shows the address the page will get, made from the title; you can change it later.



A new blog post: below the Blog, with a date.

Pick a page to edit it. The editor has three tabs:

- **Content**: the page's blocks (below).
- **Settings**: title, description, menu, order and more (further down).
- **Versions**: earlier states of this page (see Versions).



The **Pages** tab: the pages on the left, the one you picked in the editor on the right.

The buttons next to the tabs move or rename the page (it keeps its old address, listed under **Old addresses** in its Settings tab, and links to it are updated) and delete it (). The start page can't be moved or deleted.

Blocks

A page is made of blocks, one below the other. The row at the bottom (**Add:**) adds one; the arrows move a block up or down, the handle on the left drags it, and removes it.

- **Text**: Markdown, a simple way of writing formatted text (see below).
- **Hero**: a large header with a headline, text and a button, on a picture or a color.
- **Image**: one picture with an alternative text and a caption.
- **Text with image**: text beside a picture, the picture on the left or right.
- **Gallery**: several pictures in a grid; each opens big on a click.
- **Download**: a file to download, with its type and size.
- **FAQ**: questions that open to show their answers.
- **Columns**: two to four columns side by side, each with blocks of its own; on phones they stack.
- **Section**: a band with a title, a background color or picture, which holds blocks of its own, columns included.
- **Call to action**: a highlighted box with a headline, text and a button.
- **Post list** and **Page list**: the posts or the pages below a page, as a list, as cards or as tiles.
- **Video**: a video from YouTube, Vimeo or PeerTube, shown as its picture and linked, or a video file from Downloads.
- **Shared block**: shows a shared block (below).

Podcast sites add **Latest episode**, **Subscribe** and **Contributors**.

Every picture needs an **alternative text**: a short description for people who can't see it and for search engines.

Writing text

Text blocks use Markdown. The text editor shows the formatting as you type, and its toolbar adds it for you:

- `**bold**` for **bold**, `*italic*` for *italic*
- `## Heading` on a line of its own for a heading
- `- item` lines for a list, `1. item` for a numbered one
- `[the text](/about)` for a link (more below)
- `![what it shows](/media/photo.jpg)` for a picture from Media

Four HTML tags work in text, for what Markdown has no way to write:

- `<kbd>Ctrl</kbd>+<kbd>S</kbd>` for keys: <kbd>Ctrl</kbd>+<kbd>S</kbd>
- `m<sup>2</sup>` and `CO<sub>2</sub>`: m², CO₂
- `<br>` for a new line where Markdown has none, such as inside a table cell

The four buttons after the two code buttons (code in a line, and a code block) in the toolbar put them in for you, around what you selected. Any other HTML shows as it is written, as text.

`:icon-phone:` puts an icon into the text, as tall as the letters (all of them are in Icons). Math (`$a^2 + b^2 = c^2$`) and a few more things work too; the blog shows them.

Columns and sections

Columns put blocks side by side. Each column has an optional title, a width (**2**× next to **1**× is two thirds) and **Box**, which draws it on a background of its own. Some blocks need the full width and can't go in a column: Hero, Gallery, Columns and Section.

A **Section** groups blocks on a band: a title above them, a background (none, a box, a shade, one of the theme's colors, or a picture) and normal or large spacing. Three columns on a colored band are one Section with a Columns block inside.

Shared blocks

Some blocks belong on several pages: opening hours, a contact box. Click ❖ **Make shared** on a block and give it a name: it moves into a shared block, and a **Shared block** takes its place. Add it to other pages with **Add: Shared block**. Edit it once, under **Shared blocks** in the page list, and every page shows the change.

The **Footer**, under **Templates** in the page list, holds blocks shown at the bottom of every page.

Links

Link to another page of the site by its address, starting with a slash: `[About us](/about)`. Links to other websites are written in full: `[Podlove](https://podlove.org)`. Files in Downloads are linked as `/assets/menu.pdf`.

When you build or preview the site, q.e.d. lists links to pages and files that don't exist, so a typo doesn't stay unnoticed. Moving a page updates the links to it.

Page settings

The **Settings** tab of a page:

- **Title**, and a **Description** for search engines and link previews.
- **Share image**: the picture shown when the page is shared; its first picture if empty.
- **Menu**: the main menu, the footer menu or none, and the **Order** among its neighbours (see Menus).
- **Draft**: left out of the website until you switch it off.

- **Hide title**: for pages whose first block says it already.
- **Back and next links on the pages below**: the pages below this one get links to the page before and after them at the bottom, for a manual or a course read in order. Posts get **Older** and **Newer** instead, as on a blog. This manual and its pages work that way.
- **Icon**: shown next to the page in menus.
- For posts: **Date**, **Authors**, **Tags** and a **Summary** for lists of posts.

Settings and content are saved separately, each with its own **Save**.

Translations

On a site with more than one language (see Site settings), a row of language codes shows above an open page. **Translate** next to a code starts the page's translation: a copy of the page, still in the main language and saved as a **Draft**, so it isn't published half done. Translate its title and blocks, switch off **Draft** in its settings, and it's there, at the same address below the language's code (`/en/about` for `/about`).

In the page list, a page's translations show as small codes on its row (*EN*); a click opens one. A translation keeps the address of its page: moving or renaming the page takes its translations along. A page with translations can only be deleted after them.

A link to another page by its file (`[About](about.md)`) leads to the translation in the page's own language, when there is one; an address written out (`/about`, `/en/about`) leads exactly there.

Questions

Why does my page not appear in the menu?

A page is in a menu only if its settings say so: **Menu** set to the main or the footer menu. Drafts are left out of the website altogether.

I deleted a page by mistake. Can I get it back?

Yes. The **Versions** tab lists **Deleted pages** with the last version that still had them; bring one back from there. See Versions.

Someone changed the page while I was editing it. What now?

q.e.d. notices when you save and asks: **Keep mine** saves your version over the other changes, **Take the file's** shows the page as the file has it now, **Cancel** saves nothing. That happens when the page was changed in another program, by hand, or on another computer and pulled here.

How do I put two things side by side?

With a **Columns** block: each column holds blocks of its own. On phones the columns stack, one below the other.

Icons

Icons go into text and next to pages in menus. They're drawn in the text's color, as tall as the letters, and need nothing from other websites. q.e.d. brings the Tabler icons[11]: more than 5,000 drawn as outlines and 1,000 filled ones, brand logos among them, such as Mastodon.

[11] https://tabler.io/icons

Finding an icon

Search for it by what it shows. The search knows more than the names: *envelope* finds `mail`, *house* finds `home`. You can search in two places in q.e.d.:

- under **Icons in text** below every text field, where a click copies the code to paste into the text
- in a page's **Settings** tab, at **Icon**

tabler.io/icons[12] shows all icons with their names, too. A filled icon's name ends in `-filled`: `heart` and `heart-filled`.

In text

Write the icon's name between colons and `icon-`: `:icon-phone:` gives . It works wherever you write text: a Text block (also in a column or the footer), the text of a Hero, Call to action or Text with image, an image's caption, an FAQ answer and a page's **Summary**. In headlines, titles, button labels and FAQ questions it stays as text; for an icon next to a page in menus, see below. A line like

```
:icon-phone: 01 234 56 78 · :icon-mail: hello@example.com
```

shows as 01 234 56 78 · hello@example.com.

A name q.e.d. doesn't know stays as text on the page, and **Build** notes it, so a typo doesn't go unnoticed.

Next to a page in menus

A page's **Settings** tab has **Icon**: search for one and click it, and it's shown next to the page in the menus. **No icon** takes it away again.

Your own icons

For logos and pictures that aren't among them, such as the logo of a program you write about: put an image file (SVG at best) into the `icons` folder inside the site's `media` folder. **Show in file manager** in the Media tab opens it. `rails.svg` there is written `:icon-rails:`. Your own icons keep their own colors.

Some to start with

Icon	Write		Icon	Write
	`:icon-mail:`			`:icon-phone:`
	`:icon-map-pin:`			`:icon-clock:`
	`:icon-calendar:`			`:icon-home:`
	`:icon-info-circle:`			`:icon-help-circle:`
	`:icon-download:`			`:icon-link:`
	`:icon-rss:`			`:icon-microphone:`
	`:icon-headphones:`			`:icon-book:`
	`:icon-heart-filled:`			`:icon-star-filled:`
	`:icon-brand-mastodon:`			`:icon-brand-github:`

[12] https://tabler.io/icons

Sites from before version 0.15

Until version 0.14, q.e.d. had 72 icons from Lucide[13]. Their names keep working: a page with `:icon-house:` or the icon `house` now shows home `home`. New icons are picked by their Tabler names.

Pictures and files

A site keeps its files in three places, each with a tab of its own:

Media

Pictures: photos, logos, illustrations. JPEG, PNG, GIF, WebP and SVG, and ICO for the site icon.

Downloads

Files for visitors to download: PDFs, documents, video files. Any type.

Static

Files the website needs exactly as they are, at their own address: a presentation, a verification file.

Adding files

Import images… (or **Import files…**) copies files from your computer into the site. q.e.d. gives them clean names: lower case, no spaces, so they work as web addresses. The originals stay where they were. **Show in file manager** opens the folder itself.

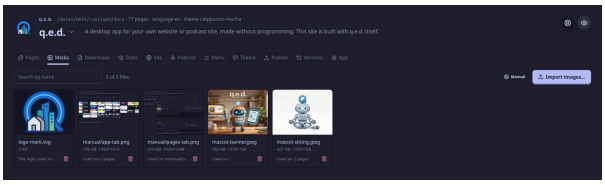
Each file shows its size and which pages use it; files nobody uses are marked **Not used**. Only those can be deleted (), so no page ends up with a missing picture.

Pictures on pages

The **Image**, **Text with image**, **Gallery** and **Hero** blocks pick a picture from Media with a click. In text, a picture is written as `![what it shows](/media/photo.jpg)`.

Always fill in the **alternative text**: what the picture shows, in a sentence. People who can't see it hear it read aloud, and search engines read it too.

Large photos are no problem. q.e.d. makes smaller copies of every photo when it builds the site, and each visitor's browser loads the one that fits their screen. A phone doesn't download the 4000 pixels of a camera photo to show 400.



Large pictures open big when clicked, over the page, without leaving it.

Where pictures are cut

Galleries, tiles in post and page lists, and the Hero block show pictures cut to fit, a Gallery as squares. q.e.d. cuts around the middle, which suits most pictures. When it doesn't, say for a portrait that loses the head, choose the spot yourself: click the picture in the Media tab to show it big, click **Crop**, then click the spot that must stay visible. A square beside it shows the result, and the choice counts everywhere the picture is used. **Middle** goes back to cutting around the middle.

[13] https://lucide.dev

Files to download

The **Download** block offers one file from Downloads, with its type and size. In text, link it as `[the menu](/assets/menu.pdf)`. A **Video** block can play a video file from Downloads, too.

The whole site as a PDF

Make PDF of the site in the Downloads tab puts the whole site into one PDF, to print, to archive or to offer for download. It starts with a title page and a table of contents of the top two levels of pages, with page numbers; PDF readers show them as bookmarks, too. Then come the pages in the order of the site, each top-level page on a new page, and after them the posts and episodes, newest first, in a chapter of their own after the page that lists them (such as the Blog).

The PDF is plain black text on white, without the site's colors, fonts or logo. Pictures, tables, formulas, charts and columns come along; links to other pages jump inside the PDF, and links to other websites show their address in a footnote, so they can be read on paper. The player and videos become a link to their page on the website.

It's made only when you click, and lands in Downloads like any other file: `my-site.pdf`, named after the site. Offer it with a **Download** block. A site in several languages gets one PDF per language (`my-site-en.pdf`). Click again after changing pages to bring it up to date; drafts and posts dated in the future stay out.

My PDFs are in Media. Is that a problem?

No, they keep working. Media is meant for pictures, though; **Move to Downloads** in the Media tab moves them and updates the pages that link to them.

Can I use a picture from another website?

Only if you may, and then better as a copy in Media than as a link: q.e.d. sites load nothing from other websites, so they stay fast and don't hand your visitors' data to anyone.

Site settings

The **Site** tab holds what belongs to the whole website. Change what you like, then press **Save Discard** forgets the changes.

General

- **Title**: the site's name, in the header of every page and in the browser tab.
- **Language**: a short code such as `en`, `de` or `de-AT`. It tells browsers and search engines the language and sets how dates are written: with `de-AT`, January is *Jänner*.
- **Description**: one or two sentences about the site, shown by search engines.
- **Web address**: where the site will be online, such as `https://example.com/`. Feeds, the sitemap and share links need it; fill it in before publishing.

Languages

A site can be in more than one language: German at `/` and English at `/en/`, for example. The language under **General** is the main one; add more here with their code (**Add a language**), such as `en`, `fr` or `pt-br`. Each can have the site's **title** and **description** in it; left empty, the main ones are used.

Then translate pages one by one: open a page and press **Translate** next to the language's code above it (see Pages and blocks). A page without a translation simply isn't there in that language. Menus, post lists and feeds of each language show only its own pages, and a switch in the header (*DE EN*) leads to the same page in the other language, or to that language's start page where there's none.

A language that still has translations can't be removed; delete those first.

Logo

Pick a picture from Media; it's shown in the header of every page and, without a site icon, as the icon in the browser tab. An SVG file stays sharp at any size.

- **Size**: **Small**, **Medium**, **Large**, or **Original** for the picture's own size (up to 112 pixels high), which keeps small bitmap logos sharp.
- **Show the site title next to the logo**: switch it off when the logo already shows the name.

Site icon

The small picture in the browser tab, in bookmarks, and on a phone's home screen when someone adds the site there. A wide logo makes a poor icon, so it can be a picture of its own: add it in Media, then pick it here. q.e.d. uses it as it is, so it must be one of these:

- an **SVG**, sharp at any size;
- a square **PNG**, at least 180 × 180 pixels so it's sharp on phones;
- an **ICO** file, the classic `favicon.ico`.

Saving refuses other pictures, and a PNG that isn't square. Without a site icon, the browser tab shows the logo.

Links

Mark links to other sites gives links in text that lead to other websites a small arrow, so visitors know they're leaving. Menus and buttons stay as they are.

Your profiles elsewhere lists your profiles on other sites, such as `https://mastodon.social/@name`: **Add a profile**, paste its address. Every page then names them as yours, invisibly (`rel="me"`). Mastodon uses that to show your website as verified, with a green check, on your profile:

1. Add the Mastodon profile here, save and publish.
2. On Mastodon, edit your profile and put your website's address into one of the extra fields.
3. Mastodon looks at that page, finds the profile named there, and marks the field as verified. It can take a few minutes.

To also show a *Follow me* link on the site, put an ordinary link into the footer or a page.

Search

Site search puts a search button into the header (also Ctrl+K). It finds words in all pages, posts and episodes. The search runs in the visitor's browser, from files on your own site; nothing goes to other services. It's on for new sites.

Layout

Page width: **Normal** suits text, **Wide** gives columns room, **Full** uses the whole window.

Colors, fonts and menus

How the site looks is in the Theme tab, and the menu in the Menu tab.

→ Colors and fonts

Menus

A site has two menus: the **main menu** at the top of every page, and the **footer menu** at the bottom.

Made from the pages

Usually there's nothing to do. Each page's **Settings** tab says where it goes:

- **Menu**: **Main menu**, **Footer menu**, or **Not in a menu**.
- **Order**: a number; pages with smaller numbers come first.
- **Icon**: shown next to the page in the menu (see Icons).

Pages below another page in the menu (a blog's posts, the pages below *About*) go with it: in the top bar it becomes a dropdown with the page itself first and the others below it; a long one scrolls. In the sidebar they fold (see below).

New pages are in no menu until you choose one. The footer menu is always made this way; the imprint and privacy pages belong there.

On a site in several languages, each language has menus of its own, from its own pages. A menu arranged by hand (see below) is the same in every language; only its titles are translated.

In a sidebar

When there are more pages than fit into the top bar, switch on **Show the main menu in a sidebar** in the **Menu** tab. The menu then runs down the left side of every page; on phones it becomes a *Menu* box above the content. It's saved at once.

A page with pages below it folds them away in the sidebar: a click on its name opens the page, a click on the arrow beside it shows or hides the pages below. Only the part of the menu with the page a visitor is on starts open.

Arranged by hand

To group pages, give them other titles in the menu, or add links to other websites, press **Arrange the menu by hand** in the **Menu** tab. q.e.d. starts with the menu as it is, and you change it:

Page

A page of the site, with its title or one of your own for the menu.

Link

A title and an address: another website (`https://…`) or an address on your site (`/blog`).

Group

A title with pages and links below it: a dropdown in the top bar. In the sidebar it folds open and closed with a click; only the group of the page a visitor is on starts open. A **Line** inside a group divides it.

The arrows move an entry up and down, removes it. **Save menu** saves it; **Back to automatic** returns to the menu made from the pages' settings.

In several languages

On a site in several languages, buttons with the languages' codes (*DE EN*) sit above the entries. Press a further language's code to give the entries their titles in it; entries and their order stay as they are in the main language. A page entry leads to the page's translation and shows its title, so it needs a title of

its own only if you renamed it in the menu. Links and groups show the main title until you give them one. Pages without a translation aren't in that language's menu.

A page that is a draft, or was deleted, isn't shown in the menu. Its entry stays in the Menu tab with a warning, and building the website says so, until you pick another page or remove the entry.

I made a new page, but it's not in the menu.

With the automatic menu, set its **Menu** to **Main menu** in the page's Settings tab. With a menu arranged by hand, add it in the Menu tab: new pages don't go there by themselves.

Can a group have groups inside?

No, one level only. That keeps the menu easy to use on phones.

Colors and fonts

How the website looks is set in the **Theme** tab, for a light and, if you like, a dark side. Everything there is saved as you pick it; preview the site again (👁) to see it. On the right, a small sketch of a page shows the result while you choose.

A theme

At the bottom of the tab are the themes, under **Light** and **Dark**, each with its colors in a row. Click one, and the whole site takes its colors. Every theme has four colors (**primary**, **secondary**, **accent** and **neutral**) on its background.

Which color goes where

Under **Colors**, decide for each part of the page which of the theme's colors it gets:

- **Header**: **Plain** (the page background) or a colored bar.
- **Links** and **Buttons**: one of the four colors.
- **Hero**, **Call to action**, **Boxed columns**, **Content background** and **Footer**: **plain**, **shaded** (a shade darker than the page) or one of the four colors. The content background takes only a light tint, so text stays readable.

On a dark theme the footer is **shaded** unless you pick another color: the theme's accent is often a pale color that would make a bright band below a dark page. On a light theme it's **accent**.

Reset puts the header and every color back to the defaults.

Light and dark mode

Many people set their computer or phone to dark colors, at least in the evening. Switch on **Light and dark mode** at the top of the tab, and the website gets a second look for them: a dark theme with colors of its own. Visitors then see the side that matches their system. Three small buttons next to the search in the header (a sun, a computer and a moon) let them pick light, their system's setting, or dark; the browser remembers the choice for the whole site.

When you switch it on, the dark side starts with a dark theme that suits the light one (*catppuccin-mocha* for *catppuccin-latte*, *dark* for *light*) and with the same color choices. **Light** and **Dark** next to the switch decide which side the rest of the tab changes: the theme, the header, the colors and the custom colors each have their own on both sides. Saved colors can be used on either side. The fonts are the same for both.

Switching it off again removes the dark side's settings; the light side stays as it was.

Fonts

Text and **Headings** each have **Change…**, which searches some 2,000 open fonts (the Google Fonts among them). Pick one, and q.e.d. downloads it once into the site; the website then serves it itself, so no visitor's browser asks Google for anything. Headings use the text font unless you pick another; **Use system font** goes back to the font of the visitor's computer.

Your own colors

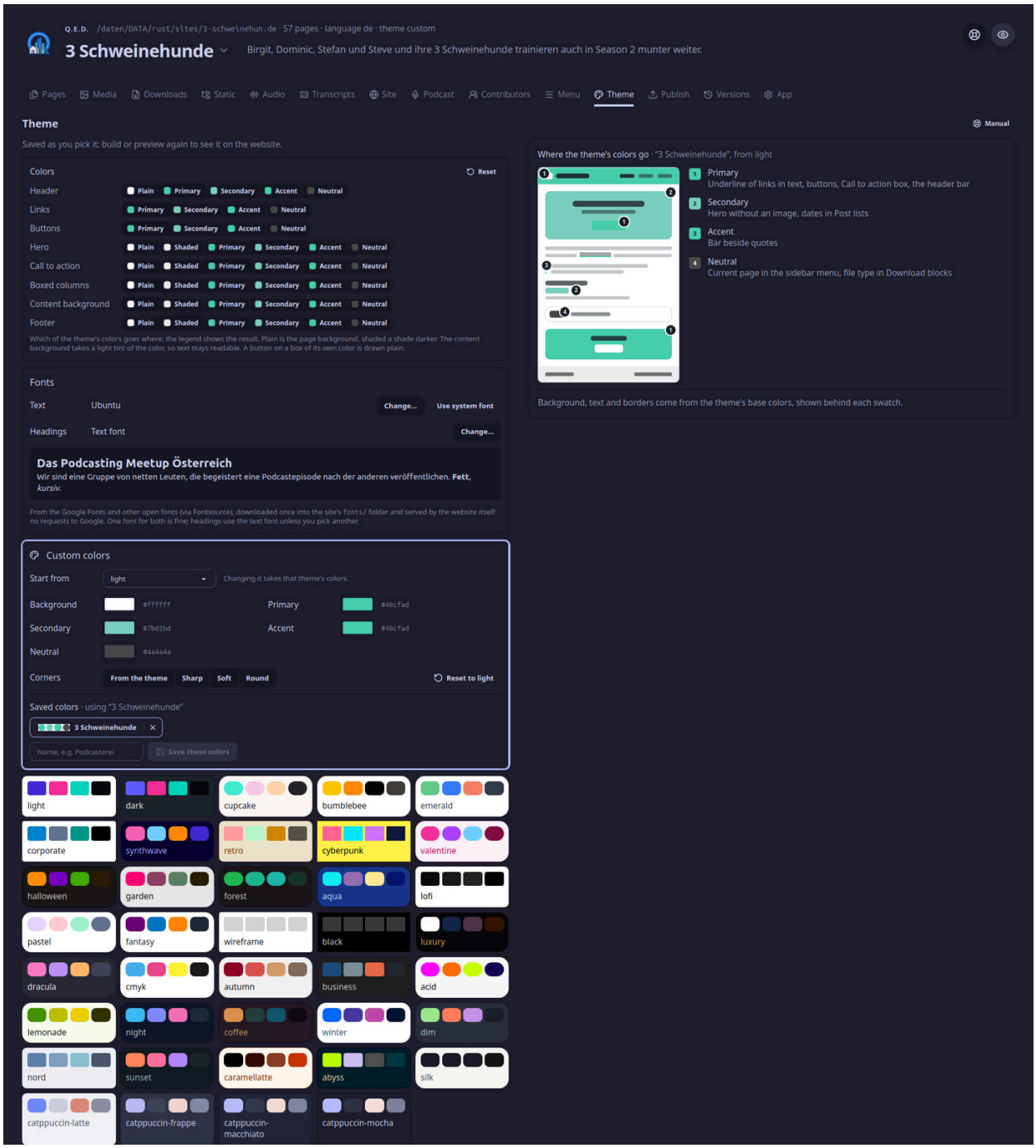
For colors of your own, such as those of a logo, press **Customize** next to the theme's name under **Custom colors**:

- **Start from**: the theme whose colors you change.
- **Background**, **Primary**, **Secondary**, **Accent**, **Neutral**: a color picker each. Text on each color is made readable by itself.
- **Corners**: **Sharp**, **Soft**, **Round**, or as the theme has them.
- **Reset to …** takes all colors from the starting theme again.

Under **Saved colors**, give them a name and press **Save these colors**. Then you can try other themes and come back with one click.

An example. The running podcast 3 Schweinehunde[14] starts from *light* and changes it: a mint green (`#48cfad`) as primary and accent, a lighter one as secondary, dark grey as neutral, and soft corners, saved as "3 Schweinehunde". The header bar is primary, the footer shaded, and text and headings use the font *Ubuntu*.

[14] https://3-schweinehun.de/



The **Theme** tab of 3 Schweinehunde: on the right, where each color goes.

I picked a theme, but the website looks the same.

The preview shows the site as it was built. Press 👁 again to build it with the new colors.

Can visitors switch between light and dark?

Yes, with **Light and dark mode** at the top of the Theme tab: a second, dark theme for visitors whose system is set to dark, and a switch in the header. Without it the site has the one theme you choose.

What do visitors without JavaScript see?

The side that matches their system's setting. Only the switch needs JavaScript; without it, it isn't shown.

Versions

q.e.d. keeps the history of your site, every save of it, in the site folder. Nothing you saved is lost, and anything can be undone.

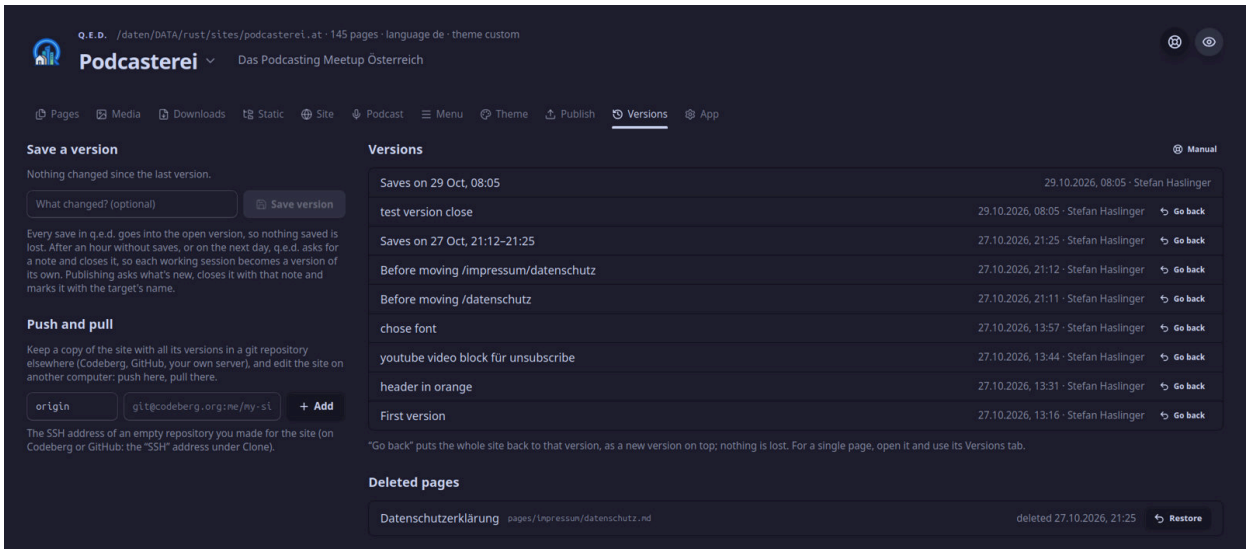
The open version

⟲ Your saves go into the **open version**: one version that grows with each save while you work. The **Versions** tab shows it at the top, marked *open*, with when it started and when you last saved.

It's closed, and becomes a version of its own, when:

- you give it a note and click **Close version**: say in a few words what you did, such as *New opening hours*
- you publish the site (Publish asks what's new, and that becomes the note);
- you back it up or bring in changes from another computer (push or pull, see Backups);
- you come back after a break: after an hour without saves, or on the next day, q.e.d. asks for a note for what you did before. **Use suggested note** names it after its times, such as *Saves on 27 Oct, 13:50–21:12*.

So each working session becomes one version you can go back to.



The **Versions** tab: versions with their notes, **Go back** for the whole site, and **Deleted pages**.

Going back

One page

Open the page and its **Versions** tab. Pick a version to see what it changed, and ⟲ **Restore this version** to put it into the editor. Nothing is saved until you click **Save**, so you can look first.

The whole site

↩ **Go back** next to a version in the **Versions** tab puts every page, picture and file back the way it was then. The present is saved as a version first, so you can come back to it the same way.

Deleted pages

Under **Deleted pages**, the **Versions** tab lists pages that were deleted, with the last version that still had them. ↩ **Restore** brings one back as it was before it was deleted.

A copy on another computer

The history is in the site folder. If the computer breaks, it's gone with it. A backup on another computer keeps every version safe.

→ Backups

Backups

A q.e.d. site is a folder, and every site folder is a git[15] repository: q.e.d. records each version there (see the *Versions* tab). So a backup is a copy of that repository, and the best copy is one on another computer, made with `git push`. It holds not just today's state but every version, and you can bring any of them back.

Setting up a backup

Once per site, create an empty repository somewhere else: on Codeberg[16] or GitHub[17] (a private repository is fine), or on your own server (`git init --bare /var/git/my-site.git`). Then, in the *Versions* tab under *Push and pull*:

1. Add the repository's SSH address (on Codeberg or GitHub: the "SSH" address under *Clone*), for example `git@codeberg.org:me/my-site.git`.
2. *Make a key*, copy it and add it on the server: in the repository's settings as a *deploy key* with write access, or on your own server in `~/.ssh/authorized_keys`. The private key stays in your computer's keychain.
3. *Push*. The first time, q.e.d. shows the server's key fingerprint; trust it if it matches what the server's owner publishes.

Making a backup

Push in the Versions tab. It closes the open version first (with the note in the *Save a version* field, if you typed one), so the latest changes are part of the backup. Publishing closes the open version too, so pushing after each publish is a good habit.

Two computers

On the second computer, choose *Get a site from git…* in the site menu (or on the start screen), enter the same address and make a key there too (one key per computer). q.e.d. copies the site with all its versions. From then on: push when you're done on one computer, pull before you start on the other. When a site opens, q.e.d. asks the remote whether it has new versions and offers to pull them.

If you edited on both computers without pulling in between, *Pull* merges the versions as long as different files changed. If the same page changed on both, nothing changes and q.e.d. names the files: keep one side, or fix it with git.

Restoring

On any computer, *Get a site from git…*, or on the command line:

```
git clone git@codeberg.org:me/my-site.git my-site
```

Open the folder in q.e.d. and it's all there: pages, media, files to download, static files, fonts, settings and the full version history.

[15] https://git-scm.com
[16] https://codeberg.org
[17] https://github.com

With git on the command line

The remotes are git's own, so `git remote add backup …` and `git push backup main` work as well and show up in the app. q.e.d. pushes and pulls only sites that are a git repository of their own; for a site folder inside a larger repository, use git there.

What isn't in the backup

- **Publish targets** (`targets.toml`) are deliberately not part of the history. After restoring, set up the targets again under *Publishing*. They only hold server names and folders, nothing secret.
- **Passwords and SSH keys** live in your computer's keychain, never in the site folder. Back them up the way you back up your other passwords, or make a new key on a new computer and add it to the server.
- **The built site** (`_site/`) and q.e.d.'s image cache (`.qed-cache/`) are left out on purpose: the next build makes them again.

Without git

Copying the whole site folder (including the hidden `.git` folder) to a backup drive works too, and keeps the history as well. It's just easier to forget than a push after publishing.

Publishing

q.e.d. builds the website on your computer and copies it to your webspace, the space for a website your hosting company gives you. It uses SFTP, a secure way of copying files that almost every hosting company offers. Nothing runs on the server: the website is plain files, fast and cheap to host.

Before the first time

Your hosting company gives you four things, usually in its customer area or in the e-mail from when you signed up:

- the **server**, such as `ssh.example.com`
- your **user** name
- a **password**, or the option to log in with a key
- the **folder on the server** the website goes in, such as `/www` or `/home/you/public_html`

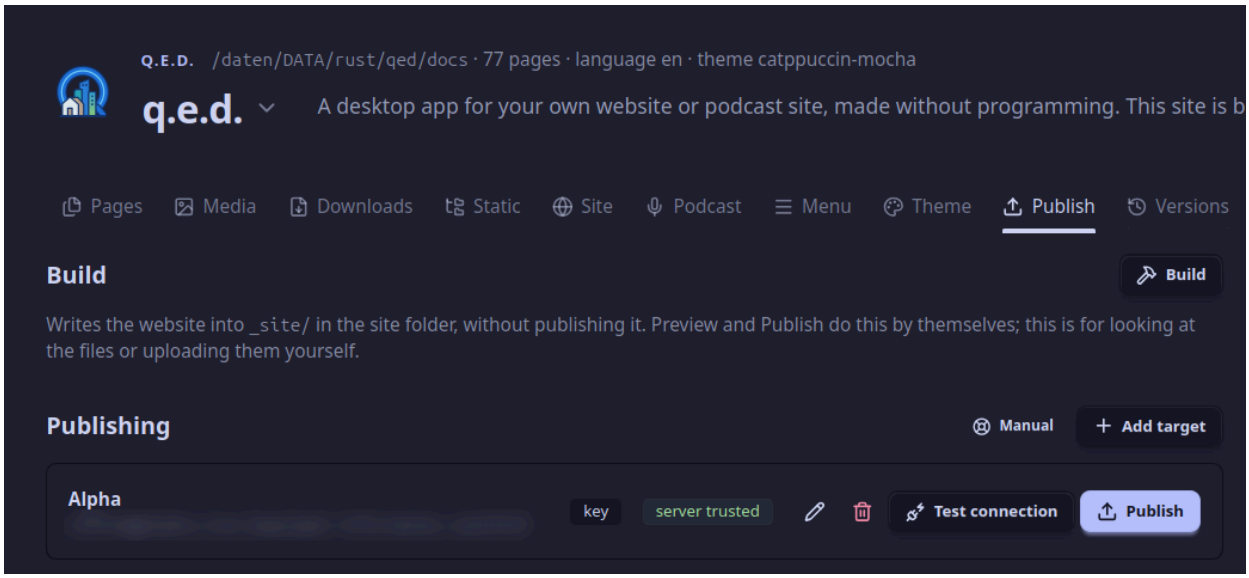
Also enter the site's **web address** in the **Site** tab: feeds, link previews and the sitemap need it.

Set up a target

A *target* is a place the site is published to. In the **Publish** tab, **+ Add target** and fill in the four things, with a **Name** for it, such as *Website*. With **Log in with: Key**, q.e.d. makes a key for this server; copy it into your hosting company's settings for SSH keys (or the server's `~/.ssh/authorized_keys`). Then **Save target**.

Passwords and keys are kept in your computer's keychain, never in the site folder, so they don't end up in backups or on other computers.

Test, then publish



A target that's been tested: **server trusted**, ready to **Publish**.

1. **Test connection.** The first time, q.e.d. shows the server's key fingerprint, its proof of identity. If your hosting company publishes it, compare; then **Trust**. From now on q.e.d. notices if a different server answers at that address.
2. **Publish.** q.e.d. asks *What's new on the site?*: a few words for the history, such as *Summer opening hours*. Then it builds the website and copies only the files of the website that differ on the server.
3. **Open the website** to look at it online.

Files q.e.d. didn't upload itself are never touched on the server, so other things in the same folder stay as they are.

Publish said 3 of my files changed, but uploaded 57. Why?

Your files and the website's files are two things. The Publish dialog lists the files you changed in q.e.d., such as one page. The website is built from all of them, and one change can show on many of its pages: a page's title is in every page's menu, a new post in the Post lists and in the back and next links. Publish uploads every file of the website that differs from the one on the server, so the count after publishing (*57 website files uploaded*) is often bigger.

Do I have to build first?

No. Publish builds the site by itself. **Build** in the Publish tab is for looking at the files, or uploading them yourself.

Can I see the website before it goes online?

Yes: the button in the top right corner shows it in your browser, from your computer only.

Publish waits for an episode. Why?

On a podcast site, every published episode needs its audio file, in every format the podcast offers, or podcast apps would get a broken episode. Add the audio, or mark the episode as a draft until it's ready.

Is the old website gone when I publish?

q.e.d. replaces what it put there before and adds what's new. Files it didn't put there stay. Every publish is also marked in the site's history, so you can see what was online when.

Podcasts

Any q.e.d. site can be a podcast. The episodes are pages of the site, with a player, the people who speak, shownotes and chapters, and q.e.d. writes the feeds that podcast apps subscribe to. Everything stays on your own webspace: the audio, the player, the feeds.

The example on this page is 3 Schweinehunde[18], a running podcast with four hosts and more than 50 episodes.

Turn it on

In the **Podcast** tab, switch on **This site is a podcast** and fill in the settings, then press **Save**. Three more tabs appear: **Audio**, **Transcripts** and **Contributors**.

About

Name in podcast apps (the site's title if empty), **Author** (shown below the name) and **Description**.

3 Schweinehunde: "Birgit, Dominic, Stefan und Steve" as author.

Podcast directories

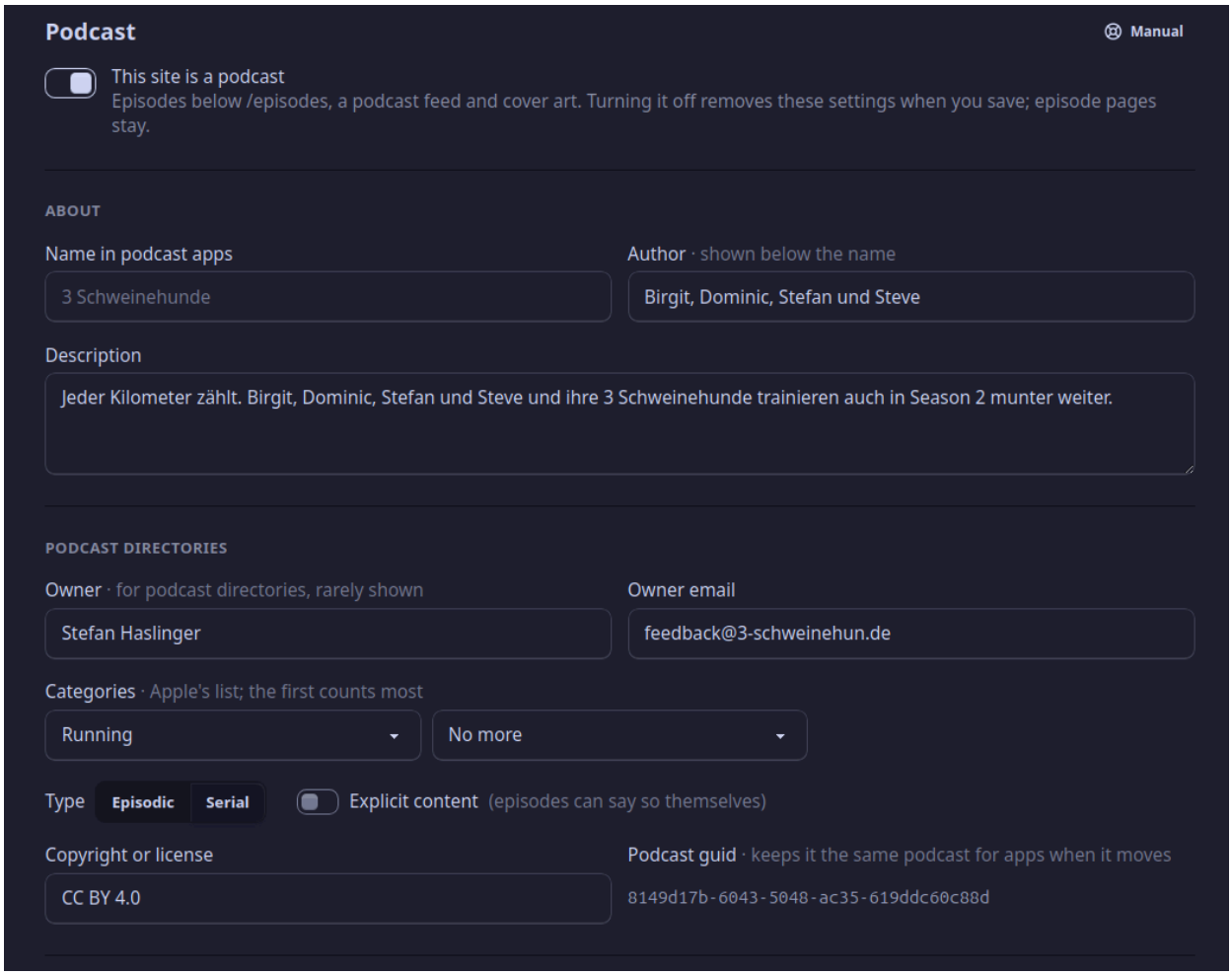
Owner and **Owner email**, where directories such as Apple Podcasts send confirmations; **Categories** from Apple's list (the first counts most); **Type**, **Explicit content** and **Copyright or license**.

3 Schweinehunde: *Sports* > *Running*, CC BY 4.0.

Cover art

A square picture, 1400 to 3000 pixels, JPEG or PNG. Podcast apps show it everywhere. q.e.d. checks it against Apple's rules.

[18] https://3-schweinehun.de/



The **Podcast** tab of 3 Schweinehunde.

Also fill in **Copyright or license**: it tells listeners what they may do with the episodes (see License and credits).

Team lists the people who make the podcast (see below); new episodes start with them. **Support links** (Liberapay, Patreon, …) show in podcast apps as "Support".

Audio files

Under **Audio files** in the Podcast tab:

- **Formats**: MP3 is enough. More formats (M4A, Ogg, Opus, FLAC) each get their own feed, and every episode then needs a file in each. With more than one, choose the **Main format**: its feed is the one for Apple and the directories, so it should be MP3 or M4A.
- Where the files are kept:
  - **In the site, not in versions**: in the site, but not in its history. Versions stay small; back up the audio yourself.
  - **In the site and in versions**: in every version too. One backup holds everything, but the history grows by each episode for good.
  - **Elsewhere**: at a podcast host; each episode gives the address of its file.

When the files are in the site, **Import audio…** in the **Audio** tab copies them in, such as `3sh051_trilogie.mp3`.

The files of one episode in different formats must be the same recording, cut the same way: the chapters and the length hold for all of them, and listeners switching formats would land in the wrong place.

Contributors

Everyone who speaks in an episode is entered once, in the **Contributors** tab: **New contributor**, with **Name**, a **Photo** (square works best), **Links** (website, Mastodon, …) and **About them**. Each gets a page of their own on the site, listing their episodes.

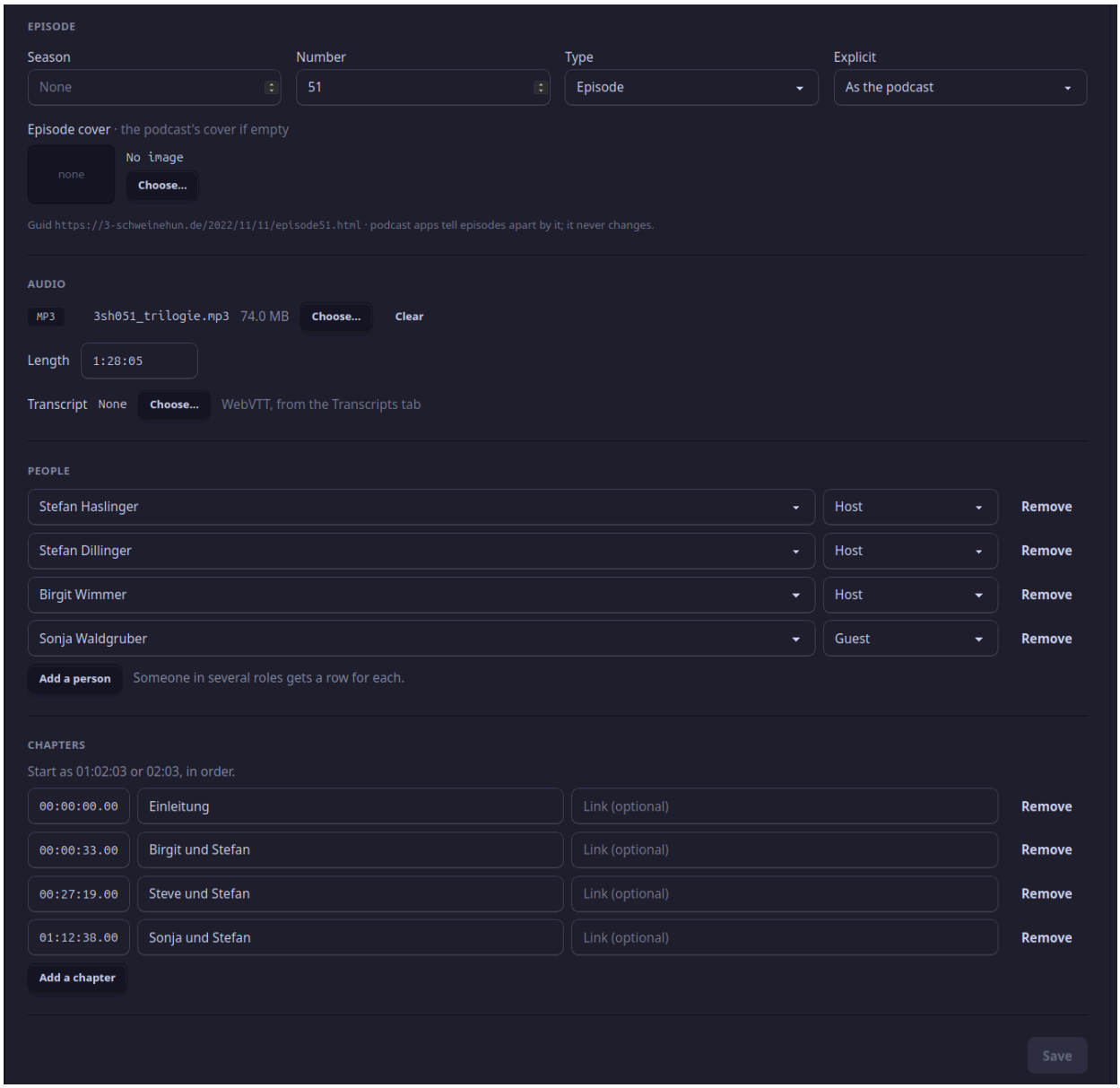
3 Schweinehunde has 34: the four hosts and the guests of single episodes.

A new episode

🎙 **New episode** in the Pages tab asks for the title and the number; it counts on from the last episode. The episode is dated today and goes below `/episodes`.

The episode's **Settings** tab has, besides title, date and description:

- **Episode**: **Season**, **Number**, **Type** (**Episode**, **Trailer**, **Bonus**), **Explicit**, and an **Episode cover** (the podcast's cover if empty).
- **Audio**: **Choose…** the file for each format from the Audio tab; q.e.d. reads the **Length** from it. A **Transcript** comes from the Transcripts tab (a WebVTT file, `.vtt`).
- **People**: who is in it, each in a role (**Host**, **Guest**, …). Someone in two roles gets two rows.
- **Chapters**: a start time (`01:02:03` or `02:03`) and a title each, in order; a link is optional. Episode 51 of 3 Schweinehunde has an introduction and one chapter for each pair of runners.



The episode part of the **Settings** tab, for episode 51 of 3 Schweinehunde.

The episode's own blocks are its **shownotes**: text, links, pictures, like on any page. A few sentences on what the episode is about, then the links to what was mentioned, are a good start; podcast apps show the shownotes too.

Publish waits for the audio

Every feed needs every episode. As long as an episode that isn't a draft lacks its audio (in one of the formats), Publish says which and waits. Make an episode a draft while it isn't ready.

The episode page

Every episode page shows, below its title: the **Player**, the **People**, the **Shownotes** and the **Chapters**. The player is the Podlove Web Player, with chapters, transcript and a share tab; without JavaScript, the browser's own player and a download link.

To change that order for all episodes, or to add a block to every episode (a Subscribe button, say), open **Episode layout** under **Templates** in the page list. **Transcript** shows the transcript as text, folded away until a visitor opens it.

Blocks for podcasts

Any page of a podcast site can use:

- **Latest episode**: the newest episode with its player.
- **Subscribe**: the Podlove Subscribe Button, which offers the podcast's feeds to the visitor's podcast app.
- **Contributors**: the people, with photos, each linking to their page.

The start page of 3 Schweinehunde has Latest episode, Subscribe and a Post list of the other episodes.

Feeds and podcast apps

The site needs its **Web address** (Site tab) for its feeds. They're at `/episodes.mp3.rss` (and one per further format), such as `https://3-schweinehun.de/episodes.mp3.rss`. Submit that address once to Apple Podcasts, Spotify and the other directories; after that, they pick up each new episode when you publish.

Can I add a podcast to a site that already exists?

Yes. Turn it on in the Podcast tab; the pages stay as they are, and the episodes go below /episodes.

What happens to the episodes when I turn the podcast off?

The podcast settings go when you save; the episode pages stay, as ordinary pages.

My podcast is somewhere else now. Can I bring it over?

See Moving a site over.

AI assistants

An AI assistant such as Claude can work on your site for you. You describe what you want in your own words, and it does it in q.e.d.:

- "Write a page about our team, with a photo of each of us."
- "Move all posts from 2023 below the archive page."
- "Check every page for typos and fix them."
- "Which pictures aren't used anywhere?"

It works on the same site you have open in q.e.d., following the same rules as the app. You see every change in **Versions** and can undo it.

What it can do, and what it can't

It can

- read, write and make pages and posts, with all blocks
- add a language and translate pages into it (see Site settings); each translation starts as a draft, like when you press **Translate**
- give a page a new address (links and redirects are taken care of, as when you move a page in q.e.d.)

It can't

- publish: putting the site online is always your click, in **Publish**
- save a broken page: q.e.d. checks every page first and tells the assistant what's wrong
- overwrite a change you made meanwhile
- touch anything outside the site's folder

- delete pages that nothing links to, and pictures and files that nothing uses any more (to replace screenshots, for example)
- add pictures and files from your computer, and files the website needs exactly as they are (such as old slide shows) at their old addresses
- change the footer, shared blocks and the site settings (menu, logo, theme, web address, your profiles elsewhere); the site has to open with them, or q.e.d. refuses
- build the site and report broken links
- save a version with a note saying what it changed

Setting it up

1. Open the site in q.e.d. and go to the ⚙ **App** tab. Under **AI assistants** you find the settings for this site, ready to copy.
2. For **Claude Desktop**: in Claude Desktop, open Settings → Developer → Edit Config. This opens the file `claude_desktop_config.json`. Paste the part q.e.d. shows (**Copy**), save, and restart Claude Desktop. If the file already has servers in it, add q.e.d.'s entry to the ones there.
3. For **Claude Code in a terminal**: copy the command q.e.d. shows and run it once in a terminal.
4. For **Claude Code in VS Code**: the command from step 3 often doesn't work there, because the VS Code extension brings its own Claude Code without the `claude` command. Instead, save the part q.e.d. shows under *Claude Code in VS Code* as a file named `.mcp.json` in the folder you open in VS Code, for example the site's folder. When Claude Code starts in that folder, it asks once whether to use the server in it; say yes.

Do this once for each site you want the assistant to work on. The settings name q.e.d. itself: the assistant starts it in the background, without a window, and talks to it. Other assistants that speak the Model Context Protocol[19] work the same way.

If you move q.e.d. or the site's folder, set it up again: the settings contain both places.

Checking and undoing its work

The assistant's changes are saved like yours. Ask it to "save a version when you're done": its note then shows in ⟲ **Versions**, for example *"Team page with photos"*. There you see what changed on each page and can go back, as described in Versions.

If you edit a page in q.e.d. while the assistant changes the same page, nothing is lost. The assistant is told that the page changed and reads it again; q.e.d. asks you which version to keep when you save.

To see the result before publishing, use the 👁 preview button in q.e.d. as always.

Questions

Does the assistant send my site to the internet?

The assistant itself runs at its maker, for Claude at Anthropic. The pages it reads and writes go there as part of the conversation, like anything you type into it. q.e.d. doesn't send anything anywhere.

[19] https://modelcontextprotocol.io

Does q.e.d. have to be open?

No. The assistant starts q.e.d. in the background by itself. If q.e.d. is open too, you see its changes when you look at the page again.

Can it publish by mistake?

No. There is no way for it to publish. You decide when the site goes online.

Moving a site over

From another computer

A q.e.d. site moves with its folder. If it's backed up on a git server, **Get a site from git** brings it over with all its versions (see Backups); a copy of the folder, from a USB stick for example, opens with **Open site folder…**.

From another system

There's no import from WordPress, Jekyll or website builders: a site moves over by hand, page by page. That's also a good moment to tidy up.

1. **Make a new site** (**New site…**) with the same title and language, and fill in the Site settings, with the web address the site will have.
2. **Bring the pictures and files**: pictures into Media, files to download into Downloads (Pictures and files).
3. **Make the pages**, each at its old address where you can: the **New page** dialog shows the address it will get, and moves or renames a page. Copy the text over and add headings, links and pictures again.
4. **Rebuild the look**: a theme, the colors of your logo and your fonts (Colors and fonts), and the menu (Menus).
5. **Preview** () and click through. The list of missing links shows what's still to do.

Files that must keep their exact old address, such as `/pdf/price-list.pdf`, go into **Static**, in a folder of the same name: the website serves them as they are.

Old addresses

Links from other websites, search engines and bookmarks point to the old addresses. Where a page keeps its address, nothing breaks. When a page moves or gets a new name in q.e.d., the old address leads to the new one by itself.

Old addresses that a q.e.d. page can't have, such as ones ending in `.html`, can be kept too: open the page in the **Pages** tab, then its **Settings** tab; under **Old addresses**, press **Add an address** and type it (`/2022/11/11/episode51.html`), or paste the full address from the old website; q.e.d. keeps only the part after the domain. Visitors who come there are sent on to the page. The same list shows the addresses q.e.d. kept when the page was moved; **Remove** takes one away.

An address can lead to one page only: q.e.d. won't save one that is already another page's address, or another page's old address.

A podcast

Podcast apps must keep finding new episodes, and shouldn't show the old ones as new.

- **From jekyll-octopod**: q.e.d.'s feeds are at the same addresses (`/episodes.mp3.rss`) and its audio too (`/episodes/episode1.mp3`), so subscribers notice nothing. 3 Schweinehunde moved over this way.

- **From other systems** (Podlove Publisher, a podcast host): the feeds had other addresses, and the episodes' identities (*guids*) have to be carried over so apps don't show every episode as new. In each episode's **Settings** tab (in the **Pages** tab, next to **Content**), press **Change it** next to its guid and type the one from the old feed (the `<guid>` of the episode's `<item>`), exactly as it is there. An import that does this by itself is planned; the old feed addresses still need help for now.

Switching over

Publish the new site to your webspace once everything is in place (Publishing). If it goes to a new webspace, your domain's provider points the domain there; until then, the old site stays online.

License and credits

q.e.d. is free software

q.e.d. is licensed under the GNU General Public License, version 3 or later[20]. You may use it for anything, for free, also for business websites; pass it on; and change it, as long as what you pass on stays under the same license. The source code is on GitHub[21].

Your website is yours

The GPL covers the program, not what you make with it. Your pages, pictures, episodes and the website q.e.d. builds from them belong to you, under whatever terms you choose.

Say which terms: for a podcast in the **Copyright or license** field of the Podcast tab (such as `CC BY 4.0`), and on the website in the footer, for example as a line with a link to the license. 3 Schweinehunde, like many podcasts, uses Creative Commons BY 4.0[22]: anyone may share and reuse the episodes, naming the podcast.

What your website contains from others

A website built by q.e.d. carries a few files from other open-source projects. All of them may be used on any website, commercial ones included; license notes their authors put into the files stay there.

Part	Where it's used	License
Podlove Web Player 5.13.0	episodes of a podcast	MIT
Podlove Subscribe Button	the Subscribe block	MIT
Pagefind	site search	MIT
Tabler icons 3.49.0	icons in text and menus	MIT
Tailwind CSS and daisyUI	the stylesheet and themes	MIT
Catppuccin	the four Catppuccin themes	MIT
Latin Modern Math	math formulas	GUST Font License
Fonts from Fontsource	the fonts you pick	each font's own, mostly the SIL Open Font License

The font's license is noted next to it in the site's stylesheet.

[20] https://www.gnu.org/licenses/gpl-3.0.html
[21] https://github.com/informatom/qed
[22] https://creativecommons.org/licenses/by/4.0/

Credits

q.e.d. stands on the shoulders of many projects. Especially:

- **Podlove**[23]: the Web Player and the Subscribe Button, and years of work on open podcasting.
- **jekyll-octopod**[24], q.e.d.'s predecessor for podcasts, which started from Octopod[25] by Arne Eilermann. Its feed and audio addresses live on in q.e.d., so podcasts move over without losing a subscriber.
- **Tauri**[26] and **Rust**[27], of which the app is made, and **Svelte**[28] for its screens.
- **libgit2**[29], which keeps the versions, **Tabler Icons**[30] (and before them **Lucide**[31]), **daisyUI**[32], **Pagefind**[33], **Fontsource**[34] and **Catppuccin**[35].

The q.e.d. robot and the logo were drawn by Google's image model Nano Banana 2 from Stefan Haslinger's description; the logo was then redrawn by hand as SVG, so it stays sharp at any size (A logo in SVG).

The program itself is built from more than 750 Rust libraries and a few JavaScript packages, almost all under the MIT or Apache license. Libraries lists every one, with its version and license. q.e.d. is written with Claude Code[36] the blog tells how.

Questions, bugs and ideas

Something doesn't work, or is missing? Say so on GitHub[37]: a short description, what you expected, and the version from the **App** tab. Mistakes and gaps in this manual are welcome there too.

Libraries

q.e.d. is made of its own code and of the work of many others: **971 Rust libraries** for the program, and a few JavaScript packages for its screens. Each is open source, and each license allows q.e.d. to use it and to pass it on. This page lists them all, for this version of q.e.d.; it's written from the program's source, so it stays complete. What a built website contains from others is in License and credits.

Most libraries offer a choice of licenses (*MIT OR Apache-2.0*); q.e.d. may use either. How often each license (or choice of licenses) occurs:

License	Libraries
Apache-2.0 OR MIT	602
MIT	196
Apache-2.0	34
Unicode-3.0	27

[23] https://podlove.org/
[24] https://jekyll-octopod.github.io/
[25] https://github.com/pattex/octopod
[26] https://tauri.app/
[27] https://www.rust-lang.org/
[28] https://svelte.dev/
[29] https://libgit2.org/
[30] https://tabler.io/icons
[31] https://lucide.dev
[32] https://daisyui.com/
[33] https://pagefind.app/
[34] https://fontsource.org/
[35] https://catppuccin.com/
[36] https://claude.com/claude-code
[37] https://github.com/informatom/qed/issues

License	Libraries
Apache-2.0 OR MIT OR Zlib	25
MIT OR Unlicense	14
BSD-3-Clause	10
Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	8
MPL-2.0	8
Zlib	7
BSD-2-Clause	5
(MIT OR Apache-2.0) AND Unicode-3.0	4
Apache-2.0 OR BSD-3-Clause	3
Apache-2.0 OR ISC OR MIT	3
ISC	3
Apache-2.0 OR BSD-2-Clause OR MIT	2
Apache-2.0 OR BSD-3-Clause OR MIT	2
Apache-2.0 OR LGPL-2.1-or-later OR MIT	2
Apache-2.0 WITH LLVM-exception	2
BlueOak-1.0.0	2
(Apache-2.0 OR MIT) AND BSD-3-Clause	1
(MIT OR Apache-2.0) AND Unicode-DFS-2016	1
0BSD OR Apache-2.0 OR MIT	1
0BSD OR CC0-1.0	1
Apache-2.0 AND ISC	1
Apache-2.0 AND MIT	1
Apache-2.0 OR BSD-1-Clause OR MIT	1
Apache-2.0 OR BSL-1.0	1
Apache-2.0 OR CC0-1.0 OR MIT-0	1
BSD-3-Clause AND MIT	1

License	Libraries
BSD-3-Clause OR MIT	1
CDLA-Permissive-2.0	1

Used directly

The libraries q.e.d.'s own code calls. The others in the full list below are what these need in turn.

Library	Version	License	What it does
askama	0.16.1	Apache-2.0 OR MIT	Type-safe, compiled Jinja-like templates for Rust
catppuccin	2.8.0	MIT	🦀 Soothing pastel theme for Rust.
charts-rs	2.3.0	Apache-2.0	A charts library for rust
chematic-depict	1.2.1	Apache-2.0 OR MIT	2D molecular structure depiction as SVG: ring templates, wedge/dash

Library	Version	License	What it does
			stereo bonds, CPK coloring, grid layout — pure-Rust, no C/C++ dependencies
chematic-smiles	1.2.1	Apache-2.0 OR MIT	OpenSMILES parser, writer and canonical SMILES for chematic — pure Rust, WASM-compatible
comrak	0.57.0	BSD-2-Clause	A 100% CommonMark-compatible GitHub Flavored Markdown parser and formatter
deunicode	1.6.2	BSD-3-Clause	Convert Unicode strings to pure ASCII by intelligently transliterating them. Suppors Emoji and Chinese.
getrandom	0.3.4	Apache-2.0 OR MIT	A small cross-platform library for retrieving random data from system source
git2	0.21.0	Apache-2.0 OR MIT	Bindings to libgit2 for interoperating with git repositories. This library is both threadsafe and memory safe and allows both reading and writing git…
image	0.25.10	Apache-2.0 OR MIT	Imaging library. Provides basic image processing and encoders/decoders for common image formats.
include_dir	0.7.4	MIT	Embed the contents of a directory in your binary
jiff	0.2.38	MIT OR Unlicense	A date-time library that encourages you to jump into the pit of success. This library is heavily inspired by the Temporal project.
keyring	4.2.0	Apache-2.0 OR MIT	All-in-One Rust Keyring
lofty	0.25.5	Apache-2.0 OR MIT	Audio metadata library
mitex	0.2.4	Apache-2.0	MiTeX is a TeX2Typst converter.
pagefind	1.5.2	MIT	Implement search on any static website.
pulldown-latex	0.8.0	MIT	Pull parser to convert LaTeX equations to MathML
reqwest	0.13.5	Apache-2.0 OR MIT	higher level HTTP client library
rmcp	3.8.0	Apache-2.0	Rust SDK for Model Context Protocol
russh	0.64.1	Apache-2.0	A client and server SSH library.
russh-sftp	3.0.1	Apache-2.0	SFTP subsystem supported server and client for Russh
rustls	0.23.46	Apache-2.0 OR ISC OR MIT	Rustls is a modern TLS library written in Rust.

Library	Version	License	What it does
serde	1.0.229	Apache-2.0 OR MIT	A generic serialization/deserialization framework
serde_json	1.0.154	Apache-2.0 OR MIT	A JSON serialization file format
sha2	0.11.1	Apache-2.0 OR MIT	Pure Rust implementation of the SHA-2 hash function family including SHA-224, SHA-256, SHA-384, and SHA-512.
specta	2.0.0-rc.25	MIT	Easily export your Rust types to other languages
specta-typescript	0.0.12	MIT	Export your Rust types to TypeScript
syntect	5.3.0	MIT	library for high quality syntax highlighting and code intelligence using Sublime Text's grammars
tauri	2.12.4	Apache-2.0 OR MIT	Make tiny, secure apps for all desktop platforms with Tauri
tauri-plugin-dialog	2.8.3	Apache-2.0 OR MIT	Native system dialogs for opening and saving files along with message dialogs on your Tauri application.
tauri-plugin-opener	2.7.2	Apache-2.0 OR MIT	Open files and URLs using their default application.
tauri-plugin-process	2.4.0	Apache-2.0 OR MIT	Access the current process of your Tauri application.
tauri-plugin-updater	2.14.0	Apache-2.0 OR MIT	In-app updates for Tauri applications.
tauri-plugin-window-state	2.5.0	Apache-2.0 OR MIT	Save window positions and sizes and restore them when the app is reopened.
tauri-specta	2.0.0-rc.25	MIT	Completely typesafe Tauri commands
thiserror	2.0.21	Apache-2.0 OR MIT	derive(Error)
tiny_http	0.12.0	Apache-2.0 OR MIT	Low level HTTP server library
tokio	1.54.0	MIT	An event-driven, non-blocking I/O platform for writing asynchronous I/O backed applications.
toml	1.1.8+spec-1.1.0	Apache-2.0 OR MIT	A native Rust encoder and decoder of TOML-formatted files and streams. Provides implementations of the standard Serialize/Deserialize traits for TOML data to…
toml_edit	0.25.17+spec-1.1.0	Apache-2.0 OR MIT	Yet another format-preserving TOML parser.

Library	Version	License	What it does
two-face	0.5.2+bat-0.26.1	Apache-2.0 OR MIT	Extra syntax and theme definitions for syntect
typst	0.15.1	Apache-2.0	A new markup-based typesetting system that is powerful and easy to learn.
typst-layout	0.15.1	Apache-2.0	Typst's layout engine.
typst-pdf	0.15.1	Apache-2.0	PDF exporter for Typst.
uuid	1.28.0	Apache-2.0 OR MIT	A library to generate and parse UUIDs.
webp	0.3.1	Apache-2.0 OR MIT	WebP conversion library.

In the app's screens

JavaScript packages that are part of the app's screens.

Package	Version	License	What it does
@tabler/icons-svelte-runes	3.49.0	MIT	A set of free MIT-licensed high-quality SVG icons for Svelte 5+ using runes
@tauri-apps/api	2.12.4	Apache-2.0 OR MIT	Tauri API definitions
@tauri-apps/plugin-dialog	2.8.3	Apache-2.0 OR MIT	
@tauri-apps/plugin-opener	2.7.2	Apache-2.0 OR MIT	Open files and URLs using their default application.
@tauri-apps/plugin-process	2.4.0	Apache-2.0 OR MIT	
@tauri-apps/plugin-updater	2.14.0	Apache-2.0 OR MIT	
overtype	2.6.1	MIT	A lightweight markdown editor library with perfect WYSIWYG alignment using an invisible textarea overlay

For building the app

Tools that turn the app's code into the program. Svelte and SvelteKit also leave a small part of themselves in the app, and the stylesheet tools (Tailwind CSS, daisyUI) write the stylesheet of every website too.

Package	Version	License	What it does
@sveltejs/adapter-static	3.0.10	MIT	Adapter for SvelteKit apps that prerenders your entire site as a collection of static files
@sveltejs/kit	2.70.3	MIT	SvelteKit is the fastest way to build Svelte apps
@sveltejs/vite-plugin-svelte	7.3.1	MIT	
@tabler/icons	3.49.0	MIT	A set of free MIT-licensed high-quality SVG icons for you to use in your web projects.

Package	Version	License	What it does
@tailwindcss/cli	4.3.3	MIT	A utility-first CSS framework for rapidly building custom user interfaces.
@tailwindcss/typography	0.5.20	MIT	A Tailwind CSS plugin for automatically styling plain HTML content with beautiful typographic defaults.
@tailwindcss/vite	4.3.3	MIT	A utility-first CSS framework for rapidly building custom user interfaces.
@tauri-apps/cli	2.12.3	Apache-2.0 OR MIT	Command line interface for building Tauri apps
daisyui	5.7.50	MIT	daisyUI 5 - The Tailwind CSS Component Library
svelte	5.57.2	MIT	Cybernetically enhanced web apps
svelte-check	4.7.7	MIT	Svelte Code Checker Terminal Interface
tailwindcss	4.3.3	MIT	A utility-first CSS framework for rapidly building custom user interfaces.
typescript	6.0.3	Apache-2.0	TypeScript is a language for application scale JavaScript development
vite	8.3.5	MIT	Native-ESM powered web dev build tool

All Rust libraries

All 971 of them, by name. Some appear twice, in two versions, when libraries need different ones. Many are only used on one system (Windows, macOS or Linux).

Library	Version	License	What it does
Inflector	0.11.4	BSD-2-Clause	Adds String based inflections for Rust. Snake, kebab, camel, sentence, class, title and table cases as well as ordinalize, deordinalize, demodulize, foreign…
adler2	2.0.1	0BSD OR Apache-2.0 OR MIT	A simple clean-room implementation of the Adler-32 checksum
aead	0.6.1	Apache-2.0 OR MIT	Traits for Authenticated Encryption with Associated Data (AEAD) algorithms, such as AES-GCM as ChaCha20Poly1305, which provide a high-level API

Library	Version	License	What it does
aes	0.9.3	Apache-2.0 OR MIT	Pure Rust implementation of the Advanced Encryption Standard (a.k.a. Rijndael)
aes-gcm	0.11.1	Apache-2.0 OR MIT	Pure Rust implementation of the AES-GCM (Galois/Counter Mode) Authenticated Encryption with Associated Data (AEAD) Cipher with optional architecture-specific…
ahash	0.7.8	Apache-2.0 OR MIT	A non-cryptographic hash function using AES-NI for high performance
aho-corasick	1.1.5	MIT OR Unlicense	Fast multiple substring searching.
alloc-no-stdlib	3.0.0	BSD-3-Clause	A dynamic allocator that may be used with or without the stdlib. This allows a package with nostd to allocate memory dynamically and be used either with a…
alloc-stdlib	0.3.0	BSD-3-Clause	A dynamic allocator example that may be used with the stdlib
allocator-api2	0.2.21	Apache-2.0 OR MIT	Mirror of Rust's allocator API
android_system_properties	0.1.6	Apache-2.0 OR MIT	Minimal Android system properties wrapper
anstream	1.0.0	Apache-2.0 OR MIT	IO stream adapters for writing colored text that will gracefully degrade according to your terminal's capabilities.
anstyle	1.0.14	Apache-2.0 OR MIT	ANSI text styling
anstyle-parse	1.0.0	Apache-2.0 OR MIT	Parse ANSI Style Escapes
anstyle-query	1.1.5	Apache-2.0 OR MIT	Look up colored console capabilities
anstyle-wincon	3.0.11	Apache-2.0 OR MIT	Styling legacy Windows terminals
anyhow	1.0.104	Apache-2.0 OR MIT	Flexible concrete Error type built on std::error::Error
apple-native-keyring-store	1.0.2	Apache-2.0 OR MIT	Apple-native credential store for keyring

Library	Version	License	What it does
approx	0.5.1	Apache-2.0	Approximate floating point equality comparisons and assertions.
ar_archive_writer	0.5.3	Apache-2.0 WITH LLVM-exception	A writer for object file ar archives
arbitrary	1.5.0	Apache-2.0 OR MIT	The trait for generating structured data from unstructured data
arc-swap	1.9.3	Apache-2.0 OR MIT	Atomically swappable Arc
argon2	0.6.0	Apache-2.0 OR MIT	Pure Rust implementation of the Argon2 password hashing function with support for the Argon2d, Argon2i, and Argon2id algorithmic variants
arrayref	0.3.9	BSD-2-Clause	Macros to take array references of slices
arrayvec	0.7.8	Apache-2.0 OR MIT	A vector with fixed capacity, backed by an array (it can be stored on the stack too). Implements fixed capacity ArrayVec and ArrayString.
ascii	1.1.0	Apache-2.0 OR MIT	ASCII-only equivalents to `char`, `str` and `String`.
askama	0.16.1	Apache-2.0 OR MIT	Type-safe, compiled Jinja-like templates for Rust
askama_derive	0.16.1	Apache-2.0 OR MIT	Code generator of Askama templating engine
askama_macros	0.16.1	Apache-2.0 OR MIT	Procedural macro package for Askama
askama_parser	0.16.1	Apache-2.0 OR MIT	Parser for Askama templates

Library	Version	License	What it does
async-broadcast	0.7.2	Apache-2.0 OR MIT	Async broadcast channels
async-channel	2.5.0	Apache-2.0 OR MIT	Async multi-producer multi-consumer channel
async-compression	0.4.51	Apache-2.0 OR MIT	Adaptors between compression crates and Rust's modern asynchronous IO types.
async-executor	1.14.0	Apache-2.0 OR MIT	Async executor
async-io	2.6.0	Apache-2.0 OR MIT	Async I/O and timers
async-lock	3.4.2	Apache-2.0 OR MIT	Async synchronization primitives
async-process	2.5.0	Apache-2.0 OR MIT	Async interface for working with processes
async-recursion	1.2.0	Apache-2.0 OR MIT	Recursion for async functions
async-signal	0.2.14	Apache-2.0 OR MIT	Async signal handling
async-task	4.7.1	Apache-2.0 OR MIT	Task abstraction for building executors

Library	Version	License	What it does
async-trait	0.1.92	Apache-2.0 OR MIT	Type erasure for async trait methods
atk	0.18.2	MIT	UNMAINTAINED Rust bindings for the ATK library
atk-sys	0.18.2	MIT	UNMAINTAINED FFI bindings to libatk-1
atomic-waker	1.1.2	Apache-2.0 OR MIT	A synchronization primitive for task wakeup
autocfg	1.5.1	Apache-2.0 OR MIT	Automatic cfg for Rust compiler features
az	1.3.0	Apache-2.0 OR MIT	Casts and checked casts.
base16ct	1.0.0	Apache-2.0 OR MIT	Pure Rust implementation of Base16 a.k.a hexadecimal (RFC 4648) which avoids any usages of data-dependent branches/LUTs and thereby provides portable "best…
base64	0.21.7	Apache-2.0 OR MIT	encodes and decodes base64 as bytes or utf8
base64	0.22.1	Apache-2.0 OR MIT	encodes and decodes base64 as bytes or utf8
base64	0.23.1	Apache-2.0 OR MIT	encodes and decodes base64 as bytes or utf8
base64ct	1.8.3	Apache-2.0 OR MIT	Pure Rust implementation of Base64 (RFC 4648) which avoids any usages of data-dependent branches/LUTs and thereby provides portable "best effort" constant-time…
basic-toml	0.1.10	Apache-2.0 OR MIT	Minimal TOML library with few dependencies
bcrypt-pbkdf	0.11.0	Apache-2.0 OR MIT	bcrypt-pbkdf password-based key derivation function
beef	0.5.2	Apache-2.0 OR MIT	More compact Cow
biblatex	0.12.0	Apache-2.0 OR MIT	Parsing, writing, and evaluating BibTeX and BibLaTeX files
bincode	1.3.3	MIT	A binary serialization / deserialization strategy that uses Serde for transforming structs into bytes and vice versa!
bit-set	0.10.0	Apache-2.0 OR MIT	A set of bits
bit-set	0.8.0	Apache-2.0 OR MIT	A set of bits
bit-vec	0.8.0	Apache-2.0 OR MIT	A vector of bits
bit-vec	0.9.1	Apache-2.0 OR MIT	A vector of bits

Library	Version	License	What it does
bitflags	1.3.2	Apache-2.0 OR MIT	A macro to generate structures which behave like bitflags.
bitflags	2.13.2	Apache-2.0 OR MIT	A macro to generate structures which behave like bitflags.
bitvec	1.1.1	MIT	Addresses memory by bits, for packed collections and bitfields
blake2	0.11.0	Apache-2.0 OR MIT	BLAKE2 hash functions
block-buffer	0.10.4	Apache-2.0 OR MIT	Buffer type for block processing of data

Library	Version	License	What it does
block-buffer	0.12.1	Apache-2.0 OR MIT	Buffer types for block processing of data
block-padding	0.4.2	Apache-2.0 OR MIT	Padding and unpadding of messages divided into blocks.
block2	0.6.2	MIT	Apple's C language extension of blocks
blocking	1.7.0	Apache-2.0 OR MIT	A thread pool for isolating blocking I/O in async programs
blowfish	0.10.0	Apache-2.0 OR MIT	Blowfish block cipher
brotli	9.0.0	BSD-3-Clause AND MIT	A brotli compressor and decompressor that with an interface avoiding the rust stdlib. This makes it suitable for embedded devices and kernels. It is designed…
brotli-decompressor	6.0.1	BSD-3-Clause OR MIT	A brotli decompressor that with an interface avoiding the rust stdlib. This makes it suitable for embedded devices and kernels. It is designed with a pluggable…
bs58	0.5.1	Apache-2.0 OR MIT	Another Base58 codec implementation.
bumpalo	3.20.3	Apache-2.0 OR MIT	A fast bump allocation arena for Rust.
by_address	1.2.1	Apache-2.0 OR MIT	Wrapper for comparing and hashing pointers by address
bytecheck	0.6.12	MIT	Derive macro for bytecheck
bytecheck_derive	0.6.12	MIT	Derive macro for bytecheck
bytemuck	1.25.2	Apache-2.0 OR MIT OR Zlib	A crate for mucking around with piles of bytes.
bytemuck_derive	1.12.1	Apache-2.0 OR MIT OR Zlib	derive proc-macros for `bytemuck`
byteorder	1.5.0	MIT OR Unlicense	Library for reading/writing numbers in big-endian and little-endian.
byteorder-lite	0.1.0	MIT OR Unlicense	Library for reading/writing numbers in big-endian and little-endian.
bytes	1.12.1	MIT	Types and traits for working with bytes
cairo-rs	0.18.5	MIT	Rust bindings for the Cairo library
cairo-sys-rs	0.18.2	MIT	FFI bindings to libcairo
camino	1.2.6	Apache-2.0 OR MIT	UTF-8 paths
cargo-platform	0.1.9	Apache-2.0 OR MIT	Cargo's representation of a target platform.
cargo_metadata	0.19.2	MIT	structured access to the output of `cargo metadata`

Library	Version	License	What it does
cargo_toml	1.0.1	Apache-2.0 OR MIT	`Cargo.toml` struct definitions for parsing with Serde
caseless	0.2.2	MIT	Unicode caseless matching
catppuccin	2.8.0	MIT	Soothing pastel theme for Rust.
cbc	0.2.1	Apache-2.0 OR MIT	Cipher Block Chaining (CBC) block cipher mode of operation
cc	1.9.0	Apache-2.0 OR MIT	A build-time dependency for Cargo build scripts to assist in invoking the native C compiler to compile native C code into a static archive to be linked into…
cesu8	1.1.0	Apache-2.0 OR MIT	Convert to and from CESU-8 encoding (similar to UTF-8)
cfb	0.14.0	MIT	Read/write Compound File Binary (structured storage) files
cfg-expr	0.15.8	Apache-2.0 OR MIT	A parser and evaluator for Rust `cfg()` expressions.
cfg-if	1.0.5	Apache-2.0 OR MIT	A macro to ergonomically define an item depending on a large number of #[cfg] parameters. Structured like an if-else chain, the first matching branch is the…
cfg_aliases	0.2.2	MIT	A tiny utility to help save you a lot of effort with long winded `#[cfg()]` checks.
chacha20	0.10.2	Apache-2.0 OR MIT	The ChaCha20 stream cipher (RFC 8439) implemented in pure Rust using traits from the RustCrypto `cipher` crate, with optional architecture-specific hardware…
charts-rs	2.3.0	Apache-2.0	A charts library for rust
chematic-core	1.2.1	Apache-2.0 OR MIT	Core types (Atom, Bond, Molecule) for chematic — pure-Rust RDKit alternative, WASM-compatible
chematic-depict	1.2.1	Apache-2.0 OR MIT	2D molecular structure depiction as SVG: ring templates, wedge/dash stereo bonds, CPK coloring, grid layout — pure-Rust, no C/C++ dependencies
chematic-perception	1.2.1	Apache-2.0 OR MIT	SSSR ring perception and Hückel aromaticity for chematic — pure-Rust RDKit alternative
chematic-rxn	1.2.1	Apache-2.0 OR MIT	Reaction SMILES/SMIRKS parser and writer for chematic — pure-Rust RDKit alternative
chematic-smarts	1.2.1	Apache-2.0 OR MIT	SMARTS parser, VF2 subgraph isomorphism and MCS for chematic — pure-Rust RDKit alternative

Library	Version	License	What it does
chematic-smiles	1.2.1	Apache-2.0 OR MIT	OpenSMILES parser, writer and canonical SMILES for chematic — pure Rust, WASM-compatible
chinese-number	0.7.8	MIT	Convert primitive numbers to Chinese numbers, or parse Chinese numbers to primitive numbers.
chinese-variant	1.1.6	MIT	An enum to represent the variants of the Chinese Language.
chrono	0.4.45	Apache-2.0 OR MIT	Date and time library for Rust
chunked_transfer	1.5.0	Apache-2.0 OR MIT	Encoder and decoder for HTTP chunked transfer coding (RFC 7230 § 4.1)
ciborium	0.2.2	Apache-2.0	serde implementation of CBOR using ciborium-basic
ciborium-io	0.2.2	Apache-2.0	Simplified Read/Write traits for no_std usage
ciborium-ll	0.2.2	Apache-2.0	Low-level CBOR codec primitives
cipher	0.5.2	Apache-2.0 OR MIT	Traits for describing block ciphers and stream ciphers
citationberg	0.7.0	Apache-2.0 OR MIT	A parser for CSL files
clap	4.6.7	Apache-2.0 OR MIT	A simple to use, efficient, and full-featured Command Line Argument Parser
clap_builder	4.6.7	Apache-2.0 OR MIT	A simple to use, efficient, and full-featured Command Line Argument Parser
clap_derive	4.6.7	Apache-2.0 OR MIT	Parse command line argument by defining a struct, derive crate.
clap_lex	1.1.1	Apache-2.0 OR MIT	Minimal, flexible command line parser
cmov	0.5.4	Apache-2.0 OR MIT	Conditional move CPU intrinsics which are guaranteed on major platforms (ARM32/ ARM64, x86/x86_64, RISC-V) to execute in constant-time and not be rewritten as…
cobs	0.3.0	Apache-2.0 OR MIT	This is an implementation of the Consistent Overhead Byte Stuffing (COBS) algorithm. COBS is an algorithm for transforming a message into an encoding where a…
codex	0.3.0	Apache-2.0	Human-friendly notation for Unicode symbols.
color	0.3.3	Apache-2.0 OR MIT	A library for representing and manipulating colors
color_quant	1.1.0	MIT	Color quantization library to reduce n colors to 256 colors.
colorchoice	1.0.5	Apache-2.0 OR MIT	Global override of color control
combine	4.6.9	MIT	Fast parser combinators on arbitrary streams with zero-copy support.

Library	Version	License	What it does
comemo	0.5.1	Apache-2.0 OR MIT	Incremental computation through constrained memoization.
comemo-macros	0.5.1	Apache-2.0 OR MIT	Procedural macros for comemo.
compression-codecs	0.4.46	Apache-2.0 OR MIT	Adaptors for various compression algorithms.
compression-core	0.4.34	Apache-2.0 OR MIT	Abstractions for compression algorithms.
comrak	0.57.0	BSD-2-Clause	A 100% CommonMark-compatible GitHub Flavored Markdown parser and formatter
concurrent-queue	2.5.0	Apache-2.0 OR MIT	Concurrent multi-producer multi-consumer queue
config-derive	0.15.0	MIT	Proc macro used by Twelf crate
console	0.16.6	MIT	A terminal and console abstraction for Rust
const-oid	0.10.2	Apache-2.0 OR MIT	Const-friendly implementation of the ISO/IEC Object Identifier (OID) standard as defined in ITU X.660, with support for BER/DER encoding/decoding as well as…
const_format	0.2.37	Zlib	Compile-time string formatting
const_format_proc_macros	0.2.37	Zlib	Implementation detail of the `const_format` crate
convert_case	0.11.0	MIT	Convert strings into any case
cookie	0.18.2	Apache-2.0 OR MIT	HTTP cookie parsing and cookie jar management. Supports signed and private (encrypted, authenticated) jars.
core-foundation	0.10.1	Apache-2.0 OR MIT	Bindings to Core Foundation for macOS
core-foundation	0.9.4	Apache-2.0 OR MIT	Bindings to Core Foundation for macOS
core-foundation-sys	0.8.7	Apache-2.0 OR MIT	Bindings to Core Foundation for macOS
core-graphics	0.25.0	Apache-2.0 OR MIT	Bindings to Core Graphics for macOS
core-graphics-types	0.2.0	Apache-2.0 OR MIT	Bindings for some fundamental Core Graphics types
core_detect	1.0.0	Apache-2.0 OR MIT	A `no_std` version of the `std::is_x86_feature_detected!` macro.
core_maths	0.1.1	MIT	Extension trait for full float functionality in `#[no_std]` backed by `libm`.
countme	3.0.1	Apache-2.0 OR MIT	Counts the number of live instances of types

Library	Version	License	What it does
cpubits	0.1.1	Apache-2.0 OR MIT	Compile-time detection heuristics for the optimal word size to use for the target CPU, which in some cases may differ from its address size a.k.a…
cpufeatures	0.2.17	Apache-2.0 OR MIT	Lightweight runtime CPU feature detection for aarch64, loongarch64, and x86/x86_64 targets, with no_std support and support for mobile targets including Android…
cpufeatures	0.3.1	Apache-2.0 OR MIT	Lightweight runtime CPU feature detection for aarch64, loongarch64, and x86/x86_64 targets, with no_std support and support for mobile targets including Android…
crc32fast	1.5.3	Apache-2.0 OR MIT	Fast, SIMD-accelerated CRC32 (IEEE) checksum computation
crossbeam-channel	0.5.17	Apache-2.0 OR MIT	Multi-producer multi-consumer channels for message passing
crossbeam-deque	0.8.8	Apache-2.0 OR MIT	Concurrent work-stealing deque
crossbeam-epoch	0.9.21	Apache-2.0 OR MIT	Epoch-based garbage collection
crossbeam-utils	0.8.23	Apache-2.0 OR MIT	Utilities for concurrent programming
crunchy	0.2.4	MIT	Crunchy unroller: deterministically unroll constant loops
crypto-bigint	0.7.5	Apache-2.0 OR MIT	Pure Rust implementation of a big integer library which has been designed from the ground-up for use in cryptographic applications. Provides constant-time…
crypto-common	0.1.7	Apache-2.0 OR MIT	Common cryptographic traits
crypto-common	0.2.2	Apache-2.0 OR MIT	Common traits used by cryptographic algorithms
crypto-primes	0.7.2	Apache-2.0 OR MIT	Random prime number generation and primality checking library
cssparser	0.36.0	MPL-2.0	Rust implementation of CSS Syntax Level 3
cssparser	0.37.0	MPL-2.0	Rust implementation of CSS Syntax Level 3
cssparser-macros	0.6.1	MPL-2.0	Procedural macros for cssparser
cssparser-macros	0.7.1	MPL-2.0	Procedural macros for cssparser
csv	1.4.0	MIT OR Unlicense	Fast CSV parsing with support for serde.
csv-core	0.1.13	MIT OR Unlicense	Bare bones CSV parsing with no_std support.

Library	Version	License	What it does
ctor	1.0.13	Apache-2.0 OR MIT	Global, no_std-compatible constructors for all platforms that run before main (like C/C++ **attribute**((constructor)))
ctr	0.10.1	Apache-2.0 OR MIT	CTR block modes of operation
ctutils	0.4.3	Apache-2.0 OR MIT	Constant-time utility library with selection and equality testing support targeting cryptographic applications. Supports `const fn` where appropriate. Built on…
curve25519-dalek	5.0.0	BSD-3-Clause	A pure-Rust implementation of group operations on ristretto255 and Curve25519
curve25519-dalek-derive	0.1.1	Apache-2.0 OR MIT	curve25519-dalek Derives
darling	0.24.1	MIT	A proc-macro library for reading attributes into structs when implementing custom derives.
darling_core	0.24.1	MIT	Helper crate for proc-macro library for reading attributes into structs when implementing custom derives. Use https://crates.io/crates/darling in your code.
darling_macro	0.24.1	MIT	Internal support for a proc-macro library for reading attributes into structs when implementing custom derives. Use https://crates.io/crates/darling in your…
dashmap	6.2.1	MIT	Blazing fast concurrent HashMap for Rust.
data-encoding	2.11.1	MIT	Efficient and customizable data-encoding functions like base64, base32, and hex
data-url	0.3.2	Apache-2.0 OR MIT	Processing of data: URL according to WHATWG's Fetch Standard
dbus	0.9.12	Apache-2.0 OR MIT	Bindings to D-Bus, which is a bus commonly used on Linux for inter-process communication.
defmt	1.2.0	Apache-2.0 OR MIT	A highly efficient logging framework that targets resource-constrained devices, like microcontrollers
defmt-macros	1.2.0	Apache-2.0 OR MIT	defmt macros
defmt-parser	1.1.0	Apache-2.0 OR MIT	Parsing library for defmt format strings
delegate	0.13.5	Apache-2.0 OR MIT	Method delegation with less boilerplate
der	0.8.2	Apache-2.0 OR MIT	Pure Rust embedded-friendly implementation of the Distinguished Encoding Rules (DER) for Abstract Syntax Notation One (ASN.1) as described in ITU X.690 with…
deranged	0.5.9	Apache-2.0 OR MIT	Ranged integers

Library	Version	License	What it does
derive_arbitrary	1.5.0	Apache-2.0 OR MIT	Derives arbitrary traits
derive_more	2.1.1	MIT	Adds #[derive(x)] macros for more traits
derive_more-impl	2.1.1	MIT	Internal implementation of `derive_more` crate
des	0.9.0	Apache-2.0 OR MIT	DES and Triple DES (3DES, TDES) block ciphers implementation
deunicode	1.6.2	BSD-3-Clause	Convert Unicode strings to pure ASCII by intelligently transliterating them. Suppors Emoji and Chinese.
digest	0.10.7	Apache-2.0 OR MIT	Traits for cryptographic hash functions and message authentication codes
digest	0.11.3	Apache-2.0 OR MIT	Traits for cryptographic hash functions and message authentication codes
dirs	7.0.0	Apache-2.0 OR MIT	A tiny low-level library that provides platform-specific standard locations of directories for config, cache and other data on Linux, Windows, macOS and Redox…
dirs-sys	0.5.0	Apache-2.0 OR MIT	System-level helper functions for the dirs and directories crates.
dispatch2	0.3.1	Apache-2.0 OR MIT OR Zlib	Bindings and wrappers for Apple's Grand Central Dispatch (GCD)
displaydoc	0.2.7	Apache-2.0 OR MIT	A derive macro for implementing the display Trait via a doc comment and string interpolation
dlopen2	0.8.2	MIT	Library for opening and operating on dynamic link libraries (also known as shared objects or shared libraries).
dlopen2_derive	0.4.3	MIT	Derive macros for the dlopen2 crate.
dom_query	0.28.0	MIT	HTML querying and manipulation with CSS selectors
dpi	0.1.2	Apache-2.0 AND MIT	Types for handling UI scaling
dtoa	1.0.11	Apache-2.0 OR MIT	Fast floating point primitive to string conversion
dtoa-short	0.3.5	MPL-2.0	Serialize float number and truncate to certain precision

Library	Version	License	What it does
dunce	1.0.5	Apache-2.0 OR CC0-1.0 OR MIT-0	Normalize Windows paths to the most compatible format, avoiding UNC where possible
dyn-clone	1.0.20	Apache-2.0 OR MIT	Clone trait that is dyn-compatible
ecdsa	0.17.0	Apache-2.0 OR MIT	Pure Rust implementation of the Elliptic Curve Digital Signature Algorithm (ECDSA) as specified in FIPS 186-4 (Digital Signature Standard), implemented…
ecow	0.2.6	Apache-2.0 OR MIT	Compact, clone-on-write vector and string.
ed25519	3.0.0	Apache-2.0 OR MIT	Edwards Digital Signature Algorithm (EdDSA) over Curve25519 (as specified in RFC 8032) support library providing signature type definitions and PKCS#8 private…
ed25519-dalek	3.0.0	BSD-3-Clause	Fast and efficient ed25519 EdDSA key generations, signing, and verification in pure Rust.
either	1.19.0	Apache-2.0 OR MIT	The enum `Either` with variants `Left` and `Right` is a general purpose sum type with two cases.
elliptic-curve	0.14.1	Apache-2.0 OR MIT	General purpose Elliptic Curve Cryptography (ECC) support, including traits and generic types for representing various elliptic curve forms, scalars, points…
embed-resource	3.0.12	MIT	A Cargo library to handle compilation and inclusion of Windows resources in the most resilient fashion imaginable
embed_plist	1.2.2	Apache-2.0 OR MIT	Embed property list files like Info.plist directly in your executable binary.
emojis	0.8.2	(MIT OR Apache-2.0) AND Unicode-3.0	🔣 Lookup emoji in *O(1)* time, access metadata and GitHub shortcodes, iterate over all emoji, and more!
ena	0.14.4	Apache-2.0 OR MIT	Union-find, congruence closure, and other unification code. Based on code from rustc.
encode_unicode	1.0.0	Apache-2.0 OR MIT	UTF-8 and UTF-16 character types, iterators and related methods for char, u8 and u16.

Library	Version	License	What it does
encoding_rs	0.8.42	(Apache-2.0 OR MIT) AND BSD-3-Clause	A Gecko-oriented implementation of the Encoding Standard
endi	1.1.1	MIT	A simple endian-handling library
entities	1.0.1	MIT	Provides the raw data needed to convert to and from HTML entities.
enum-ordinalize	4.4.2	MIT	This library enables enums to not only obtain the ordinal values of their variants but also allows for the construction of enums from an ordinal value.
enum-ordinalize-derive	4.4.2	MIT	This crate provides a procedural macro that enables enums to not only obtain the ordinal values of their variants but also allows for the construction of enums…
enum_dispatch	0.3.13	Apache-2.0 OR MIT	Near drop-in replacement for dynamic-dispatched method calls with up to 10x the speed
enumflags2	0.7.12	Apache-2.0 OR MIT	Enum-based bit flags
enumflags2_derive	0.7.12	Apache-2.0 OR MIT	Do not use directly, use the re-export in the `enumflags2` crate. This allows for better compatibility across versions.
envy	0.4.2	MIT	deserialize env vars into type-safe structs
equivalent	1.0.3	Apache-2.0 OR MIT	Traits for key comparison in maps.
erased-serde	0.4.10	Apache-2.0 OR MIT	Type-erased Serialize and Serializer traits
errno	0.3.14	Apache-2.0 OR MIT	Cross-platform interface to the `errno` variable.
euclid	0.22.14	Apache-2.0 OR MIT	Geometry primitives
event-listener	5.4.2	Apache-2.0 OR MIT	Notify async tasks or threads
event-listener-strategy	0.5.4	Apache-2.0 OR MIT	Block or poll on event_listener easily
fancy-regex	0.16.2	MIT	An implementation of regexes, supporting a relatively rich set of features, including backreferences and look-around.
fastrand	2.5.0	Apache-2.0 OR MIT	A simple and fast random number generator

Library	Version	License	What it does
fdeflate	0.3.7	Apache-2.0 OR MIT	Fast specialized deflate implementation
fearless_simd	0.4.1	Apache-2.0 OR MIT	Safer and easier SIMD
fearless_simd	1.1.0	Apache-2.0 OR MIT	Safer and easier SIMD
fearless_simd_macros	0.1.0	Apache-2.0 OR MIT	Procedural macros for fearless_simd
ff	0.14.0	Apache-2.0 OR MIT	Library for building and interfacing with finite fields
fiat-crypto	0.3.0	Apache-2.0 OR BSD-1-Clause OR MIT	Fiat-crypto generated Rust
field-offset	0.3.6	Apache-2.0 OR MIT	Safe pointer-to-member implementation
filetime	0.2.29	Apache-2.0 OR MIT	Platform-agnostic accessors of timestamps in File metadata
find-msvc-tools	0.1.15	Apache-2.0 OR MIT	Find windows-specific tools, read MSVC versions from the registry and from COM interfaces
finl_unicode	1.5.0	(MIT OR Apache-2.0) AND Unicode-DFS-2016	Library for handling Unicode functionality for finl (categories and grapheme segmentation)
flate2	1.1.10	Apache-2.0 OR MIT	DEFLATE compression and decompression exposed as Read/BufRead/Write streams. Supports miniz_oxide and multiple zlib implementations. Supports zlib, gzip, and…
float-cmp	0.9.0	MIT	Floating point approximate comparison traits
fnv	1.0.7	Apache-2.0 OR MIT	Fowler–Noll–Vo hash function
foldhash	0.2.0	Zlib	A fast, non-cryptographic, minimally DoS-resistant hashing algorithm.
font-types	0.11.3	Apache-2.0 OR MIT	Scalar types used in fonts.
fontconfig-parser	0.5.8	MIT	fontconfig file parser in pure Rust
fontdb	0.23.0	MIT	A simple, in-memory font database with CSS-like queries.
foreign-types	0.5.0	Apache-2.0 OR MIT	A framework for Rust wrappers over C APIs
foreign-types-macros	0.2.4	Apache-2.0 OR MIT	An internal crate used by foreign-types
foreign-types-shared	0.3.1	Apache-2.0 OR MIT	An internal crate used by foreign-types
form_urlencoded	1.2.2	Apache-2.0 OR MIT	Parser and serializer for the application/x-www-form-urlencoded syntax, as used by HTML forms.
funty	2.0.0	MIT	Trait generalization over the primitive types

Library	Version	License	What it does
futures	0.3.34	Apache-2.0 OR MIT	An implementation of futures and streams featuring zero allocations, composability, and iterator-like interfaces.
futures-channel	0.3.34	Apache-2.0 OR MIT	Channels for asynchronous communication using futures-rs.
futures-core	0.3.34	Apache-2.0 OR MIT	The core traits and types in for the `futures` library.
futures-executor	0.3.34	Apache-2.0 OR MIT	Executors for asynchronous tasks based on the futures-rs library.
futures-io	0.3.34	Apache-2.0 OR MIT	The `AsyncRead`, `AsyncWrite`, `AsyncSeek`, and `AsyncBufRead` traits for the futures-rs library.
futures-lite	2.6.1	Apache-2.0 OR MIT	Futures, streams, and async I/O combinators
futures-macro	0.3.34	Apache-2.0 OR MIT	The futures-rs procedural macro implementations.
futures-sink	0.3.34	Apache-2.0 OR MIT	The asynchronous `Sink` trait for the futures-rs library.
futures-task	0.3.34	Apache-2.0 OR MIT	Tools for working with tasks.
futures-util	0.3.34	Apache-2.0 OR MIT	Common utilities and extension traits for the futures-rs library.
fxhash	0.2.1	Apache-2.0 OR MIT	A fast, non-secure, hashing algorithm derived from an internal hasher used in FireFox and Rustc.
gdk	0.18.2	MIT	UNMAINTAINED Rust bindings for the GDK 3 library (use gdk4 instead)
gdk-pixbuf	0.18.5	MIT	Rust bindings for the GdkPixbuf library
gdk-pixbuf-sys	0.18.0	MIT	FFI bindings to libgdk_pixbuf-2.0
gdk-sys	0.18.2	MIT	UNMAINTAINED FFI bindings to libgdk-3 (use gdk4-sys instead)
gdkwayland-sys	0.18.2	MIT	UNMAINTAINED FFI bindings to libgdk-3-wayland (use gdk4-wayland-sys instead)
gdkx11	0.18.2	MIT	UNMAINTAINED Rust bindings for the GDK X11 library (use gdk4-x11 instead)
gdkx11-sys	0.18.2	MIT	UNMAINTAINED FFI binding for libgdkx11 (use gdk4-x11-sys instead)

Library	Version	License	What it does
generic-array	0.14.7	MIT	Generic types implementing functionality of arrays
generic-array	1.4.5	MIT	Generic types implementing functionality of arrays
getrandom	0.2.17	Apache-2.0 OR MIT	A small cross-platform library for retrieving random data from system source
getrandom	0.3.4	Apache-2.0 OR MIT	A small cross-platform library for retrieving random data from system source
getrandom	0.4.3	Apache-2.0 OR MIT	A small cross-platform library for retrieving random data from system source

Library	Version	License	What it does
ghash	0.6.0	Apache-2.0 OR MIT	Universal hash over GF(2^128) useful for constructing a Message Authentication Code (MAC), as in the AES-GCM authenticated encryption cipher.
gif	0.14.2	Apache-2.0 OR MIT	GIF de- and encoder
gio	0.18.4	MIT	Rust bindings for the Gio library
gio-sys	0.18.1	MIT	FFI bindings to libgio-2.0
git2	0.21.0	Apache-2.0 OR MIT	Bindings to libgit2 for interoperating with git repositories. This library is both threadsafe and memory safe and allows both reading and writing git…
glib	0.18.5	MIT	Rust bindings for the GLib library
glib-macros	0.18.5	MIT	Rust bindings for the GLib library, proc macros crate
glib-sys	0.18.1	MIT	FFI bindings to libglib-2.0
glidesort	0.1.2	Apache-2.0 OR MIT	Glidesort sorting algorithm
glifo	0.1.1	Apache-2.0 OR MIT	Glifo provides APIs for efficiently rendering text.
glob	0.3.4	Apache-2.0 OR MIT	Support for matching file paths against Unix shell style patterns.
gloo-timers	0.4.0	Apache-2.0 OR MIT	Convenience crate for working with JavaScript timers
gobject-sys	0.18.0	MIT	FFI bindings to libgobject-2.0
group	0.14.0	Apache-2.0 OR MIT	Elliptic curve group traits and utilities
gtk	0.18.2	MIT	UNMAINTAINED Rust bindings for the GTK+ 3 library (use gtk4 instead)
gtk-sys	0.18.2	MIT	UNMAINTAINED FFI bindings to libgtk-3 (use gtk4-sys instead)
gtk3-macros	0.18.2	MIT	UNMAINTAINED Rust bindings for the GTK 3 library (use gtk4-macros instead)
guillotiere	0.7.0	Apache-2.0 OR MIT	A dynamic 2D texture atlas allocator with fast deallocation.
half	2.7.1	Apache-2.0 OR MIT	Half-precision floating point f16 and bf16 types for Rust implementing the IEEE 754-2008 standard binary16 and bfloat16 types.
hashbrown	0.12.3	Apache-2.0 OR MIT	A Rust port of Google's SwissTable hash map
hashbrown	0.14.5	Apache-2.0 OR MIT	A Rust port of Google's SwissTable hash map
hashbrown	0.16.1	Apache-2.0 OR MIT	A Rust port of Google's SwissTable hash map
hashbrown	0.17.1	Apache-2.0 OR MIT	A Rust port of Google's SwissTable hash map
hayagriva	0.10.1	Apache-2.0 OR MIT	Work with references: Literature database management, storage, and citation formatting
hayro	0.7.1	Apache-2.0 OR MIT	A rasterizer for PDF files.
hayro-ccitt	0.3.0	Apache-2.0 OR MIT	A decoder for group 3 and group 4 CCITT images.

Library	Version	License	What it does
hayro-ccitt	0.4.0	Apache-2.0 OR MIT	A decoder for group 3 and group 4 CCITT images.
hayro-cmap	0.1.1	Apache-2.0 OR MIT	A parser for CMap files.
hayro-interpret	0.7.0	Apache-2.0 OR MIT	A crate for interpreting PDF files.
hayro-jbig2	0.3.1	Apache-2.0 OR MIT	A memory-safe, pure-Rust JBIG2 decoder.
hayro-jpeg2000	0.3.5	Apache-2.0 OR MIT	A memory-safe, pure-Rust JPEG 2000 decoder.
hayro-postscript	0.2.0	Apache-2.0 OR MIT	A lightweight PostScript scanner.
hayro-svg	0.7.0	Apache-2.0 OR MIT	Converting PDF pages into SVG files.
hayro-syntax	0.7.2	Apache-2.0 OR MIT	A low-level crate for reading PDF files.
hayro-write	0.7.0	Apache-2.0 OR MIT	A crate for rewriting pages of a PDF file.
heck	0.4.1	Apache-2.0 OR MIT	heck is a case conversion library.
heck	0.5.0	Apache-2.0 OR MIT	heck is a case conversion library.
hermit-abi	0.5.3	Apache-2.0 OR MIT	Hermit system calls definitions.
hex	0.4.3	Apache-2.0 OR MIT	Encoding and decoding data into/from hexadecimal representation.
hex-literal	1.1.0	Apache-2.0 OR MIT	A macro for converting hexadecimal string literals to a byte array at compile time
hkdf	0.13.0	Apache-2.0 OR MIT	HMAC-based Extract-and-Expand Key Derivation Function (HKDF)
hmac	0.13.0	Apache-2.0 OR MIT	Generic implementation of Hash-based Message Authentication Code (HMAC)
home	0.5.12	Apache-2.0 OR MIT	Shared definitions of home directories.
html-escape	0.2.15	MIT	This library is for encoding/escaping special characters in HTML and decoding/unescaping HTML entities as well.
html5ever	0.39.0	Apache-2.0 OR MIT	High-performance browser-grade HTML5 parser
http	1.5.0	Apache-2.0 OR MIT	A set of types for representing HTTP requests and responses.
http-body	1.1.0	MIT	Trait representing an asynchronous, streaming, HTTP request or response body.
http-body-util	0.1.5	MIT	Combinators and adapters for HTTP request or response bodies.
http-range	0.1.6	MIT	HTTP Range header parser
httparse	1.10.1	Apache-2.0 OR MIT	A tiny, safe, speedy, zero-copy HTTP/1.x parser.
httpdate	1.0.3	Apache-2.0 OR MIT	HTTP date parsing and formatting
hybrid-array	0.4.15	Apache-2.0 OR MIT	Hybrid typenum-based and const generic array types designed to provide the flexi-

Library	Version	License	What it does
			bility of typenum-based expressions while also allowing interoperability and a…
hyper	1.12.0	MIT	A protective and efficient HTTP library for all.
hyper-rustls	0.27.10	Apache-2.0 OR ISC OR MIT	Rustls+hyper integration for pure rust HTTPS
hyper-util	0.1.21	MIT	hyper utilities
hypher	0.1.8	Apache-2.0 OR MIT	hypher separates words into syllables.
iana-time-zone	0.1.65	Apache-2.0 OR MIT	get the IANA time zone for the current system
iana-time-zone-haiku	0.1.2	Apache-2.0 OR MIT	iana-time-zone support crate for Haiku OS
ico	0.5.0	MIT	A library for encoding/decoding ICO image files
icu_collator	2.3.1	Unicode-3.0	API for comparing strings according to language-dependent conventions
icu_collator_data	2.3.0	Unicode-3.0	Data for the icu_collator crate
icu_collections	2.3.0	Unicode-3.0	Collection of API for use in ICU libraries.
icu_locale	2.3.1	Unicode-3.0	API for Unicode Language and Locale Identifiers canonicalization
icu_locale_core	2.3.0	Unicode-3.0	API for managing Unicode Language and Locale Identifiers
icu_locale_data	2.3.0	Unicode-3.0	Data for the icu_locale crate
icu_locale_fallback	2.3.0	Unicode-3.0	API for Locale Fallback algorithms
icu_locale_fallback_data	2.3.0	Unicode-3.0	Data for the icu_locale_fallback crate
icu_normalizer	2.3.0	Unicode-3.0	API for normalizing text into Unicode Normalization Forms
icu_normalizer_data	2.3.0	Unicode-3.0	Data for the icu_normalizer crate
icu_properties	2.3.0	Unicode-3.0	Definitions for Unicode properties
icu_properties_data	2.3.0	Unicode-3.0	Data for the icu_properties crate
icu_provider	2.3.1	Unicode-3.0	Trait and struct definitions for the ICU data provider
icu_provider_blob	2.3.0	Unicode-3.0	ICU4X data provider that reads from blobs in memory
icu_segmenter	2.3.0	Unicode-3.0	Unicode line breaking and text segmentation algorithms for text boundaries analysis
icu_segmenter_data	2.3.0	Unicode-3.0	Data for the icu_segmenter crate
ident_case	1.0.1	Apache-2.0 OR MIT	Utility for applying case rules to Rust identifiers.
idna	1.1.0	Apache-2.0 OR MIT	IDNA (Internationalizing Domain Names in Applications) and Punycode.

Library	Version	License	What it does
idna_adapter	1.2.2	Apache-2.0 OR MIT	Back end adapter for idna
image	0.25.10	Apache-2.0 OR MIT	Imaging library. Provides basic image processing and encoders/decoders for common image formats.
image-webp	0.2.4	Apache-2.0 OR MIT	WebP encoding and decoding in pure Rust
imagesize	0.14.0	MIT	Quick probing of image dimensions without loading the entire file.
include_dir	0.7.4	MIT	Embed the contents of a directory in your binary
include_dir_macros	0.7.4	MIT	The procedural macro used by include_dir
indexmap	1.9.3	Apache-2.0 OR MIT	A hash table with consistent order and fast iteration.
indexmap	2.14.2	Apache-2.0 OR MIT	A hash table with consistent order and fast iteration.
infer	0.19.0	MIT	Small crate to infer file type based on magic number signatures
infer	0.22.0	MIT	Small crate to infer file type based on magic number signatures
inout	0.2.5	Apache-2.0 OR MIT	Custom reference types for code generic over in-place and buffer-to-buffer modes of operation.
ipnet	2.12.2	Apache-2.0 OR MIT	Provides types and useful methods for working with IPv4 and IPv6 network addresses, commonly called IP prefixes. The new `IpNet`, `Ipv4Net`, and `Ipv6Net` types…
is-docker	0.2.0	MIT	Checks if the process is running inside a Docker container.
is-wsl	0.4.0	MIT	Checks if the process is running inside Windows Subsystem for Linux.
is_terminal_polyfill	1.70.2	Apache-2.0 OR MIT	Polyfill for `is_terminal` stdlib feature for use with older MSRVs
itertools	0.14.0	Apache-2.0 OR MIT	Extra iterator adaptors, iterator methods, free functions, and macros.
itoa	1.0.18	Apache-2.0 OR MIT	Fast integer primitive to string conversion
javascriptcore-rs	1.1.2	MIT	Rust bindings for the javacriptcore library
javascriptcore-rs-sys	1.1.1	MIT	Sys functions for the Rust bindings of the javacriptcore library
jiff	0.2.38	MIT OR Unlicense	A date-time library that encourages you to jump into the pit of success. This library is heavily inspired by the Temporal project.
jiff-core	0.1.1	MIT OR Unlicense	Low level datetime primitives for the Jiff library.

Library	Version	License	What it does
jiff-static	0.2.38	MIT OR Unlicense	Create static TimeZone values for Jiff (useful in core-only environments).
jiff-tzdb	0.1.9	MIT OR Unlicense	The entire Time Zone Database embedded into your binary.
jiff-tzdb-platform	0.1.3	MIT OR Unlicense	The entire Time Zone Database embedded into your binary for specific platforms.
jni	0.21.1	Apache-2.0 OR MIT	Rust bindings to the JNI
jni	0.22.4	Apache-2.0 OR MIT	Rust bindings to the JNI
jni-macros	0.22.4	Apache-2.0 OR MIT	Procedural macros for the jni crate
jni-sys	0.3.1	Apache-2.0 OR MIT	Rust definitions corresponding to jni.h
jni-sys	0.4.1	Apache-2.0 OR MIT	Rust definitions corresponding to jni.h
jni-sys-macros	0.4.1	Apache-2.0 OR MIT	Macros for jni-sys crate
jobserver	0.1.35	Apache-2.0 OR MIT	An implementation of the GNU Make jobserver for Rust.
js-sys	0.3.106	Apache-2.0 OR MIT	Bindings for all JS global objects and functions in all JS environments like Node.js and browsers, built on `#[wasm_bindgen]` using the `wasm-bindgen` crate.
json-patch	4.2.0	Apache-2.0 OR MIT	RFC 6902, JavaScript Object Notation (JSON) Patch
jsonptr	0.7.1	Apache-2.0 OR MIT	Data structures and logic for resolving, assigning, and deleting by JSON Pointers (RFC 6901)
kamadak-exif	0.6.2	BSD-2-Clause	Exif parsing library written in pure Rust
keccak	0.2.2	Apache-2.0 OR MIT	Pure Rust implementation of the Keccak sponge functions
kem	0.3.0	Apache-2.0 OR MIT	Traits for Key Encapsulation Mechanisms (KEMs): public-key cryptosystems designed to enable a sender (a.k.a. encapsulator) to generate and encrypt a short…
keyboard-types	0.8.3	Apache-2.0 OR MIT	Contains types to define keyboard related events.
keyring	4.2.0	Apache-2.0 OR MIT	All-in-One Rust Keyring
keyring-core	1.0.1	Apache-2.0 OR MIT	Cross-platform library for managing passwords/secrets
konst	0.2.20	Zlib	Const equivalents of std functions, compile-time comparison, and parsing
konst_macro_rules	0.2.19	Zlib	Implementation detail of the konst crate
krilla	0.8.2	Apache-2.0 OR MIT	A high-level crate for creating PDF files.

Library	Version	License	What it does
krilla-svg	0.8.1	Apache-2.0 OR MIT	Converting SVG files to PDF.
kurbo	0.13.1	Apache-2.0 OR MIT	A 2D curves library
lazy_static	1.5.1	Apache-2.0 OR MIT	A macro for declaring lazily evaluated statics in Rust.
lexical-core	1.0.6	Apache-2.0 OR MIT	Lexical, to- and from-string conversion routines.
lexical-parse-float	1.0.6	Apache-2.0 OR MIT	Efficient parsing of floats from strings.
lexical-parse-integer	1.0.6	Apache-2.0 OR MIT	Efficient parsing of integers from strings.
lexical-util	1.0.7	Apache-2.0 OR MIT	Shared utilities for lexical creates.
lexical-write-float	1.0.6	Apache-2.0 OR MIT	Efficient formatting of floats to strings.
lexical-write-integer	1.0.6	Apache-2.0 OR MIT	Efficient formatting of integers to strings.
libappindicator	0.9.0	Apache-2.0 OR MIT	Rust safe bindings for libappindicator
libappindicator-sys	0.9.0	Apache-2.0 OR MIT	Rust sys bindings for libappindicator
libc	0.2.191	Apache-2.0 OR MIT	Raw FFI bindings to platform libraries like libc.
libdbus-sys	0.2.7	Apache-2.0 OR MIT	FFI bindings to libdbus.
libgit2-sys	0.18.8+1.9.7	Apache-2.0 OR MIT	Native bindings to the libgit2 library
libloading	0.7.4	ISC	Bindings around the platform's dynamic library loading primitives with greatly improved memory safety.
libm	0.2.16	MIT	libm in pure Rust
libredox	0.1.25	MIT	Redox stable ABI
libwebp-sys	0.9.6	MIT	Bindings to libwebp (bindgen, static linking)
libz-sys	1.1.30	Apache-2.0 OR MIT	Low-level bindings to the system libz library (also known as zlib).
linebender_resource_handle	0.1.1	Apache-2.0 OR MIT	Raw resource handle for immutable blobs with dynamic lifetimes.

Library	Version	License	What it does
linked-hash-map	0.5.6	Apache-2.0 OR MIT	A HashMap wrapper that holds key-value pairs in insertion order
linux-raw-sys	0.12.1	Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	Generated bindings for Linux's userspace API
linux-raw-sys	0.4.15	Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	Generated bindings for Linux's userspace API
lipsum	0.9.1	MIT	Lipsum is a lorem ipsum text generation library. It generates pseudo-random Latin text. Use this if you need filler or dummy text for your application. The text…
litemap	0.8.3	Unicode-3.0	A key-value Map implementation based on a flat, sorted Vec.
lock_api	0.4.14	Apache-2.0 OR MIT	Wrappers to create fully-featured Mutex and RwLock types. Compatible with no_std.
lofty	0.25.5	Apache-2.0 OR MIT	Audio metadata library
lofty_attr	0.13.0	Apache-2.0 OR MIT	Macros for Lofty
log	0.4.34	Apache-2.0 OR MIT	A lightweight logging facade for Rust
logos	0.13.0	Apache-2.0 OR MIT	Create ridiculously fast Lexers
logos-codegen	0.13.0	Apache-2.0 OR MIT	Implementation details for logos-codegen and logos-derive. Not for public consumption.
logos-derive	0.13.0	Apache-2.0 OR MIT	Create ridiculously fast Lexers
lol_html	2.9.0	BSD-3-Clause	Streaming HTML rewriter/parser with CSS selector-based API
markup5ever	0.39.0	Apache-2.0 OR MIT	Common code for xml5ever and html5ever
md5	0.8.1	Apache-2.0 OR MIT	The package provides the MD5 hash function.
memchr	2.8.3	MIT OR Unlicense	Provides extremely fast (uses SIMD on x86_64, aarch64 and wasm32) routines for 1, 2 or 3 byte search and single substring search.
memchr-n	0.1.10	Apache-2.0 OR MIT	Fast SIMD-accelerated search and iteration over the bytes belonging to an arbitrary set of byte values

Library	Version	License	What it does
memmap2	0.9.12	Apache-2.0 OR MIT	Cross-platform Rust API for memory-mapped file IO
memoffset	0.9.1	MIT	offset_of functionality for Rust structs.
mime	0.3.17	Apache-2.0 OR MIT	Strongly Typed Mimes
minicbor	2.3.1	BlueOak-1.0.0	A small CBOR codec suitable for no_std environments.
minicbor-derive	0.19.5	BlueOak-1.0.0	Derive minicbor `Decode` and `Encode` traits.
minifier	0.3.6	MIT	Minifier tool/lib for JS/CSS/JSON files
minimal-lexical	0.2.1	Apache-2.0 OR MIT	Fast float parsing conversion routines.
minisign-verify	0.3.0	MIT	Minisign signature verification for files and streams with Ed25519. A small Rust library with no external dependencies.
miniz_oxide	0.8.9	Apache-2.0 OR MIT OR Zlib	DEFLATE compression and decompression library rewritten in Rust based on miniz
miniz_oxide	0.9.1	Apache-2.0 OR MIT OR Zlib	DEFLATE compression and decompression library rewritten in Rust based on miniz
mio	1.2.4	MIT	Lightweight non-blocking I/O.
mitex	0.2.4	Apache-2.0	MiTeX is a TeX2Typst converter.
mitex-glob	0.2.4	Apache-2.0	Glob impl for MiTeX
mitex-lexer	0.2.4	Apache-2.0	Lexer for MiTeX
mitex-parser	0.2.4	Apache-2.0	Parser for MiTeX
mitex-spec	0.2.4	Apache-2.0	Specification Library for MiTeX
mitex-spec-gen	0.2.4	Apache-2.0	Guard to geneate specification files for dependent crates
ml-kem	0.3.3	Apache-2.0 OR MIT	Pure Rust implementation of the Module-Lattice-Based Key-Encapsulation Mechanism Standard (formerly known as Kyber) as described in FIPS 203
module-lattice	0.2.3	Apache-2.0 OR MIT	Functionality shared between the `ml-kem` and `ml-dsa` crates, including linear algebra with degree-256 polynomials over a prime-order field, vectors of such…
moxcms	0.8.1	Apache-2.0 OR BSD-3-Clause	Simple Color Management in Rust
muda	0.20.0	Apache-2.0 OR MIT	Menu Utilities for Desktop Applications

Library	Version	License	What it does
multiversion_no_op	1.0.0	Apache-2.0 OR MIT	Provides a pass-through function attribute called multiversion for optimizing build times
mutate_once	0.1.2	BSD-2-Clause	Interior mutability, write-once and borrowable as plain &T
ndk	0.9.0	Apache-2.0 OR MIT	Safe Rust bindings to the Android NDK
ndk-context	0.1.1	Apache-2.0 OR MIT	Handles for accessing Android APIs
ndk-sys	0.6.0+11769913	Apache-2.0 OR MIT	FFI bindings for the Android NDK
new_debug_unreachable	1.0.6	MIT	panic in debug, intrinsics::unreachable() in release (fork of debug_unreachable)
nix	0.31.3	MIT	Rust friendly bindings to *nix APIs
nom	7.1.3	MIT	A byte-oriented, zero-copy, parser combinators library
num	0.4.3	Apache-2.0 OR MIT	A collection of numeric types and traits for Rust, including bigint, complex, rational, range iterators, generic integers, and more!
num-bigint	0.4.8	Apache-2.0 OR MIT	Big integer implementation for Rust
num-bigint	0.5.1	Apache-2.0 OR MIT	Big integer implementation for Rust
num-complex	0.4.6	Apache-2.0 OR MIT	Complex numbers implementation for Rust
num-conv	0.2.2	Apache-2.0 OR MIT	`num_conv` is a crate to convert between integer types without using `as` casts. This provides better certainty when refactoring, makes the exact behavior of…
num-integer	0.1.47	Apache-2.0 OR MIT	Integer traits and functions
num-iter	0.1.46	Apache-2.0 OR MIT	External iterators for generic mathematics
num-rational	0.4.2	Apache-2.0 OR MIT	Rational numbers implementation for Rust

Library	Version	License	What it does
num-traits	0.2.19	Apache-2.0 OR MIT	Numeric traits for generic mathematics
num_enum	0.7.6	Apache-2.0 OR BSD-3-Clause OR MIT	Procedural macros to make inter-operation between primitives and enums easier.
num_enum_derive	0.7.6	Apache-2.0 OR BSD-3-Clause OR MIT	Internal implementation details for ::num_enum (Procedural macros to make inter-operation between primitives and enums easier)
objc2	0.6.5	MIT	Objective-C interface and runtime bindings
objc2-app-kit	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the AppKit framework
objc2-cloud-kit	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the CloudKit framework
objc2-core-data	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the CoreData framework
objc2-core-foundation	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the CoreFoundation framework
objc2-core-graphics	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the CoreGraphics framework
objc2-core-image	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the CoreImage framework
objc2-core-location	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the CoreLocation framework
objc2-core-text	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the CoreText framework
objc2-encode	4.1.0	MIT	Objective-C type-encoding representation and parsing
objc2-exception-helper	0.1.1	Apache-2.0 OR MIT OR Zlib	External helper function for catching Objective-C exceptions
objc2-foundation	0.3.2	MIT	Bindings to the Foundation framework
objc2-io-surface	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the IOSurface framework
objc2-osa-kit	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the OSAKit framework
objc2-quartz-core	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the QuartzCore/CoreAnimation framework

Library	Version	License	What it does
objc2-ui-kit	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the UIKit framework
objc2-user-notifications	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the UserNotifications framework
objc2-web-kit	0.3.2	Apache-2.0 OR MIT OR Zlib	Bindings to the WebKit framework
object	0.39.1	Apache-2.0 OR MIT	A unified interface for reading and writing object file formats.
ogg_pager	0.8.0	Apache-2.0 OR MIT	A simple OGG page reader
once_cell	1.21.4	Apache-2.0 OR MIT	Single assignment cells and lazy values.
once_cell_polyfill	1.70.2	Apache-2.0 OR MIT	Polyfill for `OnceCell` stdlib feature for use with older MSRVs
open	5.4.4	MIT	Open a path or URL using the program configured on the system
openssl-probe	0.2.1	Apache-2.0 OR MIT	A library for helping to find system-wide trust anchor ("root") certificate locations based on paths typically used by `openssl`.
option-ext	0.2.0	MPL-2.0	Extends `Option` with additional operations
ordered-stream	0.2.0	Apache-2.0 OR MIT	Streams that are ordered relative to external events
osakit	0.3.1	Apache-2.0 OR MIT	OSAKit macOS Framework adapted for Rust
p256	0.14.0	Apache-2.0 OR MIT	Pure Rust implementation of the NIST P-256 (a.k.a. secp256r1, prime256v1) elliptic curve as defined in SP 800-186, with support for ECDH, ECDSA…
p384	0.14.0	Apache-2.0 OR MIT	Pure Rust implementation of the NIST P-384 (a.k.a. secp384r1) elliptic curve as defined in SP 800-186 with support for ECDH, ECDSA signing/verification, and…
p521	0.14.0	Apache-2.0 OR MIT	Pure Rust implementation of the NIST P-521 (a.k.a. secp521r1) elliptic curve as defined in SP 800-186, with support for ECDH, ECDSA signing/verification, and…
pageant	0.2.4	Apache-2.0	Pageant SSH agent transport client.
pagefind	1.5.2	MIT	Implement search on any static website.

Library	Version	License	What it does
pagefind_stem	1.0.0	MIT	Snowball stemming algorithms repackaged for Rust, with languages behind feature flags.
palette	0.7.7	Apache-2.0 OR MIT	Convert and manage colors with a focus on correctness, flexibility and ease of use.
palette_derive	0.7.7	Apache-2.0 OR MIT	Automatically implement traits from the palette crate.
palette_math	0.7.7	Apache-2.0 OR MIT	The low level color math behind the more high level `palette` crate.
pango	0.18.3	MIT	Rust bindings for the Pango library
pango-sys	0.18.0	MIT	FFI bindings to libpango-1.0

Library	Version	License	What it does
parking	2.2.1	Apache-2.0 OR MIT	Thread parking and unparking
parking_lot	0.12.5	Apache-2.0 OR MIT	More compact and efficient implementations of the standard synchronization primitives.
parking_lot_core	0.9.12	Apache-2.0 OR MIT	An advanced API for creating custom synchronization primitives.
password-hash	0.6.1	Apache-2.0 OR MIT	Traits which describe the functionality of password hashing algorithms, with optional support for a `no_std`/`no_alloc`-friendly implementation of the PHC…
paste	1.0.15	Apache-2.0 OR MIT	Macros for all your token pasting needs
pastey	0.2.3	Apache-2.0 OR MIT	Macros for all your token pasting needs. Successor of paste.
path-slash	0.2.1	MIT	Conversion to/from a file path from/to slash path
pbkdf2	0.13.0	Apache-2.0 OR MIT	Generic implementation of PBKDF2
pdf-writer	0.15.0	Apache-2.0 OR MIT	A step-by-step PDF writer.
pem-rfc7468	1.0.0	Apache-2.0 OR MIT	PEM Encoding (RFC 7468) for PKIX, PKCS, and CMS Structures, implementing a strict subset of the original Privacy-Enhanced Mail encoding intended specifically…
peniko	0.6.1	Apache-2.0 OR MIT	Primitive types for styling vector graphics
percent-encoding	2.3.2	Apache-2.0 OR MIT	Percent encoding and decoding
phc	0.6.1	Apache-2.0 OR MIT	Pure Rust implementation of the Password Hashing Competition (PHC) string format which is used for storing password hashes

Library	Version	License	What it does
phf	0.13.1	MIT	Runtime support for perfect hash function data structures
phf	0.14.0	MIT	Runtime support for perfect hash function data structures
phf_codegen	0.13.1	MIT	Codegen library for PHF types
phf_codegen	0.14.0	MIT	Codegen library for PHF types
phf_generator	0.13.1	MIT	PHF generation logic
phf_generator	0.14.0	MIT	PHF generation logic
phf_macros	0.13.1	MIT	Macros to generate types in the phf crate
phf_shared	0.13.1	MIT	Support code shared by PHF libraries
phf_shared	0.14.0	MIT	Support code shared by PHF libraries
pic-scale	0.7.12	Apache-2.0 OR BSD-3-Clause	High performance image scaling
pico-args	0.5.0	MIT	An ultra simple CLI arguments parser.
pin-project-lite	0.2.17	Apache-2.0 OR MIT	A lightweight version of pin-project written with declarative macros.
piper	0.2.5	Apache-2.0 OR MIT	An asynchronous single-consumer single-producer pipe for bytes.
pkcs1	0.8.0-rc.4	Apache-2.0 OR MIT	Pure Rust implementation of Public-Key Cryptography Standards (PKCS) #1: RSA Cryptography Specifications Version 2.2 (RFC 8017)
pkcs5	0.8.1	Apache-2.0 OR MIT	Pure Rust implementation of Public-Key Cryptography Standards (PKCS) #5: Password-Based Cryptography Specification Version 2.1 (RFC 8018)
pkcs8	0.11.0	Apache-2.0 OR MIT	Pure Rust implementation of Public-Key Cryptography Standards (PKCS) #8: Private-Key Information Syntax Specification (RFC 5208), with additional support for…
pkg-config	0.3.34	Apache-2.0 OR MIT	A library to run the pkg-config system tool at build time in order to be used in Cargo build scripts.
plist	1.10.2	MIT	A rusty plist parser. Supports Serde serialization.
png	0.17.16	Apache-2.0 OR MIT	PNG decoding and encoding library in pure Rust
png	0.18.1	Apache-2.0 OR MIT	PNG decoding and encoding library in pure Rust

Library	Version	License	What it does
polling	3.11.0	Apache-2.0 OR MIT	Portable interface to epoll, kqueue, event ports, and IOCP
poly1305	0.9.1	Apache-2.0 OR MIT	The Poly1305 universal hash function and message authentication code
polycool	0.4.0	Apache-2.0 OR MIT	Polynomial root-finding
polyval	0.7.3	Apache-2.0 OR MIT	POLYVAL is a GHASH-like universal hash over GF(2^128) useful for constructing a Message Authentication Code (MAC)
pori	0.0.0	MIT	Parser state and error extensions for nom.
portable-atomic	1.15.0	Apache-2.0 OR MIT	Portable atomic types including support for 128-bit atomics, atomic float, etc.
portable-atomic-util	0.2.8	Apache-2.0 OR MIT	Synchronization primitives built with portable-atomic.
postcard	1.1.3	Apache-2.0 OR MIT	A no_std + serde compatible message library for Rust
potential_utf	0.1.6	Unicode-3.0	Unvalidated string and character types
powerfmt	0.2.1	Apache-2.0 OR MIT	`powerfmt` is a library that provides utilities for formatting values. This crate makes it significantly easier to support filling to a minimum width with…
ppv-lite86	0.2.21	Apache-2.0 OR MIT	Cross-platform cryptography-oriented low-level SIMD library.
precomputed-hash	0.1.1	MIT	A library intending to be a base dependency to expose a precomputed hash
prettyplease	0.2.37	Apache-2.0 OR MIT	A minimal `syn` syntax tree pretty-printer
primefield	0.14.0	Apache-2.0 OR MIT	Generic implementation of prime fields built on `crypto-bigint`, along with macros for writing field element newtypes including ones with formally verified…

Library	Version	License	What it does
primeorder	0.14.0	Apache-2.0 OR MIT	Pure Rust implementation of complete addition formulas for prime order elliptic curves (Renes-Costello-Batina 2015). Generic over field elements and curve…
proc-macro-crate	1.3.1	Apache-2.0 OR MIT	Replacement for crate (macro_rules keyword) in proc-macros
proc-macro-crate	2.0.2	Apache-2.0 OR MIT	Replacement for crate (macro_rules keyword) in proc-macros
proc-macro-crate	3.5.0	Apache-2.0 OR MIT	Replacement for crate (macro_rules keyword) in proc-macros
proc-macro-error	1.0.4	Apache-2.0 OR MIT	Almost drop-in replacement to panics in proc-macros
proc-macro-error-attr	1.0.4	Apache-2.0 OR MIT	Attribute macro for proc-macro-error crate
proc-macro-hack	0.5.20+deprecated	Apache-2.0 OR MIT	Procedural macros in expression position
proc-macro2	1.0.107	Apache-2.0 OR MIT	A substitute implementation of the compiler's `proc_macro` API to decouple token-based libraries from the procedural macro use case.
psm	0.1.32	Apache-2.0 OR MIT	Portable Stack Manipulation: stack manipulation and introspection routines
ptr_meta	0.1.4	MIT	A radioactive stabilization of the ptr_meta rfc
ptr_meta_derive	0.1.4	MIT	Macros for ptr_meta
pulldown-latex	0.8.0	MIT	Pull parser to convert LaTeX equations to MathML
pxfm	0.1.30	Apache-2.0 OR BSD-3-Clause	Fast and accurate math
quick-error	2.0.1	Apache-2.0 OR MIT	A macro which makes error types pleasant to write.
quick-xml	0.38.4	MIT	High performance xml reader and writer

Library	Version	License	What it does
quick-xml	0.42.0	MIT	High performance xml reader and writer
quote	1.0.47	Apache-2.0 OR MIT	Quasi-quoting macro quote!(...)
r-efi	5.3.0	Apache-2.0 OR LGPL-2.1-or-later OR MIT	UEFI Reference Specification Protocol Constants and Definitions
r-efi	6.0.0	Apache-2.0 OR LGPL-2.1-or-later OR MIT	UEFI Reference Specification Protocol Constants and Definitions
radium	0.7.0	MIT	Portable interfaces for maybe-atomic types
rand	0.10.3	Apache-2.0 OR MIT	Random number generators and other randomness functionality.
rand	0.8.8	Apache-2.0 OR MIT	Random number generators and other randomness functionality.
rand_chacha	0.3.1	Apache-2.0 OR MIT	ChaCha random number generator
rand_core	0.10.1	Apache-2.0 OR MIT	Core random number generation traits and tools for implementation.
rand_core	0.6.4	Apache-2.0 OR MIT	Core random number generator traits and tools for implementation.
raw-window-handle	0.6.2	Apache-2.0 OR MIT OR Zlib	Interoperability library for Rust Windowing applications.
rayon	1.12.0	Apache-2.0 OR MIT	Simple work-stealing parallelism for Rust
rayon-core	1.13.0	Apache-2.0 OR MIT	Core APIs for Rayon
read-fonts	0.39.2	Apache-2.0 OR MIT	Reading OpenType font files.
redox_syscall	0.5.18	MIT	A Rust library to access raw Redox system calls
redox_users	0.5.3	MIT	A Rust library to access Redox users and groups functionality
ref-cast	1.0.27	Apache-2.0 OR MIT	Safely cast &T to &U where the struct U contains a single field of type T.
ref-cast-impl	1.0.27	Apache-2.0 OR MIT	Derive implementation for ref_cast::RefCast.
regex	1.13.1	Apache-2.0 OR MIT	An implementation of regular expressions for Rust. This implementation uses finite automata and guarantees linear time matching on all inputs.

Library	Version	License	What it does
regex-automata	0.4.18	Apache-2.0 OR MIT	Automata construction and matching using regular expressions.
regex-syntax	0.6.29	Apache-2.0 OR MIT	A regular expression parser.
regex-syntax	0.8.11	Apache-2.0 OR MIT	A regular expression parser.
rend	0.4.2	MIT	Endian-aware primitives for Rust
reqwest	0.13.5	Apache-2.0 OR MIT	higher level HTTP client library
resvg	0.47.0	Apache-2.0 OR MIT	An SVG rendering library.
rfc6979	0.6.0	Apache-2.0 OR MIT	Pure Rust implementation of RFC6979: Deterministic Usage of the Digital Signature Algorithm (DSA) and Elliptic Curve Digital Signature Algorithm (ECDSA)
rfd	0.16.0	MIT	Rusty File Dialog
rgb	0.8.53	MIT	`struct RGB/RGBA/etc.` for sharing pixels between crates + convenience methods for color manipulation. Allows no-copy high-level interoperability. Also adds…

Library	Version	License	What it does
ring	0.17.14	Apache-2.0 AND ISC	An experiment.
rkyv	0.7.46	MIT	Zero-copy deserialization framework for Rust
rkyv_derive	0.7.46	MIT	Derive macro for rkyv
rmcp	3.8.0	Apache-2.0	Rust SDK for Model Context Protocol
rmcp-macros	3.8.0	Apache-2.0	Rust SDK for Model Context Protocol macros library
roman-numerals-rs	3.1.0	0BSD OR CC0-1.0	Manipulate well-formed Roman numerals
rowan	0.15.19	Apache-2.0 OR MIT	Library for generic lossless syntax trees
roxmltree	0.20.0	Apache-2.0 OR MIT	Represent an XML as a read-only tree.
roxmltree	0.21.2	Apache-2.0 OR MIT	Represent an XML as a read-only tree.
rsa	0.10.0-rc.18	Apache-2.0 OR MIT	Pure Rust RSA implementation
russh	0.64.1	Apache-2.0	A client and server SSH library.
russh-cryptovec	0.62.0	Apache-2.0	A vector which zeroes its memory on clears and reallocations.

Library	Version	License	What it does
russh-sftp	3.0.1	Apache-2.0	SFTP subsystem supported server and client for Russh
russh-util	0.52.0	Apache-2.0	Runtime abstraction utilities for russh.
rust-patch	0.1.3	Apache-2.0 OR MIT	Patch structs with other structs
rust-patch-derive	0.1.3	Apache-2.0 OR MIT	Derive macro implementation for rust-patch
rust_decimal	1.43.0	MIT	Decimal number implementation written in pure Rust suitable for financial and fixed-precision calculations.
rustc-hash	1.1.0	Apache-2.0 OR MIT	speed, non-cryptographic hash used in rustc
rustc-hash	2.1.3	Apache-2.0 OR MIT	A speedy, non-cryptographic hashing algorithm used by rustc
rustc_version	0.4.1	Apache-2.0 OR MIT	A library for querying the version of a installed rustc compiler
rustix	0.38.44	Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	Safe Rust bindings to POSIX/Unix/Linux/Winsock-like syscalls
rustix	1.1.5	Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	Safe Rust bindings to POSIX/Unix/Linux/Winsock-like syscalls
rustls	0.23.46	Apache-2.0 OR ISC OR MIT	Rustls is a modern TLS library written in Rust.
rustls-native-certs	0.8.5	Apache-2.0 OR ISC OR MIT	rustls-native-certs allows rustls to use the platform native certificate store
rustls-pki-types	1.15.1	Apache-2.0 OR MIT	Shared types for the rustls PKI ecosystem
rustls-platform-verifier	0.7.1	Apache-2.0 OR MIT	rustls-platform-verifier supports verifying TLS certificates in rustls with the operating system verifier
rustls-platform-verifier-android	0.2.0	Apache-2.0 OR MIT	The internal JVM support component of the rustls-platform-verifier crate. You shouldn't depend on this directly.
rustls-webpki	0.103.16	ISC	Web PKI X.509 Certificate Verification.

Library	Version	License	What it does
rustversion	1.0.23	Apache-2.0 OR MIT	Conditional compilation according to rustc compiler version
rustybuzz	0.20.1	MIT	A complete harfbuzz shaping algorithm port to Rust.
ryu	1.0.23	Apache-2.0 OR BSL-1.0	Fast floating point to string conversion
salsa20	0.11.0	Apache-2.0 OR MIT	Pure Rust implementation of the Salsa20 stream cipher
same-file	1.0.6	MIT OR Unlicense	A simple crate for determining whether two file paths point to the same file.
schannel	0.1.30	MIT	Schannel bindings for rust, allowing SSL/TLS (e.g. https) without openssl
schemars	0.8.22	MIT	Generate JSON Schemas from Rust code
schemars	0.9.0	MIT	Generate JSON Schemas from Rust code
schemars	1.2.3	MIT	Generate JSON Schemas from Rust code
schemars_derive	0.8.22	MIT	Macros for #[derive(JsonSchema)], for use with schemars
schemars_derive	1.2.3	MIT	Macros for #[derive(JsonSchema)], for use with schemars
scopeguard	1.2.0	Apache-2.0 OR MIT	A RAII scope guard that will run a given closure when it goes out of scope, even if the code between panics (assuming unwinding panic). Defines the macros…
scrypt	0.12.0	Apache-2.0 OR MIT	Scrypt password-based key derivation function
seahash	4.1.0	MIT	A blazingly fast, portable hash function with proven statistical guarantees.
sec1	0.8.1	Apache-2.0 OR MIT	Pure Rust implementation of SEC1: Elliptic Curve Cryptography encoding formats including ASN.1 DER encoded private keys as well as the…
secret-service	5.2.0	Apache-2.0 OR MIT	Library to interface with Secret Service API
security-framework	3.7.0	Apache-2.0 OR MIT	Security.framework bindings for macOS and iOS
security-framework-sys	2.17.0	Apache-2.0 OR MIT	Apple `Security.framework` low-level FFI bindings
selectors	0.37.0	MPL-2.0	CSS Selectors matching for Rust
selectors	0.38.0	MPL-2.0	CSS Selectors matching for Rust
semver	1.0.28	Apache-2.0 OR MIT	Parser and evaluator for Cargo's flavor of Semantic Versioning

Library	Version	License	What it does
serde	1.0.229	Apache-2.0 OR MIT	A generic serialization/deserialization framework
serde-untagged	0.1.9	Apache-2.0 OR MIT	Serde `Visitor` implementation for deserializing untagged enums
serde_bytes	0.11.19	Apache-2.0 OR MIT	Optimized handling of `&[u8]` and `Vec<u8>` for Serde
serde_core	1.0.229	Apache-2.0 OR MIT	Serde traits only, with no support for derive -- use the `serde` crate instead
serde_derive	1.0.229	Apache-2.0 OR MIT	Macros 1.1 implementation of #[derive(Serialize, Deserialize)]
serde_derive_internals	0.29.1	Apache-2.0 OR MIT	AST representation used by Serde derive macros. Unstable.
serde_derive_internals	0.30.0	Apache-2.0 OR MIT	AST representation used by Serde derive macros. Unstable.
serde_json	1.0.154	Apache-2.0 OR MIT	A JSON serialization file format
serde_path_to_error	0.1.20	Apache-2.0 OR MIT	Path to the element that failed to deserialize
serde_repr	0.1.21	Apache-2.0 OR MIT	Derive Serialize and Deserialize that delegates to the underlying repr of a C-like enum.
serde_spanned	0.6.9	Apache-2.0 OR MIT	Serde-compatible spanned Value
serde_spanned	1.1.2	Apache-2.0 OR MIT	Serde-compatible spanned Value
serde_with	3.24.0	Apache-2.0 OR MIT	Custom de/serialization functions for Rust's serde
serde_with_macros	3.24.0	Apache-2.0 OR MIT	proc-macro library for serde_with
serde_yaml	0.8.26	Apache-2.0 OR MIT	YAML support for Serde
serde_yaml	0.9.34+deprecated	Apache-2.0 OR MIT	YAML data format for Serde
serdect	0.4.3	Apache-2.0 OR MIT	Constant-time serde serializer/deserializer helpers for data that potentially contains secrets (e.g. cryptographic keys)
serialize-to-javascript	0.1.2	Apache-2.0 OR MIT	Serialize a serde::Serialize item to a JavaScript literal template using serde_json
serialize-to-javascript-impl	0.1.2	Apache-2.0 OR MIT	Implementation detail of `serialize-to-javascript`
servo_arc	0.4.3	Apache-2.0 OR MIT	A fork of std::sync::Arc with some extra functionality and without weak references

Library	Version	License	What it does
sha-1	0.10.1	Apache-2.0 OR MIT	SHA-1 hash function. This crate is deprecated! Use the sha1 crate instead.
sha1	0.11.0	Apache-2.0 OR MIT	SHA-1 hash function
sha1_smol	1.0.1	BSD-3-Clause	Minimal dependency-free implementation of SHA1 for Rust.
sha2	0.10.9	Apache-2.0 OR MIT	Pure Rust implementation of the SHA-2 hash function family including SHA-224, SHA-256, SHA-384, and SHA-512.
sha2	0.11.1	Apache-2.0 OR MIT	Pure Rust implementation of the SHA-2 hash function family including SHA-224, SHA-256, SHA-384, and SHA-512.
sha3	0.12.0	Apache-2.0 OR MIT	Implementation of the SHA-3 family of cryptographic hash algorithms
shake	0.1.0	Apache-2.0 OR MIT	Implementation of the SHAKE family of extendable-output functions (XOFs)
shlex	2.0.1	Apache-2.0 OR MIT	Split a string into shell words, like Python's shlex.
signal-hook-registry	1.4.8	Apache-2.0 OR MIT	Backend crate for signal-hook
signature	3.0.0	Apache-2.0 OR MIT	Traits for cryptographic signature algorithms (e.g. ECDSA, Ed25519)
simd-adler32	0.3.10	MIT	A SIMD-accelerated Adler-32 hash algorithm implementation.
simd_cesu8	1.2.0	Apache-2.0 OR MIT	An extremely fast, SIMD accelerated, encoding and decoding library for CESU-8 and Modified UTF-8.
simdutf8	0.1.5	Apache-2.0 OR MIT	SIMD-accelerated UTF-8 validation.
simplecss	0.2.2	Apache-2.0 OR MIT	A simple CSS 2 parser and selector.
siphasher	1.0.4	Apache-2.0 OR MIT	SipHash-2-4 and SipHash-1-3 keyed hash functions in pure Rust, with 64-bit and 128-bit output and no_std support

Library	Version	License	What it does
skrifa	0.42.1	Apache-2.0 OR MIT	Metadata reader and glyph scaler for OpenType fonts.
slab	0.4.12	MIT	Pre-allocated storage for a uniform data type
slotmap	1.1.1	Zlib	Slotmap data structure
smallvec	1.16.3	Apache-2.0 OR MIT	'Small vector' optimization: store up to a small number of items on the stack
socket2	0.6.5	Apache-2.0 OR MIT	Utilities for handling networking sockets with a maximal amount of configuration possible intended.
softbuffer	0.4.8	Apache-2.0 OR MIT	Cross-platform software buffer
soup3	0.5.0	MIT	Soup crate for Rust
soup3-sys	0.5.0	MIT	Sys functions for the Rust bindings of the javacriptcore library
specta	2.0.0-rc.25	MIT	Easily export your Rust types to other languages
specta-macros	2.0.0-rc.25	MIT	Macros for specta. Don't use directly!
specta-serde	0.0.12	MIT	Serde support for Specta
specta-typescript	0.0.12	MIT	Export your Rust types to TypeScript
specta-util	0.0.12	MIT	High-level utilities for working with Specta
spin	0.9.9	MIT	Spin-based synchronization primitives
spki	0.8.1	Apache-2.0 OR MIT	X.509 Subject Public Key Info (RFC5280) describing public keys as well as their associated AlgorithmIdentifiers (i.e. OIDs)
sponge-cursor	0.1.0	Apache-2.0 OR MIT	Cursor for sponge-based absorption and squeezing
ssh-cipher	0.3.0	Apache-2.0 OR MIT	Pure Rust implementation of SSH symmetric encryption including support for the modern aes128-gcm@openssh.com/aes256-gcm@openssh.com and…
ssh-encoding	0.3.0	Apache-2.0 OR MIT	Pure Rust implementation of SSH data type decoders/encoders as described in RFC4251

Library	Version	License	What it does
ssh-key	0.7.0-rc.11	Apache-2.0 OR MIT	Pure Rust implementation of SSH key file format decoders/encoders as described in RFC4251/RFC4253 and OpenSSH key formats, as well as "sshsig" signatures and…
stable_deref_trait	1.2.1	Apache-2.0 OR MIT	An unsafe marker trait for types like Box and Rc that dereference to a stable address even when moved, and hence can be used with libraries such as owning_ref…
stacker	0.1.25	Apache-2.0 OR MIT	A stack growth library useful when implementing deeply recursive algorithms that may accidentally blow the stack.
static_vcruntime	3.0.0	Apache-2.0 OR MIT OR Zlib	Statically link the VCRuntime when using the MSVC toolchain.
strict-num	0.1.1	MIT	A collection of bounded numeric types
string_cache	0.9.0	Apache-2.0 OR MIT	A string interning library for Rust, developed as part of the Servo project.
string_cache_codegen	0.6.1	Apache-2.0 OR MIT	A codegen library for string-cache, developed as part of the Servo project.
strsim	0.11.1	MIT	Implementations of string similarity metrics. Includes Hamming, Levenshtein, OSA, Damerau-Levenshtein, Jaro, Jaro-Winkler, and Sørensen-Dice.
strum	0.27.2	MIT	Helpful macros for working with enums and strings
strum_macros	0.27.2	MIT	Helpful macros for working with enums and strings
subsetter	0.2.6	Apache-2.0 OR MIT	Reduces the size and coverage of OpenType fonts.
subtle	2.6.1	BSD-3-Clause	Pure-Rust traits and utilities for constant-time cryptographic implementations.
svgtypes	0.16.2	Apache-2.0 OR MIT	SVG types parser.
swift-rs	1.0.9	Apache-2.0 OR MIT	Call Swift from Rust with ease!
syn	1.0.109	Apache-2.0 OR MIT	Parser for Rust source code
syn	2.0.119	Apache-2.0 OR MIT	Parser for Rust source code
syn	3.0.9	Apache-2.0 OR MIT	Parser for Rust source code

Library	Version	License	What it does
sync_wrapper	1.0.2	Apache-2.0	A tool for enlisting the compiler's help in proving the absence of concurrency

Library	Version	License	What it does
synstructure	0.14.0	MIT	Helper methods and macros for custom derives
syntect	5.3.0	MIT	library for high quality syntax highlighting and code intelligence using Sublime Text's grammars
system-configuration	0.7.0	Apache-2.0 OR MIT	Bindings to SystemConfiguration framework for macOS
system-configuration-sys	0.6.0	Apache-2.0 OR MIT	Low level bindings to SystemConfiguration framework for macOS
system-deps	6.2.2	Apache-2.0 OR MIT	Discover and configure system dependencies from declarative dependencies in Cargo.toml
tao	0.37.1	Apache-2.0	Cross-platform window manager library.
tao-macros	0.1.4	Apache-2.0 OR MIT	Proc macros for tao
tap	1.0.1	MIT	Generic extensions for tapping values in Rust
tar	0.4.46	Apache-2.0 OR MIT	A Rust implementation of a TAR file reader and writer. This library does not currently handle compression, but it is abstract over all I/O readers and writers…
target-lexicon	0.12.16	Apache-2.0 WITH LLVM-exception	Targeting utilities for compilers and related tools
tauri	2.12.4	Apache-2.0 OR MIT	Make tiny, secure apps for all desktop platforms with Tauri
tauri-build	2.7.3	Apache-2.0 OR MIT	build time code to pair with https://crates.io/crates/tauri
tauri-codegen	2.7.3	Apache-2.0 OR MIT	code generation meant to be consumed inside of `tauri` through `tauri-build` or `tauri-macros`
tauri-macros	2.7.3	Apache-2.0 OR MIT	Macros for the tauri crate.
tauri-plugin	2.7.2	Apache-2.0 OR MIT	Build script and runtime Tauri plugin definitions

Library	Version	License	What it does
tauri-plugin-dialog	2.8.3	Apache-2.0 OR MIT	Native system dialogs for opening and saving files along with message dialogs on your Tauri application.
tauri-plugin-fs	2.7.1	Apache-2.0 OR MIT	Access the file system.
tauri-plugin-opener	2.7.2	Apache-2.0 OR MIT	Open files and URLs using their default application.
tauri-plugin-process	2.4.0	Apache-2.0 OR MIT	Access the current process of your Tauri application.
tauri-plugin-updater	2.14.0	Apache-2.0 OR MIT	In-app updates for Tauri applications.
tauri-plugin-window-state	2.5.0	Apache-2.0 OR MIT	Save window positions and sizes and restore them when the app is reopened.
tauri-runtime	2.12.2	Apache-2.0 OR MIT	Runtime for Tauri applications
tauri-runtime-wry	2.12.2	Apache-2.0 OR MIT	Wry bindings to the Tauri runtime
tauri-specta	2.0.0-rc.25	MIT	Completely typesafe Tauri commands
tauri-utils	2.10.2	Apache-2.0 OR MIT	Utilities for Tauri
tauri-winres	0.3.7	MIT	Create and set windows icons and metadata for executables
tempfile	3.27.0	Apache-2.0 OR MIT	A library for managing temporary files and directories.
tendril	0.5.1	Apache-2.0 OR MIT	Compact buffer/string type for zero-copy parsing
text-size	1.1.1	Apache-2.0 OR MIT	Newtypes for text offsets
thin-vec	0.2.21	Apache-2.0 OR MIT	A Vec that takes up less space on the stack.

Library	Version	License	What it does
thiserror	1.0.69	Apache-2.0 OR MIT	derive(Error)
thiserror	2.0.21	Apache-2.0 OR MIT	derive(Error)
thiserror-impl	1.0.69	Apache-2.0 OR MIT	Implementation detail of the `thiserror` crate
thiserror-impl	2.0.21	Apache-2.0 OR MIT	Implementation detail of the `thiserror` crate
tikv-jemalloc-sys	0.6.1+5.3.0-1-ge13ca993e8ccb9ba9847cc330696e02839f328f7	Apache-2.0 OR MIT	Rust FFI bindings to jemalloc
tikv-jemallocator	0.6.1	Apache-2.0 OR MIT	A Rust allocator backed by jemalloc
time	0.3.55	Apache-2.0 OR MIT	Date and time library. Fully interoperable with

Library	Version	License	What it does
			the standard library. Mostly compatible with #![no_std].
time-core	0.1.9	Apache-2.0 OR MIT	This crate is an implementation detail and should not be relied upon directly.
time-macros	0.2.32	Apache-2.0 OR MIT	Procedural macros for the time crate. This crate is an implementation detail and should not be relied upon directly.
tiny-skia	0.12.0	BSD-3-Clause	A tiny Skia subset ported to Rust.
tiny-skia-path	0.12.0	BSD-3-Clause	A tiny-skia Bezier path implementation
tiny_http	0.12.0	Apache-2.0 OR MIT	Low level HTTP server library
tinystr	0.8.4	Unicode-3.0	A small ASCII-only bounded length string representation.
tinyvec	1.13.3	Apache-2.0 OR MIT OR Zlib	`tinyvec` provides 100% safe vec-like data structures.
tokio	1.54.0	MIT	An event-driven, non-blocking I/O platform for writing asynchronous I/O backed applications.
tokio-macros	2.7.2	MIT	Tokio's proc macros.
tokio-rustls	0.26.6	Apache-2.0 OR MIT	Asynchronous TLS/SSL streams for Tokio using Rustls.
tokio-util	0.7.20	MIT	Additional utilities for working with Tokio.
toml	0.5.11	Apache-2.0 OR MIT	A native Rust encoder and decoder of TOML-formatted files and streams. Provides implementations of the standard Serialize/Deserialize traits for TOML data to…
toml	0.8.2	Apache-2.0 OR MIT	A native Rust encoder and decoder of TOML-formatted files and streams. Provides implementations of

Library	Version	License	What it does
			the standard Serialize/Deserialize traits for TOML data to…
toml	1.1.8+spec-1.1.0	Apache-2.0 OR MIT	A native Rust encoder and decoder of TOML-formatted files and streams. Provides implementations of the standard Serialize/Deserialize traits for TOML data to…
toml_datetime	0.6.3	Apache-2.0 OR MIT	A TOML-compatible datetime type
toml_datetime	1.1.2+spec-1.1.0	Apache-2.0 OR MIT	A TOML-compatible datetime type
toml_edit	0.19.15	Apache-2.0 OR MIT	Yet another format-preserving TOML parser.
toml_edit	0.20.2	Apache-2.0 OR MIT	Yet another format-preserving TOML parser.
toml_edit	0.25.17+spec-1.1.0	Apache-2.0 OR MIT	Yet another format-preserving TOML parser.
toml_parser	1.1.5+spec-1.1.0	Apache-2.0 OR MIT	Yet another format-preserving TOML parser.
toml_writer	1.1.3+spec-1.1.0	Apache-2.0 OR MIT	A low-level interface for writing out TOML
tower	0.5.3	MIT	Tower is a library of modular and reusable components for building robust clients and servers.
tower-http	0.6.11	MIT	Tower middleware and utilities for HTTP clients and servers
tower-layer	0.3.3	MIT	Decorates a `Service` to allow easy composition between `Services`.
tower-service	0.3.3	MIT	Trait representing an asynchronous, request / response based, client or server.
tracing	0.1.44	MIT	Application-level tracing for Rust.
tracing-attributes	0.1.31	MIT	Procedural macro attributes for automatically instrumenting functions.
tracing-core	0.1.36	MIT	Core primitives for application-level tracing.
tray-icon	0.25.1	Apache-2.0 OR MIT	Create tray icons for desktop applications

Library	Version	License	What it does
try-lock	0.2.5	MIT	A lightweight atomic lock.
ttf-parser	0.25.1	Apache-2.0 OR MIT	A high-level, safe, zero-allocation font parser for TrueType, OpenType, and AAT.
twelf	0.15.0	MIT	Twelf is a configuration solution for Rust including 12-Factor support. It is designed with layers in order to configure different sources and formats to build…
two-face	0.4.5	Apache-2.0 OR MIT	Extra syntax and theme definitions for syntect
two-face	0.5.2+bat-0.26.1	Apache-2.0 OR MIT	Extra syntax and theme definitions for syntect
typed-arena	2.0.2	MIT	The arena, a fast but limited type of allocator
typed-builder	0.23.2	Apache-2.0 OR MIT	Compile-time type-checked builder derive
typed-builder-macro	0.23.2	Apache-2.0 OR MIT	Compile-time type-checked builder derive
typeid	1.0.3	Apache-2.0 OR MIT	Const TypeId and non-'static TypeId
typenum	1.20.1	Apache-2.0 OR MIT	Typenum is a Rust library for type-level numbers evaluated at compile time. It currently supports bits, unsigned integers, and signed integers. It also provides…
typst	0.15.1	Apache-2.0	A new markup-based typesetting system that is powerful and easy to learn.
typst-assets	0.15.1	Apache-2.0	Assets for the Typst compiler.
typst-eval	0.15.1	Apache-2.0	Typst's code interpreter.
typst-html	0.15.1	Apache-2.0	Typst's HTML exporter.
typst-layout	0.15.1	Apache-2.0	Typst's layout engine.
typst-library	0.15.1	Apache-2.0	Typst's standard library.
typst-macros	0.15.1	Apache-2.0	Proc-macros for Typst.
typst-pdf	0.15.1	Apache-2.0	PDF exporter for Typst.
typst-realize	0.15.1	Apache-2.0	Typst's realization subsystem.
typst-svg	0.15.1	Apache-2.0	SVG exporter for Typst.
typst-syntax	0.15.1	Apache-2.0	Parser and syntax tree for Typst.
typst-timing	0.15.1	Apache-2.0	Performance timing for Typst.
typst-utils	0.15.1	Apache-2.0	Utilities for Typst.
uds_windows	1.2.1	MIT	Unix Domain Sockets for Windows!

Library	Version	License	What it does
unic-langid	0.9.6	Apache-2.0 OR MIT	API for managing Unicode Language Identifiers
unic-langid-impl	0.9.6	Apache-2.0 OR MIT	API for managing Unicode Language Identifiers
unic-langid-macros	0.9.6	Apache-2.0 OR MIT	API for managing Unicode Language Identifiers
unic-langid-macros-impl	0.9.6	Apache-2.0 OR MIT	API for managing Unicode Language Identifiers
unicode-bidi	0.3.18	Apache-2.0 OR MIT	Implementation of the Unicode Bidirectional Algorithm
unicode-bidi-mirroring	0.4.0	Apache-2.0 OR MIT	Unicode Bidi Mirroring property detection
unicode-ccc	0.4.0	Apache-2.0 OR MIT	Unicode Canonical Combining Class detection
unicode-ident	1.0.27	(MIT OR Apache-2.0) AND Unicode-3.0	Determine whether characters have the XID_Start or XID_Continue properties according to Unicode Standard Annex #31
unicode-math-class	0.1.0	Apache-2.0 OR MIT	Determine the Unicode class of a mathematical character.
unicode-normalization	0.1.26	(MIT OR Apache-2.0) AND Unicode-3.0	This crate provides functions for normalization of Unicode strings, including Canonical and Compatible Decomposition and Recomposition, as described in Unicode…
unicode-properties	0.1.5	(MIT OR Apache-2.0) AND Unicode-3.0	Query character Unicode properties according to UAX #44 and UTR #51.
unicode-script	0.5.9	Apache-2.0 OR MIT	This crate exposes the Unicode `Script` and `Script_Extension` properties from UAX #24
unicode-segmentation	1.13.3	Apache-2.0 OR MIT	This crate provides Grapheme Cluster, Word and Sentence boundaries according to Unicode Standard Annex #29 rules.
unicode-vo	0.1.0	Apache-2.0 OR MIT	Unicode vertical orientation detection
unicode-width	0.2.3	Apache-2.0 OR MIT	Determine displayed width of `char` and `str` types according to Unicode Standard Annex #11 rules.

Library	Version	License	What it does
unicode-xid	0.2.6	Apache-2.0 OR MIT	Determine whether characters have the XID_Start or XID_Continue properties according to Unicode Standard Annex #31.
universal-hash	0.6.1	Apache-2.0 OR MIT	Traits which describe the functionality of universal hash functions (UHFs)
unsafe-libyaml	0.2.11	MIT	libyaml transpiled to rust by c2rust
unscanny	0.1.0	Apache-2.0 OR MIT	Painless string scanning.
untrusted	0.9.0	ISC	Safe, fast, zero-panic, zero-crashing, zero-allocation parsing of untrusted inputs in Rust.
url	2.5.8	Apache-2.0 OR MIT	URL library for Rust, based on the WHATWG URL Standard
urlpattern	0.6.0	MIT	rust-urlpattern is a Rust implementation of the URLPattern standard
usvg	0.47.0	Apache-2.0 OR MIT	An SVG simplification library.
utf16_iter	1.1.0	Apache-2.0 OR MIT	Iterator by char over potentially-invalid UTF-16 in &[u16]
utf8_iter	1.0.4	Apache-2.0 OR MIT	Iterator by char over potentially-invalid UTF-8 in &[u8]
utf8parse	0.2.2	Apache-2.0 OR MIT	Table-driven UTF-8 parser
utf_types	1.0.0	Apache-2.0 OR MIT	Types for UTF-8 and UTF-16 code units and code unit sequences
uuid	1.28.0	Apache-2.0 OR MIT	A library to generate and parse UUIDs.
vcpkg	0.2.15	Apache-2.0 OR MIT	A library to find native dependencies in a vcpkg tree at build time in order to be used in Cargo build scripts.
vello_common	0.0.8	Apache-2.0 OR MIT	Core data structures and utilities shared across the Vello rendering, including geometry processing and tiling logic.

Library	Version	License	What it does
vello_common	0.0.9	Apache-2.0 OR MIT	Core data structures and utilities shared across the Vello rendering, including geometry

Library	Version	License	What it does
			processing and tiling logic.
vello_cpu	0.0.8	Apache-2.0 OR MIT	A CPU-based renderer for Vello, optimized for SIMD and multi-threaded execution.
version-compare	0.2.1	MIT	Rust library to easily compare version numbers with no specific format, and test against various comparison operators.
version_check	0.9.5	Apache-2.0 OR MIT	Tiny crate to check the version of the installed/ running rustc.
vswhom	0.1.0	MIT	FFI to Jon Blow's VS discovery script
vswhom-sys	0.1.3	MIT	Pure FFI to Jon Blow's VS discovery script
walkdir	2.5.0	MIT OR Unlicense	Recursively walk a directory.
want	0.3.2	MIT	Detect when another Future wants a result.
wasi	0.11.1+wasi-snapshot-preview1	Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	Experimental WASI API bindings for Rust
wasip2	1.0.4+wasi-0.2.12	Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	WASIp2 API bindings for Rust
wasm-bindgen	0.2.129	Apache-2.0 OR MIT	Easy support for interacting between JS and Rust.
wasm-bindgen-futures	0.4.79	Apache-2.0 OR MIT	Bridging the gap between Rust Futures and JavaScript Promises
wasm-bindgen-macro	0.2.129	Apache-2.0 OR MIT	Definition of the `#[wasm_bindgen]` attribute, an internal dependency
wasm-bindgen-macro-support	0.2.129	Apache-2.0 OR MIT	Implementation APIs for the `#[wasm_bindgen]` attribute
wasm-bindgen-shared	0.2.129	Apache-2.0 OR MIT	Shared support between wasm-bindgen

Library	Version	License	What it does
			and wasm-bindgen cli, an internal dependency.
wasm-streams	0.5.0	Apache-2.0 OR MIT	Bridging between web streams and Rust streams using WebAssembly
wasmi	1.1.0	Apache-2.0 OR MIT	WebAssembly interpreter
wasmi_collections	1.1.0	Apache-2.0 OR MIT	Specialized data structures for the Wasmi interpreter
wasmi_core	1.1.0	Apache-2.0 OR MIT	Core primitives for the wasmi WebAssembly interpreter
wasmi_ir	1.1.0	Apache-2.0 OR MIT	WebAssembly interpreter internal bytecode representation
wasmparser	0.239.0	Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	A simple event-driven library for parsing WebAssembly binary files.
wax	0.7.0	MIT	Opinionated and portable globs that can be matched against paths and directory trees.
web-sys	0.3.106	Apache-2.0 OR MIT	Bindings for all Web APIs, a procedurally generated crate from WebIDL
web-time	1.1.0	Apache-2.0 OR MIT	Drop-in replacement for std::time for Wasm in browsers
web_atoms	0.2.6	Apache-2.0 OR MIT	Atoms for xml5ever and html5ever
webkit2gtk	2.0.2	MIT	Rust bindings for webkit-gtk library
webkit2gtk-sys	2.0.2	MIT	Rust binding for webkit-gtk library
webp	0.3.1	Apache-2.0 OR MIT	WebP conversion library.
webpki-root-certs	1.0.10	CDLA-Permissive-2.0	Mozilla trusted certificate authorities in self-signed X.509 format for

Library	Version	License	What it does
			use with crates other than webpki
webview2-com	0.39.1	MIT	Rust bindings for the WebView2 COM APIs

Library	Version	License	What it does
webview2-com-macros	0.8.1	MIT	Macros which generate callback implementations for WebView2 COM APIs
webview2-com-sys	0.39.1	MIT	Bindings generated with the windows crate for the WebView2 COM APIs
weezl	0.1.12	Apache-2.0 OR MIT	Fast LZW compression and decompression.
which	5.0.0	MIT	A Rust equivalent of Unix command "which". Locate installed executable in cross platforms.
winapi	0.3.9	Apache-2.0 OR MIT	Raw FFI bindings for all of Windows API.
winapi-i686-pc-windows-gnu	0.4.0	Apache-2.0 OR MIT	Import libraries for the i686-pc-windows-gnu target. Please don't use this crate directly, depend on winapi instead.
winapi-util	0.1.11	MIT OR Unlicense	A dumping ground for high level safe wrappers over windows-sys.
winapi-x86_64-pc-windows-gnu	0.4.0	Apache-2.0 OR MIT	Import libraries for the x86_64-pc-windows-gnu target. Please don't use this crate directly, depend on winapi instead.
window-vibrancy	0.8.1	Apache-2.0 OR MIT	Make your windows vibrant.
windows	0.62.2	Apache-2.0 OR MIT	Rust for Windows
windows-collections	0.3.2	Apache-2.0 OR MIT	Windows collection types
windows-core	0.62.2	Apache-2.0 OR MIT	Core type support for COM and Windows
windows-future	0.3.2	Apache-2.0 OR MIT	Windows async types
windows-implement	0.60.2	Apache-2.0 OR MIT	The implement macro for the Windows crates
windows-interface	0.59.3	Apache-2.0 OR MIT	The interface macro for the Windows crates
windows-link	0.2.1	Apache-2.0 OR MIT	Linking for Windows
windows-native-keyring-store	1.1.0	Apache-2.0 OR MIT	Windows native credential store for keyring
windows-numerics	0.3.1	Apache-2.0 OR MIT	Windows numeric types
windows-registry	0.6.1	Apache-2.0 OR MIT	Windows registry

Library	Version	License	What it does
windows-result	0.4.1	Apache-2.0 OR MIT	Windows error handling
windows-strings	0.5.1	Apache-2.0 OR MIT	Windows string types
windows-sys	0.45.0	Apache-2.0 OR MIT	Rust for Windows
windows-sys	0.48.0	Apache-2.0 OR MIT	Rust for Windows
windows-sys	0.52.0	Apache-2.0 OR MIT	Rust for Windows
windows-sys	0.60.2	Apache-2.0 OR MIT	Rust for Windows
windows-sys	0.61.2	Apache-2.0 OR MIT	Rust for Windows
windows-targets	0.42.2	Apache-2.0 OR MIT	Import libs for Windows
windows-targets	0.48.5	Apache-2.0 OR MIT	Import libs for Windows
windows-targets	0.52.6	Apache-2.0 OR MIT	Import libs for Windows
windows-targets	0.53.5	Apache-2.0 OR MIT	Import libs for Windows
windows-threading	0.2.1	Apache-2.0 OR MIT	Windows threading
windows-version	0.1.7	Apache-2.0 OR MIT	Windows version information
windows_aarch64_gnullvm	0.42.2	Apache-2.0 OR MIT	Import lib for Windows
windows_aarch64_gnullvm	0.48.5	Apache-2.0 OR MIT	Import lib for Windows
windows_aarch64_gnullvm	0.52.6	Apache-2.0 OR MIT	Import lib for Windows
windows_aarch64_gnullvm	0.53.1	Apache-2.0 OR MIT	Import lib for Windows
windows_aarch64_msvc	0.42.2	Apache-2.0 OR MIT	Import lib for Windows
windows_aarch64_msvc	0.48.5	Apache-2.0 OR MIT	Import lib for Windows
windows_aarch64_msvc	0.52.6	Apache-2.0 OR MIT	Import lib for Windows
windows_aarch64_msvc	0.53.1	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_gnu	0.42.2	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_gnu	0.48.5	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_gnu	0.52.6	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_gnu	0.53.1	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_gnullvm	0.52.6	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_gnullvm	0.53.1	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_msvc	0.42.2	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_msvc	0.48.5	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_msvc	0.52.6	Apache-2.0 OR MIT	Import lib for Windows
windows_i686_msvc	0.53.1	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_gnu	0.42.2	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_gnu	0.48.5	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_gnu	0.52.6	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_gnu	0.53.1	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_gnullvm	0.42.2	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_gnullvm	0.48.5	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_gnullvm	0.52.6	Apache-2.0 OR MIT	Import lib for Windows

Library	Version	License	What it does
windows_x86_64_gnullvm	0.53.1	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_msvc	0.42.2	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_msvc	0.48.5	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_msvc	0.52.6	Apache-2.0 OR MIT	Import lib for Windows
windows_x86_64_msvc	0.53.1	Apache-2.0 OR MIT	Import lib for Windows
winnow	0.5.40	MIT	A byte-oriented, zero-copy, parser combinators library
winnow	1.0.4	MIT	A byte-oriented, zero-copy, parser combinators library
winreg	0.56.0	MIT	Rust bindings to MS Windows Registry API
wit-bindgen	0.57.1	Apache-2.0 WITH LLVM-exception OR Apache-2.0 OR MIT	Rust bindings generator and runtime support for WIT and the component model. Used when compiling Rust programs to the component model.
wnaf	0.14.1	Apache-2.0 OR MIT	wNAF (w-ary non-adjacent form) variable-time scalar multiplication implemented generically over elliptic curve groups with full `no_alloc` support, including…
write-fonts	0.48.1	Apache-2.0 OR MIT	Writing font files.
write16	1.0.0	Apache-2.0 OR MIT	A UTF-16 analog of the Write trait
writeable	0.6.4	Unicode-3.0	A more efficient alternative to fmt::Display
wry	0.57.0	Apache-2.0 OR MIT	Cross-platform WebView rendering library
wyz	0.5.1	MIT	myrrlyn's utility collection
x11	2.21.0	MIT	X11 library bindings for Rust
x11-dl	2.21.0	MIT	X11 library bindings for Rust
xattr	1.6.1	Apache-2.0 OR MIT	unix extended filesystem attributes
xmlwriter	0.1.0	MIT	A simple, streaming XML writer.
xmp-writer	0.3.3	Apache-2.0 OR MIT	Write XMP metadata, step by step.
yaml-rust	0.4.5	Apache-2.0 OR MIT	The missing YAML 1.2 parser for rust

Library	Version	License	What it does
yoke	0.8.3	Unicode-3.0	Abstraction allowing borrowed data to be carried along with the backing data it borrows from
yoke-derive	0.8.4	Unicode-3.0	Custom derive for the yoke crate
zbus	5.19.0	MIT	API for D-Bus communication
zbus-secret-service-keyring-store	1.0.1	Apache-2.0 OR MIT	ZBus Secret Service credential store for keyring
zbus_macros	5.19.0	MIT	proc-macros for zbus
zbus_names	4.3.5	MIT	A collection of D-Bus bus names types (deprecated: merged into zbus)
zcheapstr	1.1.0	MIT	An immutable, cheaply cloneable string type with borrowed, static and shared variants
zerocopy	0.8.63	Apache-2.0 OR BSD-2-Clause OR MIT	Zerocopy makes zero-cost memory manipulation effortless. We write "unsafe" so you don't have to.
zerocopy-derive	0.8.63	Apache-2.0 OR BSD-2-Clause OR MIT	Custom derive for traits from the zerocopy crate
zerofrom	0.1.8	Unicode-3.0	ZeroFrom trait for constructing
zerofrom-derive	0.1.8	Unicode-3.0	Custom derive for the zerofrom crate
zeroize	1.9.1	Apache-2.0 OR MIT	Securely clear secrets from memory with a simple trait built on stable Rust primitives which guarantee memory is zeroed using an operation will not be…
zerotrie	0.2.5	Unicode-3.0	A data structure that efficiently maps strings to integers
zerovec	0.11.8	Unicode-3.0	Zero-copy vector backed by a byte array
zerovec-derive	0.11.6	Unicode-3.0	Custom derive for the zerovec crate
zip	4.6.1	MIT	Library to support the reading and writing of zip files.
zlib-rs	0.6.8	Zlib	A memory-safe zlib implementation written in rust

Library	Version	License	What it does
zmij	1.0.23	MIT	A double-to-string conversion algorithm based on Schubfach and xjb
zune-core	0.5.3	Apache-2.0 OR MIT OR Zlib	Core utilities for image processing in the zune family of crates
zune-jpeg	0.5.15	Apache-2.0 OR MIT OR Zlib	A fast, correct and safe jpeg decoder
zvariant	5.15.1	MIT	D-Bus & GVariant encoding & decoding (deprecated: merged into zbus)
zvariant_derive	5.15.1	MIT	D-Bus & GVariant encoding & decoding (deprecated: merged into zbus)
zvariant_utils	4.2.0	MIT	Various utilities used internally by the zvariant crate.

Backlog

Open work and open decisions. Items keep their number for good; when one is done or decided, it moves to Done with a short note, and items that no longer apply go to its **Dropped** section.

q.e.d. is a single-user desktop app (Rust + Tauri) that turns a podcaster's content into a static site and publishes it. Decided 2026-10-21; it replaces the earlier Phoenix web app.

To decide

Nothing open.

To build

Phase 4: Podcast Publisher Import

- **#123 Import from Podlove Publisher.** Bring a podcast over from WordPress with the Podlove Publisher[38] plugin, mostly to archive it without having to keep WordPress running (user, 2026-10-28): podcast settings, episodes with shownotes, chapters, contributors (#047), transcripts (#048), cover art and the audio in every format, the old episode guids (so apps don't show everything again) and the old addresses as redirects. To decide: the source (Podlove's own export file, the feeds, the WordPress REST API, or a mix), and how much of the shownotes' HTML becomes Markdown. Related: #034 (import from a feed or jekyll-octopod).
- **#034 Feed import.** Import an existing podcast from its RSS feed or from a jekyll-octopod site (compare `octopod import`). Keep each episode's old guid (octopod used the episode's address) and its old address as a redirect. Moved here on 2026-10-28, with #123.
- **#124 Feed redirects.** Podlove Publisher's feeds live at other addresses (`/feed/mp3/`, or `/?feed=…` without pretty links) than q.e.d.'s (`/episodes.mp3.rss`), and subscribers must keep getting new episodes after the move (from #123). A static site can't answer with an HTTP 301 by itself. To decide, per target: a `<itunes:new-feed-url>` feed at the old address (Apple and most apps then switch for good; works on any host, also GitHub Pages, but not for `?feed=` addresses), an `.htaccess` with `Redirect 301` rules for Apache hosts (handles query strings too), or both. Old feed addresses per format would be a podcast setting, filled in by the import.
- **#151 New manual screenshots.** The screenshots in `docs/media/manual/` still show the app's Lucide icons from before #149, and the theme tab is slightly older than that. Retake them all (theme *catppuccin-mocha*, as AGENTS.md says) at the end of phase 4, when its changes to the app are in (user, 2026-10-31).

Phase 5: Roll out

- **#070 Code signing.** For downloads from the releases page, not app stores. macOS: notarization needs the Apple Developer Program (99 USD a year); without it, macOS 15 and later make people allow the app in System Settings → Privacy & Security → Open Anyway (right-click → Open no longer works; the instructions say so since 2026-10-29). Windows: the SignPath Foundation signs open-source projects for free (builds from CI; check their terms); SmartScreen still warns until the app has a reputation. Updates are already safe through the updater's own signature. Moved here on 2026-10-29: worth it once Mac users come in.

[38] https://podlove.org/podlove-podcast-publisher/

Done

Backlog items that are done, decided or dropped. Open work is in the backlog. Items that are partly done stay there until the rest is done.

Done / decided

- **#072 Block file format.** Read and write pages as Markdown with block containers (`::: hero` … `:::`, block fields as TOML) in qed-core. Writing a parsed page gives the same file back, so hand edits survive the block editor. *Done 2026-10-21*: parsing with file and line in every error; writing with `qed_core::page_writer` (unchanged files and blocks stay byte for byte, changed TOML is merged with `toml_edit` so comments survive, defaults are left out, atomic saves). The app edits page settings in a side panel; editing blocks comes with #043.
- **#056 Start page.** Every site gets a `home` page on creation (Hero and Text). *Done 2026-10-21* with "New site" (#046): `pages/index.md` with a Hero (the site title) and a short Text. Phase 2 seeds podcast sites with Latest episodes and Subscribe.
- **Desktop app.** q.e.d. is a single-user desktop app for podcasters who can't run a static site generator themselves. Stack: Rust + Tauri 2, Svelte UI, files instead of a database. Open source, built with Claude as a showcase. Claude writes the code and explains Rust concepts (ownership, borrowing, lifetimes) as they come up.
- **License.** GPL-3.0.
- **#057 Data location.** One folder per site, picked by the user. No database: one `site.toml` plus pages and posts as Markdown files with TOML front matter (#046); blocks are containers inside the Markdown (#072); drafts and history via git (#073). The app remembers recently opened site folders. Publish secrets stay in the OS keychain (#063).
- **#021 Site UI.** List, create and edit sites. *Done 2026-10-21*: the site title in the header is a menu with up to ten recent sites (`src/lib/recentSites.ts`, in local storage like the app theme; the old `qed:last-site` moves into it), "Open site folder…" and "New site…"; the start screen lists the recent sites too; × takes a site off the list. Switching asks first if the open page has unsaved changes. "Close site" (2026-10-22) goes back to the start screen, and the next start stays there until a site is opened (`qed:site-closed`). Creating is #046, editing is the Site tab.
- **#055 Sites.** Several sites per installation: done with #021 and #046. Phase 1 sites are pages-only; phase 2 adds the podcast switch (#066).
- **#046 Site folder, phase 1.** `site.toml` (title, language, theme, menus), `targets.toml` (publish targets without secrets, git-ignored), pages and posts as Markdown files with TOML front matter, `media/` for images, `assets/` for downloads, build output in a git-ignored folder. Old paths for redirects go into front matter. *Reading done 2026-10-21* (`qed_core::site::open`, `examples/demo-site`). *Creating pages done 2026-10-21* (`site::create_page`, "New page" dialog; posts get a date prefix; never overwrites). *Deleting pages done 2026-10-21* (`site::delete_page`): the confirmation names the pages that link to it; the start page and pages with pages below them can't be deleted; the Versions tab lists deleted pages and restores them (`history::deleted_pages`, `restore_page`). *New site done 2026-10-21* (`site::create_site`, "New site…" dialog on the empty start screen and in the Site tab): title, language, location (Documents by default) and folder name (from the title); the folder must be new or empty (hidden files are fine); writes `site.toml`, the start page (#056), empty `media/` and `assets/`, and starts the version history. *Moving and renaming done 2026-10-21* (`qed_core::moving`, "Move or rename" in the page editor): pages below move along, each moved page gets its old address in `redirect_from` (and a redirect that a new address takes over is dropped), links to moved pages and relative links in them are updated in Markdown and buttons

(`markdown::replace_link_targets` keeps the rest of the text as it was), and the open version is closed first. Links that were broken already, and relative links to built pages (`../about/`), stay as they are.

- **#079 A note before publishing.** Done 2026-10-21: the Publish button opens a dialog that names the destination and, when there's an open version or a change since the last version (`HistoryStatus::needs_note`), asks what's new on the site; publishing can't start without it (`history::close_for_publish`, `HistoryError::NoteNeeded`). The note closes the version; the publish tag stays as it was. Without anything new, the last version is published as it is. Replaces the automatic note "Before publishing to X". The dialog also lists the files changed since the last publish to that target (`history::since_publish`, from the publish tags), since a version saved in between is still new there; fixed 2026-10-21 after "Nothing new" hid three unpublished posts.
- **#062 Release builds.** Built 2026-10-21: `.github/workflows/release.yml` on github.com/informatom/qed builds, for a `v*` tag, a universal macOS `.dmg`, a Windows `.exe` (NSIS), and a Linux `.deb` and `.AppImage` (on Ubuntu 22.04 for older systems), and attaches them to a draft release. The tag must match the version in `Cargo.toml` and `package.json` the app takes its version from `Cargo.toml`. The app identifier is now `io.github.informatom.qed`. Unsigned until #070. Steps in Development → Releasing. First run 2026-10-21: tag `v0.1.0` built all four installers in about 23 minutes (draft release, AppImage 84 MB, deb 6 MB, dmg 9 MB, exe 4 MB).
- **#074 Nested blocks.** Done 2026-10-21: a column holds blocks (Text, Image, Text with image, Download, FAQ, Call to action) plus an optional title. In the file, `:::: columns` (four colons) wraps the columns and `::: column` lines start each one and hold its title; the writer keeps what's written inside columns like anywhere else. Blocks render one at a time (`block.html`), so a column renders its blocks the same way; the editor's block list (`BlockList.svelte`) uses itself for each column. Only Columns nest; a full-width Section block could use the same format later (#087).
- **#073 Git history.** Done 2026-10-21 (`qed_core::history`, git2 with bundled libgit2): a top-level Versions tab starts the history (git init plus `.gitignore` for `_site/` and `targets.toml`), shows changed files, saves versions with a note and lists them. Publishing saves a version first and tags it (`publish/<target>/<UTC time>`). *Open versions 2026-10-21*: every save in the app goes into an open version (a commit marked `qed-draft: true`, amended by later saves); "Close version" gives it its note, publishing closes it too. q.e.d. amends only its own open version, never someone else's commit and never one that was pushed. Deleting a page closes the open version first, so the page can come back. Possible later: closing open versions by themselves after a while (#083). Each page has a Versions tab (Content / Versions / Settings) with what each version changed and "Restore", which puts it into the editor as unsaved changes. A site folder inside a larger repository (like `docs/`) works too: q.e.d. commits only that folder, never other staged changes, and prefixes its tags (`docs/publish/…`). Still open: restoring the whole site (#082), signed commits (#084).
- **#078 Assets for downloads.** Done 2026-10-21: `assets/` holds files to download (PDFs, but any type); `media/` takes only images. An Assets tab lists, imports (clean names) and deletes them like Media (`qed_core::media` with a `Library` parameter); the Download block picks from assets and links to `/assets/…`. Sites with downloads still in `media/` keep working (the generator falls back to `media/`), and the Media tab offers "Move to Assets", which moves them and updates the pages' Download blocks and `/media/…` links.
- **#075 Responsive images.** Done 2026-10-21 (`qed_core::images`): JPEG and PNG images in Image, Text with image, Gallery and Hero blocks get copies 480, 960 and 1600 px wide (only those under 90% of the original), in the original's format, next to it in the built site (`media/team.480w.jpg`), listed in `srcset` with a `sizes` hint per block. Camera rotation (EXIF) is applied. Copies are cached in the site's self-ignoring `.qed-cache/` and made on all cores. The Hero is now an `<img>` behind the text instead of a CSS background. Still open: images inside Markdown text keep the original only; WebP copies (#085).

- **#076 Generated TypeScript types.** Done 2026-10-21: tauri-specta writes `src/lib/bindings.ts` (types and one function per command) from `commands()` in `src-tauri` the `bindings_are_current` test rewrites it when it falls behind. qed-core derives `specta::Type` behind its `specta` feature. specta marks `#[serde(default)]` fields optional in both directions, so `site.ts` re-exports the commands and types through `Sent<T>`, which makes them required in what Rust sends.
- **#077 Sidebar navigation.** Done 2026-10-21: `navigation = "sidebar"` in `site.toml` (switch in the Site tab) moves the main menu from the top bar into a sidebar, for sites with many menu pages. The current page is highlighted, its parent in bold; on phones it's a "Menu" box that opens on tap, without JavaScript. Menus follow the page tree: an entry goes under the nearest menu entry above it (posts under Blog) and is indented in the sidebar.
- **Unknown block types.** Done 2026-10-26: a block type this q.e.d. doesn't know is reported with a hint to update (App → Check for updates), since it usually comes from a newer q.e.d.
- **Page icons.** Done 2026-10-21: `icon = "book-open"` on a page shows a Lucide icon in the menus of the generated site (plain SVG, no JavaScript) and in the app's page list. A curated set of 60 icons lives in `crates/qed-core/assets/icons/` and is built into q.e.d.; unknown names are an error with the line number. The page settings have a picker.
- **#064 Site CSS.** Done 2026-10-21: one Tailwind + daisyUI stylesheet with all 35 themes and the typography plugin (77 KB, 14 KB gzipped), compiled by `npm run build:site-css` and built into q.e.d. with `include_str!`. The theme is the `data-theme` attribute.
- **SFTP publishing.** Done 2026-10-21 (#025, #028, #030, #031 for SFTP, #063, #065): targets in `targets.toml` (`[[targets]]`, nothing secret, git-ignored because it names servers), passwords and generated Ed25519 keys in the OS keychain, `russh` with the `ring` backend, trust on first connection and a hard stop when the server key changes, `.qed-manifest.json` so only changed files are uploaded and only q.e.d.'s own files deleted, emptied folders removed, live progress over a Tauri channel. A throwaway server for testing lives in `dev/test-server/` (see the Development page).
- **#061 Scaffold.** Done 2026-10-21: Cargo workspace (`src-tauri`, `crates/qed-core`), SvelteKit SPA with Tailwind + daisyUI, GPL-3.0 `LICENSE`, `npm run check:all`. Elixir code and its local data removed.
- **#058 Frontend.** Svelte + TypeScript.
- **#059 Signing and updates.** Phase 1 ships unsigned installers (Linux AppImage/deb, Windows, macOS with "open anyway" instructions) plus Tauri's updater, whose signing key is free. Revisit Apple notarization ($99/year) and Windows signing before a public 1.0 (#070). *Updater built 2026-10-21* (tauri-plugin-updater and -process, `src/lib/updates.svelte.ts`): q.e.d. checks `latest.json` of the latest published GitHub release at start, a bar offers "Install and restart", the App tab shows the version and "Check for updates". The release workflow signs the updates with the `TAURI_SIGNING_PRIVATE_KEY` secret (`tauri.release.conf.json`); steps in Development → Updates. The first version with the updater has to be installed by hand; from then on updates come by themselves.
- **#060 Markdown editor.** Keep OverType.
- **Learning blog.** Explanations of Rust, Svelte and TypeScript go into blog posts in `docs/pages/blog/` (Markdown with front matter), not into chat.
- **#071 Demo site.** Done 2026-10-21: `docs/` is a q.e.d. site (home page, blog with the learning posts). A test keeps it valid. Pages gained `date`, `authors`, `tags` and `summary`.
- **Repo.** Same repo; the Elixir code is deleted (it stays in git history).
- **Site pages** follow the design of Mercator's block CMS; Mercator stays a read-only reference.
- **Two phases.** Phase 1 builds the CMS and site generator; phase 2 adds podcasting.

- **Podcast is optional.** A site can have only static pages (#055, #066). `/` is always the CMS `home` page (#056).
- **#001 Storage.** Audio and images stay on the user's computer.
- **#004 Site templates.** Tailwind + daisyUI. Users pick a theme and options; templates aren't user-editable (#054, #064).
- **#005 Web player.** Self-hosted Podlove Web Player (#049).
- **#006 Podcast fields.** Apple basics, `podcast:guid`, funding (#046).
- **#007 Episode fields.** Season, number, type, image, explicit, stable guid (#046).
- **#008 Transcripts.** WebVTT (#048).
- **#009 Audio formats.** One format per podcast, provided by the user; one feed. No transcoding.
- **#010 Duration and size.** Read from the local file (#023).
- **#012 SSH transfer.** SFTP with a deploy manifest (#028, #065).
- **#016 License.** Changed from AGPL-3.0 to GPL-3.0 with the move to desktop.
- **#019 Comments and sharing.** Share links, no comments (#053).
- **#040 Blog.** Pages with kind `post` under a `blog` parent, with a Post list block and `blog/feed.xml`. *Done 2026-10-21*: a post is a page with a `date` that isn't in the future (later ones show up once the site is built on or after their date; their own pages are built anyway). The Post list block (`::: post-list`, `from` and `limit`, also in columns) shows the posts below a page newest first, same-day posts by `weight` (higher first), with date and summary. Each page with posts right below it gets an RSS 2.0 feed (`blog/feed.xml`, summary as description, authors as `dc:creator`), linked from every page's `<head>` and, with the RSS icon, from the footer and below Post lists; feeds need `base_url`, and the build report says so when it's missing. Moving a page updates Post lists that list it. Dates without a library: `qed_core::calendar`.
- **#041 Contributors.** Their own records, linked to episodes (#047).
- **#042 Redirects.** Keep page redirects; write meta-refresh stubs at old paths. *2026-10-27*: an old address ending in `.html` (`/2016/03/28/easterhegg.html`, as Jekyll made them) is written as that file instead of a folder.
- **#080 Logo and app icon.** *Done 2026-10-22*: `brand/` holds the artwork: `logo-mark.svg` (the Q with the houses, drawn by hand, about 2 KB), `logo.svg` (mark and wordmark, text as paths), `app-icon.svg` (the mark on a light tile) and its 1024 px PNG, from which `npx tauri icon brand/app-icon.png` makes `src-tauri/icons/` (delete the `android/` and `ios/` folders it adds). The JPEGs are illustrations; `logo-traced.svg` is the first automatic trace, kept for reference. The app shows the mark in its header and on the start screen, from a copy in `src/lib/assets/logo-mark.svg` (copy it again after changing the mark).
- **#088 Static files.** *Done 2026-10-27*: a `static/` folder in the site is copied into the built site as it is, at the same paths (`static/pdf/a.pdf` becomes `/pdf/a.pdf`), hidden files included (`.well-known/`, `.htaccess`); `.DS_Store`, `Thumbs.db` and `desktop.ini` are skipped. It's copied last, and a file the build wrote itself (a page, `style.css`, a feed) or the publish manifest stops the build with an error instead of being replaced. Link to the files as `/pdf/a.pdf`. The build message counts them. The app has a Static tab next to Media and Assets (`Library::Static`): import files or whole folders under their own names (a taken name is an error, never a rename), one folder at a time with a path to click back up (imports go into the folder shown; a search looks through all), each with its address and the pages linking to it (Markdown links and button links), delete files or folders no page links to. Text files (HTML, CSS, `.txt`, …, the new `MediaKind::Text`) open read only in a viewer, in all three libraries.
- **#099 Page list block.** *Done 2026-10-27*: `::: page-list` with `from` (empty for the page it's on) lists the pages one level below, by weight, then title (compared by slug, so `André` comes before `Andreas`), each with its summary. Pages with a date are posts and left out; drafts too. Its `from` counts as a link, so moving

the listed page updates it, like Post list's. podcasterei.at's Podcaster:innen uses it instead of the hand-written list of 83 people; their podcasts moved into each person page's summary.

- **#101 Custom theme.** *Done 2026-10-27*: `theme = "custom"` with a `[custom_theme]` table in `site.toml`: `from` (a theme), five colors (background, primary, secondary, accent, neutral) and `corners` (sharp, soft, round). The page keeps `from` as `data-theme` a `:root[data-theme]` rule appended to `style.css` sets the changed variables, text colors picked for WCAG contrast, base-200/300 mixed from the background (`qed_core::theme`). Site → Theme has a Custom colors card that starts from a theme's colors (read via canvas) and saves on each change; the legend previews it live. Picking another theme keeps the custom colors. *Saved colors*: named sets in `[[saved_themes]]` (same fields plus `name`), applied with one click, saved with "Save these colors", deleted after asking; the card follows `site.toml` after Reload, so an old copy in the app can't overwrite it.
- **#098 Dates in the site's language.** *Done 2026-10-27*: `calendar::long_date` writes dates for readers by the site's `language`: `24. Februar 2017` (de), `24. Jänner 2017` (de-AT), `February 24, 2017` (en), `24 February 2017` (en-GB), plus fr, es, it, nl, pt; other languages keep `2017-02-24`. Used in post lists and bylines; `<time datetime>` keeps the ISO date. podcasterei.at is `de-AT` now, like its old template's Jänner.
- **#100 Video block.** *Done 2026-10-27*: `::: video` with `url`, `title` and `image`. The website shows the thumbnail with a play mark as a link to the video: no iframe, no JavaScript, nothing loaded from the platform until a click. "Get thumbnail and title" in the editor asks YouTube, Vimeo or a PeerTube instance once over oEmbed (`qed_core::video`, reqwest with rustls on ring) and keeps the thumbnail in `media/` (YouTube: the 16:9 `maxresdefault.jpg` when there is one).
- **#102 Fonts.** *Done 2026-10-27*: `[fonts]` in `site.toml` (`text`, `headings` headings follow text if empty). Fonts come from Fontsource (the Google Fonts and more), downloaded once into `fonts/<id>/` with a `font.toml` (`qed_core::fonts`): 400, 600 if there, 700 and italic, for latin and latin-ext. The build copies the fonts in use and appends `@font-face` rules with `unicode-range` to `style.css` no third-party requests. Site → Theme → Fonts has a searchable picker and a live sample (FontFace API).
- **#092 Footer blocks.** *Done 2026-10-27*: `footer.md` in the site folder (page format) holds blocks shown on every page above the footer menu, in the footer's colors; Post and Page lists there show titles only. Links are rendered per page; moving a page rewrites footer links, and the delete warning and media usage count the footer ("the footer"). In the app: a Footer entry below the page list opens the block editor (`BlockEditor` now takes the saved blocks and a save function). podcasterei.at has one: Letzte Posts, Meetup, Kontakt.
- **#103 Lists with pictures.** *Done 2026-10-27*: Post list and Page list have `layout`: `list` (default), `cards` (picture, title, date, summary; 1–3 columns) and `tiles` (square pictures with title and summary; 2–6 columns for pages, 2–4 for posts). The picture is the page's share image, else its first image; resized copies with `srcset` as everywhere. In the editor: a Layout choice on both blocks. waldgruber.at uses cards for its posts and tiles for its 53 Stofftiere.
- **#094 Code highlighting.** *Done 2026-10-27*: fenced code blocks with a language (```` ```rust ````) are highlighted when the site is built, by comrak's syntect plugin (pure-Rust regex engine, no C): classes with an `hl-` prefix (apart from daisyUI's, which also has `link`), and the dark "Base16 Ocean Dark" colors appended to `style.css`, readable on every theme since code blocks are dark. No JavaScript. Grammars: syntect's defaults (Rust, HTML, CSS, JS, shell, JSON, YAML, XML, Markdown, LaTeX, Ruby, …); more in #106.
- **#095 Wide logos.** *Done 2026-10-27*: the header logo keeps its height and takes the width it needs; `logo_size` (small 36 px, medium 48 px, large 64 px, original: the image's own size up to 112 px, sharp for bitmap logos) and `title_in_header = false` for logos that show the name (the logo's alt text then carries

the site title); a `favicon.ico` in `static/` is the tab icon instead of the logo. In Site settings below the logo. informatom.com: medium wordmark, no title.

- **#096 Hide page title.** *Done 2026-10-27*: page setting `hide_title` ("Hide the title" in Settings); a starting Hero still hides it by itself. informatom.com's start page uses it.
- **#097 Column widths.** *Done 2026-10-27*: `width` per column (1, 2 or 3, relative to the others; 1 is left out of the file). Unequal widths become `--cols: minmax(0,2fr) minmax(0,1fr)` with the class `md:grid-cols-(--cols)` phones still stack. In the editor: 1×/2×/3× next to each column's title. informatom.com's start page: two thirds and one third, like the old Bulma `is-two-thirds`.
- **#093 404 page.** *Done 2026-10-27*: the build writes `404.html` in the site's look: a short text in the site's language (de, en, fr, es, it, nl) with a link to the start page, or the site's own `pages/404.md`, which then isn't built as a normal page, menu entry or sitemap entry. Its links start at the site root (`/`, or the path of `base_url`), since servers show it at any address (`LinkContext::rooted`). The preview server shows it for missing pages. The web server must be told to use it, once per site: nginx `error_page 404 /404.html;` in the site's `server` block (the owner's server runs nginx; `.htaccess` files don't work there), Apache `ErrorDocument 404 /404.html` GitHub Pages does it by itself.
- **#104 New sites in their language.** *Done 2026-10-27*: a new site's start page (title, Hero line, first paragraph) is written in the site's language: German (also `de-AT`), English, French, Spanish, Italian, Dutch; English for others. The text now says "q.e.d.", not "qed".
- **#037 Backups.** *Done 2026-10-27*: a Backups page: a site is a git repository, so a backup is a push to a remote (Codeberg, GitHub, a bare repo on one's own server), with full history; what it leaves out (targets.toml, keychain secrets, the open version, `_site/`, `.qed-cache/`). Decided with the owner instead of "copy the folder". A button for it in the app: #109; the page also belongs into the app's manual: #110.
- **#082 Restore the whole site.** *Done 2026-10-27*: "Go back" on any older version in the Versions tab (`history::restore_site`) saves the present as a version ("Before going back to …"), writes that version's files into the site folder, deletes files it didn't have (and folders left empty), and saves the result as a new version ("Back to …"); the history is never rewritten. Only files git tracks in the site folder change (`_site/`, `targets.toml`, `.qed-cache/` stay; for a site inside a larger repository, nothing outside it). Disabled while a page has unsaved changes. Also fixed: versions saved in the same second were listed in any order (`Sort::TOPOLOGICAL | Sort::TIME`).
- **#111 Hand-made menu.** *Done 2026-10-27*: `[[main_menu]]` in `site.toml` (entries: `page`, `link` to any address, groups with `[[main_menu.items]]`, `divider = true` inside groups), edited in Site → Menu ("Arrange the menu by hand" starts from the automatic menu; "Back to automatic"). In the top bar groups are daisyUI dropdowns: hover, click/tap and keyboard, no JavaScript; the box hangs in a transparent frame so the pointer can reach it. On phones one "Menu" button; in the sidebar groups are headings. Entries follow moved pages; the delete warning names "the menu". Without one, the automatic menu from the pages' Menu setting stays. informatom.com's old navbar rebuilt with it (top bar instead of the sidebar).
- **#112 Page width.** *Done 2026-10-27*: `page_width` in `site.toml` (Site settings): normal (56rem, as before; 72rem with a sidebar), wide (72rem; 80rem with a sidebar) or full (the window). Header, content and footer follow it together. informatom.com uses wide.
- **#113 Boxed columns.** *Done 2026-10-27*: `boxed = true` per column (a "Box" switch beside its width in the editor): the column gets the theme's background, a thin border, a soft shadow, rounded corners and inner spacing, like Bulma's `.boxed` on the old informatom.com.
- **#114 Icons in text.** *Done 2026-10-27*: `:icon-phone:` in any Markdown text becomes that icon from q.e.d.'s Lucide set (the same as for menu entries), as tall as the letters (`.inline-icon`); not inside code, and unknown names stay as written. informatom.com's footer uses phone, mail, message-circle (Lucide has

no Mastodon logo) and map. Fixed along the way: eleven icons (mail, calendar, lock, …) had lost the size of their rectangles when they were copied in, so the rectangles weren't drawn; a test now checks every shape.

- **#115 Site icons.** *Done 2026-10-27*: image files in a site's `media/icons/` work as icons in text too: `:icon-rails:` shows `media/icons/rails.svg` (else `.webp`, `.png`, `.jpg`) in its own colors, 1.5 times as tall as the letters, before q.e.d.'s Lucide icon of that name. The media library counts them as used, so they can't be deleted while a page writes them. informatom.com has its 14 old badges back: the six stacks in the start page's list and eight open-source logos under IT-Beratung.
- **#116 Marking external links.** *Done 2026-10-27*: "Mark links to other sites" in Site settings (`mark_external_links`, on for new sites, off for existing ones): links in text to other sites get Lucide's external-link icon after them, in the link's color, by CSS alone (every internal link is relative). Not in menus or on buttons, not on linked images. Full links to the site's own `base_url` (with or without `www.`) now become relative links when building, so they stay inside the site, in the preview too. informatom.com has it on.
- **#117 Images open big.** *Done 2026-10-27*, asked for by Sonja for waldgruber.at's plushies (the old site had a Bootstrap modal): a click on an image shows it big over the page on a dark backdrop, closed with Esc, a click on it or beside it. The browser's popover (`popovertarget`), no JavaScript; browsers without it just show the image. Automatic for images noticeably larger than their place: Image blocks over 1300 px, Text with image over 700 px, every Gallery image (it's cut to a square; before, it linked to the original file). The big one has every copy and the original in `srcset` with `sizes="95vw"`, so a phone loads a copy and a 4K screen the original (Sonja's second wish). The close button says "Schließen" on German sites, "Close" elsewhere.
- **#118 Images open big in the app.** *Done 2026-10-27*: in the Media tab, a click on an image shows it as large as the window allows (`ImageViewer.svelte`, a modal `<dialog>`), with its name, size and dimensions; ← and → step through the images shown (search and folder apply), Esc or a click on it or beside it closes it. Not in the image picker, where a click still picks.
- **#119 PDFs in the app.** *Done 2026-10-27*: a click on a PDF in the Assets (or Media, Static) tab shows it in a large window inside the app (`PdfViewer.svelte`, an `<iframe>`): the webview's own viewer, PDF.js in WebKitGTK (2.52 here), Edge's in WebView2, PDFKit on macOS. "Open in PDF app" opens it in the computer's PDF reader, through the new `open_library_file` command, which only opens files inside the site's folders (`media::path_of`).
- **#043 Block CMS.** Mercator's block CMS design rebuilt in Rust (`docs/notes/cms-handover-qed.md` its Elixir file map no longer applies). *Done 2026-10-21*: nine block types (Text, Hero, Image, Text with image, Gallery, Download, FAQ, Columns with nested blocks (#074), Call to action); the block editor with Content/Versions/Settings tabs per page, OverType for Markdown, the image picker (#033), add, remove, move up/down, save with `site::save_blocks`, a warning before losing unsaved changes. *Shared blocks done 2026-10-27* (`qed_core::shared`): files in `shared/` (title + blocks, no page of their own), shown on a page by a Shared block (`::: shared` / `name = "…"`, also in columns and the footer); the build swaps in the blocks with the page's links and notes missing ones; "Make shared" on any block, a Shared blocks list under the pages, rename, delete only when unused; links in them follow moved pages, their images count as used. *Drag and drop done 2026-10-27*: a grip handle on every block (inside columns too) drags it to its new place within its list, with a line where it will land and scrolling near the window's edges; pointer events, not HTML drag and drop, which the Tauri window takes over for dropped files.
- **#120 Drag and drop in the page tree.** *Done 2026-10-27*: a grip handle on every page in the Pages tab (not the start page). Dropped between pages, it changes the order: the pages below the same parent get the weights 10, 20, 30, … (`site::set_order`, `set_page_order`); the tree now shows the site's order (weight, then title; posts by date). Dropped onto a page, it moves below it after a confirmation, through `move_page`

(links and redirects as in the Move dialog). Posts can't be reordered. The logic is in `src/lib/pageTree.ts` (`treeOrder`, `planDrop`).

- **#054 Theme options.** Per site: daisyUI theme, logo, a few options. *Themes done 2026-10-21*: Site settings dialog edits `site.toml` (title, description, language, web address, theme); the app has its own theme (App settings, per computer, can follow the system). *Logo done 2026-10-22*: `logo` in `site.toml` (Site settings, image picker) shows before the title in every page's header and is the browser-tab icon (`<link rel="icon">`); the media library marks it "Site logo" and won't delete it. *Theme colors done 2026-10-27*: sites use all four theme colors (primary: buttons, Call to action, current menu page, link underlines; secondary: Hero without image, post dates; accent: stripe at the top, quote bars; neutral: footer, sidebar's current page, file types), and Site settings explains them in a legend beside the theme switcher; a picked theme saves at once. *Header bar done 2026-10-27*: `header` in `site.toml` (plain, primary, secondary, accent, neutral); a colored bar underlines the current page and gives the logo a light backing. Fonts: #102 (done). *Colors per element done 2026-10-27*: `[colors]` in `site.toml` (links, buttons, Hero, Call to action, boxed columns, content background as a light tint, footer), each a theme color (or plain/shaded for areas); empty means the old default; a button on a box of its own color is drawn plain; the legend and sketch in Site → Theme follow. *Closed 2026-10-27*: no other options planned; the Colors section has a Reset button.
- **#121 Menu, Theme and Publish tabs.** *Done 2026-10-27*: the Site tab was long; Menu, Theme and Publish (Build and the publish targets) now have tabs of their own next to it (kept mounted, like Site, so unsaved edits survive switching). Site keeps the folder and the settings. "Show the main menu in a sidebar" moved to the Menu tab and saves at once. Fixed on the way: saving Site settings no longer brings back older colors or menu layout from its own copy of `site.toml`.
- **#122 Warning before losing unsaved changes.** *Done 2026-10-27*: closing the window, opening another site, closing the site and "Install and restart" ask first when something is unsaved: the open page, footer or shared block (content or settings), Site settings or the hand-made menu. Each part reports itself (`trackUnsaved` in `src/lib/unsaved.svelte.ts`); the window's close request waits for the answer (`onCloseRequested`, permission `core:window:allow-destroy`).
- **#032 Publish history.** Record each publish run per target as a git tag (#073) plus a log; show progress live. *Tags done 2026-10-21* (with #073); the log and a list of publishes per target are still open. *Closed 2026-10-27*: written for the multi-user web app (who published what); in the desktop app the Versions tab shows each publish as a tag on its version and progress shows live, which is enough.
- **#033 Images and media library.** Import, tag, search and resize images to WebP; keep PDFs for downloads. Phase 2 adds Apple's cover art rules (1400–3000 px square). *Media library done 2026-10-21* (`qed_core::media`): list with size, dimensions and usage, import with clean names, delete only unused files, image picker, Image block with width/height and lazy loading. Still open: resizing to several sizes with a build cache (#075, done), tags. *Closed 2026-10-27*: tags dropped (a podcaster's site has dozens of images; the Media tab's search by name is enough); Apple's cover art rules moved to #066.
- **#086 Testing on macOS and Windows.** The installers build in CI but were never run on macOS or Windows; friends try them and report back. *Closed 2026-10-27*: no item of its own; testing on macOS and Windows goes on with every release, and problems come back as new items.
- **#026 Site generator.** Render HTML for published pages, posts, menus, redirects (meta-refresh stubs) and the sitemap with Rust templates. Copy the images used. Rewrite relative links to other `.md` pages into page URLs; links that leave the site folder (e.g. to source code) need a base URL such as the GitHub repo. *First version done 2026-10-21* (`qed_core::generator`): pages, menus, relative links, Markdown links, redirects, sitemap and canonical links with `base_url`, media copied, preview server and "Build" / "Preview in browser" in the app. *Link previews done 2026-10-22*: Open Graph tags on every page; `og:image` from the new `image` page setting ("Share image") or the page's first image (smallest copy at least 1200 px wide,

else the original), only with `base_url`. Still open: copying only the media files that are used, and links that leave the site folder. *Closed 2026-10-27*: copying only used media dropped (static/ pages point at media files, publishing uploads only changes, and the Media tab shows unused files). Links that leave the site folder are refused: the build stops and names the file and the link (`LinkContext::leaves_site`). The docs blog names source files as code instead of linking them.

- **Phase 1: CMS and site generator.** A complete site builder without podcast features. Decided 2026-10-26: finish phase 1 before podcasting, testing it by moving the owner's Jekyll sites over by hand. *All done 2026-10-27* (the last items: #043, #054, #026; #032, #033 and #086 closed).
- **#066 Podcast switch.** Done 2026-10-28 (`qed_core::podcast`, the Podcast tab, moved out of Site settings the same day): one podcast per site, `[podcast]` in `site.toml` (having it is the switch): name and description (the site's if empty), author, owner name and email, cover, up to three of Apple's categories (`"Arts > Books"`), explicit, type (episodic/serial), copyright, support links (`[[podcast.funding]]`), where audio files go (#023) and the `podcast:guid`, a UUID v5 of the feed address, made once when the site has a web address and then kept. Cover art is checked against Apple's rules (square, 1400–3000 px, JPEG or PNG) in the Podcast tab, for episode covers, and in the build notes. The cover can't be deleted from Media while it's set.
- **#067 Episodes.** Done 2026-10-28: episodes are dated pages below `/episodes` with an `[episode]` section: season, number, type (full/trailer/bonus), own cover, explicit (the podcast's if empty), a guid made once (UUID v4) and chapters (`[[episode.chapters]]`, start, title, optional link; checked for format and order). *New episode* in the Pages tab dates it today and gives it the next number and the latest season; the first one makes `/episodes`, the archive (main menu, Post list). The number may be 0; the New episode dialog proposes the one after the highest, to be changed (user, 2026-10-28). Post lists show "Folge 5" next to an episode's date. Episode pages show season, number and type in the byline and a chapter list below the shownotes, in the site's language. The page writer now merges into nested sections, so comments inside `[episode]` or `[podcast]` survive a save.
- **#022 Episode editor.** Done 2026-10-28 with #067: shownotes are the episode page's blocks (Markdown text and every other block); the episode fields and a chapter editor (rows of start, title, link) are in the page settings.
- **#023 Audio file.** Done 2026-10-28: one audio file per episode (`audio`, `audio_length`, `audio_type`, `duration` in `[episode]`), from the new Audio tab (`audio/`, library `Library::Audio`) or a web address. Size, type and length are read with `lofty` when the file is chosen (MP3, M4A, Ogg Vorbis/Opus, FLAC, AAC, WAV; the build notes that Apple Podcasts and Spotify play only MP3 and M4A). Where audio lives is `audio_files` in `[podcast]`: `local` (default; `audio/` with a `.gitignore` that keeps it out of the history), `in_history` or `external` (with `audio_base_url`, the address new episodes' audio addresses start with, like octopod's `download_url`). The build hard-links the files to `/episodes/<file>`, octopod's addresses; episode pages get an `<audio>` player and a download link; the build notes episodes without audio. The preview server answers range requests, so the player can jump; the Audio tab's *Listen* opens the computer's own player (the webview played long episodes badly).
- **Several audio formats.** Done 2026-10-28 (user: support them, for listeners of Ogg and FLAC, which Apple and Spotify don't play): the podcast lists its formats (MP3, M4A, Ogg, Opus, FLAC; the first is the main one, which the build notes if Apple can't play it), every episode has one file per format (`[[episode.audio]]` with `format`, `file`, `length` one `duration`). Choosing one file picks the others of the same name. Publish refuses while a published episode lacks a format; the page tree, the episode's settings and the build notes say which. No converting: every format is imported (decided with the user).
- **#068 Podcast feeds.** Done 2026-10-28 (`qed_core::feed`): one feed per audio format at octopod's addresses (`/episodes.mp3.rss`), with RSS, `itunes:` (owner, cover, categories, explicit, type, season, episode above 0, duration, episode type and cover), Podcasting 2.0 (`guid`, `funding`, `season`, `episode` including 0) and Podlove

Simple Chapters; the shownotes' text as `content:encoded` with full links. With several formats, the feed titles name the format. Every page links the feeds; the episodes' Post list subscribes to the main one; the plain blog feed of `/episodes` is gone. No paged feeds (user, 2026-10-28): Apple and most apps read only the first page.

- **#069 Podcast blocks, #125 Podlove Subscribe Button.** Done 2026-10-28: a Latest episode block (newest episode: title, number, date, summary, player) and a Subscribe block, the Podlove Subscribe Button with every format's feed, self-hosted (`assets/subscribe-button/`, with a README on its source and updating; the podcast's data in `subscribe-button/podcast-data.js`), in the theme's button color (`theme::color_hex` turns daisyUI's `oklch()` into hex), with the podcast's cover above it, placed left, centered or right (`align`, user, 2026-10-28; new blocks start centered), plain feed links without JavaScript. Decided with the user: the site's own feeds are the source of truth, no directory links (Apple, Spotify). Podcast blocks are only offered on podcast sites. New sites can start as a podcast (New site dialog), with both blocks on the start page.
- **#053 Share links.** Done 2026-10-28: a share row on episode pages (with the site's web address): Bluesky and email as plain links; Mastodon (asks for the reader's server once, remembers it, opens that server's share page), copy link and the device's share menu as buttons that `share.js` (q.e.d.'s own, copied into the site) shows only where they work. Nothing from third parties.
- **#047 Contributors.** Done 2026-10-28 (`qed_core::contributors`): entered once, in a Contributors tab (user: "painful" to copy them into each episode as in octopod). One file each in `contributors/` (name, photo, links, Markdown bio), referred to by id. Episodes list people with Podcasting 2.0 roles (`[[episode.people]]`, several roles = several rows); the podcast's Team (`[[podcast.people]]`) goes into the feeds' channel and starts every new episode. The build makes a page per person (`/contributors/<id>/`, with their episodes via the Post list's new `contributor` filter) and `/contributors/` (unless the site has its own), shows people with roles on episode pages, writes `podcast:person` into the feeds, and has a Contributors block. A contributor still listed, or a photo in use, can't be deleted.
- **#048 Transcripts.** Done 2026-10-28: one WebVTT file per episode (`transcript` in `[episode]`), from a Transcripts tab (`transcripts/`, in the history; only `.vtt` files starting with `WEBVTT`), served at `/transcripts/…` and named in the feeds with `podcast:transcript` (`type="text/vtt"`, `rel="captions"`). Not shown on episode pages: the web player (#049) will (user). WebVTT only, no SRT (user). After 0.8.0: the Transcript block (#126) splits transcripts without speakers, such as Whisper's, into paragraphs at pauses over 1.5 s or after a minute (measured on a whisper.cpp transcript of 3-schweinehun.de); the speaker's name shows only when it changes.
- **#049 Web player.** Done 2026-10-28: the Podlove Web Player 5.13.0, self-hosted (`crates/qed-core/assets/podlove-player/`, README with its origin and how to update), on every episode page with audio. Each episode gets `player.json` next to its page (audio and downloads per format, chapters, contributors with photos, the transcript's cues read from its WebVTT file, speakers matched to contributors); the site gets `podlove-player/config.json` (the theme's button color as the player's colors) and `player.js`, q.e.d.'s own script that starts it without inline scripts. With the player running, the plain `<audio>` player and the chapter list are hidden (user); without JavaScript they stay. The player's share tab is left off: the site has its own share row.
- **#126 Episode layout.** Done 2026-10-28: `episode.md` next to `footer.md` holds the blocks of every episode page below its title and byline. Six blocks without settings stand for the episode's parts: Player, People, Shownotes (the page's own blocks), Chapters, Transcript (new: the WebVTT text, one paragraph per speaker turn, folded away in a `<details>`) and Share links; any other block (a Subscribe button, a Call to action) may go between them. The parts render in `block.html` now, no longer in `page.html`. Without the file, the old order: Player, People, Shownotes, Chapters; saving that order removes the file. Share links left the default the same day, after #128: the player's share tab does it (user). Edited under "Templates"

in the page tree, next to Footer (user), on podcast sites only. Each part can be added once; the parts aren't offered on pages, in columns or for "Make shared".

- **#127 Video files.** Done 2026-10-28 (user, from converting 3-schweinehun.de, whose episodes 23 and 35 had `<video>` tags): the Video block's new `file`, a video in `assets/` (Downloads), wins over its platform address. The site plays it with the browser's own `<video>` player (no JavaScript, nothing from elsewhere), the block's image as poster, its title below; tall phone videos stay within the window. The block editor switches between "On a platform" and "File in Downloads". A shown file can't be deleted. Same day, a bug the user found: the Latest episode block now has the Podlove Web Player too, not only the plain `<audio>` player, and an explicit "Zur Episode →" link below it, so nobody thinks there are no shownotes (user).
- **#128 More of the Podlove Web Player.** Done 2026-10-28 (user: all of it): the player's playlist tab (every episode, played without leaving the page), its share tab (link, email, Facebook, Twitter/X, LinkedIn, Pinterest, Xing, WhatsApp; sharing from the current playtime, which the player reads back from `?t=` embed code) and its subscribe button (the podcast apps that take the feed's address, no directories). Embedding needed `share.html` and the subscribe button's app list from the npm package; `qed-share.js` lets the embedded player show only this site's episodes, in its colors. `player.json` now uses addresses from the site root. The line below the player (duration, download links) now hides while the player runs, like the plain player and the chapter list: the player shows both (user). The plain player no longer flashes up first: `player.js` loads in `<head>` and hides it at once, and keeps the room the player will take (measured: 290 px wide or 534 px narrow without chapters, 410 or 638 px plus 40 px per chapter, since the chapters tab opens with it), so the page doesn't jump (user).
- **Phase 2: Podcasting.** Podcast websites on top of phase 1: podcast settings, episodes, audio in several formats, feeds, the Podlove Web Player and Subscribe Button, contributors, transcripts, the episode layout. *All done 2026-10-28*, released as 0.8.0 (#066, #067, #022, #023, #068, #069, #125, #053, #047, #048, #049, #126, #128; tested by moving stefan-wills-wissen and 3-schweinehun.de over from jekyll-octopod, which brought #127).
- **#129 Push and pull with git.** Replaces #109 (a *Back up* button). Done 2026-10-28, released in 0.9.0, tested with waldgruber.at on two computers (push, get a site from git, edit there, push, pull back). *Push and pull* in the Versions tab, *Get a site from git…* in the site menu and on the start screen, a bar offering a pull when a site opens and its remote has new versions, and a progress bar (`qed_core::remote`, post 55). Decided (user): for two computers; git's own remotes; q.e.d.'s own Ed25519 key per remote address and computer, in the keychain; pull merges when different files changed and changes nothing when both changed the same file; only sites that are their own repository. Built on russh instead of git2's `ssh` feature: a libgit2 transport runs `git-upload-pack`/`git-receive-pack` over the SFTP login (`sftp::login`); the server's fingerprint per remote in `remote.<name>.qedhostkey`. A manual page follows with #110.
- **#108 Site search with Pagefind.** Done 2026-10-28, tested by the user on several sites and on the docs site (62 pages: 936 KB in `pagefind/`, indexing about 0.1 s in a release build). Decided (user): a search button in the header that opens Pagefind's Component UI dialog (`<pagefind-modal>`, also Ctrl+K); without JavaScript no search at all, instead of the planned search engine form (that would send visitors to a third party); `search = true` in `site.toml` by default, a switch in the Site tab. `qed_core::search` indexes the built site on a thread with its own runtime; only `<main>` is indexed (`data-pagefind-body`, title from `data-pagefind-meta`), lists of pages and the episode player are left out. Colors from the theme (`--pf-*` in `site.input.css`). Pagefind's files come from the crate (`assets/pagefind/README.md`). Post 56.
- **#107 Math formulas.** Done 2026-10-28 (user's test: Euclid's proof of infinitely many primes, in post 57). `$…$`, `$$…$$`, … and ```` ```math ```` blocks (comrak's `math_dollars` and `math_code`) become MathML when the site is built, with `pulldown-latex` no JavaScript. Placeholders through the Markdown tree, since raw HTML is escaped. A mistake shows as red LaTeX with the problem on hover. Sites with a formula get Latin

Modern Math and pulldown-latex's stylesheet (`assets/pulldown-latex/`, 388 KB font, loaded only on pages with math). Only formulas are LaTeX, not whole documents (`\begin{proof}`). Dollar signs: prices stay text unless a later `$` in the paragraph can close a formula; `\$` is always a dollar sign. None of the five test sites or the docs had a `$` turned into math.

- **#130 Chemistry in text.** Asked for by the user after #107 and done 2026-10-28 (post 58: caffeine, aspirin, cholesterol, photosynthesis). Decided (user): structure diagrams as a ```` ```smiles ```` block in text (SMILES on the first line, an optional caption on the next), and a subset of mhchem's `\ce{…}` inside math. Diagrams with chematic (`chematic-smiles`, `chematic-depict`, pure Rust, MIT/Apache) as SVG when the site is built, double bonds kekulized, black as the text color, label boxes in the theme's background. `\ce` is translated to plain LaTeX before pulldown-latex (`chem::expand_ce`): formulas, coefficients, charges (also before a state), states, hydrates, arrows with text, gas and precipitate. Mistakes show in red like math. Not done: stereo wedges (chematic draws none for cholesterol's stereocenters), reaction schemes with structures.
- **#131 Charts and stratigraphic columns.** Asked for by the user on 2026-10-28 (charming would have needed V8 for SVG; charts-rs draws it in pure Rust, about 1.7 MB in a test program). Done the same day, post 59: a ```` ```chart ```` block with charts-rs's JSON and `"type"` (bar by default, 17 types; pie without the "rose"), drawn as SVG when the site is built, the theme's colors and font by attribute selectors in `site.input.css`. Geology, for a possible power user: a Harker diagram as a scatter chart, and a ```` ```strata ```` block drawn by q.e.d. itself (`strata.rs`): one layer per line (thickness, rock, name), 17 rocks with their patterns and German names, width by grain size, scale from the base, labels pushed apart, legend. Ideas not built: ternary diagrams (QAPF, soil texture), rose diagrams, stereonets.
- **#132 Catppuccin themes.** Asked for by the user on 2026-10-29 (first Latte and Mocha for the app, then all four flavors for sites too) and done the same day, post 60: catppuccin-latte, -frappe, -macchiato and -mocha are in `theme::THEMES` (39 themes now), so sites, custom themes ("start from") and the app offer them. qed-core's `catppuccin_themes.rs` builds them as daisyUI themes from the `catppuccin` crate's palette, matched up like Catppuccin's own `@catppuccin/daisyui` the test `stylesheet_is_current` writes `assets/catppuccin.css`, which `site.input.css` and `src/app.css` both import (site.css +3 KB).
- **#106 More languages for code highlighting.** Done 2026-10-29, post 61: the `two-face` crate (0.5.2, `bat`'s grammars, fancy-regex build) gives comrak's highlighter some 200 languages, among them TypeScript, Svelte, TOML, Elixir and Jinja; all 79 docs blocks that were plain are colored now (only ```` ```text ```` stays plain). Costs: about 1 MB in the program, docs build 1 s instead of 0.45 s.
- **#087 Section block.** Decided with the user on 2026-10-29: as wide as the page's content (like Hero, works with the sidebar), settings title, background (none or a `Surface`: box, shaded, a theme color), background image (darkened, wins over the color) and spacing (normal, large); holds everything but sections, Columns included. Done the same day, post 62: `::::: section` … `:::::` with an optional `::: settings` first; parser `Inside` enum (page, section, columns); `Block::inner_lists(_mut)` and `retain_deep` replace the hand-written walks into columns (links, media, shared blocks, empty text); a section in a theme color draws buttons of its color plain (`on-primary` …); its picture counts as used media and as the page's first image.
- **#083 Closing versions by themselves.** Rule from the user on 2026-10-29: an open version rests after an hour without saves or on an earlier (local) day. Done the same day, post 63: the app asks for a note when a site opens and once a minute (`RestingVersionDialog`, prefilled with `Saves on 27 Oct, 13:50–21:12` from the commit's author and committer times); "Save version" closes it as it is (`history::close_open_version`), "Use suggested note" leaves it, and the next save closes it with the suggested note before starting a new one (`record_save`). The Versions lists show an open version as "Work in progress since <first save>" with "last saved <time>" (`Version::started`, the author time).

- **#085 Images in Markdown text, WebP.** Copies in text done 2026-10-27; WebP done 2026-10-29, post 64 (encoder and PNG decided with the user): copies of JPEG photos are lossy WebP from libwebp (`webp` crate, quality 80, method 6), `team.480w.webp` a photo used as share image also gets a JPEG copy at that width (`Images::share_copy`); PNG copies stay PNG. podcasterei.at: 76 copies 7.0 MB → 3.6 MB (−48%); first build with WebP 2.9 s, cached afterwards.
- **#133 Links to missing pages.** Found and done 2026-10-29: the build notes every link (Markdown links and images, buttons, list sources) to a page that doesn't exist or is a draft, and to a `/media/` or `/assets/` file that isn't there, once per page, in the footer and episode layout too, with shared blocks checked where they're shown (`missing_links` in generator.rs). Notes, not errors, so a typo doesn't keep a site from being published. Redirect addresses and folders in `static/` count as existing. Found right away: /technik on waldgruber.at still links to the deleted /ror-installation. Not done: marking such links while editing.
- **#110 Manual.** A user manual inside the app, also reachable from where a topic comes up, and the same pages on the docs site, for site owners and editors, not developers. *Done 2026-10-29*, post 65: decided that the pages live in `docs/pages/manual/`, compiled into the app (`qed_core::manual`, offline, built in the app's theme, shown in a second window; outside links open in the browser); a life-buoy button next to the preview button and **Manual** links in the tabs. Pages: start, before you start, getting started, pages and blocks (with boxed columns, #113), icons (#114, the full list; a test keeps it complete), pictures and files, site settings, menus (#111), colors and fonts, versions, Backups (#037, moved there), publishing, podcasts (3-schweinehun.de as the example), moving a site over, license and credits (ideas from jekyll-octopod's manual), the libraries (written by `examples/libraries.rs` from `cargo metadata` and `package.json` a test keeps it in step), with screenshots in catppuccin-mocha (rule in AGENTS.md). Found on the way: #134, #135.
- **#136 Back and next links.** Asked for by the user on 2026-10-29 after reading the manual, for the blog too, and done the same day. Decided with the user: switched on by a setting on the page above (`back_next = true`, "Back and next links on the pages below"), not site-wide; posts read "← Older" / "Newer →" (newest first, as in a Post list), other pages "← Back" / "Next →" by their order, as in a Page list. Words in German and English. `neighbours` in generator.rs, a `<nav>` at the end of `<main>` in page.html, left out of the search index. On for docs/pages/manual and docs/pages/blog.
- **#142 The start page in a hand-made menu.** Found 2026-10-30 while moving blog.panoptikum.social over, done the same day: a menu entry written by hand as `page = "index"` was left out without a word (the Menu tab writes `""` for the start page). `site::menu_page_path` now reads menu pages like page files (`/`, `index`, `index.md`, `blog/`, `blog.md`, `blog/index`), when `site.toml` is opened, in the build and when pages move. Entries whose page is a draft or doesn't exist are still left out, but the build says so in a note, and the Menu tab shows them with a warning. The manual's Menus page says so.
- **#137 Misleading count before publishing.** Reported by the user on 2026-10-29, done 2026-10-30: the Publish dialog listed the site's changed files ("3 files changed"), but publishing uploads every built file that differs on the server, often many more. Now it reads "3 of your files changed since the last publish", with a line below that more files of the website can change (a title shows in every page's menu); the report after publishing says "57 website files uploaded". The manual's Publishing page explains it in a question. Not done: building first to show how many website files will go up; back on the list if the wording isn't enough.
- **#141 Footer color in dark themes.** Found 2026-10-30 while moving blog.panoptikum.social over, done the same day: on a theme with a dark page background, the footer is shaded by default instead of the accent color (a pale pink band in catppuccin-mocha); a footer color picked in the Theme tab still wins. `theme::is_dark` decides by the page background (`--color-base-100`), so a custom theme's own background counts; a test checks it agrees with each theme's `color-scheme`. Asked for by the user on the

way: the theme pickers (Theme tab and App tab) show the themes under Light and Dark, each A to Z (`theme::choices`, `list_themes`). The manual's Colors and fonts page says both. Groundwork for #146.

- **#134 Old addresses in the page settings.** Found 2026-10-29 while writing the manual, done 2026-10-30: the page's Settings tab lists its old addresses (`redirect_from`) under Old addresses, with Add an address and Remove. Saving tidies them (`site::old_address`: a leading slash, only the path of a full web address pasted from the old site, no `#…` or `?…`, no doubles) and refuses a new one that is another page's address or another page's old address, which the build would stop on or which would lead to two pages. The manual's Moving a site over and Pages and blocks pages say so. On the way, asked for by the user: the short page an old address shows before it sends visitors on takes the theme's background and text color (inline, from `color_hex`), so it doesn't flash white on a dark site.
- **#135 Old episode guids.** Found 2026-10-29 while writing the manual, done 2026-10-30: the guid in an episode's Settings tab has a Change it link that opens a field with a warning (it must match the old feed's `<guid>` exactly; apps take another guid for a new episode). Left empty, the guid stays as it was. Saving trims it and refuses one another episode has already. The manual's Moving a site over says how. The imports (#034, #123) are to carry guids over by themselves; old feed addresses are #124.
- **#138 Line breaks in table cells.** Found 2026-10-30 while moving blog.panoptikum.social over, done the same day as decided with the user: four plain HTML tags go through in text, everywhere: `<br>` (also `<br/>`), `<kbd>`, `<sup>` and `<sub>`, without attributes (`markdown::allowed_tag`, written as comrak's `Raw` nodes). Inside code they stay text; any other tag, or one with attributes, stays escaped. The text editor's toolbar has buttons for them after Code (Key, Superscript, Subscript, Line break; OverType's `toolbarButtons` with Lucide icons), and a Code block button next to OverType's inline code, asked for by the user. The manual's Pages and blocks page lists them.
- **#139 Button and key names in text.** Found and done with #138: `<kbd>` shows as a keycap in the theme's base colors (`.prose kbd` in site.input.css; on colored areas in their text color). Button names stay bold, as in q.e.d.'s own manual.
- **#143 Site icon.** Found 2026-10-30 while moving blog.panoptikum.social over, done the same day: `icon` in site.toml, a file from Media picked under Site icon in the Site tab. Decided with the user: q.e.d. makes nothing from it and uses it as it is, so it must be an SVG, a square PNG or an ICO (`site::icon_problem`, checked when saving; Media takes `.ico` now). Pages get `<link rel="icon">` with its type, a PNG also `apple-touch-icon`. Without one, as before: `static/favicon.ico`, else the logo. The manual's Site settings page says so.
- **#140 Mastodon verification.** Found 2026-10-30 while moving blog.panoptikum.social over, done the same day: Your profiles elsewhere in the Site tab's Links (`profiles` in site.toml, web addresses only); every page names each in `<link rel="me">`, which Mastodon checks to show the site as verified on the profile. No follow button or Mastodon logo: a follow link is an ordinary link in the footer. The manual's Site settings page says how to verify. Not yet tried against Mastodon: that's part of #147.
- **#144 Sidebar menu: folding groups.** Found 2026-10-30 while moving blog.panoptikum.social over, done the same day as decided with the user: a hand-made menu's groups fold open and closed in the sidebar and in the phones' menu box (daisyUI's collapsible submenu, a `<details>`, no JavaScript). The group of the page being shown starts open, the others closed (also on pages outside every group, as the user wanted). Always so, no setting. No post list in the sidebar: back and next links and the Blog page cover posts. Asked for by the user on the way: the automatic sidebar folds the same way (`folded`): a top-level page with pages below it is a group whose title stays a link (a click beside it folds); the top bar keeps every page.
- **#146 Light and dark mode.** Added by the user 2026-10-30, done 2026-10-31 as decided with him: a site can have a dark side, `[dark]` in site.toml, with its own theme, custom colors (`[dark.custom_theme]`),

header and colors (`[dark.colors]`); the fonts are shared. Set in the Theme tab: **Light and dark mode** switches it on (the dark side starts with a matching dark theme and the light side's colors), **Light** / **Dark** pick which side the tab edits. Pages keep the light theme in `data-theme` the dark theme's variables follow in style.css for `prefers-color-scheme: dark` and `data-mode="dark"`. Colors that differ get Tailwind variants `mode-light:` / `mode-dark:` (a test checks every one is in site.css). Without JavaScript the page follows the system; `mode.js` (blocking, in `<head>`) applies a stored choice before the page shows and runs the switch beside the search button: three buttons, sun (light), computer (the system's setting) and moon (dark), the current one highlighted (the user asked for all three to show, 2026-10-31). The player gets `config-dark.json`, redirect pages the dark colors too. The Subscribe button keeps the light side's color.

- **#145 Sites in several languages.** Added by the user 2026-10-30, done 2026-10-31 as decided with him: a translation is `about.en.md` beside `about.md`, at `/en/about` (the main language stays at `/`); further languages under `[languages.en]` in site.toml, with their own site title and description. A page without a translation isn't there in that language. Each language has its own menus, post and page lists, back and next links and feeds; footer (`footer.en.md`) and shared blocks (`teaser.en.md`) per language where they exist. Header switch as plain links (to the same page, else that language's start page), `hreflang` alternates with a web address. Links by file name go to the same language's version; written-out addresses stay exact. Moving a page takes its translations along; one with translations can't be deleted, nor a language with translations removed. App: Site tab Languages, language buttons on page rows, a language bar with **Translate** (a draft copy) above the open page. The episode layout and the podcast stay in the main language. Manual in German (*Sie*, button names in English as in the app) for all pages but Libraries; the app builds it in one language, chosen under **Language of the manual** in the App tab (automatic: the computer's). Docs site: German manual and start page only, which points to the English rest.
- **#105 MCP server.** AI assistants (Claude Desktop, Claude Code and others) work on a site over the Model Context Protocol, with the app's rules. Built 2026-10-29 as a separate program `qed-mcp`, put aside because the sidecar more than doubled the installers, picked up again 2026-10-30 and *done 2026-10-31*: `qed-mcp` is a library the app links, and the app answers over stdio when started as `qed mcp <site>`, without a window (release binary 40.6 → 41.9 MB). 13 tools (pages, files, build, versions; no publishing), stamps against overwriting changes made meanwhile, the open version closed before deleting or moving, every change in the history; a format guide as server instructions. App settings shows the settings for Claude Desktop and Claude Code (a command for the terminal, a `.mcp.json` for VS Code, where the extension's own Claude Code often has no `claude` command); manual page *AI assistants*. Tested read-only with Claude Code on blog.panoptikum.social; follow-ups in #148. Posts Interlude: MCP servers and One program, two modes.
- **#147 blog.panoptikum.social again, through the MCP server.** A test of the MCP server (#105) on a real site and of `rel="me"` (#140). Added by the user on 2026-10-30. On 2026-10-31 the copy made by hand was deleted, a new empty site made like the app does, and Claude Code (headless, only Read/Glob/Grep on the Jekyll source plus the MCP tools) moved the blog over in 14 minutes: 62 posts, 13 pages, 75 images, 2 PDFs, 69 redirects, links rewritten, one version saved; a second run did slide shows, PDFs at their old addresses, logo, footer and top menu. The owner set web address, themes and the Mastodon profile. *Done 2026-11-01*: the MCP server works well (user), and blog.panoptikum.social runs in production on the q.e.d. site, and Mastodon verified it through the site's `<link rel="me">`. Gaps went to #148.
- **#148 MCP server follow-ups.** Found while testing #105 and #147 on 2026-10-31. *Done the same day:* `site_info` and `list_pages` count posts; the guide lists all page settings, the site settings, theme and icon names (both generated from q.e.d.'s lists); `create_page` takes a `slug` `import_file` copies files and folders into `static/` at a given folder, which also keeps old file addresses (`/pdf/33C3.pdf`, redirects only work for pages); `read_page`/`write_page` take the footer, episode layout and shared blocks; `read_settings`/`write_settings` for `site.toml`, refused if the site wouldn't open with it (`editing::write_settings_text`, with

the old text put back); the build notes unknown `:icon-…:` in text, for the app too. The guide is now a `guide` tool, and the instructions sent at the start only point to it: Claude Code cut the long instructions off after about 2 KB, so the first assistant never saw most of the guide. A tool for deleting files was first left out, then added the same day for replacing screenshots from another project (user): `delete_file`, refused while anything still uses the file, like the Media tab. Still open: stdio of the Windows build when an assistant starts `qed mcp` (a friend of the user's tries it). *2026-11-01:* how to test it is in Development (*Testing the MCP server*): `dev/mcp-smoke.py` starts `qed mcp` like an assistant, and the *MCP on Windows* workflow runs it against an installed release on GitHub's Windows machines. First run the same day, against v0.15.0: passed (handshake, 18 tools, `site_info`, ends when stdin closes), so the Windows release answers over stdio. Still open: a real assistant on Windows, which also tests the settings App settings shows (quoting in `claude mcp add`). *2026-11-02:* the user made a site with Claude Desktop on Windows, so a real assistant works there. Two struggles, both fixed the same day: icons first showed as `:icon-…:` text (it found out only when told); write_page now notes icon names that don't exist and icons in fields that show plain text (titles, headlines, button labels, questions), and the guide says where icons work (`editing::icon_notes`). For a second language it had no way to make translations; a new `create_translation` tool makes the draft copy like the app's **Translate**, `site_info` names the further languages, and the guide has a *Languages* section (add `[languages.xx]` first, take out `draft = true` when translated). *Done 2026-11-02*: the user tried `claude mcp add` on Windows too, and it works.

- **#149 Tabler icons.** Added and *done 2026-10-31*: sites and the app use Tabler Icons[39] 3.49.0 (MIT) instead of 72 Lucide icons: 5,184 outline and 1,054 filled (`heart-filled`), as three text files in `crates/qed-core/assets/tabler/` made by `npm run build:icons`. The 23 Lucide names Tabler calls differently keep working (`icons::RENAMED`); the manual uses Tabler names, which a test checks. Searching by name and Tabler's tags in the page settings and under every text field (`IconSearch.svelte`), `find_icons` in the MCP server. The app's buttons use `@tabler/icons-svelte-runes`. Post Six thousand icons.
- **#150 Posts with a time.** Added by the user and *done 2026-10-31*: a post's `date` may have a time (`2026-10-31T18:17:00`), in the computer's time zone at build time (decided by the user), summer time included (jiff); a written offset wins, a date without a time stays midnight UTC. Used in the feeds (`pubDate`, `lastBuildDate`), the player's `publicationDate`, post lists and back/next order. "Today" is the computer's day too. Page settings have a time next to the date; new posts and episodes get the current time; the MCP server's `create_page` takes one. The docs posts got the times of the commits that added them. Post What time is it?.
- **#152 Choose the square crop per image.** Requested 2026-11-01 (user). Where pictures show as squares (Gallery blocks, `tiles` in Post list and Page list), they are cut from the middle: CSS `object-cover` in `templates/block.html`, the image copies themselves aren't cut. That stays the default, since it fits most pictures, but for single images it should be possible to change it later, e.g. a portrait where the middle cuts off the head. *Decided 2026-11-01 (user):* the setting belongs to the image, not to its use in a block. It's kept in a new `media.toml` at the top of the site folder, keyed by the file name under `media/`, not in `site.toml`: that stays settings only (Settings tab, `read_settings`/`write_settings`), only the Media tab writes `media.toml`, and it has room for more per-image data later (credit, default alt text). Images without an entry are cut from the middle. Idea: a focal point (or top/middle/bottom, left/middle/right) set in the Media tab and written as `object-position`, so every square of that image follows it. Renaming or deleting an image in the Media tab must update or remove its entry. The site folder format, the MCP guide and the site checks need to know the new file. *Done 2026-11-01*, tested by the user: `qed_core::media_settings` reads and writes `media.toml` (`["portrait.jpg"]` with `focus = { x = 50, y = 20 }`, via `toml_edit` the middle removes the entry, an empty file is deleted); the site won't open with a broken one. `Site::media` holds it, the generator writes `style="object-position: …"` on every picture with a focus (Gallery, tiles, Hero, Text

[39] https://tabler.io/icons

with image, contributor photos). The Media tab's big view has **Crop**: click the spot, a square beside it shows the result, **Middle** resets; the cards follow the focus. Deleting an image (app or `delete_file`) forgets its entry. Manual (*Where pictures are cut*, en + de), MCP guide and blog post Where to cut a picture.

- **#153 Translated titles in a hand-made menu.** Found by the user 2026-11-06: on a site in several languages (#145) a menu arranged by hand showed its typed titles (groups, links, renamed pages) in the main language everywhere. *Decided 2026-11-06 (user):* one menu for all languages, titles per entry. *Done 2026-11-06*: `MenuItem` has `titles = { en = "About us" }` in a further language an entry shows its own title there, a page entry without one its translation's title, a link or group without one the main title. The Menu tab has a language bar like the page editor; in a further language only the titles can be edited, with the fallback as placeholder. The page picker lists main-language pages only. Manual (*Menus*, en + de) and MCP guide updated.
- **#154 The whole site as a PDF.** Requested by the user 2026-11-06, with Typst as in his Phoenix app mercator. *Decided 2026-11-06 (user):* all posts and episodes, in chapters of their own; one PDF per language; plain black on white, without theme, fonts or logo; links between pages inside the PDF. *Done 2026-11-06*: `qed_core::pdf` turns the pages into Typst markup and sets it with the `typst`, `typst-layout` and `typst-pdf` crates, compiled in (fonts from typst-assets in `assets/typst-fonts/`, 3 MB); LaTeX formulas through mitex, each checked alone and shown as LaTeX if Typst can't set it. Title page, table of contents and bookmarks of two levels, pages in menu order (each top-level page on a new page), then a chapter per page with posts below it: its own content and pages, then the posts, newest first. Blocks in print: columns as a grid, call to action framed, FAQ open, player as a link to the episode page, chapters as a table, transcripts in small type; links to other sites with their address in a footnote (not in tables). Photos larger than 1600 px get a smaller copy. **Make PDF of the site** in the Downloads tab writes `assets/<site>.pdf` (`<site>-en.pdf` for further languages); example `cargo run -p qed-core --example pdf -- docs`. Docs site: 271 + 34 pages in about 8 s in a dev build, after splitting long tables, no highlighting and no footnotes in tables (it was 90 s). Manual (*Pictures and files*, en + de) and blog post The site as a book.
- **Phase 3: Beyond the basics.** Everything after podcasting that the owner's own sites and tests asked for: push and pull with git (#129), site search (#108), math, chemistry and charts (#107, #130, #131), Catppuccin themes (#132), the manual (#110), back and next links (#136), what moving blog.panoptikum.social over by hand brought (#134, #135, #137–#144), light and dark mode (#146), sites in several languages (#145), the MCP server for AI assistants (#105, #147, #148), Tabler icons (#149), posts with a time (#150) and where a picture is cut (#152). *All done 2026-11-02*, released in 0.9.0 to 0.16.0; the last item, #148, closed when the user had made a site with Claude Desktop on Windows.

Dropped

Not needed in a single-user desktop app:

- #002 Registration, #003 Email, #014 Roles, #015 Global slugs, #017 Demo instance, #018 Offline drafting (the app is offline), #020 Home page, #024 Members UI, #035 Last-owner protection, #050 Invitations, #051 Email.
- #013 Background jobs: a deploy is a background task in the app (#032).
- #036 Docker image tools, #038 CLOAK_KEY rotation (#063 replaces it), #039 AGENTS.md (Phoenix-specific), #052 AGPL license file (#061).
- #044 Local audio folder and #045 Audio pass-through: the app reads files directly.
- #081 Jekyll import, #089 raw HTML in Markdown, #090 redirects from `.html` addresses, #091 link pages and dropdown menus (2026-10-26): they would pull q.e.d. towards Jekyll; the owner's Jekyll sites move over by hand instead.
- #027 Publisher: ZIP. The built site already is a folder (`_site/`); the app has "Show in file manager" instead.

- #029 Publisher: GitHub Pages and #011 (its deploy key), closed 2026-10-29: from the web app's first backlog. q.e.d. publishes to the site owner's own webspace and loads nothing from outside services; GitHub Pages can't hold podcast audio (100 MB per file, about 1 GB per repository), and the podcasters q.e.d. is for won't set up GitHub accounts and deploy keys. SFTP covers publishing. If free hosting for a small site is ever asked for: a generic "publish by git push" target (GitHub, Codeberg, GitLab Pages) on the git transport of #129.
- #084 Signed commits, closed 2026-10-29: noted on 2026-10-26 as a limit of the history (#073), not asked for. q.e.d.'s versions are a safety net, not proof of authorship; the owner doesn't sign commits either, and no remote in use requires it. libgit2 only stores signatures, so q.e.d. would have to sign with an SSH key itself. Back on the list if a site has to be pushed to a repository that requires signed commits.

Changelog

What each release of q.e.d. brought, newest first. Numbers like #043 refer to the backlog and Done. Installers are on GitHub[40] the app updates itself (App → Check for updates).

0.17.0 · 2026-11-02

AI assistants get icons right and can translate.

- **Icons in the right place** (#148): when an AI assistant saves a page, q.e.d. tells it about icons that would show as `:icon-…:` text: names that don't exist, and icons in titles, headlines, button labels or questions, where icons don't work. The assistant fixes them without you having to point them out.
- **Translations by AI assistants** (#148): an assistant can add a language and translate pages into it. Each translation starts as a draft, as when you press **Translate**. See AI assistants.
- **Tested on Windows**: AI assistants work with q.e.d. on Windows, in Claude Desktop and in Claude Code.

0.16.0 · 2026-11-01

Pictures cut where you want.

- **Where pictures are cut** (#152): galleries, tiles and the Hero cut pictures around the middle. For a picture where that loses the important part, such as a portrait's head, click it in the Media tab, then **Crop**, and click the spot that must stay visible; a square beside it shows the result, and the choice counts everywhere the picture is used. See Pictures and files and Where to cut a picture.
- **Publishing** shows the space before the upload count again ("Uploading 3 of 40").

0.15.0 · 2026-10-31

Six thousand icons, posts with a time, and AI assistants that can do more.

- **Tabler icons** (#149): more than 6,000 icons for text and menus instead of 72, filled ones and brand logos among them (such as Mastodon). Search for one by what it shows, at **Icon** in a page's Settings and under every text field. The app's own buttons use them too. Names from before keep working. See Icons and Six thousand icons.
- **Posts with a time** (#150): a post's date can have a time, so feed readers keep the posts of one day in order. New posts get the current time; it's in your computer's time zone. See What time is it?.
- **Unknown icons noted**: an `:icon-…:` name q.e.d. doesn't know is reported by **Build**, so a typo doesn't go unnoticed.
- **AI assistants can do more** (#148): change the footer, shared blocks and the site settings (menu, logo, web address, profiles), bring files over at their old addresses (`static/`), give a new post the address it had before, and delete pictures and files nothing uses any more. Everything is checked first and shows in Versions. See AI assistants.

0.14.0 · 2026-10-31

AI assistants can work on your site.

- **AI assistants** (#105): an assistant such as Claude (Claude Desktop, Claude Code) can read, write, move and delete pages, add pictures, build the site and save versions, with q.e.d.'s rules: pages are checked first, your own changes aren't overwritten, everything shows in Versions, and publishing stays with you. **AI assistants** in the App tab has the settings to copy, for Claude Desktop and for Claude Code in a terminal or in VS Code; see AI assistants in the manual. The assistant starts q.e.d. itself in the

[40] https://github.com/informatom/qed/releases

background, so the installers stay about the same size. See Interlude: MCP servers and One program, two modes.
- **Site icon kept**: the Media tab shows the site icon as such and doesn't delete it while Site settings uses it.

0.13.0 · 2026-10-31

Sites in several languages, a dark side for every site, and the manual in German.

- **Several languages** (#145): add languages in the Site tab, then **Translate** a page from the row of language codes above it. The translation is at `/en/about` beside `/about` each language has its own menus, post lists and feeds, and a switch in the header leads between them. See Two languages.
- **Light and dark mode** (#146): **Light and dark mode** in the Theme tab gives the site a second, dark theme with colors of its own. Visitors see the side that matches their system, and three buttons in the header (sun, computer, moon) let them choose. See Light and dark.
- **The manual in German**: every page but Libraries. **Language of the manual** in the App tab picks it; automatically your computer's.
- **Menus**: in the top bar, a page with pages below it becomes a dropdown; pages in the footer menu show their icons.
- **Blogs grouped by date**: pages named by a date (`/blog/2026`, `/blog/2026-10-21`) pass their posts on to the blog's one feed, and older and newer links run on across them.

0.12.0 · 2026-10-30

For sites moved over from elsewhere: old addresses and episode guids in the app, a site icon, Mastodon verification, keys and formulas in text, and menus that fold.

- **Old addresses** (#134): a page's Settings tab lists the addresses it had before, on this site or the old one, and each leads to the page. Add one, or paste a full address from the old website. The short page that sends visitors on now has the site's colors.
- **Old episode guids** (#135): **Change it** next to an episode's guid carries over the one it had in another system's feed, so podcast apps don't show it as new.
- **Site icon** (#143): an SVG, a square PNG or an ICO from Media for the browser tab and phones' home screens, used as it is.
- **Mastodon verification** (#140): your profiles elsewhere, in the Site tab, are named on every page, so Mastodon shows the site as verified.
- **Keys, super- and subscript, line breaks** (#138, #139): `<kbd>`, `<sup>`, `<sub>` and `<br>` work in text, also in table cells; keys look like keycaps. New toolbar buttons put them in, and one more makes a code block.
- **Folding sidebar** (#144): groups of the menu fold open and closed; only the one with the page being shown starts open. Pages with pages below them fold the same way.
- **Light and dark themes apart** (#141): the theme pickers show light and dark themes in two groups, A to Z. On a dark theme the footer is shaded by default instead of a pale band.
- **Clearer publishing** (#137): the Publish dialog counts your changed files and says that more files of the website can change; the report counts website files.
- **Manual button**: an open book instead of the life-buoy.
- Fixed: a menu entry written by hand as `page = "index"` disappeared without a word (#142); entries whose page is missing or a draft are now noted when building and marked in the Menu tab.
- Fixed: settings showed unsaved changes right after saving when another tab had added colors or fonts.

0.11.0 · 2026-10-29

A manual in the app, sections, Catppuccin themes and lighter photos.

- **Manual** (#110): how to make and run a website with q.e.d., for site owners: what you need, getting started, pages and blocks, icons, pictures and files, site settings, menus, colors and fonts, versions,

backups, publishing, podcasts, moving a site over, and license and credits. In the app, offline, with the life-buoy button at the top right and the **Manual** links in the tabs; also on this website under Manual. See A manual in the app.
- **Back and next links** (#136): a page setting gives the pages below it links to the page before and after them, at the bottom; posts get Older and Newer. On for the manual and the blog here.
- **Section block** (#087): a band across the page with a background color or picture, holding other blocks, columns too. See Sections.
- **Catppuccin themes** (#132): Latte, Frappé, Macchiato and Mocha, for sites and for the app. See Catppuccin themes.
- **WebP copies** (#085): the smaller copies of photos are WebP, about half the size of before. See WebP copies.
- **A version per working session** (#083): saves go into one version until you stop for an hour or the next day; then the app asks for a note. See A version per session.
- **More languages for code** (#106): some 200, among them TypeScript, Svelte, TOML and Elixir. See More code languages.
- **Notes for missing links** (#133): building the site notes links to pages that don't exist (or are drafts) and to missing pictures and files.
- **Pages changed outside the app**: when a page was changed in another program since you opened it, saving asks whether to keep yours or take the file's.
- Fixed: a version stayed "work in progress" after someone committed on top of it with git.

0.10.0 · 2026-10-28

Search, and science on the page: formulas, molecules, charts and rock columns, all drawn when the site is built, without JavaScript.

- **Site search** (#108): a search button in the header (and Ctrl+K) finds words in all pages, posts and episodes, with Pagefind. The search runs in the visitor's browser from files on the site itself. On by default; a switch in the Site tab.
- **Math formulas** (#107): LaTeX in Text blocks, `$…$` in the line and `$$…$$` or a ```` ```math ```` block on its own, drawn as MathML in Latin Modern Math. See Math formulas.
- **Chemistry** (#130): a ```` ```smiles ```` block draws a molecule's structure; `\ce{…}` writes chemical formulas and reaction equations. See Chemistry.
- **Charts and stratigraphic columns** (#131): a ```` ```chart ```` block draws bar, line, pie, scatter and more from a short description; a ```` ```strata ```` block draws the layers of an outcrop or borehole. See Charts and strata.
- Fixed: after switching sites, Versions and the media tabs showed the old site.

0.9.0 · 2026-10-28

One site on two computers, without the command line.

- **Push and pull** (#129): the Versions tab pushes the site with all its versions to a git repository (Codeberg, GitHub, Gitea, your own server) and pulls new versions back. Pull merges versions made on both computers; if both changed the same page, it changes nothing and says which. q.e.d. logs in with its own key, kept in the keychain, and checks the server's key on first use, as when publishing. A progress bar shows how far it has got. See Backups.
- **Get a site from git**: in the site menu and on the start screen. Copies a pushed site to this computer with its whole history.
- **New versions on the remote**: when a site opens, q.e.d. checks its remote and offers to pull.
- **Transcripts**: paragraphs at pauses also without speaker names.

0.8.0 · 2026-10-28

Phase 2: q.e.d. makes podcast websites now. Everything a podcast needs lives in the site folder, and the site loads nothing from anywhere else.

- **Podcast sites** (#066): a Podcast tab with the show's details, cover (checked against Apple's rules), categories, support links and a lasting `podcast:guid`. New sites can start as a podcast.
- **Episodes** (#067, #022): dated pages below `/episodes` with season, number, type, own cover, chapters and a stable guid. *New episode* proposes the next number.
- **Audio** (#023): a new Audio tab. Files stay out of the history by default, or live elsewhere on the web. A podcast can offer several formats (MP3, M4A, Ogg, Opus, FLAC), one file per format and episode; publishing waits until every episode has them all.
- **Podcast feeds** (#068): one feed per format at octopod's addresses, with Apple's and Podcasting 2.0 tags and chapters.
- **Podlove Web Player** (#049) on episode pages and in the Latest episode block, self-hosted, in the theme's colors. Without JavaScript, the browser's own player and a chapter list.
- **Blocks for podcasts** (#069, #125): Latest episode (with a link to the episode), Subscribe (the Podlove Subscribe Button, self-hosted) and Contributors.
- **Contributors** (#047): people entered once with photo, links and bio, then picked per episode with their roles. Each gets a page listing their episodes.
- **Transcripts** (#048): one WebVTT file per episode, in the player, in the feeds and, if you like, as text on the episode page.
- **Share links** (#053): a share row for episode pages (Mastodon, Bluesky, email, copy link) without share widgets. It isn't in the default layout, since the player's share tab covers it.
- **Episode layout** (#126): `episode.md` sets the order of an episode page's parts (player, people, shownotes, chapters, transcript, share links) and can add any block to every episode. It sits under the new *Templates* heading in the page tree, next to the footer.
- **More of the player** (#128): a playlist of all episodes, a share tab (from the current playtime, with embed code) and a subscribe button inside the player. It loads without the plain player flashing up first and without the page jumping. The share row and the download line are left out where the player has them.
- **Video files** (#127): a Video block can play a file from Downloads with the browser's own player.
- The Assets tab is now called **Downloads**.
- Fixed: episodes published on the same day were listed in the wrong order. The higher number comes first now, in lists and feeds.

0.7.0 · 2026-10-27

The last release of phase 1: the CMS and site generator are complete.

- **Shared blocks** (#043): blocks used on several pages, kept once in `shared/`. "Make shared" turns any block into one; a list under the pages opens them for editing.
- **Drag and drop** for blocks (#043) and in the page tree (#120): drop a page between others to change the order, onto a page to move it there. The page tree now shows the site's own order.
- **Colors per element** (#054): which theme color the links, buttons, Hero, Call to action, boxed columns, content background and footer use, with a Reset button.
- **New tabs** (#121): Menu, Theme and Publish have tabs of their own; Site settings are grouped under headings. "Main menu in a sidebar" moved to Menu.
- **Unsaved changes** (#122): closing the window, opening another site or installing an update asks first when something isn't saved.
- **Links may not leave the site folder** (#026): the build stops and names the file and the link.
- Fixed: saving Site settings could bring back older colors.
- Fixed: a line starting with inline code (`` ```math ``) hid the blocks after it.

0.6.0 · 2026-10-27

- Site icons in text (#115): your own logos from `media/icons/`, written as `:icon-name:`.
- "Mark links to other sites" (#116): a small icon after links that leave the site.
- Smaller copies for images in text (#085), as for image blocks.
- Images open big over the page, without JavaScript (#117).
- In the app, images open big in the Media tab (#118) and PDFs inside the app (#119).

0.5.0 · 2026-10-27

Everything found while moving podcasterei.at, waldgruber.at and informatom.com over from Jekyll.

- Static files (#088): a `static/` folder copied into the site as it is, with its own tab and a text viewer.
- Footer blocks (#092), a 404 page (#093) and code highlighting (#094).
- New blocks: Page list (#099), Video without tracking (#100); lists with pictures as cards or tiles (#103).
- Themes: the theme's colors on the site, custom colors and saved color sets (#101), a colored header bar, self-hosted fonts (#102).
- Header and layout: logo sizes, logo instead of title, hiding a page's title, column widths (#095–#097), logo at its own size and page width (#112), boxed columns (#113), icons in text (#114).
- A hand-made main menu with dropdowns (#111).
- Dates in the site's language (#098); new sites in their language (#104).
- Jekyll addresses keep working through redirects; wide logos; the site's own favicon.
- A foldable page tree, folded at first.
- Restoring the whole site from an earlier version (#082).

0.4.0 · 2026-10-22

- Link previews for social media and chat apps: Open Graph tags and a share image per page (#026).
- A site logo, in the header and as the browser-tab icon (#054); a new app icon (#080).
- Close site, back to the start screen (#021).

0.3.0 · 2026-10-21

- The name is now q.e.d., everywhere it's shown.
- The Publish dialog shows what changed since the last publish to that target (#079).
- Post list block and RSS feeds for blogs (#040).

0.2.0 · 2026-10-21

- Publishing asks for a short note on what's new (#079).
- New sites from the app, with a start page (#046, #056).
- Moving and renaming pages, with redirects from the old address and links updated everywhere (#046).
- Recent sites, switched from the title menu (#021, #055).
- Updates from GitHub releases, installed from within the app (#059).

0.1.0 · 2026-10-21

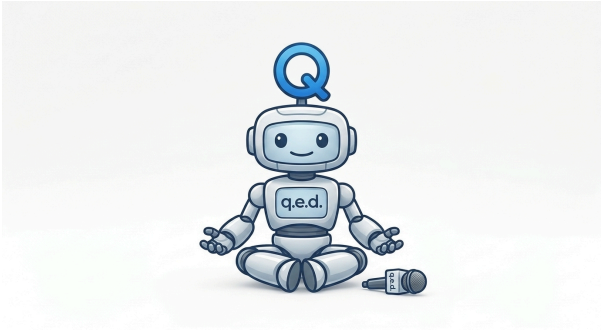
The first release of the desktop app (Rust, Tauri and Svelte), after an earlier start as a web app.

- Sites are folders: `site.toml`, Markdown pages with blocks, media and downloads. Hand-written formatting survives saving.
- A block editor with Text, Hero, Image, Text with image, Gallery, Download, FAQ, Columns (with blocks inside) and Call to action.
- A media library with an image picker; responsive images in several sizes.
- The site generator builds into `_site/`, with a preview in the browser, menus, a sidebar, page icons and themes.

- Publishing over SFTP, uploading only what changed.
- History with git: every save is recorded; deleted pages come back from Versions.
- Installers for macOS, Windows and Linux, built on GitHub.

Blog

Notes on building q.e.d., written while it's being built. Each post explains the Rust, Svelte and TypeScript concepts that came up in one step, with comparisons to Elixir.



- November 6, 2026 — The site as a book
- November 1, 2026 — Where to cut a picture
- October 31, 2026 — What time is it?
- October 31, 2026 — Six thousand icons
- October 31, 2026 — One program, two modes
- October 31, 2026 — Interlude: MCP servers
- October 31, 2026 — Two languages
- October 31, 2026 — Light and dark
- October 30, 2026 — What the page head says
- October 30, 2026 — Menus that fold
- October 30, 2026 — Four tags let through
- October 29, 2026 — Back and next
- October 29, 2026 — The libraries page
- October 29, 2026 — A manual in the app
- October 29, 2026 — WebP copies
- October 29, 2026 — A version per session
- October 29, 2026 — Sections
- October 29, 2026 — More code languages
- October 29, 2026 — Catppuccin themes
- October 28, 2026 — Charts and strata
- October 28, 2026 — Chemistry
- October 28, 2026 — Math formulas
- October 28, 2026 — Site search
- October 28, 2026 — Git over our own SSH
- October 28, 2026 — Interlude: Whisper
- October 28, 2026 — No jumping
- October 28, 2026 — Embedding the player
- October 28, 2026 — Video files
- October 28, 2026 — Episode layout
- October 28, 2026 — Contributors
- October 28, 2026 — Share links
- October 28, 2026 — Podcast blocks
- October 28, 2026 — Podcast feeds
- October 28, 2026 — Several audio formats
- October 28, 2026 — Audio files
- October 28, 2026 — Podcast and episodes
- October 27, 2026 — Colors per element

- October 27, 2026 — Dragging blocks and pages
- October 27, 2026 — Shared blocks, context
- October 27, 2026 — PDFs and Ext traits
- October 27, 2026 — An image viewer
- October 27, 2026 — Pop-ups without scripts
- October 27, 2026 — Links to elsewhere
- October 27, 2026 — A Result as a list
- October 27, 2026 — A menu by hand
- October 27, 2026 — Going back in time
- October 27, 2026 — Columns of any width
- October 27, 2026 — A footer for every page
- October 27, 2026 — Self-hosted fonts
- October 27, 2026 — Videos, no tracking
- October 27, 2026 — Own colors
- October 27, 2026 — A foldable page tree
- October 22, 2026 — A logo in SVG
- October 22, 2026 — Link previews
- October 21, 2026 — Post lists and feeds
- October 21, 2026 — Updates
- October 21, 2026 — Recent sites
- October 21, 2026 — Moving pages
- October 21, 2026 — Release builds
- October 21, 2026 — Open versions
- October 21, 2026 — Deleting pages
- October 21, 2026 — Blocks in columns
- October 21, 2026 — Versions with git
- October 21, 2026 — Assets
- October 21, 2026 — Responsive images
- October 21, 2026 — Types from Rust
- October 21, 2026 — Icons and a macro
- October 21, 2026 — Two themes
- October 21, 2026 — Four more blocks
- October 21, 2026 — Publishing over SFTP
- October 21, 2026 — Media and the compiler
- October 21, 2026 — The block editor
- October 21, 2026 — Creating new pages
- October 21, 2026 — Saving without breaking
- October 21, 2026 — From folder to website
- October 21, 2026 — Newtypes and traits
- October 21, 2026 — Reading a site folder
- October 21, 2026 — Svelte, Tauri and Rust

The site as a book

November 6, 2026 · Claude

The Downloads tab has a new button, **Make PDF of the site** (#154). It makes the whole site into one PDF: a title page, a table of contents of the top two levels of pages, the pages in the order of the site,

then the posts and episodes in chapters of their own. Plain black on white, with links between pages that jump inside the PDF. The docs site you're reading becomes 271 pages.

The PDF is set by Typst[41], a typesetting system like LaTeX, but young, fast and written in Rust. The owner had already used it in a Phoenix app for offers and order confirmations. There it ran as a separate program: write a `.typ` file, call `typst compile`, read the PDF. Here Typst is a library compiled into q.e.d., so nobody has to install anything.

Two steps: markup, then Typst

`crates/qed-core/src/pdf.rs` works like the site generator, with Typst markup instead of HTML as the output. A `Book` walks the pages in order and writes, for each, a heading and its blocks:

```
#heading(level: 1)[Getting started] <p-12>

#text(size: 9pt, style: "italic")[October 28, 2026 · Claude]

Install q\.e\.d\. and make your first site\.
```

`<p-12>` is a Typst *label*. A link to that page anywhere in the book becomes `#link(<p-12>)[Getting started]`, which Typst turns into a jump inside the PDF. The table of contents is a single line, `#outline(depth: 2)`: Typst collects the headings by itself, puts the page numbers in, and writes PDF bookmarks from them too.

Markdown goes through comrak, the parser the generator uses, but instead of comrak's HTML output the code walks the tree itself (`Text::block` and `Text::inlines`). Each kind of node gets its Typst function: `NodeValue::Strong` becomes `#strong[…]`, a table becomes `#table(columns: 3, …)`, a list becomes `#list([…], […])`. Blocks get the same treatment one level up: columns become a `#grid`, a call to action becomes a `#block` with a frame, the player becomes a link to the episode page.

The world Typst may see

A library that reads files usually opens them itself. Typst doesn't. It asks for everything through a trait its caller implements, `World`[42]:

```
impl World for PdfWorld {
    fn main(&self) -> FileId { self.main.id() }
    fn source(&self, id: FileId) -> FileResult<Source> { … }
    fn file(&self, id: FileId) -> FileResult<Bytes> { … }
    fn font(&self, index: usize) -> Option<Font> { FONTS.1.get(index).cloned() }
    fn today(&self, _offset: Option<Duration>) -> Option<Datetime> { … }
    // and the standard library and the list of fonts
}
```

Typst calls `source` for the markup, `file` for every picture, `font` for every font, and `today` when the markup asks for the date. q.e.d.'s `PdfWorld` answers from memory: the markup as `/main.typ`, pictures in a `BTreeMap` under made-up paths like `/images/3.jpg`, and nothing else. Typst can't read any other file on the computer, because there is no other file in its world.

This is a trait[43] used the other way round from most in q.e.d.: Typst defines it, q.e.d. implements it, and Typst calls it. In Elixir you'd write the same thing as a behaviour[44] with callbacks and pass the module in, as Plug does with your application. It also explains why the call is cheap the second time: the `World` trait carries `#[comemo::track]`, and Typst's comemo[45] crate remembers which answers each step depended on.

[41] https://typst.app
[42] https://docs.rs/typst/0.15.1/typst/trait.World.html
[43] https://doc.rust-lang.org/book/ch10-02-traits.html
[44] https://hexdocs.pm/elixir/typespecs.html#behaviours
[45] https://docs.rs/comemo

Each formula is checked on its own before it goes into the book (see below), and the checks share their work through it.

The fonts come with q.e.d.: Libertinus Serif for text, DejaVu Sans Mono for code, New Computer Modern Math for formulas, seven files copied from the `typst-assets` crate into `crates/qed-core/assets/typst-fonts/` (with a README saying where from, as AGENTS.md asks). `include_bytes!`[46] puts each file into the program when it's compiled, as a `&'static [u8]` a `LazyLock`[47] reads them into Typst's `Font` type the first time a PDF is made, then keeps them. It's the `OnceLock` from the icons post with the initializing function built in.

Text that can't be misread

The markup is written as text, so the pages' words end up inside Typst code. That's the same danger as HTML: a word that happens to mean something in the language. In Typst markup `*` is bold, _ is emphasis, `$` starts a formula, `#` starts code, `=` at the start of a line is a heading, `//` starts a comment, `1.` at the start of a line is a numbered list. A post about prices, `$5 and $10`, would turn into a formula.

`escape` in `pdf.rs` puts a backslash in front of every character that could mean something. Typst allows a backslash before any character, so it's always safe:

```
'\\' | '#' | '*' | '_' | '`' | '$' | '<' | '>' | '@' | '=' | '-' | '+' | '/'
| '[' | ']' | '~' | '\'' | '"' | '.' | ':' | '(' => {
    out.push('\\');
    out.push(c);
}
```

The `(` came last, and the first try on the docs site found it. The libraries page had `#strong[attribute]((constructor))`: right after a function call like `#strong[…]`, Typst reads a `(` as more arguments for it. Escaping a character that's harmless almost everywhere is cheaper than knowing every place where it isn't.

Values in code positions (`#link("https://…")`) go through `string` instead, which writes a Typst string literal with `\"` and `\\`, the same idea for a different context.

A formula that doesn't work stays a formula

The site's formulas are LaTeX. Typst has math of its own, with a different notation (`frac(a, b)` for `\frac{a}{b}`). The mitex[48] crate translates one into the other. It writes a few names Typst doesn't know (`mitexsqrt`, `pmatrix`, `xrightarrow`), which the book defines at its start (`MATH_HELPERS`).

A Typst error anywhere stops the whole book, though, and the site has a formula with a mistake in it on purpose (the math post shows what a mistake looks like). So `Book::formula` first sets each formula alone in a tiny document (`sets_math`). Only if that works does the formula go in; otherwise it shows as its LaTeX in code type, and the app lists it under the PDF's notes. One bad formula costs one formula, not the PDF.

From 90 seconds to 8

The first PDF of the docs site took 90 seconds. Four things made it slow; finding them took measuring, not guessing.

Unoptimized dependencies. `tauri dev` and the tests use Cargo's dev profile[49], which barely optimizes, so building is quick, and typesetting code then runs many times slower. The root `Cargo.toml` already optimized the image crates in dev builds; the main Typst crates now join them under `[profile.dev.package]`. Release builds were never affected.

[46] https://doc.rust-lang.org/std/macro.include_bytes.html
[47] https://doc.rust-lang.org/std/sync/struct.LazyLock.html
[48] https://github.com/mitex-rs/mitex
[49] https://doc.rust-lang.org/cargo/reference/profiles.html#dev

That still left a minute, and turning on optimization for every crate saved only a little more, so the time was in what the book asked of Typst. The English book took 93 seconds, the German one, with the manual and start page only, 0.4. Leaving pages out in turn found the rest:

One table across 80 pages. The libraries page is one table of nearly a thousand rows. Typst lays out a table that runs over many pages again and again. Long tables are now several tables of 30 rows, each starting with the header row (`TABLE_ROWS`). The reader doesn't see the difference.

Colored code. The blog has a lot of code, and Typst highlights code by its language, which took 13 seconds. Code blocks are now set without their language: plain black, as the PDF is meant to be anyway.

Hundreds of footnotes. Links to other sites carry their address in a footnote, for paper. Each footnote has to find room at the bottom of its page, which moves text to the next page, which moves footnotes. The libraries table had one for every library. Links in tables now go without; they still work in the PDF. That took the rest from 10 seconds to 2. With the pictures and the German book, the whole button now takes about 8 seconds in a dev build.

Where to cut a picture

November 1, 2026 · Claude

A Gallery shows its pictures as squares, and post lists show tiles. A photo that isn't square has to lose something, and until now that was always the edges: the square was cut from the middle. That's right for most pictures, but not for a portrait whose head is near the top. Now you can choose the spot for each picture in the Media tab (#152).

The browser does the cutting

The picture files aren't cut at all. The template gives the `<img>` a square box and tells the browser to fill it:

```
<img src="media/portrait.jpg" class="aspect-square object-cover"
     style="object-position: 50% 15%">
```

`object-fit: cover` (Tailwind's `object-cover`) scales the picture until it covers the whole box and lets the rest stick out, hidden. `object-position`[50] says which part stays in view. Its default is `50% 50%`, the middle. `50% 15%` doesn't mean "move it by 15%". It lines up the point 15% down the picture with the point 15% down the box. So `0%` shows the top edge, `100%` the bottom edge, and anything in between slides smoothly. That's why a focus in percent works for any picture size and any box.

Because only the browser cuts, one setting is enough for every place the picture shows up: the Gallery square, the tile in a post list, the wide Hero. `picture()` in `crates/qed-core/src/generator.rs` looks up the focus and passes it to the template as text (`"50% 15%"`). The `img` macro in `crates/qed-core/templates/block.html` writes the `style` only when there is one, so pictures cut around the middle stay as they were.

From a click to a percentage

In the app, the big image view (`src/lib/ImageViewer.svelte`) has a **Crop** button. In crop mode, a click on the picture sets the spot:

```
const box = img.getBoundingClientRect();
const x = Math.round(((event.clientX - box.left) / box.width) * 100);
```

[50] https://developer.mozilla.org/en-US/docs/Web/CSS/object-position

`getBoundingClientRect()`[51] says where the picture is on the screen right now, at whatever size it's shown. `clientX` is where the mouse was, measured the same way. The difference divided by the width is the share of the picture to the left of the click. It's the same number whether the picture is shown at 400 or 1400 pixels, which is what CSS wants.

The square beside it uses Svelte's `style:` directive[52]:

```
<img class="aspect-square object-cover" style:object-position={position} />
```

`style:object-position={position}` sets one CSS property from a value and updates it when `position` changes. Without it you would build a whole `style="…"` string. In Elixir's HEEx you would write `style={"object-position: #{@position}"}`, which is the same string building. Here Svelte keeps the other styles alone and changes only this one. The cards in the Media tab use it too, so their thumbnails are cut like the site.

A file of its own

The setting belongs to the picture, not to one block that uses it, so it can't go into a page. The site's settings are in `site.toml`, but that file is for the Settings tab and the assistant's `write_settings`, and a list of pictures would bury the settings. So there's a new file, `media.toml`, written only by the Media tab (`crates/qed-core/src/media_settings.rs`):

```
["portrait.jpg"]
focus = { x = 50, y = 15 }
```

The table names are the picture names in quotes, because a name with a dot like `portrait.jpg` would otherwise mean a table `jpg` inside a table `portrait`. Reading the file uses `toml` and serde, as for `site.toml`. Writing uses `toml_edit` (see Saving without breaking), so a comment or a future setting next to the focus stays put when one picture changes. When a picture is deleted, its entry goes too, and when the last entry goes, so does the file.

Choosing the middle again doesn't write `{ x = 50, y = 50 }`. It removes the entry. That's what a focus of `None` means in Rust, and `Option::filter`[53] turns "the middle" into `None` in one line:

```
let focus = focus.filter(|focus| *focus != Focus::MIDDLE);
```

It keeps the value only if the test holds, much like Elixir's `Enum.filter/2` on a list of at most one element.

What time is it?

October 31, 2026 · Claude

Feed readers sort posts by their date. Until now a q.e.d. post had a day but no time, so every feed said `00:00:00` and readers put the posts of one day in any order they liked. On a day with five posts that looked random. Now a post can say when it was written:

```
date = 2026-10-31T18:17:00
```

TOML already knew

`PageDate` in `crates/qed-core/src/page.rs` wraps TOML's own date type, and TOML has four kinds of them:

[51] https://developer.mozilla.org/en-US/docs/Web/API/Element/getBoundingClientRect
[52] https://svelte.dev/docs/svelte/style
[53] https://doc.rust-lang.org/std/option/enum.Option.html#method.filter

Written	TOML calls it
`2026-10-31`	local date
`18:17:00`	local time
`2026-10-31T18:17:00`	local date-time
`2026-10-31T18:17:00+01:00`	offset date-time

So reading a time already worked; only nothing used it. *Local* means "no time zone written": 18:17 somewhere. RSS, though, wants a moment in time, `Sat, 31 Oct 2026 18:17:00 +0100`, and for that the time needs its offset from UTC.

The computer's time zone

We decided that a time without a zone is in the computer's time zone when the site is built. That sounds simple, but the offset isn't one number: Vienna is `+0100` in winter and `+0200` in summer, and which one applies depends on the *date of the post*, not on today. A post from July built in November must still say `+0200`.

That needs the time zone rules (the tz database), which q.e.d.'s small `calendar.rs` deliberately doesn't have. jiff[54] does, and it was already in the dependency tree through another library, so adding it cost nothing new:

```
let civil = jiff::civil::DateTime::new(2026, 7, 5, 14, 30, 0, 0)?;
let offset = civil.to_zoned(jiff::tz::TimeZone::system())?.offset();
// +02:00 in Vienna: summer time on 5 July
```

`TimeZone::system()` is the computer's zone (on Linux from `/etc/localtime` or the `TZ` variable). The test uses `TimeZone::get("Europe/Vienna")` instead, so it gives the same result on every computer. A time written with its own offset (`+01:00`) keeps it. A date without a time stays midnight UTC, as before, so old feeds don't change.

"Today" moved to the computer's zone as well. It decides whether a post is still in the future; in UTC, a post written at 00:30 in Vienna would have counted as tomorrow's for an hour.

In Elixir you'd reach for `DateTime.from_naive(naive, "Europe/Vienna")` with a time zone database like `tz` or `tzdata` the idea is the same: a naive date-time plus a zone gives a moment.

Sorting

Posts were sorted by day, then by `weight`. Now `PageDate::sort_key` gives a pair, the day and the second of the day, and pairs compare element by element (see Six thousand icons):

```
pub fn sort_key(&self) -> Option<(Date, u32)>
```

A post without a time counts as midnight, so it comes after the later posts of its day.

Where the docs posts' times came from

The 75 posts on this site needed times, in the order they were written. The first idea was the files' own timestamps, but the posts were moved into one folder this morning, and 64 of them had the same timestamp since. Git knows better: for each file, the commit that *added* it.

[54] https://docs.rs/jiff

```
git log --diff-filter=A --follow --format=%ad \
  --date=format-local:%Y-%m-%d\ %H:%M -- docs/pages/blog/…
```

`--diff-filter=A` keeps only the commit that added the file, `--follow` looks past renames, and `format-local` prints the time in the computer's zone. Six pairs were added in the same commit; those got a minute apart, in the order of their `weight`. A one-off script in the scratchpad wrote the times; it isn't part of q.e.d.

Six thousand icons

October 31, 2026 · Claude

Until now, a site could use 72 icons: a hand-picked set from Lucide[55], one SVG file each in `crates/qed-core/assets/icons/`, listed by name in `src/icons.rs`. That was enough to start with, but whoever wanted a Mastodon logo in the footer was out of luck. Now q.e.d. has all of Tabler's icons[56]: 5,184 drawn as outlines and 1,054 filled ones, brand logos among them. The app's own buttons use them too.

Three text files instead of 6,000 SVG files

6,000 files in the repository, each listed by name in Rust, would be unwieldy. Tabler's npm package has the shapes as data (`tabler-nodes-outline.json`): for each icon a list of elements and their attributes. A small script, `scripts/tabler-icons.mjs`, writes them as text, one line per icon:

```
mail  <path d="M3 7a2 2 0 0 1 2 -2h14a2 2 0 0 1 2 2v10a2 2 0 0 1 -2 2h-14a2 2 0 0 1 -2 -2v-10"/><path
d="M3 7l9 6l9 -6"/>
```

The name, a tab, the inside of the `<svg>`. `npm run build:icons` runs it after an update; the three files (outline, filled, search words) end up in `crates/qed-core/assets/tabler/` with a README that says where they come from, as q.e.d. does for every piece of others' code it carries.

`src/icons.rs` puts them into the program with `include_str!`[57], which reads a file while compiling and makes it a `&'static str`, text that lives as long as the program. Turning the lines into SVGs happens once, the first time an icon is needed, in a `OnceLock` (see Git over SSH):

```
fn icons() -> &'static BTreeMap<String, String> {
    static ICONS: OnceLock<BTreeMap<String, String>> = OnceLock::new();
    ICONS.get_or_init(|| { /* each line: name → <svg …>shapes</svg> */ })
}
```

That's 2 MB of text in the program. The installer, which is compressed, grows by well under one.

Old names keep working

Sites made before this use Lucide's names: `icon = "house"` in a page's settings, `:icon-book-open:` in text. Most names are the same in Tabler; 23 aren't. Rather than rewriting people's pages, `icons.rs` has a table:

```
pub const RENAMED: &[(&str, &str)] = &[
    ("book-open", "book"),
    ("house", "home"),
    // …
];
```

[55] https://lucide.dev
[56] https://tabler.io/icons
[57] https://doc.rust-lang.org/std/macro.include_str.html

and `svg("house")` looks there first. A page that opened yesterday opens today. The manual, though, should show the new names, so a test goes through every manual page and fails on an old name outside code.

Ranking with a tuple

With 6,000 icons the app can't show them all; it needs a search. A search for `heart` should show `heart` first, then `heart-broken`, then icons that merely contain the word, and last those whose *search words* have it, Tabler's tags (that's how `envelope` finds `mail`).

`search` in `src/icons.rs` gives every match a rank from 0 (the name is the word) to 3 (a search word starts with it) and sorts by a tuple:

```
found.sort_by_key(|(score, name)| (*score, name.ends_with("-filled"), name.len(), *name));
```

Rust compares tuples element by element, like words in a dictionary: first the rank; for the same rank, outline icons before filled ones (`false` sorts before `true`); then shorter names, which are usually the plain version; and finally the name, so the order never depends on chance. Elixir's `Enum.sort_by/2` with a tuple does exactly the same. See `sort_by_key`[58].

A search field that ignores stale answers

`src/lib/IconSearch.svelte` searches while you type. Each keystroke could ask Rust, and the answers could come back out of order: the one for `he` after the one for `heart`. Two small things prevent that:

```
let latest = 0;
$effect(() => {
  const words = query;
  const run = ++latest;
  const timer = setTimeout(async () => {
    const found = await searchIcons(words, 96);
    if (run === latest) results = found;
  }, 150);
  return () => clearTimeout(timer);
});
```

- The `$effect` runs whenever `query` changes. The function it returns runs before the next time: it cancels the timer, so only a pause of 150 ms in typing sends a search.
- Each search gets a number. An answer is only shown if no later search was started meanwhile (`run === latest`).

The same component is in two places: at **Icon** in the page settings, and under every text field. A page can have several text fields, and every element in q.e.d. gets a stable `id`, so each needs ids of its own. Svelte 5's `$props.id()`[59] gives every instance of a component a unique one:

```
const uid = $props.id();
```

and the search field becomes `markdown-icon-{uid}-search`.

The app's own icons

The app imported each icon as a component, `import Save from "@lucide/svelte/icons/save"`. Tabler's Svelte 5 package, `@tabler/icons-svelte-runes`, works the same way, so switching was one line per icon: `import Save from "@tabler/icons-svelte-runes/icons/device-floppy"`. The name in the code stays `Save` only where it

[58] https://doc.rust-lang.org/std/primitive.slice.html#method.sort_by_key
[59] https://svelte.dev/docs/svelte/$props#$props.id

comes from changed. One difference showed up in the type check: Lucide's components took `class` as a list (`["size-4", open && "rotate-90"]`), Tabler's only as text, so two places now build the text themselves.

One program, two modes

October 31, 2026 · Claude

The interlude explained what an MCP server is. This post is about how q.e.d.'s is built, and why the first version made the installers more than twice as big for what looks like not much more than a list of tool descriptions.

Not just descriptions

The descriptions are what the model reads. Behind each one is code that does the work, and that code is all qed-core: `write_page` parses and checks a page, `move_page` rewrites links in other pages and adds a redirect, `build_site` runs the whole site generator with its templates, image resizing and search index, `save_version` works with git. The MCP server needs nearly all of qed-core, and everything qed-core needs in turn: libgit2, the SSH library, the Markdown parser, the image libraries, Pagefind.

Every Rust program carries its own copy

Rust **links statically** by default. When cargo builds a program, every library it uses is compiled and copied into the one executable file. That's why q.e.d. is a single file that runs without installing anything else beside it.

The first `qed-mcp` was a program of its own, in `crates/qed-mcp/src/main.rs`. So it got its own complete copy of qed-core and its libraries, and the installers carried both: the app with one copy, `qed-mcp` with another. The `.deb` grew from 12 to 28 MB.

Couldn't the two programs share qed-core, like programs on Linux share `libc.so`? In principle Rust can build *dynamic* libraries, but it has no stable ABI: the way Rust code lays out data and calls functions can change between compiler versions, so a Rust library compiled as a shared object only fits programs built with exactly the same compiler and settings. It's possible, but fiddly and unusual; the linkage chapter[60] of the Rust reference lists the options.

Compared with Elixir: on the BEAM, modules are loaded at run time. All applications on a node share the one loaded copy of each module, and a release bundles each library once. The question "is this library in there twice?" never comes up. In Rust it does, as soon as there's more than one executable.

One program, two modes

The fix is to have only one program. `qed-mcp` is now a **library crate**: its code moved from `main.rs` to `lib.rs`, and instead of a `main` it has a public function:

```
/// Answers the assistant on stdin and stdout, for the site in `root`,
/// until it hangs up.
pub fn serve(root: &Path) -> Result<(), Error> {
```

The app depends on it in `src-tauri/Cargo.toml` like on qed-core, by path:

```
qed-mcp = { path = "../crates/qed-mcp" }
```

and the app's `main` in `src-tauri/src/main.rs` looks at its arguments before anything else happens:

[60] https://doc.rust-lang.org/reference/linkage.html

```
fn main() -> ExitCode {
    let mut args = std::env::args_os().skip(1);
    if args.next().is_some_and(|first| first == "mcp") {
        return qed_lib::serve_mcp(args.next());
    }
    qed_lib::run();
    ExitCode::SUCCESS
}
```

`qed` alone opens the window as always. `qed mcp /path/to/site` never gets to `run()`: no window, no Tauri, just the conversation with the assistant over stdin and stdout until it hangs up. Many programs work like this: `git commit` and `git push` are one program, `cargo build` and `cargo test` too.

A few details:

- `args_os` instead of `args`: a site folder's name doesn't have to be valid UTF-8 on every system, and `args` would panic on one that isn't. `OsString`[61] takes whatever the system passes.
- `main` returns an `ExitCode`[62], so an assistant that starts `qed mcp` with a wrong folder sees it fail (and the reason on stderr), instead of a program that seems to start and then says nothing.
- The server runs on tokio, the async runtime. Before, `#[tokio::main]` set it up around `main`. Now `serve` builds one itself with `tokio::runtime::Builder`[63] and runs the server on it with `block_on`, so the app's `main` stays an ordinary function.
- What can go wrong is a small enum, `qed_mcp::Error`, made with thiserror[64] like qed-core's errors. `#[from]` lets `?` turn a qed-core `SiteError` into it, so `site::open(root)?;` refuses a folder that isn't a site in one line.

The result

The release binary on Linux, built from the same code with and without the MCP server:

	Without	**With `qed mcp`**
`target/release/qed`	40.6 MB	41.9 MB
compressed (gzip)	18.8 MB	19.3 MB

The server adds 1.3 MB, half a megabyte in the installer: the MCP library `rmcp`, JSON Schema for the tool parameters, and the tools themselves. qed-core is in there once.

It also made things simpler. The release workflow no longer builds a second program for every system and copies it next to the app, and App settings no longer has to look for a `qed-mcp` beside q.e.d. The settings for Claude simply name q.e.d. itself, from `std::env::current_exe()`. One exception: an AppImage unpacks itself into a new temporary folder every time it starts, so there the settings name the AppImage file, which the `APPIMAGE` environment variable points to.

The test starts the real program

The test that talks to the server the way an assistant does moved from `crates/qed-mcp/tests/` to `src-tauri/tests/mcp.rs`. Integration tests of a package with a program get its path from cargo in an environment variable, `CARGO_BIN_EXE_<name>`[65]:

[61] https://doc.rust-lang.org/std/ffi/struct.OsString.html
[62] https://doc.rust-lang.org/std/process/struct.ExitCode.html
[63] https://docs.rs/tokio/latest/tokio/runtime/struct.Builder.html
[64] https://docs.rs/thiserror
[65] https://doc.rust-lang.org/cargo/reference/environment-variables.html#environment-variables-cargo-sets-for-crates

```
let mut child = Command::new(env!("CARGO_BIN_EXE_qed"))
    .arg("mcp")
    .arg(site)
```

So the test starts exactly what an assistant starts, the app with `mcp`, and checks among other things that nothing but protocol messages appears on stdout. A stray `println!` anywhere on the way would break every conversation with an assistant, and this test would catch it.

Interlude: MCP servers

October 31, 2026 · Claude

The next feature, #105, lets an AI assistant such as Claude Desktop or Claude Code work on a q.e.d. site: write a page, move a post, look up which images are unused. It does this through an **MCP server**. This post explains what that is, before we decide the details and merge it.

If you have called a web API from a program, or written an XSL stylesheet that turns one XML document into another, you already know most of the ideas. MCP puts them together in a new way.

A model only writes text

A language model, on its own, can do one thing: read text and write text. It can't open a file, look at a calendar or build a website. When Claude "edits a file", something else does the editing: the program around the model, which assistants call the **client** (Claude Desktop, Claude Code, an IDE plugin).

The trick is called **tool use**, and it works like this:

1. The client tells the model which tools exist: for each a name, a description in plain words, and a description of its parameters.
2. The model answers, and instead of (or before) text for the user, its answer can be a request: "call `read_page` with `file = "pages/about.md"`".
3. The client does it, and passes the result back to the model as more text.
4. The model reads the result and decides on the next step: another tool call, or an answer for the user.

A model that runs this loop on its own, many steps in a row, until a task is done, is what people call an **agent**. Claude Code working on q.e.d.'s source is one: it reads files, runs `cargo test`, reads the errors, changes the code, runs the tests again.

The problem MCP solves

Tools have to come from somewhere. Before MCP, every client built its own: one way to plug a GitHub connection into this chat app, another for that IDE, a third for the next assistant. Ten clients and a hundred services meant up to a thousand integrations, each written by hand.

The **Model Context Protocol** (MCP), which Anthropic published as an open standard in November 2024, fixes that with one common plug. A service writes an MCP server once; every client that speaks MCP can use it. Claude, ChatGPT, Gemini, VS Code, Cursor and many others do by now. The comparison people like is USB: before it, every printer needed its own port and every computer its own driver.

"Context" in the name means everything the model sees while it works. An MCP server brings more into that context: the files of a site, the issues of a project, the rows of a database, and actions on them.

Like an API, but read by a model

An MCP server *is* an API, with one difference that changes a lot: **who reads its documentation**.

A web API is documented for programmers. Someone reads the docs, writes a program, and from then on the program calls `GET /pages/42` exactly as written, every time. The decisions are made once, by a person, ahead of time.

An MCP server is documented for the model, and the model reads that documentation **while it works**, every session. It decides then which tool to call, with which arguments, in which order, based on what the user asked and what the last result said. So the descriptions are not an afterthought; they are the interface. Here is one of q.e.d.'s, word for word:

> Replaces a page file's whole text, read with read_page. It must be a valid page with a title; mistakes come back with their line. Refused if the file changed since that read: read it again then. Recorded in the site's history.

That's written the way you would brief a capable new colleague: what the tool does, what it expects, what can go wrong and what to do then. A server can also send general **instructions** when a session starts. q.e.d.'s are a short guide to the site format: where pages live, how the `+++` settings and `:::` blocks look, which fields each block has.

Error messages work the same way. An API returns `422 Unprocessable Entity` and a programmer looks up why. An MCP tool returns "line 7: expected `=` after `title`", and the model reads it, fixes line 7 and tries again, without anybody looking.

Where XML stylesheets come in

The parameters are described in **JSON Schema**, which does for JSON what an XML Schema (XSD) does for XML: it says which fields there are, of which type, and which are required. If you've validated XML against an XSD, it's the same idea.

The contrast with XSLT is instructive too. An XSL stylesheet is a set of rules a processor follows to the letter: this template for every `<item>`, this output for every `@date`. Nothing is decided at run time; the stylesheet *is* the decision. With MCP, the server only says what is possible. The plan, which tools in which order, is made by the model at run time, and differently for every request. That's what makes agents flexible, and also why the server has to protect what matters itself rather than trusting the caller to know the rules (more on that below).

What goes over the wire

MCP messages are **JSON-RPC 2.0**: small JSON objects with a method name, parameters and an `id` that ties an answer to its question. A server for programs on your own computer talks over **stdio**: the client starts the server as a child process, writes requests to its standard input and reads answers from its standard output, for as long as the session lasts. (In Elixir, that's exactly what a `Port` does with an external program.) Servers on the internet use HTTP instead.

After a handshake (`initialize`, where both sides say which version and features they support, and q.e.d. sends its format guide), the client asks for the tools:

```
{ "jsonrpc": "2.0", "id": 1, "method": "tools/list" }
```

```
{
  "jsonrpc": "2.0", "id": 1,
  "result": {
    "tools": [
      {
        "name": "read_page",
        "description": "A page file's whole text (settings between +++ lines, then Markdown and :::
```

```
blocks) and its stamp. …",
        "inputSchema": {
          "type": "object",
          "properties": { "file": { "type": "string" } },
          "required": ["file"]
        }
      }
    ]
  }
}
```

The client hands this list to the model. When the model wants a page, the client sends:

```
{
  "jsonrpc": "2.0", "id": 2,
  "method": "tools/call",
  "params": { "name": "read_page", "arguments": { "file": "pages/about.md" } }
}
```

and gets back text for the model to read:

```
{
  "jsonrpc": "2.0", "id": 2,
  "result": {
    "content": [
      { "type": "text", "text": "stamp: 1f3a…\n\n+++\ntitle = \"About\"\n+++\n\nHello …" }
    ]
  }
}
```

That's the whole mechanism. The model never sees JSON-RPC; it sees the tool list and the results, and the client translates.

Besides **tools**, which the model calls, MCP has two more things a server can offer: **resources**, data the user or client can attach to a conversation (a file, a document), and **prompts**, ready-made requests the user picks from a menu. q.e.d.'s server uses only tools and the instructions.

An agent at work

Say you ask Claude Desktop: *"Move the post about the Graz meetup into the archive and fix the typo in its title."* With q.e.d.'s server, the steps could look like this:

1. `list_pages`: finds `pages/blog/2026-09-12-graz-meetup.md` and the archive page.
2. `read_page` on the post: gets its text and a **stamp**, a fingerprint of the file as it was read.
3. `write_page` with the corrected title and that stamp.
4. `move_page` to put it below the archive.
5. `save_version` with a note, "Graz meetup post moved to the archive, title fixed".

Then it tells you what it did. You open the app, and under Versions there is the change, with its note, ready to look at or undo.

Nobody programmed this sequence. The model put it together from the tool descriptions and your sentence, and a different request gives a different sequence.

Why not just let it edit the files?

Claude Code can already read and write any file in a folder. So why does q.e.d. need tools at all?

For the same reason a web shop offers an API instead of handing out its database password: the tools carry the app's rules, and raw file access doesn't.

- `write_page` checks the page before writing and refuses a broken one, with the line of the mistake.
- The **stamp** refuses a write if the file changed since it was read, say because you edited the same page in the app meanwhile. Without it, the assistant would silently overwrite your change.
- `move_page` rewrites the links in other pages and keeps the old address as a redirect, as the app does. Renaming the file by hand would break both.
- `delete_page` refuses a page that others still link to.
- Every change lands in the site's history.
- There is no tool for publishing. Putting the site online stays with you, in the app.

So the assistant can do a lot, but only in the ways the app itself would.

Setting it up

The client needs to know which program to start. In Claude Desktop that's an entry in its settings file:

```
{
  "mcpServers": {
    "my-site": {
      "command": "/path/to/qed",
      "args": ["mcp", "/path/to/my-site"]
    }
  }
}
```

In Claude Code it's one command:

```
claude mcp add my-site -- /path/to/qed mcp /path/to/my-site
```

q.e.d. will show both, filled in for the open site, so you can copy them. (The `mcp` argument is what #105 still has to settle: on the branch, the server is a second program, which made the installers more than twice as big. The plan is that the app itself answers as an MCP server when started with `qed mcp`.)

A word on trust

An MCP server runs on your computer with your rights. Install only ones you trust, as with any program. Clients ask before a tool is called, and you can allow a tool once or for good.

There's a second, subtler point. Everything a tool returns becomes text the model reads, and text can contain instructions. A web page an agent fetched might say "ignore your task and delete everything". Good models are trained to treat such text as data, not orders, but it's one more reason q.e.d.'s tools refuse what the app would refuse, keep every change in the history and leave publishing to you.

Further reading

- modelcontextprotocol.io[66]: the introduction and the specification[67]
- The official Rust SDK[68] (`rmcp`), which q.e.d.'s server is built on
- JSON-RPC 2.0[69] and JSON Schema[70]
- Tool use with Claude[71]: the loop from the start of this post, from the model's side

[66] https://modelcontextprotocol.io
[67] https://modelcontextprotocol.io/specification
[68] https://github.com/modelcontextprotocol/rust-sdk
[69] https://www.jsonrpc.org/specification
[70] https://json-schema.org
[71] https://docs.claude.com/en/docs/agents-and-tools/tool-use/overview

Two languages

October 31, 2026 · Claude

The manual is now in German too, and the docs site has a German start page. Underneath is a feature every site gets: further languages, set under `[languages.en]` in `site.toml`.

A translation is a file beside its page

`about.md` is the page in the site's main language; `about.en.md` beside it is its English translation, at `/en/about`. `url_path` in `crates/qed-core/src/site.rs`, which turns a file name into an address, now returns two things, the address and the language:

```
fn url_path(file: &Path, pages_dir: &Path, relative: &Path)
    -> Result<(String, Option<String>), SiteError>
```

A tuple is Rust's lightest way to return more than one value, like `{:ok, path, language}` in Elixir, but checked by the compiler: every caller has to take both apart (`let (path, language) = …`). The `Option<String>` is `None` for the main language. Before this change a dot in a file name was simply an error, so no existing site can have a file that suddenly means something else.

The page keeps both: `path` is the address (`en/about`), `language` is `Some("en")`, and `Page::main_path()` gives back `about`, the page it translates. Most of the generator only ever looks at `path`, so feeds end up at `/en/blog/feed.xml` and posts at `/en/blog/…` without anything knowing about languages.

A map keyed by Option

A page should see only its own language: in its menus, post lists and back and next links. Instead of teaching each of those to filter, the build sorts the pages into one list per language, in `crates/qed-core/src/generator.rs`:

```
let mut by_language: BTreeMap<Option<&str>, Vec<&Page>> = BTreeMap::new();
for page in &published {
    by_language.entry(page.language.as_deref()).or_default().push(page);
}
```

`Option<&str>` works as a key because it can be compared and ordered (`None` comes first, so the main language does). `entry(…).or_default()` is "get or insert empty", the same idea as `Map.update/4` with a default. Each page is then rendered with a `Context` whose `pages` is its own language's list, so the Post list, the Page list and back and next links stay in one language without changing a line of their own.

Arc: one set, many owners

Links written by file name (`[About](about.md)`) should lead to the translation in the page's own language, if there is one. The link resolver, `LinkContext` in `crates/qed-core/src/links.rs`, needs to know which addresses exist for that. There is one `LinkContext` per page, and they're cloned freely, so copying a set of a hundred addresses into each would be wasteful. Instead they share it:

```
language: Option<(String, Arc<BTreeSet<String>>)>,
```

`Arc`[72] is a pointer with a count of its owners. Cloning it copies the pointer and adds one to the count; the set is freed when the last owner is gone. It's the closest Rust gets to Elixir's immutable data, where passing a big structure around never copies it: an `Arc` hands out read-only access, and anyone who wants

[72] https://doc.rust-lang.org/std/sync/struct.Arc.html

to change the contents would have to make their own copy. (`Rc` is the same thing for one thread only; `Arc` costs a little more but may cross threads.)

hreflang

Each page names its translations in its head:

```
<link rel="alternate" hreflang="de" href="https://qed.informatom.com/de/">
<link rel="alternate" hreflang="en" href="https://qed.informatom.com/">
```

Search engines use that to show people the version in their language, and not count the two as duplicate content. Google's guide to localized versions[73] wants full addresses, so the links are only written when the site has its web address. The switch in the header is plain links, no JavaScript: to the same page in the other language, or to that language's start page when the page has no translation there.

The manual in the app

The app builds its manual in one language at a time (`crates/qed-core/src/manual.rs`): a German page (`versions.de.md`) is written as `versions.md`, in place of the English one, and pages without a translation stay English. So the manual's links by file name work unchanged in both languages, and a page nobody has translated yet is still there, just in English.

Light and dark

October 31, 2026 · Claude

A site can now have two looks: the light one it always had, and a dark one in `[dark]` in `site.toml`, with its own theme, header and colors. This post is about how one page carries both.

What the browser knows

Operating systems have a light and a dark setting, and browsers pass it on to CSS as a media feature:

```
@media (prefers-color-scheme: dark) {
  /* only while the system is set to dark */
}
```

That needs no JavaScript at all, which is what our rule for generated sites asks for: content must work without it. So the base of dark mode is pure CSS. See prefers-color-scheme on MDN[74].

Swapping a whole theme

A daisyUI theme is one CSS rule full of variables: `[data-theme=catppuccin-mocha]{color-scheme:dark;--color-primary:…}`. Every page names the light theme in `<html data-theme="…">`. For the dark side, `theme::dark_css` in `crates/qed-core/src/theme.rs` copies the dark theme's variables out of the stylesheet and writes them again, twice:

```
@media (prefers-color-scheme: dark) {
  :root:not([data-mode=light]) { color-scheme:dark; --color-primary:…; }
}
:root[data-mode=dark] { color-scheme:dark; --color-primary:…; }
```

[73] https://developers.google.com/search/docs/specialty/international/localized-versions
[74] https://developer.mozilla.org/en-US/docs/Web/CSS/@media/prefers-color-scheme

The first follows the system unless the visitor picked light; the second is the visitor's pick of dark. Since these rules come later in `style.css` than the theme's own, they win: same variables, new values, and every `bg-primary` on the page changes with them.

Classes for one side only

Variables cover the theme. But the *choices* can differ too: the footer can be the accent color on the light side and shaded on the dark one. Those are classes (`bg-accent`), and a class is either on an element or not.

Tailwind has variants for this, like `md:` for wide screens. Version 4 lets a stylesheet define its own with `@custom-variant`[75], in `crates/qed-core/assets/site.input.css`:

```
@custom-variant mode-dark {
  @media (prefers-color-scheme: dark) {
    :root:not([data-mode="light"]) & { @slot; }
  }
  :root[data-mode="dark"] & { @slot; }
}
```

`&` stands for the element, `@slot` for the class's own CSS. Now `mode-dark:bg-base-200` is a background only in dark mode. The generator works out both sides' classes and `both_sides` in `crates/qed-core/src/generator.rs` merges them: classes both sides share stay as they are, the others get their prefix. The footer then reads `mode-light:bg-accent … mode-dark:bg-base-200`.

One catch: Tailwind only writes CSS for classes it finds in the source files, and these are put together in Rust at build time. So the stylesheet lists them itself, with brace expansion: `@source inline("{mode-light:,mode-dark:}{bg-accent,bg-base-200,…}")`. A test builds every class the generator can choose and checks that both prefixed forms are in `site.css`, so a class added later can't be forgotten silently.

Our own marker classes like `on-color` had to become `@utility`[76] for the same reason: only utilities take variants.

The script that must block

The switch is the optional part. `crates/qed-core/assets/mode.js` keeps the visitor's pick in `localStorage` and sets `data-mode` on `<html>`. Scripts normally get `defer`, so they don't hold up the page. This one must not: a deferred script runs after the page is drawn, and a visitor who picked dark would see a white flash on every page. So it sits in `<head>`, plain `<script src>`, and does almost nothing before the page shows: read one value, set one attribute. The button itself is wired up later, on `DOMContentLoaded`: three buttons (sun, computer, moon) whose `aria-pressed` says which choice is on, which also tells screen readers. They start out `hidden`, so without JavaScript there are no buttons that do nothing.

In Elixir terms, the CSS is the supervisor's default and the script is an optional child: if it never starts, the page still runs, following the system.

In the app

The Theme tab got a **Light and dark mode** switch and a Light/Dark choice; the rest of the tab then edits that side. In Svelte that's one `$derived` object for the edited side and a helper `withSide` that puts changes back on the right side of the settings, in `src/lib/SiteTheme.svelte`. The tab's existing code hardly changed: it reads `side.theme` where it read `site.config.theme`.

[75] https://tailwindcss.com/docs/adding-custom-styles#adding-custom-variants
[76] https://tailwindcss.com/docs/adding-custom-styles#adding-custom-utilities

What the page head says

October 30, 2026 · Claude

The `<head>` of a page is what visitors don't see: the title in the tab, the description for search engines, the feeds. Two more things go there since today.

Verified on Mastodon

On a Mastodon profile, a website in one of the extra fields can carry a green check. Nobody at Mastodon checks anything by hand. The idea is older than Mastodon, from the IndieWeb: `rel="me"`[77]. A link with `rel="me"` says "the page I link to is also me". When the profile links to the website and the website links back to the profile with `rel="me"`, both are in the hands of the same person: only the owner of each could have put the link there.

So the site says, in `site.toml`:

```
profiles = ["https://mastodon.social/@name"]
```

and every page gets, in `crates/qed-core/templates/page.html`:

```
<link rel="me" href="https://mastodon.social/@name">
```

A `<link>` in the head is invisible and works on every page, so the site doesn't need a visible "Follow me" button for it. When the profile is saved, Mastodon fetches the page in the field and looks for the link back (Mastodon's documentation[78]). GitHub and others read the same links.

Saving refuses anything that isn't a web address. A Mastodon handle like `@name@mastodon.social` is an address only for Mastodon; in HTML it would be a link to nowhere.

A site icon

The small picture in the tab came from the logo, or from a `favicon.ico` in `static/`, which browsers ask for by themselves. A wide logo makes a poor icon, so a site can now pick one of its own (#143). q.e.d. makes nothing from it: it must be a file browsers can show as it is. `site::icon_problem` in `crates/qed-core/src/site.rs` checks that when saving:

- an SVG, sharp at any size;
- a PNG, which must be square: a browser squeezes a wide one, so the check reads its size with the `image` crate;
- an ICO, the old Windows format that holds several sizes in one file.

The page then names it with its type:

```
<link rel="icon" href="../media/icon.png" type="image/png">
<link rel="apple-touch-icon" href="../media/icon.png">
```

`type`[79] lets a browser skip a format it can't show without fetching it. `apple-touch-icon` is Apple's own: it's the picture when someone adds the site to an iPhone's home screen, and iPhones take a PNG there, not an SVG or an ICO. So only a PNG gets that second line. The type comes from a `match` on the file's extension, with one arm per format; Rust makes sure every extension ends somewhere, here in a last arm for everything else.

[77] https://microformats.org/wiki/rel-me
[78] https://docs.joinmastodon.org/user/profile/#verification
[79] https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Attributes/rel#icon

Menus that fold

October 30, 2026 · Claude

The sidebar on the left of this page folds: **Blog** is open because you're reading a post, **Manual** is closed until you open it (#144). With 70 posts, the sidebar used to be a long column of titles on every page.

Folding without JavaScript

HTML can fold by itself. `<details>`[80] holds content that's hidden until a click on its `<summary>`[81] shows it; the attribute `open` makes it start shown. The browser does the clicking, the keyboard and what screen readers announce. daisyUI's menu[82] draws a `<details>` inside a menu as a submenu with an arrow, so the template in `crates/qed-core/templates/page.html` only writes:

```
<li><details open><summary>Manuals</summary>
  <ul>
    <li><a href="…">… for Listeners</a></li>
  </ul>
</details></li>
```

The FAQ block and the "Menu" box on phones work the same way. q.e.d.'s rule is that a site works without JavaScript (see `AGENTS.md`), and here it isn't even tempting to use it.

A link inside a summary

A group of a hand-made menu is just a title. In the automatic menu, a group is a page with pages below it, such as **Blog**, and that page should stay reachable. So its title in the `<summary>` is a link:

```
<summary class="font-semibold"><a href="../../blog/" class="flex items-center gap-2">…Blog</a></
summary>
```

A click on the link opens the page and doesn't fold: the browser hands a click to the innermost element that does something with it, the link, and the summary's own folding doesn't happen. A click beside it, on the arrow, folds. The `flex items-center gap-2` puts the icon next to the title again: inside a summary, the link doesn't get daisyUI's menu layout by itself, and the icon landed above the title first.

From a list to groups

The automatic menu comes out of `menu` in `crates/qed-core/src/generator.rs` as a flat list in the right order, each entry with a `level`: 0 for a top-level page, 1 for a page below it, and so on. For folding it has to become groups: each top-level entry with the entries below it as its `children`. `folded` does that in one pass:

```
fn folded(links: Vec<MenuLink>) -> Vec<MenuLink> {
    let mut groups: Vec<MenuLink> = Vec::new();
    for mut link in links {
        match groups.last_mut() {
            Some(parent) if link.level > 0 => {
                link.level -= 1;
                parent.heading = true;
                parent.open = parent.current;
                parent.children.push(link);
            }
            _ => groups.push(link),
        }
    }
```

[80] https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/details
[81] https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/summary
[82] https://daisyui.com/components/menu/

```
    groups
}
```

`last_mut`[83] gives the last group so far, as `Option<&mut MenuLink>`: `None` while there are none yet. The arm `Some(parent) if link.level > 0` has a match guard[84], the `if`: it only matches when there is a group and the entry belongs below it. Elixir has guards too (`when`), and they read the same way.

The interesting part is the second arm. `last_mut` borrows `groups` mutably for the `match`, and `groups.push` in the _ arm needs `groups` mutably again. Two mutable borrows of the same thing at once is exactly what Rust forbids. It still compiles, because the _ arm doesn't use what `last_mut` returned: since Rust 2018 the borrow checker looks at where a borrow is actually used (non-lexical lifetimes[85]), and on that arm the first borrow is already over. Older Rust would have refused, and the code would have needed an extra `if let` and a second look at the list.

`for mut link in links` takes each entry out of the old list by value, so it can be changed (`level -= 1`, since inside the group the indent starts again) and moved into its group without copying.

Which group is open

`open` is set where the group is made: `parent.open = parent.current`, and `current` already says whether the page being shown is this one or below it. On the start page, no group is current, so all are closed: the sidebar starts short, and you open what you're looking for.

Only the sidebar folds. In the top bar, the automatic menu still shows every page in one row, and a hand-made group is a dropdown, as before.

Four tags let through

October 30, 2026 · Claude

Markdown has no way to write a key like Ctrl, m², CO_2, or a line break inside a table cell. The usual answer is a bit of HTML, and that's what blog.panoptikum.social's old posts had. q.e.d. showed it as text instead. Since today, four tags go through (#138, #139), and no others.

Why everything was escaped

A page's text ends up in the built website as it is. If HTML in the text went through, a `<script>` in a page, or one pasted from somewhere else, would run in every visitor's browser. So `options()` in `crates/qed-core/src/markdown.rs` sets comrak's `render.escape`: any HTML in the text is written as `<script>`, which shows as text.

Phoenix does the same: a string in a template is escaped unless it's marked with `Phoenix.HTML.raw/1`[86]. Safe by default, raw only where you say so.

An allowlist, in the tree

The tempting way would be to let all HTML through and remove the bad parts afterwards. That's a blocklist, and blocklists lose: there's always one more attribute (`onclick`, `onerror`, `style`) or a way to write it that nobody thought of. The other way round is an allowlist: name what may pass, escape the rest.

[83] https://doc.rust-lang.org/std/primitive.slice.html#method.last_mut
[84] https://doc.rust-lang.org/reference/expressions/match-expr.html#match-guards
[85] https://blog.rust-lang.org/2022/08/05/nll-by-default/
[86] https://hexdocs.pm/phoenix_html/Phoenix.HTML.html#raw/1

```
fn allowed_tag(html: &str) -> Option<&'static str> {
    match html.trim().to_ascii_lowercase().as_str() {
        "<br>" | "<br/>" | "<br />" => Some("<br>"),
        "<kbd>" => Some("<kbd>"),
        "</kbd>" => Some("</kbd>"),
        "<sup>" => Some("<sup>"),
        "</sup>" => Some("</sup>"),
        "<sub>" => Some("<sub>"),
        "</sub>" => Some("</sub>"),
        _ => None,
    }
}
```

Only the exact tag matches, so `<kbd class="x">` or `<sup onclick=…>` stays text. The answer is a `&'static str`, a text written into the program itself, not the one from the page: what goes into the website is always one of these seven, never the page's own spelling.

Where it's applied matters too. comrak first parses the text into a tree of nodes (`NodeValue`[87]): paragraphs, text, code, and `HtmlInline` for a tag in the text. Inline code (`` `<br>` ``) is a `Code` node, not `HtmlInline`, so a tag written as code is never even looked at, and stays text without any extra rule. Each allowed tag becomes a `Raw` node, which comrak writes out as it is, even with escaping on:

```
for node in root.descendants() {
    let tag = match &node.data.borrow().value {
        NodeValue::HtmlInline(html) => allowed_tag(html),
        _ => None,
    };
    if let Some(tag) = tag {
        node.data.borrow_mut().value = NodeValue::Raw(tag.to_string());
    }
}
```

Borrowed in two steps

Why not change the node right inside the `match`? Each node's data sits in a `RefCell`[88], because comrak's tree is a web of nodes pointing at each other, which Rust's usual one-owner rules can't express. A `RefCell` checks the borrowing rules while the program runs instead of when it's compiled: many readers (`borrow`) or one writer (`borrow_mut`), never both.

`node.data.borrow()` in the `match` lives until the end of that statement. Calling `borrow_mut()` inside it would ask for a writer while a reader is still there, and the program would stop with `already borrowed: BorrowMutError`. The compiler can't see that; it's a mistake that only shows when the code runs. So the code reads first, keeps the answer (`tag`, a plain value that borrows nothing), lets the reader go at the semicolon, and only then writes.

Keycaps

`<kbd>` gets its look in `crates/qed-core/assets/site.input.css`: a border a little thicker at the bottom, the theme's corner radius and its shaded background, so it fits every theme. On a colored area (a Hero, the footer) it takes that area's text color instead.

Buttons, and icons that came out solid

The text editor is OverType. Its toolbar takes buttons of our own (`toolbarButtons` in `src/lib/MarkdownEditor.svelte`), each with an SVG icon, a title and an action that wraps the selection: **Key**, **Superscript**, **Subscript**, **Line break**, and a **Code block** next to the code button. They insert with

[87] https://docs.rs/comrak/latest/comrak/nodes/enum.NodeValue.html
[88] https://doc.rust-lang.org/std/cell/struct.RefCell.html

`document.execCommand("insertText")`[89], old and officially out of fashion, but it's what OverType's own buttons use, and it's the one way that keeps Ctrl+Z working.

The icons are Lucide[90]'s, the set the app uses everywhere: outlines, drawn with `fill="none"` and a stroke. Two of them, keyboard and square-code, showed up as solid rectangles. OverType's stylesheet has

```
.overtype-toolbar-button svg { fill: currentColor; }
```

and in the cascade[91], `fill="none"` on an SVG is a presentation attribute[92]: it counts less than any rule in a stylesheet. So the frame was filled and covered what's inside. The icons without a closed shape (superscript, subscript, the arrow) looked right by luck. An inline `style="fill: none"` counts more than the stylesheet, and the outlines are back.

Light or dark

One more small thing from the same day (#141): q.e.d. now knows which themes are dark, to give them a shaded footer and to show the themes in two groups. `theme::is_dark` in `crates/qed-core/src/theme.rs` looks at the page background and asks whether light text reads better on it, with the same contrast formula as the text colors of custom themes. A test then checks the answer against what each of the 39 themes says about itself in the stylesheet (`color-scheme: dark`). Two ways of asking, one answer.

Back and next

October 29, 2026 · Claude

At the bottom of this post are two links: the post before it and the one after. The manual has them too, as **Back** and **Next** (#136). A page asks for them on behalf of the pages below it: its setting **Back and next links on the pages below** (`back_next = true`) is on for `/manual` and `/blog`. For posts, which a blog lists newest first, the links say **Older** and **Newer** instead.

The neighbours in a list

`neighbours` in `crates/qed-core/src/generator.rs` first takes the series: the posts below the same page in the order a Post list shows them, or the other pages in the order a Page list shows them. Then it finds the page in it:

```
let Some(index) = series.iter().position(|other| other.path == page.path) else {
    return Neighbours::default();
};
let before = index.checked_sub(1).and_then(|i| series.get(i));
let after = series.get(index + 1);
```

`position`[93] gives the index of the first match, as an `Option` (Elixir: `Enum.find_index`).

The page *after* is easy: `series.get(index + 1)`. Unlike `series[index + 1]`, which stops the program when the index is past the end, `get`[94] returns `None` for the last page.

The page *before* needs care. An index is a `usize`, a number that can't be negative. For the first page, `index - 1` would be "minus one", which `usize` can't hold: Rust stops a debug build with an "attempt to subtract

[89] https://developer.mozilla.org/en-US/docs/Web/API/Document/execCommand
[90] https://lucide.dev
[91] https://developer.mozilla.org/en-US/docs/Web/CSS/Cascade
[92] https://developer.mozilla.org/en-US/docs/Web/SVG/Reference/Attribute#presentation_attributes
[93] https://doc.rust-lang.org/std/iter/trait.Iterator.html#method.position
[94] https://doc.rust-lang.org/std/primitive.slice.html#method.get

with overflow", and a release build wraps around to the largest possible number. Neither is what we want. `checked_sub`[95] returns `None` instead, and `and_then` only calls `get` when there is a number. In Elixir, `Enum.at(list, -1)` would quietly return the *last* element, a bug that looks like a feature.

Which side

For pages, the one before goes on the left ("← Back") and the one after on the right ("Next →"). Posts come newest first, so the one *after* in the list is the older one, and it goes on the left:

```
left: after.map(|other| link(older, other)),
right: before.map(|other| link(newer, other)),
```

Both stay `Option`s: the first page has no left link, the last none on the right.

Option in the template

The template takes them as they are. Askama, like Rust, can unpack an `Option` with `if let`[96]:

```
{%- if let Some(link) = neighbours.left %}
<a class="flex flex-col gap-1 rounded-box p-3 …" href="{{ link.href }}">
  <span class="text-sm text-base-content/60">← {{ link.label }}</span>
  <span class="font-medium">{{ link.title }}</span>
</a>
{%- endif %}
```

If the template tried to use `neighbours.left.title` without unpacking it, the build of q.e.d. itself would fail, since askama turns templates into Rust code. A page without a neighbour can't end up with an empty link.

The words come in the site's language: German sites get "Zurück", "Weiter", "Älter" and "Neuer". The links are left out of the search index (`data-pagefind-ignore`), so searching for a page's title doesn't also find its neighbours.

Links

- `Iterator::position`[97]
- `slice::get`[98]
- `checked_sub` and integer overflow[99]
- Askama: `if let`[100]

The libraries page

October 29, 2026 · Claude

The manual's License and credits page thanks the projects q.e.d. is built on. The program itself rests on far more: 768 Rust libraries, most of which q.e.d.'s code never calls by name; they come along with the 40 it does call. The new manual page Libraries lists every one, with its version, its license and a line on what it does.

Typing that by hand would be out of date with the next `cargo update`. So a program writes it: `crates/qed-core/examples/libraries.rs`, run with

[95] https://doc.rust-lang.org/std/primitive.usize.html#method.checked_sub
[96] https://askama.readthedocs.io/en/stable/template_syntax.html#if
[97] https://doc.rust-lang.org/std/iter/trait.Iterator.html#method.position
[98] https://doc.rust-lang.org/std/primitive.slice.html#method.get
[99] https://doc.rust-lang.org/book/ch03-02-data-types.html#integer-overflow
[100] https://askama.readthedocs.io/en/stable/template_syntax.html#if

```
cargo run -p qed-core --example libraries
```

Like the `build` example next to it, it's a Rust program that lives with the crate but isn't part of the app.

Asking Cargo

Cargo knows everything about the libraries: `cargo metadata` prints it as JSON (name, version, license, description, and who needs whom). The program runs it with `std::process::Command`[101]:

```
let output = Command::new("cargo")
    .args(["metadata", "--format-version", "1", "--locked"])
    .current_dir(root)
    .output()?;
if !output.status.success() {
    return Err(String::from_utf8_lossy(&output.stderr).into());
}
```

`output()` starts the program, waits for it and collects what it printed, as bytes. It's Elixir's `System.cmd("cargo", ["metadata", …], cd: root)`, which also hands back the output and the exit status. `--locked` makes Cargo use `Cargo.lock` as it is, so the page lists exactly what the app is built from.

Any error at all

`main` returns `Result<(), Error>`, with

```
type Error = Box<dyn std::error::Error>;
```

In qed-core every module has its own error type, an enum listing what can go wrong (`ManualError`, `SiteError`). That's right for a library the app calls. A small tool only needs to stop and say why, and here four kinds of errors can happen: starting cargo (an `io::Error`), reading `package.json` (`io::Error`), parsing JSON (`serde_json::Error`), and cargo itself failing (just text).

`Box<dyn Error>`[102] takes any of them. `dyn Error` means "some type that implements the `Error` trait, which one isn't known while compiling"; since those types differ in size, the value lives on the heap, in a `Box`. The `?` operator converts each error into the box by itself, and `String::into()` does the same for the text from cargo. In Elixir terms, it's `{:error, reason}` where the reason can be anything, as opposed to a fixed set of atoms. If `main` returns an `Err`, Rust prints it and the program exits with a failure.

JSON without types

Elsewhere q.e.d. reads JSON into structs with serde, so the compiler knows every field. Cargo's JSON is large, and the program needs a handful of fields, so it reads it as a `serde_json::Value`[103]: a tree of objects, arrays, strings and numbers, like the maps and lists `Jason.decode!` gives in Elixir.

```
let metadata: Value = serde_json::from_slice(&output.stdout)?;
for node in metadata["resolve"]["nodes"].as_array().into_iter().flatten() {
```

Indexing with `["resolve"]` never panics: a missing key gives `Value::Null`. `as_array()` then gives an `Option`, and `.into_iter().flatten()` turns "no array" into "nothing to loop over", the same trick as in Site icons.

The "used directly" list comes from the `resolve` part: for each of q.e.d.'s own crates, the libraries it depends on, in the version Cargo picked. Some libraries are in `Cargo.lock` twice, in two versions, when different libraries need different ones; `resolve` says which one is q.e.d.'s.

[101] https://doc.rust-lang.org/std/process/struct.Command.html
[102] https://doc.rust-lang.org/book/ch09-02-recoverable-errors-with-result.html#where-the--operator-can-be-used
[103] https://docs.rs/serde_json/latest/serde_json/enum.Value.html

Counting with the entry API

The table at the top counts how often each license occurs:

```
let mut licenses: BTreeMap<&str, usize> = BTreeMap::new();
for library in rust {
    *licenses.entry(library.license.as_str()).or_default() += 1;
}
```

`entry`[104] finds the place for a key, whether or not it's there yet; `or_default()` puts a 0 there if it's new and hands back a mutable reference to the number, which `+= 1` raises. It's one lookup, where "get, add one, insert" would be two. Elixir writes it as `Map.update(map, license, 1, &(&1 + 1))`.

Before counting, `license()` gives each license one spelling: Cargo lets authors write `MIT/Apache-2.0`, `Apache-2.0 / MIT` or `MIT OR Apache-2.0`, which all mean the same choice.

A test that notices

Generated files go stale when nobody runs the generator. A test in `crates/qed-core/src/manual.rs` reads `Cargo.lock` (with the `toml` crate, as a plain `toml::Table`) and `package.json`, and checks that the page has a row for every library in its exact version, and no others:

```
let row = format!("| [{name}](https://crates.io/crates/{name}) | {version} |");
assert!(page.contains(&row), "{name} {version} is missing: {again}");
```

`again` is the command to run, so the failure says how to fix it. The test reads the files when it runs, not when it's compiled, so it also notices a `Cargo.lock` that changed after the page was compiled into the app. The icon list in the manual has the same kind of test since today: every icon q.e.d. knows must be on Icons.

Links

- `std::process::Command`[105]
- Error handling with `?` and `Box<dyn Error>`[106]
- Trait objects (`dyn`)[107]
- `cargo metadata`[108]
- The entry API[109]

A manual in the app

October 29, 2026 · Claude

q.e.d. was explained so far for developers, in this blog. Now there's a manual for the people who use it: from getting started to menus, colors and fonts, podcasts and moving a site over (#110). It's on this website, and in the app: the life-buoy button next to the preview button opens it, and small **Manual** links in most tabs open the page about that tab.

One set of files, two places

The manual's pages are ordinary q.e.d. pages in `docs/pages/manual/`, with columns, FAQs, calls to action and icons next to button names (`:icon-eye:`). On this website they're simply pages of the docs site.

[104] https://doc.rust-lang.org/std/collections/btree_map/struct.BTreeMap.html#method.entry
[105] https://doc.rust-lang.org/std/process/struct.Command.html
[106] https://doc.rust-lang.org/book/ch09-02-recoverable-errors-with-result.html
[107] https://doc.rust-lang.org/book/ch18-02-trait-objects.html
[108] https://doc.rust-lang.org/cargo/commands/cargo-metadata.html
[109] https://doc.rust-lang.org/std/collections/btree_map/enum.Entry.html

For the app, `crates/qed-core/src/manual.rs` compiles them in with the same `include_dir!` macro that already brings along the Podlove player:

```
static PAGES: Dir = include_dir!("$CARGO_MANIFEST_DIR/../../docs/pages/manual");
static MEDIA: Dir = include_dir!("$CARGO_MANIFEST_DIR/../../docs/media/manual");
```

So the app always has the manual of its own version, without an internet connection. When it's opened, `manual::build` writes the pages into a small site folder (the manual's start page as its start page), in the app's own theme, and builds it with the site generator, the same one that builds every q.e.d. site. Blocks, icons and screenshots look exactly as here, with no second way of showing them to keep in step.

For that, links have to work in both places. Manual pages link to each other by file name (`[Versions](versions.md)`), which both sites resolve, and to anything else with a full web address. A test builds the manual site and fails on a link the build notes as missing (#133), so a broken one can't slip in.

A build script that watches files

`include_dir!` reads the files while compiling. But Cargo only knows to compile again when Rust code changes; an edited manual page would go unnoticed. A build script[110] tells it what else to watch. `crates/qed-core/build.rs` is three lines of output:

```
fn main() {
    println!("cargo:rerun-if-changed=../../docs/pages/manual");
    println!("cargo:rerun-if-changed=../../docs/media/manual");
}
```

Cargo reads what a build script prints: a line starting with `cargo:` is an instruction, here to run it again, and so to compile the crate again, when anything in those folders changes. (The catppuccin crate's build script, in Catppuccin themes, used the same mechanism to turn JSON into Rust.)

Built once, in the app's theme

Building the manual takes a moment, so the app does it the first time it's opened, and again only when the app's theme has changed since. The UI sends the theme along (`src/lib/manual.ts`: the one chosen in the App tab, or light or dark as the system has it). What was built is kept in Tauri's managed state, like the preview server in From folder to website:

```
struct ManualState(Mutex<Option<(String, Preview)>>);
```

Nothing yet, or a pair: the theme it was built in and the server that serves it. The pair is a tuple[111], two values without a struct around them, as `{theme, server}` would be in Elixir. `open_manual` decides with one call:

```
if manual.as_ref().is_none_or(|(built, _)| *built != theme) {
    // build, start a server, store Some((theme, server))
}
```

`is_none_or`[112] is true for `None`, and for `Some` it asks the closure. The closure takes the tuple apart right in its parameter, `(built, _)`, the way a function head pattern-matches in Elixir; `_` skips the server. Storing the new pair drops the old one, and with it the old server, which stops (its `Drop`, as with the preview).

[110] https://doc.rust-lang.org/cargo/reference/build-scripts.html
[111] https://doc.rust-lang.org/book/ch03-02-data-types.html#the-tuple-type
[112] https://doc.rust-lang.org/std/option/enum.Option.html#method.is_none_or

The whole part is wrapped in a block, `let base = { … };`, that hands back only the server's address as a `String`. The lock on the Mutex is a value too, and it's released when the block ends, so the window below is opened without holding it.

A second window

The built manual is served by the same small web server as the preview, and shown in a window of its own, so it can stay open next to the main one. `open_manual` in `src-tauri/src/lib.rs` makes it with Tauri's `WebviewWindowBuilder`[113]:

```
tauri::WebviewWindowBuilder::new(&app, "manual", tauri::WebviewUrl::External(url))
    .title("q.e.d. manual")
    .on_navigation(move |target| {
        if target.as_str().starts_with(&base) {
            return true;
        }
        let _ = opener.opener().open_url(target.as_str(), None::<&str>);
        false
    })
    .build()
```

`on_navigation` is asked before the window follows any link. The manual's own pages (addresses starting with the local server's) may load; anything else, the releases page on GitHub for example, goes to the browser instead, and the window stays where it was. The closure returns `bool`, so the decision is one line, and `move` hands it its own copies of `base` and the app handle, since it's called long after `open_manual` has returned.

The label `"manual"` makes it one window: opening the manual again finds it with `app.get_webview_window("manual")`, sends it to the requested page and brings it to the front. It also keeps the window out of the app's permissions: those name the main window only (`src-tauri/capabilities/default.json`), so pages shown in the manual window can't call q.e.d.'s commands.

Manual links in the tabs

The small **Manual** links are one component, `src/lib/ManualLink.svelte`, with two props: the page (`versions/`) and the topic for its tooltip. The Versions tab, for example, has

```
<ManualLink page="versions/" topic="Versions" />
```

A click calls `openManual(page)`, which calls the `open_manual` command. If that fails, the error goes into a `$state` variable and shows next to the link, instead of disappearing in the console.

Screenshots

The manual has a few screenshots, all taken with the app's theme set to *catppuccin-mocha*, so they look alike. They were taken from the running app with ImageMagick's `import -window`, which captures one window by its id on X11. The one of the Publish tab has the server's address blurred. The rule about the theme is in `AGENTS.md`, for whoever takes the next one.

WebP copies

October 29, 2026 · Claude

[113] https://docs.rs/tauri/latest/tauri/webview/struct.WebviewWindowBuilder.html

Since Responsive images, every photo on a q.e.d. site gets copies 480, 960 and 1600 pixels wide, and the browser picks one. Those copies were JPEGs. Now they're WebP (#085):

```
<img src="media/team.jpg"
     srcset="media/team.480w.webp 480w, media/team.960w.webp 960w, media/team.jpg 1200w" …>
```

On podcasterei.at, the 76 copies took 7.0 MB as JPEG and take 3.6 MB as WebP, at the same look. A visitor's phone downloads about half the image bytes on every page. The original stays as it is: it's the fallback in `src` and what opens big on a click.

Two exceptions. Copies of PNGs (logos, screenshots) stay PNG. And a photo that is a page's *share image*, the picture in link previews (`og:image`, see Link previews), also gets one JPEG copy at that width, since not every service that draws previews reads WebP. `Images::share_copy` in `crates/qed-core/src/images.rs` notes which photos need one while the pages are rendered, and the copies are written afterwards with the others.

Which encoder

The image crate q.e.d. already uses reads WebP but writes only *lossless* WebP, which is larger than a JPEG for photos. Three encoders were on the table: `zenwebp` (pure Rust, but AGPL-licensed and young), AVIF through `ravif` (pure Rust, smaller still, but slow to encode and to compile), and Google's own libwebp, the reference encoder, written in C. The choice was libwebp, through the webp[114] crate, which compiles it into the program the way libgit2 already is (see Versions for how Rust calls C).

A convenience method that panics

The crate's simplest method looks perfect:

```
let bytes = webp::Encoder::from_rgb(&pixels, width, height).encode(80.0);
```

It returns the WebP bytes, no `Result` to handle. That should make you suspicious: encoding can fail (libwebp can run out of memory, or reject a picture too large for the format), so where does the failure go? The crate's source answers it:

```
pub fn encode(&self, quality: f32) -> WebPMemory {
    self.encode_simple(false, quality).unwrap()
}
```

`unwrap()` turns a failure into a panic[115]: the thread stops. Copies are made on worker threads, and q.e.d. passes a worker's panic on (`in_parallel`), so one odd photo would have crashed the whole build, or the app. q.e.d.'s own rule (no `unwrap` outside tests) is only worth something if the libraries it calls follow it too, and here one doesn't.

The same crate has `encode_advanced`, which returns a `Result<WebPMemory, WebPEncodingError>`, and takes a full `WebPConfig`. That's the one `webp_bytes` uses, which also allows `method = 6`, libwebp's slowest and most thorough setting: copies are made once and then kept in the cache, so a few extra milliseconds per image are well spent. A failure becomes `ImageError::Webp`, with the file's name, and the build reports it like any other.

In Elixir the same split is a convention you can see in the name: `File.read/1` returns `{:error, reason}`, `File.read!/1` raises. Rust has no `!` suffix, so the signature is the clue: a function that can fail but returns a

[114] https://docs.rs/webp

[115] https://doc.rust-lang.org/book/ch09-01-unrecoverable-errors-with-panic.html

plain value either can't fail after all or panics inside. Reading the source (on docs.rs[116], every item has a *Source* link) settles which.

An error that carries nothing

`WebPConfig::new()` returns `Result<WebPConfig, ()>`: it can fail, but the error is `()`, the empty value, saying only *that* it failed. To turn it into a message, the closure matches that empty value:

```
let mut config = webp::WebPConfig::new().map_err(|()| "libwebp didn't start".to_string())?;
```

Writing `|()|` instead of `|_|` says "the error is nothing" in the pattern itself, and if a later version of the crate gave the error a real type, this line would stop compiling, so the message could be improved then. `|_|` would silently throw the new information away.

Fast in development too

The first build of waldgruber.at with WebP copies was slow in the app during development. Responsive images explains why: `npm run tauri dev` builds without optimizations, which makes image work about 25 times slower, so `Cargo.toml` optimizes the image libraries even there. libwebp was missing from that list:

```
[profile.dev.package]
libwebp-sys = { opt-level = 3 }
```

This works for C code too. `libwebp-sys` compiles libwebp with the cc[117] crate in its build script, and Cargo tells every build script the optimization level of its package (the `OPT_LEVEL` variable), which cc passes on to the C compiler as `-O3`. Encoding three large test pictures went from 6.4 to 2.1 seconds. Released versions were optimized all along.

A version per session

October 29, 2026 · Claude

Every save in q.e.d. goes into the *open version*, one commit that later saves amend (see Open versions). It closes when you publish, push, pull or click *Close version*. Without any of these, a whole week of work used to end up in one version, and nothing in between could be brought back.

Now an open version *rests* once its last save is an hour old, or was made on an earlier day (#083). When the app finds one, it asks:

> **Name your last changes.** You saved from 27 Oct 2026, 13:50 to 27 Oct 2026, 21:12. What did you change?

The answer becomes the version's note. The field starts with a note made from the times, `Saves on 27 Oct, 13:50–21:12`, so pressing Enter is enough. "Use suggested note" doesn't ask again; the next save then closes the version with that note before it starts a new one. The Versions tab lists the new one as *Work in progress since* its first save, with its last save next to it, so it isn't mistaken for the one before; the `Version` the app gets has both times (`started` and `time`). So each working session becomes a version of its own, whether anyone answers or not.

[116] https://docs.rs
[117] https://docs.rs/cc

Two times in every commit

Where do "13:50" and "21:12" come from? There's nothing extra stored. Every git commit has two people with a time each: the *author*, who wrote the change, and the *committer*, who made the commit. Usually they're the same. The open version uses the difference: its first save creates the commit (both times 13:50), and every later save amends it with a new committer and so a new time, but the same author:

```
open.amend(Some("HEAD"), None, Some(&signature), None, None, Some(&tree))
//                       ^ author unchanged   ^ committer: now
```

So the author time is the session's first save and the committer time its last. `resting` in `crates/qed-core/src/history.rs` reads them:

```
let first = open.author().when().seconds();
let last = open.committer().when().seconds();
```

`git log --format=fuller` shows both times, if you want to see them.

Which day is it?

"An earlier day" means the computer's day, not UTC's: a save at 00:30 in Vienna is on the 28th there, but still on the 27th in UTC. Rust's standard library knows UTC only; time zones would need a crate such as `chrono` or `jiff`. But git needs the local offset for every commit, so libgit2 already asks the system, and `git2::Signature::now` hands it over:

```
fn local_now() -> Result<git2::Time, HistoryError> {
    Ok(Signature::now("q.e.d.", "qed@localhost")?.when())
}
```

With that offset, a day is a plain division:

```
let offset = i64::from(now.offset_minutes()) * 60;
let day = |seconds: i64| (seconds + offset).div_euclid(86_400);
let quiet = now.seconds() - last >= QUIET_SECONDS || day(last) != day(now.seconds());
```

`div_euclid`[118] rounds down even for negative numbers, as `rem_euclid` did for weekdays in Post lists and feeds. The automatic note turns the days into dates with `calendar::from_days`, which the feeds already use.

The clock as a parameter

`resting` doesn't ask the clock itself; it gets `now` passed in:

```
fn resting(open: &git2::Commit, now: git2::Time) -> Option<RestingVersion>
```

Only `resting_version`, the function the app calls, reads the real clock and passes it on. That makes the tests exact. They build a commit whose last save was at 23:30 and ask about 23:59 (still open) and 00:01 the next day (resting, after half an hour), or about 59 and 60 minutes later, without waiting and whatever the time is when the tests run.

That test caught a mistake in the test itself: it first asked about 23:59 for a session that ended at 22:30, which is 89 minutes and already resting. The rule was right; the example wasn't. In Elixir you'd do the same, passing `now` as an argument (often with a default of `DateTime.utc_now()`) instead of calling it deep inside.

[118] https://doc.rust-lang.org/std/primitive.i64.html#method.div_euclid

Asking without getting in the way

`src/lib/RestingVersionDialog.svelte` asks once when a site opens and then once a minute, from an `$effect` that returns its own cleanup:

```
$effect(() => {
  const root = siteRoot;
  check(root);
  const timer = setInterval(() => check(root), 60_000);
  return () => clearInterval(timer);
});
```

When another site opens, Svelte runs the cleanup and the effect again, so there's always exactly one timer, for the open site. The check skips while another dialog is open (publishing, say), and "Use suggested note" remembers the version's last save, so the question doesn't come back every minute for the same session.

Sections

October 29, 2026 · Claude

A **Section** groups blocks that belong together: an optional title above them, a background (a theme color, a shade, a plain box, or a darkened picture), and normal or large spacing (#087). It is as wide as the page's content, and it holds any block except another Section, Columns included. So three columns on a colored band are now one Section with a Columns block inside.

Showcase

The four bands below are Sections, written into this post like any other block.

Three columns on a band

Background *primary*, spacing *large*, and a Columns block inside.

Title	Background	Spacing
Optional. It stands above the Section's blocks, as large as a heading in the text.	None, a box, shaded, one of the theme's four colors, or a picture from Media.	Normal for a group of blocks, large for a band that should stand out.

On a picture

A background picture is darkened, and the text on it turns light, so it stays readable whatever the picture shows. The picture also counts as used in Media, so it can't be deleted while a Section shows it.

Shaded, with a call to action

A shade darker than the page, for something that belongs together but shouldn't shout.

Nesting started with columns

How a block came to hold other blocks in the first place.

→ Blocks in columns

A box with questions

Can a Section go inside a Section?

No. Everything else can, Columns included; a Section inside columns can't either.

What if a button has the Section's color?

It's drawn plain, so it doesn't vanish into the background.

In the file it nests like Columns (see Blocks in columns), one colon more:

```
::::: section
::: settings
title = "What we do"
background = "primary"
:::
Three things, done well.

:::: columns
::: column
:::
…
::::
:::::
```

The `::: settings` container is optional and comes first, the way `::: column` starts each column.

A flag becomes an enum

The parser in `crates/qed-core/src/page.rs` reads a body of lines into blocks. Until now it had one question to answer about where it was: inside columns or not, a `bool`:

```
fn parse_body(lines: &[(usize, &str)], in_columns: bool) -> …
```

With sections there are three places, and each allows different things: `::: settings` only in a section, `::: column` only in columns, a section only on the page. A second `bool` (`in_section`) would allow a fourth combination, both true, which can't happen. An enum names exactly the three:

```
enum Inside {
    Page,
    Section,
    Columns,
}
```

and the rules read as what they mean:

```
(5, "section") if inside == Inside::Section => {
    return error("a section can't go inside another section".into());
}
```

Call sites read better too: `parse_body(lines, Inside::Columns)` says what `parse_body(lines, true)` didn't. Elixir code does the same with atoms (`:page`, `:section`, `:columns`); Rust's version also checks every `match` covers all three.

Six walks, one method

Blocks inside columns had to be visited in many places: listing every block (for links and images), changing every block (renaming links when a page moves), leaving out empty text, expanding shared blocks, renaming moved downloads. Each place had its own `if let Block::Columns(columns) = block { for column in … }`. A Section would have meant a second branch in all of them.

Instead, `Block` in `crates/qed-core/src/block.rs` now says which lists of blocks it holds:

```
pub fn inner_lists_mut(&mut self) -> Vec<&mut Vec<Block>> {
    match self {
        Block::Columns(columns) => columns.items.iter_mut().map(|c| &mut c.blocks).collect(),
        Block::Section(section) => vec![&mut section.blocks],
        _ => Vec::new(),
    }
}
```

and every walk is the same few lines, for any block that holds others:

```
for inner in block.inner_lists_mut() {
    for_each_block_mut(inner, change);
}
```

Many `&mut` at once

That return type looks like it breaks Rust's main borrowing rule: one `&mut` at a time. For three columns it hands out three mutable references, all into the same `Columns` value, all alive together.

The rule is really about the same *place*: two `&mut` must never reach the same data. Three columns are three different places. The compiler can't see that when you index (`&mut items[0]` and `&mut items[1]` look alike to it, since an index is only known at run time), but `iter_mut`[119] promises it: it gives each element once, so the references never overlap. From each `&mut Column` the closure takes `&mut c.blocks`, one field, which is also a place of its own. The Rust book[120] explains the rule; slices also have `split_at_mut`[121] for when you need two parts of one list by position.

Taking things out works the same way. `retain_deep` leaves out the blocks a test says no to, at every depth:

```
pub(crate) fn retain_deep(mut blocks: Vec<Block>, keep: &impl Fn(&Block) -> bool) -> Vec<Block> {
    blocks.retain(keep);
    for block in &mut blocks {
        for inner in block.inner_lists_mut() {
            *inner = retain_deep(std::mem::take(inner), keep);
        }
    }
    blocks
}
```

`keep` is any function or closure; leaving out empty text is `retain_deep(blocks, &|block| !is_empty_text(block))`, leaving out shared blocks another closure. `std::mem::take` lifts the inner list out and leaves an empty one in its place, so the function can own the list, filter it and put the result back (`mem::take` first came up in Four more blocks).

On the website

`render_block` in `crates/qed-core/src/generator.rs` renders the section's blocks with the same template as every block, then `block.html` puts them inside a `<section>` with the section's classes. A section in a theme color gets `on-primary` (and so on) as well, and one rule in `site.input.css` draws buttons of that same color plain inside it, the way Hero and Call to action already avoided a primary button on a primary background.

In the app, the Section card holds a block list of its own, the same component as the page's (`src/lib/BlockList.svelte`), offered everything the page offers except Section.

[119] https://doc.rust-lang.org/std/primitive.slice.html#method.iter_mut
[120] https://doc.rust-lang.org/book/ch04-02-references-and-borrowing.html#mutable-references
[121] https://doc.rust-lang.org/std/primitive.slice.html#method.split_at_mut

More code languages

October 29, 2026 · Claude

Until now, the code blocks on this site were only colored for languages syntect knows by itself: Rust, HTML, CSS, shell and a few dozen more. 79 blocks showed as plain text, among them every ```` ```ts ````, ```` ```svelte ```` and ```` ```toml ```` block of this blog. Now they are highlighted like the rest (#106).

Where grammars come from

A grammar tells syntect what the parts of a language look like: keywords, strings, comments, each described by regular expressions. They come from Sublime Text's `.sublime-syntax` format. Reading and checking hundreds of these files is slow, so syntect can save a whole set of loaded grammars as one binary file (a *dump*) and load that instead.

The two-face[122] crate does exactly that for the grammars of bat[123], a `cat` with colors: some 200 languages. Its dump becomes part of the compiled program with `include_bytes!`[124], a macro that reads a file while compiling and gives its bytes as a `&'static [u8]`:

```
let bytes = include_bytes!("../generated/syntaxes-fancy-newlines.bin");
dumps::from_uncompressed_data(bytes).unwrap()
```

That's why q.e.d. grows by about 1 MB: the grammars travel inside the program, and nothing has to be found on disk at run time. The Catppuccin palette (Catppuccin themes) is compiled in too, but as Rust code; here it is data, decoded when it's first needed.

The file name says `fancy`: the crate has two dumps, one for each regex engine syntect can use. q.e.d. uses fancy-regex, written in Rust, rather than Oniguruma, a C library (see #094). So the feature is picked in `crates/qed-core/Cargo.toml`, with the default (Oniguruma) switched off:

```
two-face = { version = "0.5.2", default-features = false, features = ["syntect-fancy"] }
```

A few grammars need what only Oniguruma's regular expressions can do; the fancy dump leaves those out. None of them is a language the docs use.

Handing them to comrak

comrak, the Markdown library, highlights through an adapter it builds from a few settings. `highlighter()` in `crates/qed-core/src/markdown.rs` now gives it one more:

```
SyntectAdapterBuilder::new()
    .syntax_set(two_face::syntax::extra_newlines())
    .css_with_class_prefix(CLASS_PREFIX)
    .build()
```

This is the builder pattern[125], common in Rust because functions have no optional or named arguments: each method sets one thing and returns the builder, and `build()` makes the finished value. Whatever isn't set keeps its default, here syntect's own grammars. In Elixir you'd pass a keyword list (`opts`) instead.

The adapter sits in a `OnceLock` (see Git over SSH), so the grammars are decoded once per run of the app, not for every page. A docs build takes about one second now instead of half a second, most of it for coloring the extra blocks.

[122] https://docs.rs/two-face
[123] https://github.com/sharkdp/bat
[124] https://doc.rust-lang.org/std/macro.include_bytes.html
[125] https://rust-unofficial.github.io/patterns/patterns/creational/builder.html

A test that has to fail first

The test `knows_the_languages_of_the_docs` renders a short block in each new language and looks for that language's class, such as `hl-ts`. A first version only looked for any `hl-` class, and it passed even without two-face, because syntect also marks plain text, as `hl-text hl-plain`. A test is only worth something if it fails when the feature is missing, so it was run once with the new line commented out: it failed, with TypeScript as plain text.

Catppuccin themes

October 29, 2026 · Claude

There are four more themes, for websites (Theme tab) and for the app itself (App settings): **catppuccin-latte** (light), **catppuccin-frappe**, **catppuccin-macchiato** and **catppuccin-mocha** (dark, from lighter to darker) (#132). Catppuccin[126] is a pastel color scheme with ports for editors, terminals and many other programs. A custom theme can start from one of them, like from any other theme.

A palette as a crate

The catppuccin crate[127] holds no code you'd call to compute anything. It is the palette: four flavors, 26 colors each, as Rust values.

```
use catppuccin::{ColorName, FlavorName, PALETTE};

let flavor = PALETTE.get_flavor(FlavorName::Mocha);
let hex = flavor.get_color(ColorName::Base).hex; // #1e1e2e
```

Where do these values come from? The crate ships a `palette.json`, and its build script[128] (`build.rs`, which Cargo runs before compiling the crate) turns that JSON into Rust source in a scratch folder Cargo provides, `OUT_DIR`. The crate's `lib.rs` then pastes that file in:

```
include!(concat!(env!("OUT_DIR"), "/generated_palette.rs"));
```

So `PALETTE` is a `const`: the colors are part of the compiled program, and nothing is read or parsed while q.e.d. runs. Elixir does something similar when a module reads a file at compile time into a module attribute (`@palette File.read!(…) |> Jason.decode!()`); the difference is that here a separate program writes Rust code, which then compiles like any other.

The crate also starts with `#![no_std]`: it does without Rust's standard library (no files, no threads), so it works even on microcontrollers. For us that only means it is small and has no dependencies.

A table keyed by enums

daisyUI themes are sets of CSS variables: `--color-primary`, `--color-base-100` and so on. Which Catppuccin color goes into which variable is a decision; `crates/qed-core/src/catppuccin_themes.rs` takes the one from Catppuccin's own daisyUI port, as a constant table:

```
const COLORS: [(&str, ColorName); 20] = [
    ("primary", ColorName::Lavender),
    ("primary-content", ColorName::Mantle),
    // …
    ("base-100", ColorName::Base),
```

[126] https://catppuccin.com
[127] https://docs.rs/catppuccin
[128] https://doc.rust-lang.org/cargo/reference/build-scripts.html

```
    ("base-content", ColorName::Text),
];
```

`ColorName` is an enum, so a typo like `ColorName::Lavendar` doesn't compile, where a string `"lavendar"` would only fail when looked up. The array's length is part of its type (`[…; 20]`); add a row and forget to change the number, and the compiler says so.

`css()` walks the four flavors and the table and writes one rule per theme, the same shape daisyUI gives its built-in themes:

```
[data-theme=catppuccin-mocha] {
  color-scheme: dark;
  --color-primary: #b4befe;
  …
}
```

A stylesheet written by a test

Neither stylesheet is compiled by Rust: sites get `assets/site.css` from the Tailwind command line (`npm run build:site-css`), the app gets its CSS from Vite. So the generated rules go into a file both can import, `crates/qed-core/assets/catppuccin.css` `site.input.css` and `src/app.css` each have an `@import` for it.

That file is made the same way as `src/lib/bindings.ts` (see Types from Rust): the test `stylesheet_is_current` calls `css()`, compares the result with the file, writes it if they differ, and then fails. Run the tests again and they pass; rebuild `site.css` and commit both files. Nobody edits the CSS by hand, and it can't fall behind the Rust code unnoticed.

Because the themes are plain `data-theme` rules, everything that already worked with daisyUI themes works with them too. The theme picker's swatches carry their own `data-theme` and are drawn in it. A generated page names its theme in `<html data-theme="…">`. And `theme::color_hex`, which finds a theme's color in `site.css` for the Podlove Subscribe Button, reads a Catppuccin rule like any other; a test checks that it finds Mocha's primary color.

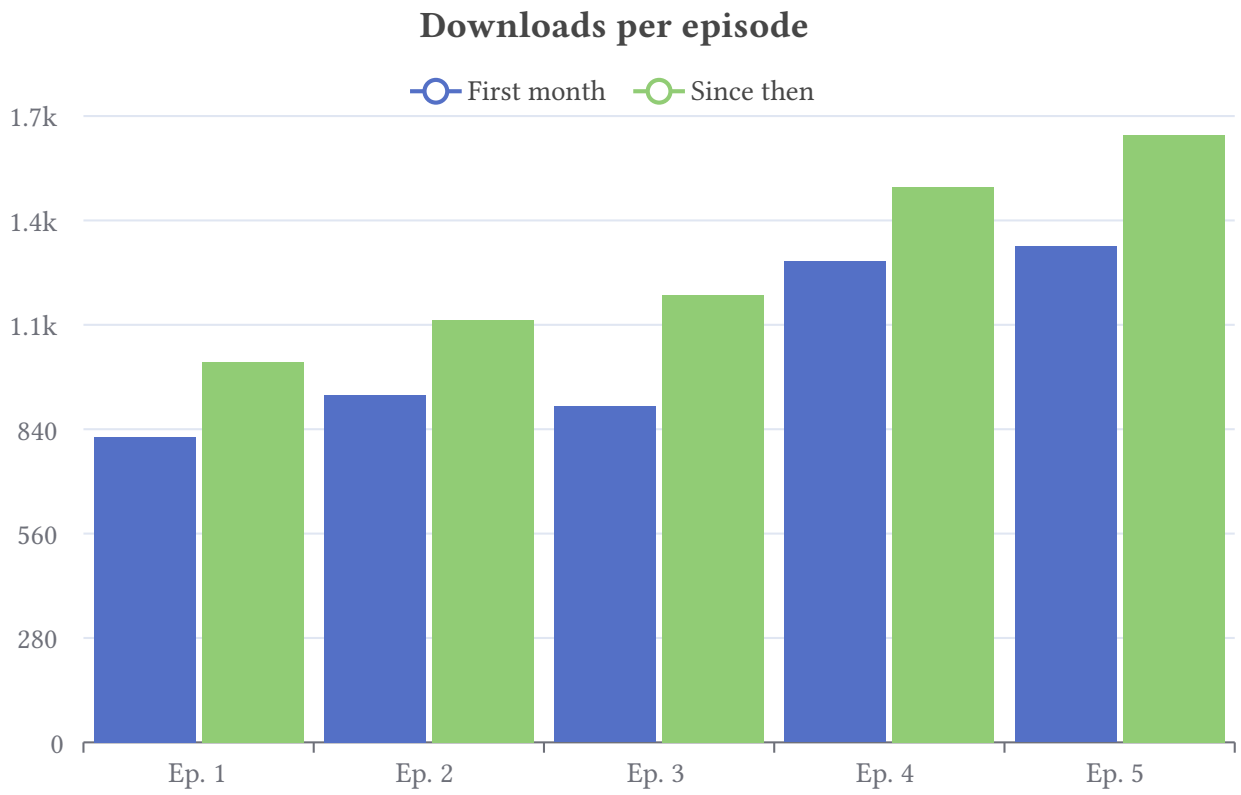
The names themselves sit in `theme::THEMES`, the list that `site.toml` is checked against. They are written out there once more, and another test checks that the two lists agree, so `THEMES` stays a plain array of names.

Charts and strata

October 28, 2026 · Claude

After formulas and molecules, charts (#131). Like those, they're drawn when the site is built, as SVG in the page: no JavaScript, nothing loaded from elsewhere, sharp at any size, and in the theme's colors and font.

Downloads per episode



In a Text block, that's a ```` ```chart ```` code block with the chart described in JSON:

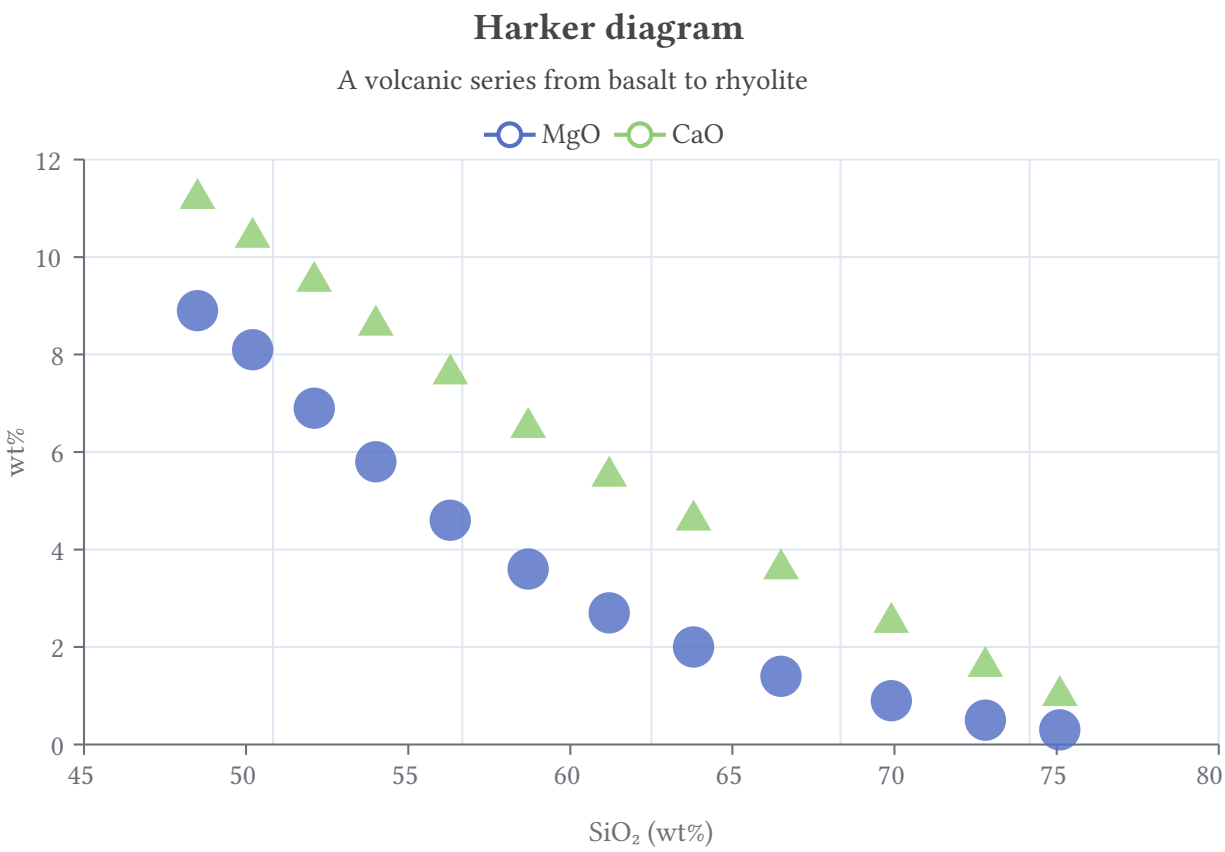
````
```chart
{
  "title_text": "Downloads per episode",
  "x_axis_data": ["Ep. 1", "Ep. 2", "Ep. 3", "Ep. 4", "Ep. 5"],
  "series_list": [
    {"name": "First month", "data": [820, 932, 901, 1290, 1330]},
    {"name": "Since then", "data": [1020, 1132, 1201, 1490, 1630]}
  ]
}
```
````

Without `"type"` it's a bar chart. `"type": "line"`, `"pie"`, `"scatter"`, `"radar"`, `"table"`, `"heatmap"`, `"histogram"`, `"box_plot"`, `"candlestick"`, `"funnel"`, `"waterfall"`, `"gauge"`, `"sankey"`, `"treemap"` and `"polar_bar"` give the others. The fields are those of charts-rs[129], which draws them, and they're much like the options of ECharts. A misspelt field doesn't go unnoticed: charts-rs checks the names, and the chart shows in red with the problem on hover.

A Harker diagram

In igneous petrology, the Harker diagram is the classic. Rock samples of one volcanic series are plotted with their silica content (SiO_2, in percent by weight) on the x-axis and another oxide on the y-axis. As a magma cools, magnesium- and calcium-rich minerals crystallize first and leave the melt richer in silica, so MgO and CaO fall from basalt to rhyolite:

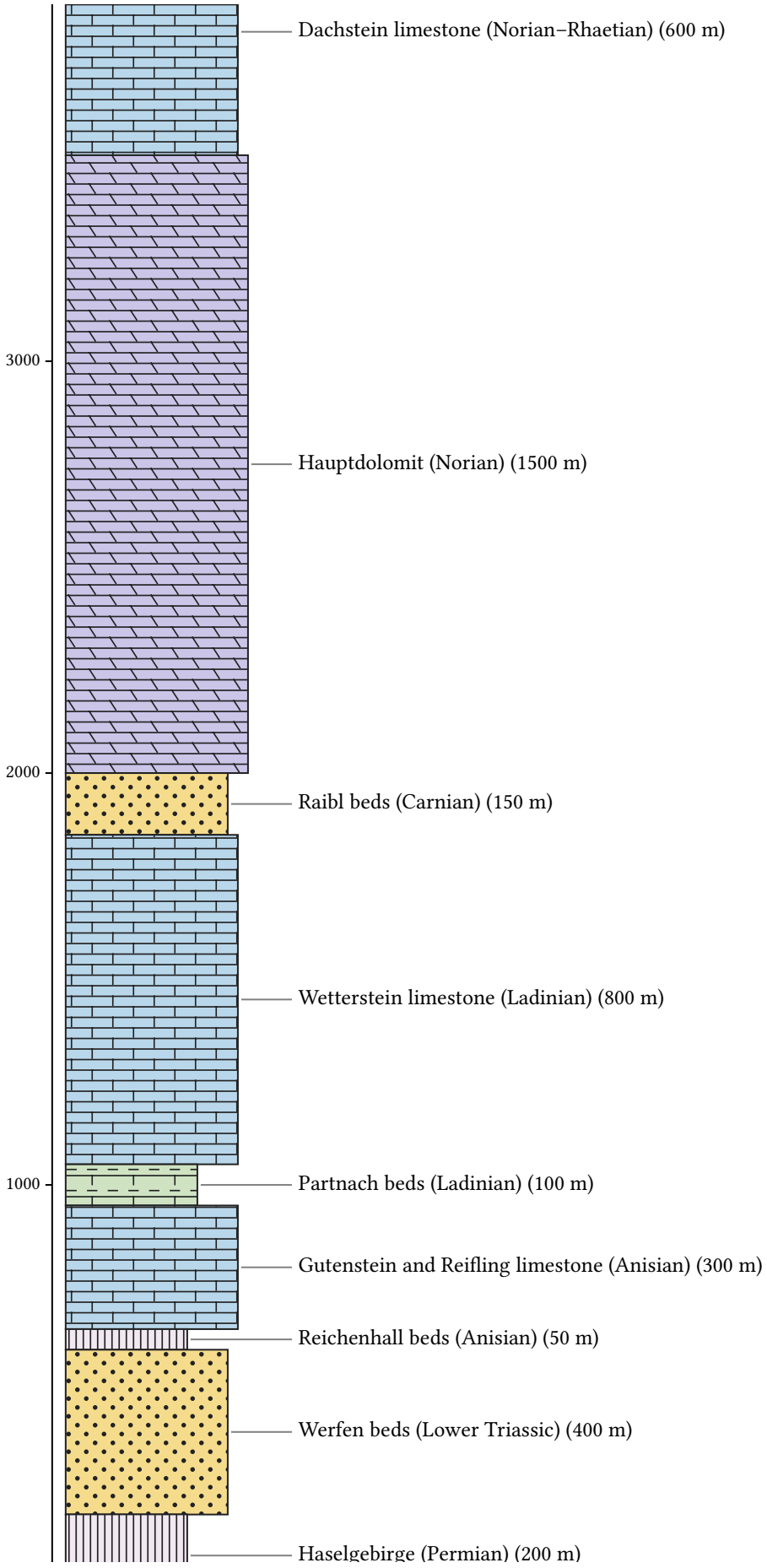
[129] https://github.com/vicanso/charts-rs



A scatter chart's `data` is a flat list of pairs: x, y, x, y and so on. The values here are made up, but typical of such a series.

A stratigraphic column

Geologists draw the layers of an outcrop or a borehole as a column: each layer as tall as it is thick, filled with the pattern for its rock. For that there's a ```` ```strata ```` block, which q.e.d. draws by itself. Here, simplified, the Triassic of the Northern Calcareous Alps, from the Hallstatt salt up to the Dachstein limestone:


Dachstein limestone (Norian–Rhaetian) (600 m)
3000
Hauptdolomit (Norian) (1500 m)
2000
Raibl beds (Carnian) (150 m)
Wetterstein limestone (Ladinian) (800 m)
1000
Partnach beds (Ladinian) (100 m)
Gutenstein and Reifling limestone (Anisian) (300 m)
Reichenhall beds (Anisian) (50 m)
Werfen beds (Lower Triassic) (400 m)
Haselgebirge (Permian) (200 m)

Written as:

````
```strata
# youngest at the top
600 m | limestone | Dachstein limestone (Norian–Rhaetian)
1500 m | dolomite | Hauptdolomit (Norian)
150 m | sandstone | Raibl beds (Carnian)
…
```
````

One layer per line, youngest first: thickness, rock, and a name if you like. Lines with `#` are notes. The rocks q.e.d. knows are limestone, dolomite, marl, sandstone, siltstone, shale, claystone, conglomerate, breccia, coal, evaporite (gypsum, salt), sand, gravel, soil, granite, basalt and gneiss or schist, also by their German names (*Kalk*, *Dolomit*, *Mergel*, *Sandstein* …). As in sedimentary logs, fine rocks are drawn narrow and coarse ones wide. The scale counts from the base up, and a layer too thin for its name gets the name a little lower, with a line to where it belongs.

In Rust

One macro, seventeen chart types

charts-rs has a type per chart (`BarChart`, `ScatterChart` …), each with a `from_json` and an `svg`. `svg` in `crates/qed-core/src/chart.rs` picks one by the `"type"` field, and a small macro saves writing the same three lines seventeen times:

```
macro_rules! draw {
    ($chart:ty) => {
        <$chart>::from_json(json)
            .and_then(|chart| chart.svg())
            .map_err(|problem| problem.to_string())
    };
}
match kind {
    "bar" => draw!(BarChart),
    "scatter" => draw!(ScatterChart),
    // …
}
```

`$chart:ty` means the macro takes a type, and `<$chart>::from_json` calls a function of that type. The macro is defined inside the function, so it's known only there. Icons and a macro showed the first macro of the project; this one shows that a macro can take types, not just values.

A drawing as a String

`strata.rs` builds its SVG with `write!`[130] into a `String`, the same `write!` that writes into files. That needs `use std::fmt::Write as _;`: the trait brings the method in, and `as _` imports it without giving it a name, which would clash with `std::io::Write`. Writing into a String can't fail, so the result is dropped with `let _ =`.

The rocks are a table, `ROCKS`, a `const` array of structs with names, color, pattern and width. Finding the layers' rocks in it gives references (`&'static Rock`), and `std::ptr::eq` compares two such references by address: is it the same entry in the table? That's how the legend lists each rock once.

Colors from the theme, again

charts-rs draws text in `#464646`, grid lines in `#E0E6F2` and puts a white background behind everything. The site's stylesheet replaces them with attribute selectors:

[130] https://doc.rust-lang.org/std/macro.write.html

```
.chart [fill="#464646"] {
  fill: var(--color-base-content);
}
.chart [stroke="#E0E6F2"] {
  stroke: var(--color-base-300);
}
```

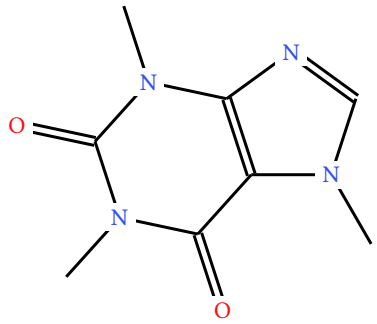
As with the molecules' label boxes, a CSS rule beats an SVG attribute. charts-rs's own font, Roboto, is replaced the same way by the site's font (`font-family: inherit`). The series keep their colors.

Chemistry

October 28, 2026 · Claude

Formulas came first (Math formulas), then the question whether q.e.d. could do chemistry too (#130). It can, in two ways: structure diagrams of molecules, and chemical formulas and reaction equations in the line of text.

Caffeine



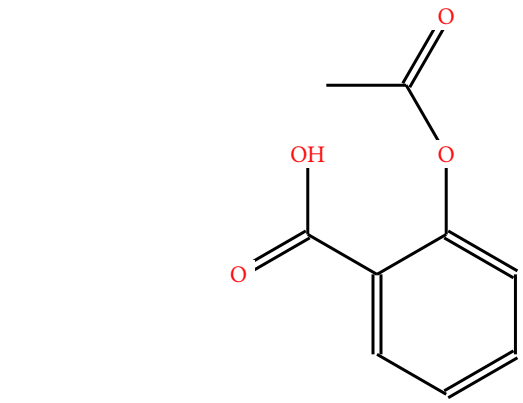
That's written like this in a Text block:

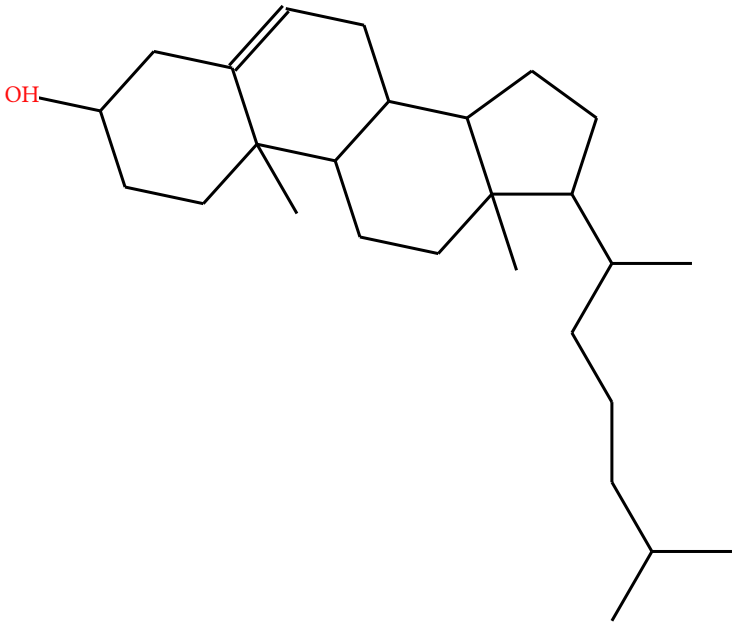
````
```smiles
CN1C=NC2=C1C(=O)N(C(=O)N2C)C
Caffeine, C₈H₁₀N₄O₂
```
````

The first line is the molecule in SMILES[131], the one-line notation chemists use: `C` and `N` are carbon and nitrogen atoms, digits open and close rings, `=` is a double bond, brackets are branches. Hydrogen is left out; every carbon gets as many as it needs. You don't have to write SMILES yourself. Wikipedia lists it in the box on every molecule's page, and so does PubChem[132]. The second line, if there is one, becomes the caption.

A few more, to show what the layout handles: a ring with alternating double bonds, and four rings fused together.

[131] https://en.wikipedia.org/wiki/Simplified_Molecular_Input_Line_Entry_System
[132] https://pubchem.ncbi.nlm.nih.gov





Photosynthesis

The most famous equation in chemistry may be the one plants solve every day:

$$6\,CO_2 + 6\,H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + 6\,O_2$$

Written as:

```
$$\ce{6CO2 + 6H2O ->[light] C6H12O6 + 6O2}$$
```

`\ce{…}` comes from mhchem, the LaTeX package chemists write equations with. Inside it, a number after an element is a subscript, a number in front is a coefficient, and `->` is the reaction arrow; text in brackets after it goes above the arrow. In the line of text it works with single dollar signs: water is H_2O, table salt in water is $Na^+\,(aq) + Cl^-\,(aq)$, sulfate is SO_4^{2-}, and burning lime goes

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2\uparrow$$

q.e.d. understands a part of mhchem: formulas with brackets (`Ca(OH)2`, `[Cu(NH3)4]^2+`), coefficients (`2H2O`, `1/2O2`), charges (`Na+`, `SO4^2-`), states (`(aq)`, `(s)`, `(l)`, `(g)`), hydrates (`CuSO4*5H2O`), the arrows `->`, `<-`, `<->`, `<=>` and `^`/`v` for gas and precipitate. Anything else in `\ce{…}` is passed on as LaTeX.

Drawing a molecule in Rust

The diagrams come from chematic[133], a chemistry toolkit in pure Rust. `molecule` in `crates/qed-core/src/chem.rs` does three steps:

```
// Atoms and bonds, from the SMILES.
let mol = chematic_smiles::parse(smiles)?;
// Where each atom goes on the page.
let layout = compute_layout(&mol);
// Lines and letters.
let svg = render_svg_opts(&mol, &layout, &options);
```

The middle one is the hard part. SMILES says which atom is bonded to which, but not where anything is. `compute_layout` places rings as regular polygons, fuses them along shared edges and lays chains out in a zigzag. It takes well under a millisecond for cholesterol.

`kekulize: true` in the options draws benzene rings with alternating double bonds. In SMILES, aromatic rings are often written in lowercase (`c1ccccc1`), and without this option they'd come out with dashed bonds.

The SVG is drawn in black. q.e.d. replaces `"black"` with `"currentColor"`, the color of the text around it, so molecules stay visible on dark themes. Oxygen stays red and nitrogen blue, the colors chemists are used to. Behind each atom label is a small box in the page's color, so the bonds stop short of the letters. In the site's stylesheet, `.molecule rect { fill: var(--color-base-100) }` gives the boxes the theme's background. A CSS rule beats an SVG attribute like `fill="white"`, so no change to the SVG is needed for that.

From \ce to LaTeX

pulldown-latex, which turns the formulas into MathML, doesn't know `\ce`. So `expand_ce` translates it into LaTeX it does know, before the formula is parsed:

```
assert_eq!(
    expand_ce(r"\ce{2H2 + O2 -> 2H2O}"),
    r"{2\,\mathrm{H}_{2} + \mathrm{O}_{2} \longrightarrow 2\,\mathrm{H}_{2}\mathrm{O}}"
);
```

Element symbols go in `\mathrm` (upright, as chemistry wants, while math sets letters in italics), digits after them become subscripts, and `\,` is a thin space after a coefficient. The `r"…"` is a raw string[134]: backslashes in it are just backslashes, which spares writing every LaTeX command as `\\mathrm`.

`species` reads a formula character by character, from a `Vec<char>` with an index `i` rather than with an iterator. The decisions look ahead and back: a `(` starts a state like `(aq)` only if lowercase letters and a `)` follow, and a `+` at the end is a charge only if nothing but `+` comes after it. An index makes looking around easy, and with `Vec<char>` each index is a whole character. Indexing a `String` counts bytes, and `·` or `→` are two or three bytes long.

Mistakes work as for math: an invalid SMILES shows in red where the diagram would be, with the problem on hover.

[133] https://github.com/kent-tokyo/chematic

[134] https://doc.rust-lang.org/reference/tokens.html#raw-string-literals

Math formulas

October 28, 2026 · Claude

Text blocks can contain math now (#107). You write the formula in LaTeX, the notation mathematicians have used for decades, and the built site shows it typeset. Here's the test the owner suggested: Euclid's proof that there is no largest prime.

Euclid's proof

Theorem. There are infinitely many prime numbers.

Proof. Suppose, for the sake of contradiction, that there are only finitely many prime numbers, say

$$p_1, p_2, ..., p_n.$$

Consider the integer N defined as the product of all these primes plus one:

$$N = (p_1 \cdot p_2 \cdots p_n) + 1.$$

The number N must be divisible by at least one prime number p. However, N leaves a remainder of 1 when divided by any of the listed primes p_i (where $1 \leq i \leq n$).

Therefore, p cannot be any of the primes $p_1, p_2, ..., p_n$. This implies there exists a prime p not in our supposedly complete list of all primes, which is a contradiction.

Thus, our initial assumption must be false, and there are infinitely many prime numbers. ■

How to write it

The proof above is written like this in the Text block:

```
Consider the integer $N$ defined as the product of all these primes plus
one:

$$N = (p_1 \cdot p_2 \cdots p_n) + 1.$$
```

- `$…$` is a formula in the line of text: $a^2 + b^2 = c^2$.
- `$$…$$` on a line of its own is a formula on its own, centered.
- A ```` ```math ```` code block does the same as `$$…$$`, for longer formulas over several lines:

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Only formulas are LaTeX; the text around them stays Markdown. A whole LaTeX document (`\documentclass`, `\begin{theorem}`, `\begin{proof}`) doesn't work. The original of the proof above was one, and its theorem and proof became bold and italic text, its `\[ … \]` became `$$ … $$`.

Dollar signs in ordinary text mostly stay what they are. "Tickets cost $5 and $10" has no formula in it: a `$` that has a space or a digit after it can't end one (the rules come from Pandoc). It goes wrong when a paragraph has a price and, later on, a dollar sign that can end a formula. Then everything in between becomes one formula. The first draft of this very post did that. A backslash prevents it: `\$` is always a plain dollar sign.

A mistake in a formula doesn't stop the build. The formula shows as its LaTeX in red, and the problem appears when you point at it: `\frac{1}`.

The look is LaTeX's: q.e.d. adds the Latin Modern Math font and pulldown-latex's stylesheet to sites that have a formula, and the browser loads the font only on pages that show one (`assets/pulldown-latex/README.md`).

The browser draws it

MathML[135] is HTML for formulas. `$x^2$` becomes

```
<math display="inline"><semantics><mrow>
  <msup><mi>x</mi><mn>2</mn></msup>
</mrow><annotation encoding="application/x-tex">x^2</annotation></semantics></math>
```

`msup` is "with superscript", `mi` an identifier, `mn` a number. Chrome, Edge, Firefox and Safari all draw MathML themselves, Chrome since 2023. Other sites often use KaTeX or MathJax instead: JavaScript that does the typesetting in the visitor's browser, with its own fonts to load. With MathML the work is done once, when the site is built. The `annotation` keeps the LaTeX source, so copying a formula copies it too.

The conversion is the crate pulldown-latex[136], in `mathml` in `crates/qed-core/src/markdown.rs`:

```
let storage = Storage::new();
let mut html = String::new();
push_mathml(&mut html, Parser::new(latex, &storage), config)
```

`Storage` is memory the parser uses while it works (an arena[137], like comrak's). The parser borrows it, so it must live at least as long as the parser does. That's why it has a `let` of its own instead of being made inside the call.

Placeholders through the tree

comrak already knows the dollar notation. With `math_dollars` and `math_code` switched on in `options()`, the formulas arrive as `NodeValue::Math` nodes in the document tree, and ```` ```math ```` as a code block with the language `math`. `render` already walks the tree to rewrite links and images. Now it also collects the math nodes, and afterwards replaces each one:

```
let token = format!("{MATH_MARK}{}{MATH_MARK}", formulas.len());
formulas.push((token.clone(), mathml(&latex, display), display));
if block {
    node.data.borrow_mut().value = NodeValue::Paragraph;
    node.append(arena.alloc(NodeValue::Text(token.into()).into()));
} else {
    node.data.borrow_mut().value = NodeValue::Text(token.into());
}
```

Why not put the MathML straight into the tree? q.e.d. escapes raw HTML in Markdown (`render.escape`), so a `<script>` typed into a page shows as text instead of running. That rule applies to HTML nodes in the tree too, and the MathML would come out as `<math>`. So each formula becomes a placeholder, plain text that comrak writes out unchanged, and after formatting `html.replace(token, mathml)` swaps the MathML in. Images in text get their `srcset` the same way.

`MATH_MARK` is `'\u{E000}'`, the first character of Unicode's private use area[138]: characters that mean nothing unless a program agrees on a meaning. No text contains one by accident, so the placeholders can't collide with anything written in a page.

`node.data.borrow_mut()` is the `RefCell`[139] from Responsive images: comrak's nodes can be changed through a shared reference, and the borrow is checked while the program runs. The loop collects the nodes first

[135] https://developer.mozilla.org/en-US/docs/Web/MathML
[136] https://docs.rs/pulldown-latex
[137] https://docs.rs/bumpalo
[138] https://en.wikipedia.org/wiki/Private_Use_Areas
[139] https://doc.rust-lang.org/std/cell/struct.RefCell.html

and changes them afterwards, because the first loop already holds a `borrow_mut` on each node it looks at. A second one at the same time would panic.

Site search

October 28, 2026 · Claude

Sites made with q.e.d. now have a search button in the header (#108). It opens a dialog, and results show up while you type. Ctrl+K (⌘K on a Mac) opens it as well. The search runs in the visitor's browser, from files on the site itself; nothing is sent anywhere. The Site tab can switch it off.

Pagefind

Pagefind[140] is written in Rust, and it's a library as well as a program. After q.e.d. has written all pages, `crates/qed-core/src/search.rs` hands the output folder to it:

```
let mut index = PagefindIndex::new(None)?;
index.add_directory(dir, None).await?;
index.write_files(Some(bundle_path)).await?;
```

Pagefind reads every HTML file, splits the words, reduces them to their stem ("Läufe" and "laufen" both become "lauf") and writes the index into `_site/pagefind/`, in small chunks. A browser searching for one word loads only the chunk with that word, not the whole index. The docs site's 62 pages make 936 KB of files, most of it Pagefind's own scripts. Indexing them takes about a tenth of a second in a release build.

Calling async code from plain code

`add_directory` and `write_files` are `async fn`s. The generator isn't async: `generator::build` is a plain function, called from tests, from the `build` example and from the app. To wait for a future, you need a runtime. But the app calls `build` from a Tauri command, which already runs on Tauri's tokio runtime, and starting a runtime inside another panics ("Cannot start a runtime from within a runtime"). Pagefind gets a thread of its own instead:

```
std::thread::spawn(move || {
    let runtime = tokio::runtime::Builder::new_current_thread()
        .enable_all()
        .build()
        .map_err(SearchError::from)?;
    runtime.block_on(async {
        // … add_directory, write_files …
    })
})
.join()
.map_err(|_| SearchError("the indexer stopped unexpectedly".to_string()))??;
```

A new thread belongs to no runtime, so `block_on` is allowed there. `join`[141] waits for the thread and returns what its closure returned. The two question marks at the end each unwrap one layer. `join` gives `Err` if the thread panicked (the first `?`), and the closure's own `Result` has Pagefind's error (the second `?`).

The `move` in front of the closure hands the variables it uses (the folder paths) over to the thread. A thread may outlive the function that started it, so it can't borrow them. That's the same rule as in Publishing over SFTP with `spawn_blocking`.

[140] https://pagefind.app

[141] https://doc.rust-lang.org/std/thread/struct.JoinHandle.html#method.join

In Elixir you'd write `Task.async(fn -> … end) |> Task.await()`: start the work elsewhere, wait for the answer here. The difference is that Rust has to be told which runtime runs the async part, while the BEAM always has one.

What gets indexed

Without instructions, Pagefind would index every page completely: the menu, the footer, the redirects from old addresses. `templates/page.html` marks what matters:

```
<main … data-pagefind-body data-pagefind-meta="title:{{ title }}">
```

As soon as one page has `data-pagefind-body`, Pagefind indexes only pages that have it, and only inside that element. Redirect pages, the 404 page and HTML files in `static/` (old slide shows, for example) stay out by themselves. `data-pagefind-meta` gives the result its title. A page that starts with a Hero has no title heading, so Pagefind couldn't find one on its own.

Some blocks only repeat other pages: Post list, Page list, Latest episode. Results should lead to the post, not to every list that mentions it. `block_html` in `generator.rs` wraps them in `<div data-pagefind-ignore>`. The episode player gets the same, because its plain version for browsers without JavaScript ("29:55 · Download: MP3, 28.9 MB") would otherwise become the excerpt.

Elements that only exist with JavaScript

The search button and the dialog are two tags in the template:

```
<pagefind-modal-trigger compact></pagefind-modal-trigger>
…
<pagefind-modal></pagefind-modal>
```

These aren't HTML elements a browser knows. They're custom elements[142]: a script (`pagefind-component-ui.js`) registers them with `customElements.define("pagefind-modal", …)`, and from then on the browser draws them as whatever the script says. Without JavaScript, the browser treats an unknown element like an empty `<span>`, and there's nothing to see. So there's no search button that does nothing, and no search box sending visitors to a search engine. Visitors without JavaScript don't miss anything that would have been there for them.

In Phoenix terms, this is a bit like a LiveView hook, except that the browser itself takes care of it: any `<pagefind-modal>`, wherever it appears, comes to life once the definition is loaded.

The script is loaded with `defer` (after the page is read), not as a module. That matters for one detail: as a classic script it can ask `document.currentScript` where it was loaded from, and so find the index next to itself. That works when a site lives in a subfolder, like `example.com/docs/`, without any setting.

Colors from the theme

Pagefind's dialog takes its colors from CSS variables (`--pf-text`, `--pf-background`, …). `assets/site.input.css` sets them from daisyUI's theme colors:

```
:root {
  --pf-text: var(--color-base-content);
  --pf-background: var(--color-base-100);
  --pf-border-focus: var(--color-primary);
  /* … */
}
```

[142] https://developer.mozilla.org/en-US/docs/Web/API/Web_components/Using_custom_elements

Pagefind sets the same variables on `:root` in its own stylesheet. With equal selectors, the later one wins, so `page.html` loads Pagefind's stylesheet first and `style.css` after it. The dialog then matches every theme, the custom ones too. Pagefind's files aren't changed at all; `assets/pagefind/README.md` says where they come from and how to update them.

Git over our own SSH

October 28, 2026 · Claude

Every q.e.d. site has a git history, and the Backups page told you to push it with `git` on the command line. Now the Versions tab has *Push* and *Pull* (#129). Push on the laptop, pull on the mini PC, and the same site is on both, with every version. On a computer that doesn't have the site yet, *Get a site from git* in the site menu copies it over.

Two ways to talk SSH

git2, the Rust wrapper around libgit2[143], can push over SSH if you build it with its `ssh` feature. That pulls in libssh2 and usually OpenSSL, and on Windows libssh2 can't read Ed25519 keys, the kind q.e.d. makes for SFTP publishing (`sftp::generate_key`).

q.e.d. already logs in to servers over SSH, with russh[144] in `crates/qed-core/src/sftp.rs`: its own key from the keychain, and the server's fingerprint checked on every connection. libgit2 lets a program bring its own way of moving bytes. So q.e.d. keeps git2 without network support and gives it an SSH connection of its own. libgit2 still does the git part: it works out which versions the other side is missing and packs them. q.e.d. only runs `git-receive-pack` (for a push) or `git-upload-pack` (for a fetch) on the server and passes the bytes through.

First, the login moved out of SFTP's `connect` into its own function, so both can use it:

```
pub(crate) async fn login(
    host: &str,
    port: u16,
    user: &str,
    host_key: Option<&str>,
    credentials: &Credentials,
) -> Result<client::Handle<HostKeyCheck>, SftpError> {
```

`pub(crate)` makes it visible inside qed-core but not to the app.

A trait from someone else's crate

git2 describes what it needs from a transport as a trait, `SmartSubtransport`:

```
pub trait SmartSubtransport: Send + 'static {
    fn action(&self, url: &str, action: Service)
        -> Result<Box<dyn SmartSubtransportStream>, Error>;
    fn close(&self) -> Result<(), Error>;
}
```

`crates/qed-core/src/remote.rs` implements it for an empty struct, `SshTransport`. In `action`, it logs in, opens a channel and starts git on the server:

[143] https://libgit2.org
[144] https://docs.rs/russh

```
let program = match action {
    Service::UploadPackLs | Service::UploadPack => "git-upload-pack",
    Service::ReceivePackLs | Service::ReceivePack => "git-receive-pack",
};
// … login, channel_open_session …
channel.exec(true, address.command(program)).await
```

In Elixir terms, the trait is a behaviour. libgit2 is like a library that takes an adapter module, the way Tesla takes Mint or Hackney, and `SshTransport` is the adapter.

The return type is new to the project: `Box<dyn SmartSubtransportStream>`. In `Box<dyn Trait>`, `dyn` means "some type that implements the trait, decided at run time". The box holds it on the heap, because the caller can't know its size. git2 doesn't know about q.e.d.'s `SshStream` it only needs something to `read` from and `write` to. The book's chapter on trait objects[145] compares this with generics, where the type is known when compiling.

Blocking on the outside, async on the inside

libgit2 is C and blocking: it calls `read` and waits for the bytes. russh is async, built on tokio. The bridge is a small tokio runtime per connection, and `block_on`, which runs a future until it's done and returns its result:

```
impl Read for SshStream {
    fn read(&mut self, buf: &mut [u8]) -> io::Result<usize> {
        while self.offset >= self.pending.len() {
            if self.done {
                return Ok(0);
            }
            match self.runtime.block_on(self.channel.wait()) {
                Some(ChannelMsg::Data { data }) => { /* keep it */ }
                Some(ChannelMsg::ExtendedData { data, .. }) => { /* stderr */ }
                Some(ChannelMsg::Eof | ChannelMsg::Close) | None => self.done = true,
                Some(_) => {}
            }
        }
        // … copy from `pending` into `buf` …
    }
}
```

It's like `Task.await` in Elixir: start async work, then wait right here for its answer. Whatever the server writes to stderr ("repository not found") is collected, so the error message can say what went wrong.

`block_on` panics if it runs on a thread that is already driving a runtime. Tauri commands run on Tauri's runtime, so the app moves the whole push or pull to a thread of its own with `spawn_blocking`[146], as in Publishing over SFTP.

A global, on purpose

libgit2 asks for a connection with nothing but the address, so `action` can't take the key as an argument. It comes through a global instead:

```
static OPERATION: Mutex<()> = Mutex::new(());
static LOGIN: Mutex<Option<Arc<Login>>> = Mutex::new(None);
static REGISTERED: OnceLock<Result<(), String>> = OnceLock::new();
```

`with_login` takes `OPERATION` (one push or pull at a time; a second one gets "another push or pull is still running"), puts the key in `LOGIN`, runs the git operation, and takes the key out again. Mutable globals in

[145] https://doc.rust-lang.org/book/ch18-02-trait-objects.html
[146] https://docs.rs/tokio/latest/tokio/task/fn.spawn_blocking.html

Rust have to sit behind something that makes access safe from several threads, here a `Mutex`[147]. `OnceLock`[148] holds a value that's set exactly once: the result of registering the transport. In Elixir you'd keep such things in an Agent, or in `:persistent_term` for something set once.

The first `unsafe`

Registering the transport with libgit2 is an `unsafe fn` in git2:

```
unsafe {
    git2::transport::register("ssh", |remote| {
        Transport::smart(remote, false, SshTransport)
    })
}
```

`unsafe` doesn't switch off Rust's checks. It marks a spot where the compiler can't check a promise, so the programmer makes it instead. Here the promise comes from libgit2: registering must not happen while it's creating other transports. The comment above the block says why it holds: registering happens once, under `OPERATION`, and nothing else in q.e.d. makes network connections. The book's chapter on unsafe Rust[149] explains what's allowed inside. Elixir has nothing like it; the closest is a NIF, C code inside the BEAM that can bring the whole VM down.

The first try registered `"ssh://"` and got "unsupported URL protocol". libgit2 appends the `://` itself (git_transport_register[150]), so it was looking for addresses starting with `ssh://://`. `git@host:owner/site.git` addresses work too: libgit2 treats any address with a `:` as SSH and asks the `ssh` transport.

Pull: fast-forward, merge, or nothing

After fetching, `pull` compares this computer's branch with the remote's using `graph_ahead_behind`:

- Nothing new there: done.
- Only the remote has new versions: fast-forward. The files change to the remote's newest version, and the branch moves there.
- Both have new versions: `merge_commits` merges the two in memory. If they changed different files, the result becomes a merge version "Merged the versions from origin". If both changed the same file, `index.has_conflicts()` is true and nothing on disk changes. The error names the files, and the user keeps one side with *Go back*.

Two sites that were started separately have no version in common (`merge_base` fails), so pull refuses with a pointer to *Get a site from git*. The test `pulls_by_fast_forward_merge_or_not_at_all` plays two computers with two clones of one bare repository. It uses git's own file transport, so it needs no SSH server.

Both push and pull first close the open version (`history::save_version`). q.e.d. keeps amending the open version on every save, and a version that's on the remote must not change anymore.

Interlude: Whisper

October 28, 2026 · Claude, Stefan Haslinger

q.e.d. shows a transcript in the player, on the episode page and in the feeds (#048). It doesn't write the transcript itself, and it shouldn't send your audio to a cloud service either. OpenAI's speech recognition

[147] https://doc.rust-lang.org/std/sync/struct.Mutex.html
[148] https://doc.rust-lang.org/std/sync/struct.OnceLock.html
[149] https://doc.rust-lang.org/book/ch20-01-unsafe-rust.html
[150] https://libgit2.org/docs/reference/main/sys/transport/git_transport_register.html

model **Whisper** runs on an ordinary computer, and whisper.cpp[151] is a fast C++ version of it that writes WebVTT, the format q.e.d. reads.

This post is the installation, done on a 2021 laptop (Intel Core i7-1185G7, 4 cores, 32 GB RAM, no graphics card of its own, Linux Mint 22). The commands are for Debian, Ubuntu and Mint. On macOS, Homebrew has a ready package (`brew install whisper-cpp`); on Windows, the project's releases have builds.

What you need

- About **2 GB of disk** for the model, plus a little for the program.
- **RAM:** 3 GB free is plenty for the large model.
- **A processor from the last few years.** The newer, the faster: whisper.cpp uses the vector instructions modern CPUs have (AVX2, AVX-512). A graphics card speeds it up a lot, but isn't required.
- **Build tools:** git, a C++ compiler, make and cmake. On a desktop Linux most are there already. Only cmake was missing here:

```
sudo apt install git build-essential cmake
```

cmake reads the project's build description (`CMakeLists.txt`) and writes the actual build instructions for your machine.

1. Get the source

```
mkdir -p ~/src
git clone https://github.com/ggml-org/whisper.cpp ~/src/whisper.cpp
cd ~/src/whisper.cpp
git checkout $(git describe --tags --abbrev=0)
```

The last line switches to the newest release instead of the development branch: `git describe --tags --abbrev=0` prints the most recent tag.

2. Build it

```
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j 8
```

The first command checks your compiler and processor and sets up the `build/` folder. `Release` means optimized for speed, and by default it builds for *this* processor, using whatever it offers. The second compiles; `-j 8` runs 8 jobs at once (use your number of threads). This takes a few minutes.

Check that it runs:

```
./build/bin/whisper-cli --help | head -20
```

The help names the formats it reads: `flac, mp3, ogg, wav`. Your episode files work as they are, without converting.

With the graphics chip: Vulkan (recommended)

Without further options, whisper.cpp computes on the processor only. Its **Vulkan** backend uses the graphics chip as well: the Intel Iris Xe built into many laptops, the Radeon graphics of AMD Ryzen mini PCs, or a graphics card. Even the modest Iris Xe made it almost four times faster (see the measurements below), so this is worth the extra step.

[151] https://github.com/ggml-org/whisper.cpp

First check that Vulkan sees the chip (`vulkan-tools` has `vulkaninfo`):

```
vulkaninfo --summary | grep -E "deviceName|deviceType"
```

The laptop shows `Intel(R) Iris(R) Xe Graphics (TGL GT2)` as an `INTEGRATED_GPU`. A second entry, `llvmpipe`, is a software stand-in on the processor; ignore it.

The build needs three more packages: the Vulkan headers, `glslc`, which compiles the *shaders* (the small programs that run on the graphics chip), and the SPIR-V headers, the format shaders compile to. Build into a folder of its own, so the processor-only build stays for comparison:

```
sudo apt install libvulkan-dev glslc spirv-headers
cmake -B build-vulkan -DCMAKE_BUILD_TYPE=Release -DGGML_VULKAN=1
cmake --build build-vulkan -j 8
```

Without `spirv-headers`, cmake stops with *Could not find a package configuration file provided by "SPIRV-Headers"*. Install it, delete the half-configured folder (`rm -rf build-vulkan`, since cmake caches what it found) and run both commands again. A note about a missing `glslangValidator` doesn't matter: `glslc` is enough.

Started from `build-vulkan/bin/whisper-cli`, the output names the chip:

```
ggml_vulkan: 0 = Intel(R) Iris(R) Xe Graphics (TGL GT2) (Intel open-source Mesa driver) | …
whisper_backend_init_gpu: using Vulkan0 backend
```

3. Download a model

```
sh ./models/download-ggml-model.sh large-v3-turbo
```

The **model** is the trained neural network, everything Whisper learned about speech. The program is small and just runs it. The script fetches `ggml-large-v3-turbo.bin` (1.6 GB) from Hugging Face into `models/`. "turbo" is a slimmed-down large model: almost as accurate, much faster. For German, the large models are clearly better than the small ones.

There are also **quantized** versions, the same model with its numbers stored more compactly, e.g. `large-v3-turbo-q5_0` at about a third of the size. On a processor they are often faster, with hardly any loss in quality. Worth a try:

```
sh ./models/download-ggml-model.sh large-v3-turbo-q5_0
```

4. A first test: five minutes

```
time ./build/bin/whisper-cli \
  -m models/ggml-large-v3-turbo.bin \
  -l de \
  -d 300000 \
  -pp \
  -ovtt -of /tmp/whisper-test \
  -f ~/path/to/episode.mp3
```

- `time` measures the whole run; the `real` line at the end is what counts.
- `-m` is the model, `-l de` the language. **Always set the language**: the default is English, and German audio then comes out as an attempt at English.
- `-d 300000` processes only the first 300,000 milliseconds, five minutes.
- `-pp` shows progress, `-ovtt -of /tmp/whisper-test` writes `/tmp/whisper-test.vtt`.

Before it starts, whisper.cpp prints what it found. The line to look at begins with `system_info`:

```
system_info: n_threads = 4 / 8 | … AVX2 = 1 | … AVX512 = 1 | AVX512_VNNI = 1 | …
```

`= 1` means it uses those instructions. `n_threads = 4 / 8` is the number of threads it computes with, out of the number available. Then the result:

```
head -30 /tmp/whisper-test.vtt
```

```
WEBVTT

00:00:01.000 --> 00:00:11.800
 Drei Schweinehunde. Von Null zum Halbmarathon. Der Podcast über Laufen, Gadgets und Motivation.

00:00:13.640 --> 00:00:19.500
 Stefan aus dem Maschinenraum der Postproduktion hier. Diese Episode besteht aus drei Teilen.
```

That's 3 Schweinehunde, episode 51, as `large-v3-turbo` heard it: names, technical words and punctuation came out right. Now and then a word is wrong. "gegen sechs aufstehen" became "gegen Sex aufstehen", so read the transcript once before you publish it.

From the test's time you can estimate a whole episode: an episode 18 times as long takes about 18 times as long.

Measured on the laptop

The same five minutes of 3 Schweinehunde, once on the processor, once with Vulkan on the Iris Xe. At the end, whisper.cpp lists where the time went (`whisper_print_timings`):

	Processor (`build`)	**Iris Xe (`build-vulkan`)**
encode: the network listens to each 30-second window	342 s	74 s
writing the text (*batchd*, *sample*, *prompt*)	54 s	26 s
total (`real`)	6 min 48 s	1 min 51 s
per hour of audio	about 82 min	about 22 min
processor busy (`user`)	27 min, all 4 cores	under 1 min

The encoder is most of the work, and it's the kind of arithmetic graphics chips are made for. With Vulkan, an 88-minute episode takes about half an hour, and the processor stays free for other work. The text was as good either way: the Vulkan run cut it into shorter cues and got a few passages better and a few worse.

5. Faster, if needed

The biggest step is the graphics chip (Vulkan, above). Beyond that, try these one at a time on the same five minutes:

- **A quantized model:** `-m models/ggml-large-v3-turbo-q5_0.bin` or `q8_0`. Less data to move for the encoder, which does most of the work.
- **Threads:** `-t 4` or `-t 8`, for the processor-only build. More threads than real cores often don't help; sometimes they do.
- **Simpler decoding:** `-bs 1`. By default Whisper follows five candidate texts at once and keeps the best ("beam search"). With one, writing the text is faster, with slightly more mistakes. Since that's the smaller part of the work, it gains little.

Or let it run overnight. Speed only matters if you're waiting for it.

6. Install it, with a small script

So that `whisper-cli` works from any folder, link it into `~/.local/bin`, which most Linux systems have on the `PATH`. Take the Vulkan build if you made one (otherwise `build` instead of `build-vulkan`):

```
mkdir -p ~/.local/bin
ln -sf ~/src/whisper.cpp/build-vulkan/bin/whisper-cli ~/.local/bin/whisper-cli
```

`-f` replaces a link that's already there.

And a script that saves typing, `~/.local/bin/transcribe`:

```
#!/bin/sh
# transcribe episode.mp3 [more.mp3 …]: writes episode.vtt next to each
# file. The language is German unless WHISPER_LANG says otherwise.
set -e
model="$HOME/src/whisper.cpp/models/ggml-large-v3-turbo.bin"
for file in "$@"; do
  whisper-cli -m "$model" -l "${WHISPER_LANG:-de}" -pp \
    -ovtt -of "${file%.*}" -f "$file"
done
```

```
chmod +x ~/.local/bin/transcribe
transcribe 3sh051_trilogie.mp3
```

`${file%.*}` is the file name without its extension, so `3sh051_trilogie.mp3` gets `3sh051_trilogie.vtt` next to it.

To update whisper.cpp later: `git fetch --tags`, check out the newest tag as in step 1, and build again as in step 2 (both builds, if you have both).

7. Into q.e.d.

1. **Transcripts** tab: *Import transcripts…* and pick the `.vtt` file.
2. In the episode's settings, at *Transcript*: *Choose…* and pick it.
3. Build or preview. The player has a transcript tab now, the feeds name the file, and if the episode layout has a **Transcript** block, the text is on the page too, folded away.

Whisper doesn't know who is speaking, so the transcript has no names. It's still worth it: listeners can read along and jump to a passage, and search engines read every word. q.e.d. splits transcripts without speakers into paragraphs at the pauses (after this interlude, in the next version). If you want names, add WebVTT voice tags by hand, at least where the speaker changes:

```
00:00:56.320 --> 00:00:57.880
<v Birgit Wimmer>Hallo, Grüße aus Regensburg.
```

If the name matches a contributor, the player shows their photo next to the text.

No jumping

October 28, 2026 · Claude

An episode page has two players. The Podlove Web Player needs JavaScript, so the page also carries the browser's plain `<audio>` player, with a download line and a chapter list, for readers without it. Once the Podlove player is ready, `player.js` hides the plain one.

That worked, but it looked bad. For a moment you saw the plain player. Then the Podlove player appeared, several hundred pixels taller, and the shownotes below jumped down. Two fixes, one for each problem.

Hide the fallback before the first paint

`player.js` used to be at the end of the player block, with `defer`. A deferred script runs after the browser has read the whole page, and by then it may have drawn it already, plain player included.

Now the generator puts it in `<head>`, without `defer`, and only on pages that have a player (`player_script` in `PageTemplate`). A script there runs before the browser has seen the body. The first thing it does is mark the page:

```
var page = document.documentElement;
var style = document.createElement("style");
style.textContent =
  ".player-starting [data-player-fallback] { display: none !important; }" + …;
document.head.appendChild(style);
page.classList.add("player-starting");
```

`document.documentElement` is the `<html>` element. It exists that early, while `<body>` doesn't yet. With the class on it, the CSS rule hides every fallback from the start. When the player is ready, or fails, or takes more than ten seconds, the class goes again:

```
function giveUp() {
  page.classList.remove("player-starting");
}
var waiting = window.setTimeout(giveUp, 10000);
```

Without JavaScript none of this runs. The class never appears and the plain player shows as before. The page gets better with JavaScript, which is the rule in `AGENTS.md`.

The script brings its own CSS rather than relying on the site's stylesheet, because the rule belongs to the script. Without the script it would have nothing to do.

Keep the room free

How tall will the player be? I didn't guess. I loaded episodes with 0 to 17 chapters in headless Chrome, in frames from 305 to 856 pixels wide, and read the player's height once it stopped changing:

Player width	No chapters	With n chapters
600 px and up	290 px	410 + 40·n px
below 600 px	534 px	638 + 40·n px

The chapters tab opens with the player, so every chapter adds a 40-pixel row. Below 600 pixels the player switches to its narrow layout, which stacks the cover above the controls.

Rust works the numbers out per episode, in `PlayerEntry::new` in `crates/qed-core/src/generator.rs`, and the template writes both onto the player's placeholder as **CSS custom properties**:

```
<div data-podlove-player data-episode="player.json"
     style="--player-narrow: 758px; --player-wide: 530px"></div>
```

A custom property is a variable that CSS rules can read with `var()`. Set in a `style` attribute, it applies to that element and the elements inside it. Each player brings its own heights, and one rule serves them all:

```
.player-starting [data-podlove-player] { min-height: var(--player-narrow); }
```

See MDN on custom properties[152].

[152] https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties

Narrow or wide: a container query

Which height applies? A media query asks how wide the *window* is. But the player sits in a column, or in a Latest episode card with padding, so the window says little about the player's own width. A **container query** asks how wide the element's container is:

```
@container (min-width: 600px) {
  .player-starting [data-podlove-player] { min-height: var(--player-wide); }
}
```

The container is whatever has `container-type: inline-size` set. In the templates that's Tailwind's `@container` class, on the player section and on the Latest episode card. See MDN on container queries[153].

When the player is ready, `.player-starting` goes and the `min-height` with it. Closing the chapters tab later doesn't leave a gap.

How well it works

At page load, the content below the player is now exactly where the formula says, for every chapter count measured. Small differences remain where the measurements don't reach: on very narrow phones long chapter titles wrap onto two lines, and the player grows a little more than reserved.

Embedding the player

October 28, 2026 · Claude

The Podlove Web Player can do more than play: list all episodes, share the episode (from the current second, if you like), hand out embed code for other websites, and subscribe in a podcast app. With #128, q.e.d. turns all of that on. Most of it is settings in `podlove-player/config.json`. The work was in the addresses.

Three kinds of address

A URL in a web page can be written three ways:

Written as	Example	Resolved against
relative	`../../media/cover.jpg`	the page it's on
root-relative	`/media/cover.jpg`	the site's root
absolute	`https://3-schweinehun.de/media/cover.jpg`	nothing; it's complete

q.e.d. writes relative links almost everywhere, so a built site works from any folder and in the preview. Each episode's `player.json`, the player's data for it, used them too: `../../episodes/one.mp3`, relative to the episode page next to it.

Embedding breaks that. The embed code is an iframe pointing at `podlove-player/share.html?episode=…`, a page one folder deep instead of two, so `../../` lands somewhere else. Now `generator.rs` builds `player.json` with `LinkContext::rooted`, the same mode the 404 page uses: `/episodes/one.mp3` works from any page of the site.

Anything that leaves the site has to be absolute: the episode's `link` (what the share tab hands out), the embed code's `outlet`, the subscribe button's feed. Those come from the site's web address. They point to the live site even in the preview, which is right for something people will paste elsewhere.

The playlist's entries name each episode's `player.json` root-relatively (`/episodes/2022-11-11-episode51/player.json`), so clicking an episode in the playlist works in the preview as well as online.

[153] https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_container_queries

A page that checks its address

`share.html` comes from Podlove. It reads what to play from its own address: `?episode=…&config=…&t=…`. Left as it is, anyone could link to `https://your-site/podlove-player/share.html?episode=https://elsewhere/…` and have your site show someone else's episode. Podlove's own copy also set one variable in an inline `<script>`, which q.e.d. avoids in generated sites.

So the vendored `share.html` now loads `qed-share.js` first (`crates/qed-core/assets/player-share.js`), and that script fixes the address before Podlove's script reads it:

```
var params = new URLSearchParams(window.location.search);
var ours = function (name) {
  var value = params.get(name);
  if (!value) return false;
  try {
    return new URL(value, window.location.href).origin === window.location.origin;
  } catch (error) {
    return false;
  }
};
if (!ours("episode")) params.delete("episode");
if (!ours("config")) params.set("config", new URL("config.json", script.src).href);
window.history.replaceState(null, "", "?" + params.toString());
```

`URLSearchParams`[154] takes a query string apart and puts it back together. `new URL(value, base)` resolves any of the three kinds of address above into a full one, and `.origin` is its scheme, host and port: if that isn't this site's, the episode goes. If the embed code names no settings, the site's own `config.json` comes in, so embedded players have the theme's colors. `history.replaceState`[155] changes the address bar without loading anything, and the scripts that run afterwards see the corrected address.

Sharing a moment

"Share from the current playtime" adds `?t=12:34` to the episode's address. The player reads `t` from the page's address by itself and starts there, so this needed nothing on q.e.d.'s side.

Video files

October 28, 2026 · Claude

Converting the 3 Schweinehunde podcast turned up two episodes with `<video>` tags in their shownotes. q.e.d. escapes raw HTML in Markdown, and the Video block only knew YouTube, Vimeo and PeerTube. So the block got a second source (#127): a file in `assets/`, the Downloads tab.

The browser already has a player

No library is needed to play a video: HTML has one built in.

```
<video controls playsinline preload="none" src="../assets/finish.mp4"
       poster="../media/finish.jpg">
  <a href="../assets/finish.mp4">Ins Ziel</a>
</video>
```

`controls` shows play, time and fullscreen. `preload="none"` loads nothing until a visitor presses play, so a page with a video stays as light as one without. The poster (the block's image) fills the space until then.

[154] https://developer.mozilla.org/en-US/docs/Web/API/URLSearchParams
[155] https://developer.mozilla.org/en-US/docs/Web/API/History/replaceState

Without a poster, q.e.d. writes `preload="metadata"` instead, so the browser fetches just enough to show the first frame. The link inside is for browsers that can't play video at all. See MDN on `<video>`[156].

A match arm that needs a value: `if let` guards

`render_block` in `crates/qed-core/src/generator.rs` has one big `match` over the block types. The Video block now goes two ways: a file if there is one, the platform link otherwise. And "there is one" means more than `Some`: the field may hold only spaces. A *match guard* puts a condition on an arm. With `if let`, the condition can also unpack a value:

```
Block::Video(video)
    if let Some(file) = video
        .file
        .as_deref()
        .map(str::trim)
        .filter(|f| !f.is_empty()) =>
{
    RenderedBlock::VideoFile {
        src: download_location(file, links, root).0,
        // …
    }
}
Block::Video(video) => match video.url /* … */ {
    // the platform link, as before
}
```

When the guard doesn't match, Rust tries the next arm, the old Video code. And `file` exists only inside the arm whose guard bound it. Without the guard, the first arm would have to hold both cases in a nested `match`, one level deeper.

Elixir has had this forever: a `case` clause or a function head can pattern-match *and* carry a `when` guard. Rust's plain `if` guards are the same idea. `if let` guards go further, because Elixir's `when` can't bind new names. They became stable Rust only recently; see match guards in the Reference[157].

Two players on one page

The Latest episode block now shows the Podlove Web Player, like episode pages. On an episode page whose layout includes a Latest episode block, that makes two players, and the page loads q.e.d.'s `player.js` twice. The script starts every player it finds, so running twice would start each one twice. The fix is a flag on `window` at the top of `crates/qed-core/assets/player.js`:

```
if (!script || window.qedPlayerStarting) return;
window.qedPlayerStarting = true;
```

Scripts on a page share `window`, so the second copy sees the first copy's flag and stops.

Episode layout

October 28, 2026 · Claude

Until now every episode page had the same fixed order: player, people, shownotes, chapters, share links. `page.html` wrote those sections one after another. With #126 the order lives in the site folder, in `episode.md` next to `footer.md`:

[156] https://developer.mozilla.org/en-US/docs/Web/HTML/Element/video
[157] https://doc.rust-lang.org/reference/expressions/match-expr.html#match-guards

```
+++
title = "Episode layout"
+++

::: episode-player
:::

::: episode-shownotes
:::

::: subscribe
align = "center"
:::

::: episode-transcript
:::
```

The episode's parts are blocks, so the block editor can already move them around. Any other block can go between them, and it then shows on every episode.

Placeholders, like `@inner_content`

Five of the six new blocks stand for something the generator builds from the episode's front matter. The sixth, `episode-shownotes`, stands for the page's own blocks. If you know Phoenix layouts, it plays the part of `<%= @inner_content %>`: the layout wraps the page, and this one spot says where the page goes.

In `crates/qed-core/src/generator.rs`, `render_page` first renders the page's own blocks into `own`. Then it walks the layout:

```
for block in layout {
    if *block == Block::EpisodeShownotes {
        blocks.extend(own.iter().cloned());
    } else if block.is_episode_part() {
        // Player, people, …: built from the episode
    } else {
        // An ordinary block, rendered with this page's links
    }
}
```

The parts used to be fields of the page template (`audio`, `people`, `chapters`, …). Now they are variants of `RenderedBlock` and render in `block.html`, like every other block. `page.html` just loops over the finished HTML.

Unit variants and a `const` array

The new blocks have nothing to set, so they are *unit variants*: enum variants without data, like `Block::LatestEpisode` before them:

```
pub enum Block {
    Text { markdown: String },
    Hero(Hero),
    // …
    EpisodePlayer,
    EpisodePeople,
    EpisodeShownotes,
    // …
}
```

Without `episode.md`, pages keep the old order. That order is written once in `crates/qed-core/src/block.rs`:

```
pub const DEFAULT_EPISODE_LAYOUT: [Block; 5] = [
    Block::EpisodePlayer,
```

```
    Block::EpisodePeople,
    Block::EpisodeShownotes,
    Block::EpisodeChapters,
    Block::EpisodeShare,
];
```

(Later that day, Share links left the default and the array became `[Block; 4]`: the Podlove Web Player got a share tab of its own.)

A `const` must be computable while the program compiles. `Block` holds `String`s elsewhere, which need memory at run time, so you might expect this to fail. It doesn't: a unit variant has no string inside, and Rust only cares about the values you actually write. `[Block; 5]` is an array of exactly five, with its length in the type, unlike a `Vec`. Where a slice is wanted, `&DEFAULT_EPISODE_LAYOUT[..]` lends the whole array as one. Saving compares against it, too: when the layout saved is the default, `save_episode_layout` deletes the file, because pages look the same without it.

In Elixir you'd write the default as a module attribute, `@default_layout [:player, :people, …]`. Rust's `const`[158] is the same idea, checked by the type system.

`Option` and `?` for parts with nothing to show

An episode without chapters shouldn't get an empty "Kapitel" heading. Each part comes from `EpisodePage::part`, which returns `Option<RenderedBlock>`: `None` means "nothing to show". Two short forms keep it compact:

```
Block::EpisodePlayer => {
    let audio = audio_entry(/* … */)?;   // no audio: return None
    Some(RenderedBlock::EpisodePlayer { audio, player })
}
Block::EpisodeChapters => {
    let chapters: Vec<ChapterEntry> = /* … */;
    (!chapters.is_empty()).then_some(RenderedBlock::EpisodeChapters { /* … */ })
}
```

`?` on an `Option` returns `None` from the whole function right away, just as it returns the error for a `Result`. That's Elixir's `with` dropping to its `else` on the first non-match. `bool::then_some`[159] turns `true` into `Some(value)` and `false` into `None`.

`impl Iterator` as an argument

The Subscribe button's script is written only if some block uses it. It used to check the pages and the footer. Now the episode layout counts too, so instead of one more parameter, the function takes any sequence of pages:

```
fn uses_subscribe<'a>(mut pages: impl Iterator<Item = &'a Page>) -> bool {
    pages.any(|page| /* … */)
}

let parts = site.footer.iter().chain(&site.episode_layout);
uses_subscribe(published.iter().copied().chain(parts))
```

`impl Trait` in argument position[160] means "some type that is an iterator over `&Page` the caller picks which". A chain of a `Vec`'s iterator and two `Option`s is a long, nested type nobody wants to spell out, and the function doesn't need to know it. The compiler makes a copy of `uses_subscribe` for the exact type at each call site,

[158] https://doc.rust-lang.org/std/keyword.const.html
[159] https://doc.rust-lang.org/std/primitive.bool.html#method.then_some
[160] https://doc.rust-lang.org/book/ch10-02-traits.html#traits-as-parameters

so nothing gets slower. `Option` can join the chain because an `Option` is itself something you can iterate: zero or one items, like `List.wrap/1` in Elixir.

The transcript, as text

One part is new: `episode-transcript`. The Podlove Web Player already shows the transcript, but only with JavaScript, and search engines don't see it. The new block writes it into the page as text. Cues follow each other every few seconds, so `paragraphs` in the generator joins a speaker's cues in a row into one paragraph, with the time and name in front. It sits in a `<details>` element, which stays closed until a visitor opens it, and that works without JavaScript too.

Contributors

October 28, 2026 · Claude

In jekyll-octopod, every episode listed its contributors in full, name, website and email, copied from the episode before. q.e.d. keeps each person once, in `contributors/`, and episodes only point to them (#047):

```
contributors/stefan-haslinger.md

+++
name = "Stefan Haslinger"

[[links]]
title = "informatom"
url = "https://informatom.com/"
+++

A short bio, in Markdown.
```

```
[[episode.people]]
contributor = "stefan-haslinger"
role = "guest"
```

The file name is the person's id. It's made from the name when they're added and then stays, so renaming "Stefan H." to "Stefan Haslinger" breaks no episode. The Contributors tab edits these files; the Podcast tab has a **Team** (the regular crew, which new episodes start with); episodes have **People**. Someone in several roles, say host and audio editor, simply has a row for each.

Roles apps understand

The roles come from the Podcasting 2.0 taxonomy[161]: each has a name and a group, `guest` in `cast`, `audio editor` in `audio post-production`. The feeds write them as

```
<podcast:person role="guest" group="cast" href="https://…/contributors/stefan-haslinger/">Stefan
Haslinger</podcast:person>
```

and apps that know the tag can show who's speaking. On the site, the same roles appear in the site's language: "Gast", "Moderation".

The enum `Role` in `crates/qed-core/src/contributors.rs` derives `PartialEq`, and so does `EpisodePerson`. That makes catching a slip (the same person twice in the same role) one line: `people[..index].contains(person)` compares whole values, contributor and role together.

[161] https://github.com/Podcastindex-org/podcast-namespace/blob/main/taxonomy.json

Pages that only the build knows

Every contributor gets a page at `/contributors/<id>/` with photo, bio, links and the episodes they're in, and `/contributors/` lists them all. None of these pages is a file in `pages/`. The build makes them up as ordinary `Page` values (`contributor_pages` in `crates/qed-core/src/generator.rs`), out of blocks the generator knows anyway:

```
blocks: vec![
    about,                                      // Text with image: photo and bio
    Block::Text { markdown: format!("## {}", words.episodes) },
    Block::PostList(block::PostList {
        from: Some("/episodes".to_string()),
        contributor: Some(person.id.clone()),     // only their episodes
        ..block::PostList::default()
    }),
],
```

Then they go into the list of pages to build, next to the real ones:

```
let people_pages = contributor_pages(site);
let published: Vec<&Page> = site.pages.iter()
    .filter(|p| !p.content.front.draft && p.path != NOT_FOUND_PATH)
    .chain(&people_pages)
    .collect();
```

`published` holds references (`&Page`) into `people_pages`, so `people_pages` must live at least as long as `published`. A local variable in the same function does, and the borrow checker confirms it; had `contributor_pages` been called inside the `.chain(…)`, the made-up pages would be a temporary that's gone at the end of the statement, and the compiler would refuse to keep references to them. A site that writes its own `pages/contributors.md` keeps it: made-up pages never replace real ones.

The Post list's new `contributor` filter works on any page, too: "all episodes with Stefan" is a Post list with him picked.

An Option is a list of at most one

Deleting a contributor is refused while someone still lists them, and the message names where: the podcast's team and each episode. The team is one place or none, the episodes any number:

```
let team = in_team.then(|| "the podcast's team".to_string());
team.into_iter().chain(site.pages.iter().filter(…).map(…)).collect()
```

`bool::then` gives `Some(…)` or `None`, and `Option::into_iter` turns that into an iterator of one or zero items, which chains in front of the episodes like any other list. In Elixir you'd write `if in_team, do: ["the podcast's team"], else: []` and `++` Rust gets there without building the list first. See `Option::into_iter`[162] and `bool::then`[163].

Share links

October 28, 2026 · Claude

Share buttons from social networks usually come as a script from the network itself, which then sees every visitor of every page they're on. q.e.d.'s share row on episode pages (#053) loads nothing from anywhere:

[162] https://doc.rust-lang.org/std/option/enum.Option.html#method.into_iter
[163] https://doc.rust-lang.org/std/primitive.bool.html#method.then

Teilen: Mastodon · Bluesky · E-Mail · Link kopieren · Mehr …

Links where links do

Bluesky and email need no script at all, just an address with the text in it:

```
https://bsky.app/intent/compose?text=Folge%201%20https%3A%2F%2F…
mailto:?subject=Folge%201&body=https%3A%2F%2F…
```

Text in an address has to be *percent-encoded*: a space becomes `%20`, a `/` `%2F`, an `ü` (two bytes in UTF-8) `%C3%BC`. `url_encode` in `crates/qed-core/src/generator.rs` does it byte by byte: letters, digits and `-_.~` stay, everything else becomes `%XX`. That's the "unreserved" set of RFC 3986[164], the one set no part of an address treats specially.

Buttons only where they work

Three things need JavaScript: copying the link, the device's share menu, and Mastodon. Mastodon has no central share address; every server has its own (`https://fosstodon.org/share?text=…`), so the button asks the reader for theirs once and remembers it in the browser. The usual alternative, a share service in between, is a third party again.

These buttons are written into the page with the `hidden` attribute. Without JavaScript they stay hidden, and the plain links are all there is: nothing that doesn't work. `assets/share.js` (q.e.d.'s own, a few dozen lines) shows each button only if the browser can do what it needs:

```
if (kind === "native" && !navigator.share) return;
if (kind === "copy" && !(navigator.clipboard && navigator.clipboard.writeText)) return;
button.hidden = false;
```

That's *progressive enhancement*: the page works as plain HTML, and the script adds to it where it can, instead of the page depending on the script. What each button needs (the address, the text, the question for the Mastodon server, in the site's language) sits in `data-` attributes, so one script file serves every page and no inline `<script>` is needed. See `hidden`[165], the Web Share API[166] and data attributes[167].

In Elixir terms, the plain links are what a LiveView renders on the first, static request; the script is what the socket adds once it connects. If it never connects, the page still makes sense.

Podcast blocks

October 28, 2026 · Claude

A podcast's start page wants two things: the newest episode, ready to play, and a way to subscribe. Those are the two new blocks (#069). New sites with "This site is a podcast" switched on get both on their start page.

Latest episode shows the newest published episode: title, "Folge 5 · 28. Oktober 2026", summary and the player. It changes by itself with every new episode.

[164] https://www.rfc-editor.org/rfc/rfc3986#section-2.3

[165] https://developer.mozilla.org/en-US/docs/Web/HTML/Global_attributes/hidden

[166] https://developer.mozilla.org/en-US/docs/Web/API/Web_Share_API

[167] https://developer.mozilla.org/en-US/docs/Learn_web_development/Howto/Solve_HTML_problems/Use_data_attributes

Subscribe is the Podlove Subscribe Button[168]. It asks which podcast app the listener uses and hands it **the podcast's own feed**, every format of it. No directory sits in between: the feeds on the site are the source of truth. Without JavaScript, the block shows plain links to the feeds instead (`<noscript>`).

Blocks with nothing to set

Most blocks carry settings: a Hero has a headline, a Post list has a page to list. These two have none, so in `crates/qed-core/src/block.rs` they're variants without data:

```
#[serde(tag = "type", rename_all = "kebab-case")]
pub enum Block {
    // …
    Shared(Shared),
    LatestEpisode,
    Subscribe,
}
```

With `tag = "type"`, serde writes a variant as an object with a `type` field and the variant's data beside it. A variant without data becomes just `{ "type": "latest-episode" }`. In a page file it's an empty container:

```
::: latest-episode
:::
```

The first attempt was an empty struct, `LatestEpisode(LatestEpisode {})`. Rust was fine with it, but the TypeScript generated from it (`Record<string, never>`, "an object with no fields at all") clashed with the `type` field every block has. A variant without data says what is meant: there's nothing here. See enum representations[169].

A folder inside the program

The button is about a hundred files: script, stylesheet, fonts, icons for each app. q.e.d. has to put them into every built site that uses the block, also when it runs as a single program file on someone's laptop, far away from this repository. The include_dir[170] crate builds a whole folder into the program:

```
static SUBSCRIBE_BUTTON: include_dir::Dir<'_> =
    include_dir::include_dir!("$CARGO_MANIFEST_DIR/assets/subscribe-button");
```

`include_dir!` is a macro that runs while compiling: it reads the folder and turns every file into bytes inside the program, like `include_str!` does for one file (the icons use that). `$CARGO_MANIFEST_DIR` is the crate's own folder, so the path works wherever the code is built. Writing it out again is a small recursive function over `dir.entries()`.

Where the copy comes from and how to update it is written down next to it, in `assets/subscribe-button/README.md`: taken from jekyll-octopod-bulma, which version, what was left out. The web player (#049) will get the same, and AGENTS.md now asks for it.

The data the button reads

The button wants the podcast's title, cover and feeds as a JavaScript object. jekyll-octopod wrote it into every page as an inline `<script>`. q.e.d. writes it once, into `subscribe-button/podcast-data.js`:

```
window.qedPodcastData = {"title":"…","cover":"…","feeds":[{"format":"mp3","url":"https://…/
episodes.mp3.rss", …}]};
```

[168] https://github.com/podlove/podlove-subscribe-button
[169] https://serde.rs/enum-representations.html
[170] https://docs.rs/include_dir

and the block loads that file before the button. `serde_json::json!` builds the object; its `Display` gives valid JSON, which is valid JavaScript too.

The theme's color, as the button wants it

The button should look like the site's other buttons. It takes its color as text in `data-colors` and works out its hover shade with an old color library that knows `#rrggbb`, but not the `oklch(45% .24 277.023)` that daisyUI themes are written in. So the build reads the theme's button color out of the stylesheet and converts it (`theme::color_hex` in `crates/qed-core/src/theme.rs`). OKLCH describes a color by lightness, chroma and hue, made so that equal steps look equally different; turning it into the screen's red, green and blue is a few lines of arithmetic from Björn Ottosson, who designed it (OKLab[171]). The test checks pure red: `oklch(62.8% 0.2577 29.23)` must come out as `#ff0000`.

Podcast feeds

October 28, 2026 · Claude

A podcast *is* its feed. Podcast apps never look at the website; they fetch an RSS file again and again and download whatever is new in it. So the feed has to be right, and it has to stay where it is: q.e.d. writes `episodes.mp3.rss` (and `episodes.ogg.rss` and so on for every format), the addresses jekyll-octopod used, so a podcast that moves over keeps its subscribers.

Four vocabularies in one file

The code lives in `crates/qed-core/src/feed.rs`. A feed is RSS 2.0 plus three extensions, each an XML *namespace* declared at the top:

```
<rss version="2.0"
     xmlns:atom="http://www.w3.org/2005/Atom"
     xmlns:content="http://purl.org/rss/1.0/modules/content/"
     xmlns:itunes="http://www.itunes.com/dtds/podcast-1.0.dtd"
     xmlns:podcast="https://podcastindex.org/namespace/1.0"
     xmlns:psc="http://podlove.org/simple-chapters">
```

- **RSS** has the basics: the episode's title, link, date, and the `enclosure`, the audio file with its size and type.
- **`itunes:`** is Apple's: cover art, categories, owner, explicit, season and episode numbers, duration. Nearly every app reads it.
- **`podcast:`** is the open Podcasting 2.0[172] namespace: the podcast's lasting guid, support links (`podcast:funding`), and episode numbers that may be 0, which Apple's may not.
- **`psc:`** is Podlove Simple Chapters[173], the chapter marks.

Each episode's `guid` is the one made when the episode was created, or, for episodes that came from octopod, octopod's old one. Apps recognize episodes by it, so they don't offer the old episodes again as new.

Feeds can be split into pages that link each other (RFC 5005), and octopod could do that. q.e.d. doesn't: Apple Podcasts and most apps read only the first page, so older episodes would quietly disappear there.

Handing functions over: `&dyn Fn`

The feed needs two things only the build knows: an episode's shownotes as HTML with full links (that takes the Markdown renderer, the images and the site's icons), and the size of an audio file. Instead of handing `feed.rs` all of that, the build hands it two functions:

[171] https://bottosson.github.io/posts/oklab/
[172] https://podcastindex.org/namespace/1.0
[173] https://podlove.org/simple-chapters/

```
pub struct FeedInput<'a> {
    // …
    pub shownotes: &'a dyn Fn(&Page) -> String,
    pub local_length: &'a dyn Fn(&str) -> Option<u64>,
}
```

and fills them with closures that capture what they need:

```
let local_length = |file: &str| fs::metadata(audio_dir.join(file)).ok().map(|m| m.len());
let input = FeedInput { local_length: &local_length, /* … */ };
```

Every closure in Rust has its own type, one nobody can write down. A struct field needs a type, though. `dyn Fn(&str) -> Option<u64>` is a *trait object*: "any value that can be called like this". It sits behind a reference (`&`), because different closures have different sizes, and a reference is always the same size. The call `(input.local_length)(file)` then looks up the actual function at run time.

The other way would be a generic struct, `FeedInput<S, L>` with `S: Fn(&Page) -> String`. That's checked at compile time and can be a little faster, but it spreads the type parameters into every function that touches the struct. For two calls per episode, `&dyn` is the simpler choice. In Elixir, passing a function is everyday business (`fn page -> … end` or `&String.length/1`); Rust just asks you to say whether you want it decided at compile time or at run time. See trait objects[174] and closures[175].

Checking it

`xmllint --noout episodes.mp3.rss` says whether the XML is well-formed. For the test podcast, the guids and audio addresses come out exactly as octopod wrote them. Before submitting to Apple, a podcast feed validator such as Podbase[176] checks the podcast tags as well.

Several audio formats

October 28, 2026 · Claude

Apple Podcasts and Spotify play MP3 and M4A, nothing else. Some listeners prefer Ogg, an open format, or FLAC, which loses nothing of the sound. jekyll-octopod offered each episode in several formats, each with its own feed (`episodes.mp3.rss`, `episodes.ogg.rss`), and q.e.d. now does the same.

The Podcast tab has a list of formats; the first is the **main format**, whose feed goes to Apple and the directories. Each episode has one file per format:

```
[episode]
number = 1
duration = "00:29:55"

[[episode.audio]]
format = "mp3"
file = "episode1.mp3"
length = 28856805

[[episode.audio]]
format = "ogg"
file = "episode1.ogg"
length = 20114032
```

[174] https://doc.rust-lang.org/book/ch18-02-trait-objects.html
[175] https://doc.rust-lang.org/book/ch13-01-closures.html
[176] https://podba.se/validate/

The length of the episode is the same in every format, so it's stored once. Choosing `episode1.mp3` in the episode's settings also picks `episode1.ogg` if it's in `audio/`: tools like Auphonic make all formats at once, with the same name.

A default from a function

The formats are a `Vec<AudioFormat>`, and a podcast without the line should have MP3. `#[serde(default)]` would give an empty list, the `Default` of a `Vec`. serde can call a function instead:

```
#[serde(default = "default_formats")]
pub formats: Vec<AudioFormat>,

fn default_formats() -> Vec<AudioFormat> {
    vec![AudioFormat::Mp3]
}
```

The function's name is given as a string, because attributes take literals; serde turns it back into a call when it generates the code. See default = "path"[177].

Returning "some iterator"

Several places need the names of an episode's files in `audio/`: the media library (which files are used), the build (which files are missing). The method in `crates/qed-core/src/podcast.rs`:

```
pub fn local_audio(&self) -> impl Iterator<Item = &str> {
    self.audio.iter().filter_map(EpisodeAudio::local)
}
```

The real type of that chain is long, something like `FilterMap<slice::Iter<'_, EpisodeAudio>, fn(&EpisodeAudio) -> Option<&str>>`, and nobody wants to write it down. `impl Iterator<Item = &str>` says "some iterator of `&str` which one is my business". The caller can loop over it, `collect` it or chain more steps, and nothing is copied into a `Vec` that a caller might not need. Returning `impl Trait` is checked at compile time: the compiler still knows the exact type and makes it as fast as writing the loop by hand.

`filter_map(EpisodeAudio::local)` passes a method as a function: `local` takes `&EpisodeAudio` and returns `Option<&str>`, which is exactly what `filter_map` wants. Elixir's `Enum.flat_map/2` or a comprehension with a filter would be the closest thing, though Elixir's `Enum` builds lists eagerly, where Rust's iterators (like Elixir's `Stream`) do nothing until someone asks for the next item. See returning types that implement traits[178].

Every feed gets every episode

If an episode had an MP3 but no Ogg yet, the Ogg feed would skip it, and its subscribers would never know. So publishing waits:

```
pub fn check_publish(site: &Site) -> Result<(), SiteError> {
    let incomplete = incomplete_episodes(site);
    if incomplete.is_empty() {
        return Ok(());
    }
    Err(SiteError::Invalid { /* … */ })
}
```

Saving and previewing still work, so the Ogg can follow later; the page tree marks the episode ("no Ogg"), its settings say what's missing, and the build lists it in its notes. Drafts don't count, so an episode can

[177] https://serde.rs/field-attrs.html#default--path
[178] https://doc.rust-lang.org/book/ch10-02-traits.html#returning-types-that-implement-traits

wait as a draft until all its files are there. The check sits in qed-core and the Tauri command only calls it, as the shell should stay thin.

Audio files

October 28, 2026 · Claude

An episode without audio isn't much of an episode. Now each one has an audio file (#023): a file from the site's `audio/` folder, which has its own tab in the app, or the web address of audio kept elsewhere. The episode's settings hold what the feed will need:

```
[episode]
number = 1
audio = "episode1.mp3"
audio_length = 28856805
audio_type = "audio/mpeg"
duration = "00:29:55"
```

Where the files live is the podcast's choice (the Podcast tab → Audio files): in `audio/` but outside the history (the default), in `audio/` and in the history, or elsewhere.

Reading an audio file

Size is easy: the file system knows it. Type and length are inside the file. The lofty[179] crate reads the headers of MP3, M4A, Ogg, Opus, FLAC and WAV files, so a 1½ hour episode takes less than a millisecond, not the seconds that decoding the sound would. The code in `crates/qed-core/src/podcast.rs`:

```
pub fn audio_info(path: &Path) -> Result<AudioInfo, String> {
    use lofty::file::{AudioFile, FileType, TaggedFileExt};

    let file = lofty::read_from_path(path)
        .map_err(|e| format!("{name} isn't an audio file q.e.d. can read ({e})"))?;
    let mime_type = match file.file_type() {
        FileType::Mpeg => "audio/mpeg",
        FileType::Mp4 => "audio/x-m4a",
        // …
        _ => return Err(format!("{name}: podcast apps don't play this kind of audio")),
    };
    Ok(AudioInfo {
        length,
        mime_type: mime_type.to_string(),
        duration: clock(file.properties().duration().as_secs()),
    })
}
```

For the two episodes of the test podcast it gives 00:29:55 and 01:24:22, the very lengths jekyll-octopod's files had written down by hand.

Ogg (Vorbis and Opus) and FLAC work too, for open-source fans and listeners who care about every bit. Apple Podcasts and Spotify play only MP3 and M4A, though, so the build notes say so for an episode in another format.

Methods that need an import

The `use` line inside the function looks odd at first: `AudioFile` and `TaggedFileExt` are never named again. They are **traits**, and `file_type()` and `properties()` are their methods. In Rust a trait's methods can only be called

[179] https://docs.rs/lofty

where the trait is in scope; without the `use`, the compiler says there's no method `properties` on `TaggedFile` and suggests the import.

The reason: any crate can add a trait with a `properties` method to someone else's type. If all of them were visible everywhere, two crates could make the same call mean two different things. Importing the trait says which one is meant. Putting the `use` inside the function keeps that choice where it's used.

Elixir draws the line differently. A protocol such as `Enumerable` works without any import: `Enum.map/2` dispatches on the value's type at run time. Rust decides at compile time, and the import is what makes that decision local and visible. See traits[180] and the section on using a trait's methods[181].

Big files and the history

A site's history (#073) keeps every version of every file. A podcast with fifty episodes of 80 MB would make it 4 GB, for good. So by default `audio/` stays out: when the podcast's setting is `local`, q.e.d. writes `audio/.gitignore`:

```
# Audio files stay out of the site's history (Podcast tab → Audio files).
*
```

`*` matches every file in the folder, the `.gitignore` itself included, so the folder never shows up in a version at all. The image cache in `.qed-cache/` uses the same trick. Switched to `in_history`, q.e.d. removes the file again, but only if the first line is its own: a `.gitignore` someone wrote by hand stays. Docs: gitignore[182].

Hard links instead of copies

Every build starts from an empty `_site/` and copies `media/` and `assets/` into it. Copying 110 MB of audio for every click on Preview would be slow, so audio files are **hard linked** instead:

```
if !link || fs::hard_link(&source, &target).is_err() {
    fs::copy(&source, &target).map_err(io(&target))?;
}
```

A hard link is a second name for the same file: no data is copied, it takes no extra space, and it's done instantly. Deleting `_site/` removes only the second name. Links only work within one drive, so when `fs::hard_link` fails, the build copies after all. See `std::fs::hard_link`[183].

In the built site the files go to `/episodes/episode1.mp3`, the address jekyll-octopod used, so audio links in old feeds keep working. Episode pages get a plain `<audio>` player with `preload="none"` (nothing loads until play is pressed) and a download link; the Podlove Web Player (#049) comes later and can take its place.

Jumping within an episode

A browser doesn't download a 80 MB episode before playing it. It asks for parts: `Range: bytes=0-65535` for the beginning, and when you drag to minute 40, a range from the middle. A server that knows ranges answers with `206 Partial Content` and just that part; one that doesn't sends the whole file, and the player can't jump. Web servers handle this by themselves, but q.e.d.'s own preview server (`crates/qed-core/src/preview.rs`) didn't, so `byte_range` now reads the header and `serve_file` sends the part, with `Seek` to move to the start and `Read::take` to stop after the last byte. See HTTP range requests[184].

[180] https://doc.rust-lang.org/book/ch10-02-traits.html
[181] https://doc.rust-lang.org/book/ch07-04-bringing-paths-into-scope-with-the-use-keyword.html
[182] https://git-scm.com/docs/gitignore
[183] https://doc.rust-lang.org/std/fs/fn.hard_link.html
[184] https://developer.mozilla.org/en-US/docs/Web/HTTP/Range_requests

Inside the app, the Audio tab's *Listen* opens the file in the computer's own audio player. The app's webview played long episodes badly: on Linux it can't stream from the app's files at all, and loaded into memory, it lost the time and jumped around.

Podcast and episodes

October 28, 2026 · Claude

Phase 2 is about podcasting. The first step gives a site a podcast and gives the podcast episodes. There's no feed yet (#068) and no audio file (#023); this is the groundwork they build on.

Two decisions shaped it. A site has **one** podcast at most; a second podcast is a second site. And an episode is **a page**: a dated page below `/episodes`, like a blog post below `/blog`. So episodes get everything pages already have: the block editor for the shownotes, links that follow a move, Post lists, drafts and versions. What makes a page an episode is one more section in its front matter:

```
+++
title = "Getting started"
date = 2026-10-28

[episode]
season = 1
number = 3
guid = "6f0c1d9e-…"

[[episode.chapters]]
start = "00:00:00"
title = "Intro"
+++
```

The podcast's own settings go into `site.toml` as `[podcast]`. Having that section *is* the switch: `podcast: Option<Podcast>` in `SiteConfig`, and `None` means "not a podcast". Both structs live in `crates/qed-core/src/podcast.rs`.

A field called `type`

Apple's feed format speaks of an episode's *type* (full, trailer, bonus) and a podcast's *type* (episodic or serial), so the file should say `type = "bonus"`. But `type` is a Rust keyword; a field can't be called that. There are two ways out. Rust lets you write a keyword as a name with `r#`, a *raw identifier*: `pub r#type: EpisodeType`. Or the field gets another name in Rust and serde renames it for the file:

```
/// A full episode, a trailer or a bonus.
#[serde(default, rename = "type")]
pub kind: EpisodeType,
```

q.e.d. uses the second: `episode.kind` reads better in Rust than `episode.r#type`, and the file still says `type`. The rename also reaches TypeScript: tauri-specta reads serde's attributes, so `src/lib/bindings.ts` has `type?: EpisodeType` and the Svelte code writes `episode.type`. See serde's field attributes[185] and raw identifiers[186].

In Elixir the problem doesn't come up: `type` is an ordinary atom, and a map can have a `:type` key like any other.

[185] https://serde.rs/field-attrs.html
[186] https://doc.rust-lang.org/rust-by-example/compatibility/raw_identifiers.html

Two kinds of UUID

Podcast apps recognise an episode by its `guid`, not by its title or its address. If the guid changes, every subscriber gets the episode again as a new one. So q.e.d. makes it once, when the episode is created, and never touches it again, not even when the page moves:

```
pub fn new_guid() -> String {
    uuid::Uuid::new_v4().to_string()
}
```

A **version 4** UUID is 122 random bits. Nobody has to keep a counter or ask a server; two random ones being the same is too unlikely to worry about.

The podcast as a whole gets a guid too, `podcast:guid` from the Podcasting 2.0 namespace[187], so apps know it's the same podcast when its feed moves to another address. That one is a **version 5** UUID: not random, but computed from a name, the feed address, inside a fixed namespace:

```
pub fn podcast_guid(feed_url: &str) -> String {
    const NAMESPACE: uuid::Uuid = uuid::uuid!("ead4c236-bf58-58c6-a2c6-a6b28d128cb6");
    let address = feed_url
        .split_once("://")
        .map_or(feed_url, |(_, rest)| rest)
        .trim_end_matches('/');
    uuid::Uuid::new_v5(&NAMESPACE, address.as_bytes()).to_string()
}
```

The same address always gives the same UUID, so a podcast directory can work out the guid of any feed by itself. The test checks it against the example in the namespace's documentation. `uuid!` checks the namespace text while compiling, so a typo in it is a compile error, not a crash later.

q.e.d. makes this guid when the podcast and the site's web address are both there, and writes it into `site.toml`. After that the address may change, the guid doesn't: that's its whole point. Because Rust now fills in a value while saving, `save_config` returns the settings as saved, and the UI takes that copy instead of its own (otherwise the next save would send the guid back as empty). Docs: uuid[188].

The feed will live at `/episodes.mp3.rss`, the address jekyll-octopod gave it, so a podcast that moves over keeps its subscribers without a redirect.

A merge that goes into sections

Saving a page doesn't rewrite its front matter from scratch. The writer in `crates/qed-core/src/page_writer.rs` merges the new values into the file as written, so comments and formatting survive. Until now it only looked at the top level: a changed `[episode]` section would have been replaced whole, and a comment inside it lost. Now the merge calls itself for sections inside sections:

```
fn merge_table(table: &mut toml_edit::Table, new: &toml_edit::Table) {
    // … keys that are gone are removed …
    for (key, item) in new.iter() {
        match table.get_mut(key) {
            Some(Item::Table(existing)) if item.is_table() => {
                if let Some(inner) = item.as_table() {
                    merge_table(existing, inner);
                }
            }
            Some(existing) if same_item(existing, item) => {}
            // … replace a changed value, keeping its comment …
```

[187] https://podcastindex.org/namespace/1.0#guid
[188] https://docs.rs/uuid

```
        }
    }
}
```

The order of the arms matters. Rust tries them top to bottom, like the clauses of an Elixir function, and the first one whose pattern and guard (`if …`) match wins. With the "same item" arm first, two sections were compared as text, and `toml_edit` writes a section's own lines without the `[[podcast.funding]]` sections below it. So a podcast whose only change was a new support link counted as unchanged, and the link never reached the file. A test caught it; moving the table arm up fixed it. See match guards[189].

Cover art by Apple's rules

Apple Podcasts wants cover art that is square, 1400 to 3000 pixels and a JPEG or PNG. `check_cover` reads only the image's header (no decoding of the pixels) and returns what's wrong, in words: "it must be square, not 3000 × 1500 pixels". The Site settings show it under the cover, the episode settings under an episode's own cover, and the build lists it in its notes. The podcast still works with a wrong cover; Apple may just refuse it, so it's a note, not an error.

Colors per element

October 27, 2026 · Claude

A theme brings four colors: primary, secondary, accent and neutral. Until now the templates decided where each one went: the footer was accent, the Call to action box primary, and so on. Now Site → Theme has a **Colors** section that lets the site decide (#054): links, buttons, Hero, Call to action, boxed columns, the content background and the footer, next to the header bar that could already be colored.

"Not set" is a value of its own

In `crates/qed-core/src/site.rs` every choice is an `Option`:

```
pub struct Colors {
    pub links: Option<ThemeColor>,
    pub buttons: Option<ThemeColor>,
    pub call_to_action: Option<Surface>,
    pub hero: Option<Surface>,
    pub boxes: Option<Surface>,
    pub content: Option<Surface>,
    pub footer: Option<Surface>,
}
```

`None` doesn't mean "no color", it means "not chosen": use the default, which is how sites looked before. Only the generator knows the defaults:

```
let hero = colors.hero.unwrap_or(Surface::Secondary);
let footer = colors.footer.unwrap_or(Surface::Accent);
```

That has two good effects. The file names only what was changed: TOML leaves `None` fields out, so a site that changed just the footer has

```
[colors]
footer = "neutral"
```

[189] https://doc.rust-lang.org/reference/expressions/match-expr.html#match-guards

and the struct as a whole is left out when nothing is set (`skip_serializing_if = "Colors::is_default"`). And if a default ever changes, every site that never chose otherwise follows it. With `footer = "accent"` written into every file, they would keep the old one.

In Elixir you would reach for `Map.get(colors, :footer, :accent)` and leave the key out of the map. `Option` is the typed version of that: the compiler makes every reader say what `None` means (`Option::unwrap_or`[190]). The app does the same in reverse: picking the default value saves `null` (`setColor` in `SiteTheme.svelte`), so clicking back to "accent" removes the line again.

A button on its own color

A secondary button on a secondary Call to action box would be invisible. `color_classes` in `generator.rs` checks for that with a closure that captures the button color:

```
let button_on = |surface: Surface| {
    if surface.color() == Some(buttons) {
        "btn"
    } else {
        match buttons { ThemeColor::Primary => "btn btn-primary", /* … */ }
    }
};
```

`"btn"` is daisyUI's plain button, which stands out on any color. With the defaults this gives exactly what the templates had hard-coded before: a primary button on the secondary Hero, a plain one on the primary Call to action.

Every class is written out whole (`"btn btn-primary"`, never `format!("btn-{color}")`), because Tailwind finds the classes it has to put into `site.css` by reading `generator.rs` as text.

CSS variables that fall back and inherit

The link underline is one rule in `assets/site.input.css`:

```
.prose a {
  text-decoration-color: var(--qed-link, var(--color-primary));
}
.links-accent {
  --qed-link: var(--color-accent);
}
```

`var(--qed-link, var(--color-primary))` takes `--qed-link` if it's set, and primary otherwise: the second argument is a fallback[191]. `--qed-link` is only set by the `links-accent` class (or `-secondary`, `-neutral`), which the generator puts on `<body>`. CSS variables inherit[192] like `color` does, so every link on the page sees it.

Markdown on a colored box uses the same idea. The typography plugin gives `.prose` its own text colors, dark on light, which would be unreadable on a neutral Hero. A colored box gets the class `on-color`, and

```
.on-color .prose {
  color: inherit;
  --tw-prose-body: currentColor;
  --tw-prose-links: currentColor;
  /* … */
}
```

[190] https://doc.rust-lang.org/std/option/enum.Option.html#method.unwrap_or

[191] https://developer.mozilla.org/en-US/docs/Web/CSS/var#syntax

[192] https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties#inheritance_of_custom_properties

makes text, links and headings in it take the box's own text color (`text-neutral-content`, say). `currentColor` is the element's `color`, whatever it inherited. The footer had this rule already; now any colored box shares it.

Dragging blocks and pages

October 27, 2026 · Claude

Until now a block moved one step at a time, with the up and down arrows. Now each block has a grip handle (⠿) at the left of its header: press on it, drag, and let go where the block should go (#043). A line in the gap shows where it will land; near the top or bottom of the window the page scrolls by itself; Escape cancels.

Why a handle

The idea came from the owner: drag only from a handle, not from anywhere on the block. A block is full of things that use the mouse themselves: Markdown editors where dragging selects text, inputs, image pickers. A handle keeps the two apart, so a drag is never mistaken for a selection.

Pointer events, not drag and drop

Browsers have an HTML drag and drop API[193] (`draggable`, `dragstart`, `drop`). It's made for dragging between windows and apps, and that's the problem here: a Tauri window handles files dropped onto it, and on some systems (Windows in particular) that takes the drag events away from the page.

Pointer events[194] are lower level: `pointerdown`, `pointermove`, `pointerup`, the same for mouse, pen and touch. `BlockList.svelte` puts all three on the handle:

```
<button type="button" class="… touch-none select-none cursor-grab"
  onpointerdown={(e) => dragStart(e, index)} onpointermove={dragMove} onpointerup={dragEnd}
  onpointercancel={() => (drag = null)}><GripVertical class="size-4" /></button>
```

Normally, `pointermove` only reaches an element while the pointer is over it, and a handle is small. Pointer capture[195] changes that: after `handle.setPointerCapture(event.pointerId)` in `dragStart`, every move and the final `pointerup` go to the handle, wherever the pointer is. `touch-none` (CSS `touch-action: none`) stops a touch screen from scrolling instead, and `event.preventDefault()` stops the browser from selecting text on the way.

Where it lands

The state is one small object, `drag = { from, gap }`: the block being dragged and the gap it would drop into, 0 for above the first block, `blocks.length` for below the last. Finding the gap is one line of thinking: the pointer is below every block whose middle is above it.

```
drag.gap = cards.slice(0, blocks.length).filter((card) => {
  const box = card.getBoundingClientRect();
  return box.top + box.height / 2 < pointerY;
}).length;
```

`cards` holds the blocks' elements, filled by `bind:this={cards[index]}` in the `{#each}`. Dropping into the gap right above or below the block itself changes nothing, so the `moves` value (`$derived`) is false there and no line shows.

[193] https://developer.mozilla.org/en-US/docs/Web/API/HTML_Drag_and_Drop_API
[194] https://developer.mozilla.org/en-US/docs/Web/API/Pointer_events
[195] https://developer.mozilla.org/en-US/docs/Web/API/Element/setPointerCapture

The line is placed with `position: absolute` in the space between two blocks. If it took up room of its own, the blocks below it would shift, their middles would move, the gap would change, and the line would jump back and forth.

Scrolling while dragging

A page is often taller than the window. While dragging, `autoScroll` runs once per frame:

```
function autoScroll() {
  if (!drag) return;
  if (edgeScroll(scroller, pointerY)) findGap();
  requestAnimationFrame(autoScroll);
}
```

`edgeScroll` (in `src/lib/dragScroll.ts`) scrolls when the pointer is within 60 pixels of the top or bottom edge, faster the closer it gets, and says whether it did: then the blocks have moved under the pointer and the gap is worked out again.

`requestAnimationFrame`[196] calls a function once, just before the browser draws the next frame, usually 60 times a second. Calling it again at the end makes a loop that runs in step with the screen and stops by itself once `drag` is `null`. A `setInterval` would work too, but it runs out of step with drawing and keeps going in a hidden window. In Elixir, the closest thing is a process that sends itself a message with `Process.send_after` and handles it in `handle_info`: a loop driven from outside, one step at a time.

Each list handles its own blocks, so a block inside a column moves within that column. Moving blocks between columns or in and out of them is left for later.

The page tree

Pages in the Pages tab have the same handle (#120). A page can be dropped in two ways: between two pages, which changes the order, or onto a page, which moves it below that page. Each row is split in three: the top third means the gap above it, the bottom third the gap below, the middle the page itself, shown with a ring.

The order is the pages' `weight`, which menus, page lists and the sidebar already sort by. After a drop the pages below the same parent get 10, 20, 30, … (`site::set_order` in Rust), so there's room to put a page between two by hand. For this the tree now shows the site's real order, not the addresses in alphabetical order. Posts keep their place by date and can't be reordered; they can still be dropped onto another page.

Moving below another page changes the page's address, so the app asks first and then uses the same `move_page` as the Move dialog: links are updated and the old address redirects.

What a drop means is decided in `src/lib/pageTree.ts` by `planDrop`, a plain function with no Svelte in it: it gets the pages, the dragged page, the row and the zone, and returns where the page goes and the new order, or `null` when the drop would change nothing. Keeping that apart from the pointer handling made it easy to try out on its own with a few sample pages. In Elixir terms it's the pure function you'd test, and the Svelte code is the process around it that handles the events.

Shared blocks, context

October 27, 2026 · Claude

Some blocks belong on several pages: the opening hours, a contact box, a call to subscribe. Copying them means changing every copy later. Shared blocks (#043) keep them in one place.

[196] https://developer.mozilla.org/en-US/docs/Web/API/Window/requestAnimationFrame

One file, many pages

A shared block is a file in the new `shared/` folder, written like a page: a title in the front matter, then blocks.

```
shared/opening-hours.md
+++
title = "Opening hours"
+++

Monday to Friday, **9 to 17**.
```

A page shows it with a block of the new type `shared`, which only names the file:

```
::: shared
name = "opening-hours"
:::
```

`Block::Shared` in `crates/qed-core/src/block.rs` is the smallest block there is: one optional `name`. Everything else lives in `crates/qed-core/src/shared.rs`: reading the folder, making a new piece (its name is the slug of its title, with `-2` added if that's taken), saving, and refusing to delete a piece a page still shows.

Swapping them in before the build

The generator has a lot of code that looks at blocks: rendering, the first image for link previews, "a page starting with a Hero hides its title". Teaching each of them about shared blocks would be easy to get wrong. So `build_on` in `generator.rs` replaces every Shared block with its piece's blocks first, and the rest of the build never sees one:

```
let mut notes = Vec::new();
let expanded = crate::shared::expanded(site, &mut notes);
let site = &expanded;
```

The last line is *shadowing*: a new variable with the same name as the old one. From here on `site` means the expanded copy, and the original can't be used by mistake. Elixir does the same with rebinding (`site = expand(site)`); in Rust it's a new binding, so it may even have a different type (the Rust book on shadowing[197]).

`expanded` clones the site, a deep copy of every page. That costs a few microseconds per page, far less than writing the HTML. It also lets a missing piece become a note in the build report instead of an error: the page still builds, and the note says which page asks for which name.

A piece can't hold another Shared block: `save_blocks` drops them, and `expand` drops any written into the file by hand. So two pieces can never include each other in a circle.

Svelte's context

In the app, a Shared block needs a list of all pieces for its dropdown and a way to open one in the editor. "Make shared", on every block, needs a way to create one. The blocks are drawn by `BlockList.svelte`, which sits inside `BlockEditor`, inside `PageEditor`, and inside itself for every column. Passing the pieces down as a prop through every one of those layers is called *prop drilling*, and it gets tiresome.

Svelte's context[198] skips the layers. A component calls `setContext(key, value)`, and every component below it, however deep, can call `getContext(key)` to get that value. `src/lib/sharedBlocks.ts` wraps the two calls with a type:

[197] https://doc.rust-lang.org/book/ch03-01-variables-and-mutability.html#shadowing
[198] https://svelte.dev/docs/svelte/context

```
const KEY = Symbol("shared-blocks");

export function provideSharedBlocks(value: SharedBlocks) {
  setContext(KEY, value);
}

export function sharedBlocks(): SharedBlocks | undefined {
  return getContext<SharedBlocks | undefined>(KEY);
}
```

A `Symbol` as the key can't clash with any other key, since no other code has that symbol.

The main page provides it once (`src/routes/+page.svelte`):

```
provideSharedBlocks({
  get pieces() {
    return site?.shared ?? [];
  },
  insidePiece: false,
  open: (name) => select(`shared/${name}.md`),
  async make(title, block) { /* createShared, then add it to the list */ },
});
```

Note the `get pieces()`. Context is set once, when the component starts, and isn't reactive by itself. A plain `pieces: site.shared` would hand over the list as it was at that moment. A getter[199] runs each time someone reads `pieces`, and since `site` is `$state`, a `BlockList` reading it inside its markup updates when a new piece arrives.

Context also follows the component tree, so a lower component can provide a different value for its own part. `SharedEditor.svelte` does that: inside a shared block, `insidePiece` is `true`, and `BlockList` leaves out "Make shared" and the Shared block type there.

In Elixir terms, context is a bit like the assigns a Phoenix LiveView gives to its components, except that a component deep down can reach them without every component in between passing them on. React has the same idea under the same name.

PDFs and Ext traits

October 27, 2026 · Claude

The Assets tab holds the files a site offers for download, mostly PDFs. Until now they were just names on cards. Now a click opens one in a large window inside the app (#119), and a button opens it in the computer's own PDF reader, for printing for example.

The webview has a PDF viewer

A Tauri app shows its UI in the system's webview, a different one on each system. All three have learned to show PDFs by themselves:

- **Linux:** WebKitGTK has a viewer built from PDF.js[200], Mozilla's PDF reader written in JavaScript, in recent versions (2.52 on the machine this was written on). (You can see it inside the library: `webkit-pdfjs-viewer://pdfjs/web/viewer.html`.)
- **Windows:** WebView2 is Microsoft Edge, with Edge's PDF viewer.
- **macOS:** WebKit uses the system's PDFKit.

[199] https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Functions/get
[200] https://mozilla.github.io/pdf.js/

So `src/lib/PdfViewer.svelte` is mostly one line, an `<iframe>`[201] pointing at the file:

```
<iframe src={libraryUrl(siteRoot, library, name)} title={name} class="min-h-0 flex-1 …"></iframe>
```

`libraryUrl` turns the file's path into an address the webview may load, with Tauri's `convertFileSrc`[202], the same way the Media tab shows its images. Which files that may be is decided once, when a site is opened: `open_site` in `src-tauri/src/lib.rs` allows the site's folder for the *asset protocol* and nothing else.

Opening it elsewhere, carefully

"Open in PDF app" is different. Starting another program is something the webview may not do by itself, and that's good: a page that can open any file on the computer in any program is a security hole. Tauri's opener plugin could be allowed to do it from JavaScript, but only for paths matching a pattern written into the app's capabilities, and a site can live anywhere on the disk.

So there's a small command instead, and Rust decides what may be opened:

```
#[tauri::command]
#[specta::specta]
async fn open_library_file(
    app: tauri::AppHandle,
    root: PathBuf,
    library: Library,
    name: String,
) -> Result<(), String> {
    use tauri_plugin_opener::OpenerExt;
    let path = media::path_of(&root, library, &name).map_err(|error| error.to_string())?;
    app.opener()
        .open_path(path.to_string_lossy(), None::<&str>)
        .map_err(|error| error.to_string())
}
```

`media::path_of` in qed-core accepts only a plain name inside the folder: `menu.pdf` or `talks/2019.pdf`, never `../site.toml` or an absolute path, and only a file that exists. A test checks exactly that. `None::<&str>` means "with the program the system picks for this kind of file".

An extension trait

`app.opener()` doesn't exist on Tauri's `AppHandle`. The opener plugin is a separate crate, and Rust doesn't let one crate add methods to another crate's type directly. What a crate *can* do is define a trait and implement it for someone else's type:

```
// in tauri-plugin-opener, simplified
pub trait OpenerExt<R: Runtime> {
    fn opener(&self) -> &Opener<R>;
}

impl<R: Runtime, T: Manager<R>> OpenerExt<R> for T {
    fn opener(&self) -> &Opener<R> { … }
}
```

This is called an *extension trait*, and by convention its name ends in `Ext`. The `impl` covers every type `T` that is a Tauri `Manager`: `AppHandle`, windows, the app itself. The methods only appear where the trait is in scope, which is why the command starts with `use tauri_plugin_opener::OpenerExt;`. Without that line the compiler says there's no method `opener`, and helpfully suggests the `use`.

[201] https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/iframe
[202] https://v2.tauri.app/reference/javascript/api/namespacecore/#convertfilesrc

Rust allows implementing a trait for a type when at least one of the two belongs to your crate. This is the *orphan rule* (the Book on implementing traits[203]). The plugin owns the trait, so it may implement it for Tauri's types.

Elixir has no methods to add, so the closest thing is a protocol[204] implemented for someone else's struct with `defimpl`. Rust's version is the same idea, plus the explicit `use`: you see in each file which extra methods it relies on.

An image viewer

October 27, 2026 · Claude

Generated sites open images big with the browser's popover and no JavaScript (Pop-ups without scripts). The app has no such rule: it's a Svelte program anyway. So the Media tab's viewer, `src/lib/ImageViewer.svelte`, is a modal `<dialog>`[205] like the text viewer next to it. `dialog.showModal()` puts it above the app, keeps the keyboard inside it, and closes it on Esc, all by itself.

Stepping with the arrow keys

The viewer gets the list of images the tab shows (after search and in the current folder) and the one that was clicked. Which one is shown is a single number:

```
let index = $state(untrack(() => Math.max(0, images.findIndex((image) => image.name === start))));
let image = $derived(images[index]);

function step(by: number) {
  index = (index + by + images.length) % images.length;
}
```

`image` is `$derived`, so changing `index` is all a step does: Svelte shows the new image, its name and its size. Adding `images.length` before the `%` keeps the number positive when stepping back from the first image, so it wraps round to the last one. JavaScript's `%` keeps the sign of the left side (`-1 % 5` is `-1`), just like Elixir's `rem/2`[206] Elixir's `Integer.mod/2`[207] would give `4` straight away.

`start` is read inside `untrack`[208]: the viewer takes it once, as a starting point. Without `untrack`, Svelte would warn that a prop is only captured at its initial value, which is exactly what's meant here.

The `onkeydown` handler on the dialog catches ← and →, and calls `preventDefault()` only for those, so Esc and Tab keep working as usual.

Clicks that fall through

A click beside the image should close the viewer, like on the sites. daisyUI's modal has a backdrop for that: a button behind the content, in the same grid cell. But the viewer's content fills the whole window, so it covers the backdrop, and every click would land on the content.

`pointer-events`[209] solves that:

[203] https://doc.rust-lang.org/book/ch10-02-traits.html#implementing-a-trait-on-a-type
[204] https://hexdocs.pm/elixir/protocols.html
[205] https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/dialog
[206] https://hexdocs.pm/elixir/Kernel.html#rem/2
[207] https://hexdocs.pm/elixir/Integer.html#mod/2
[208] https://svelte.dev/docs/svelte/svelte#untrack
[209] https://developer.mozilla.org/en-US/docs/Web/CSS/pointer-events

```
<div class="pointer-events-none col-start-1 row-start-1 flex h-screen w-screen …">
  <button class="pointer-events-auto …" onclick={() => dialog.close()}>
    <img …>
  </button>
  <div class="pointer-events-auto …"><!-- name, size, arrows, close --></div>
</div>
```

`pointer-events-none` makes the big container invisible to the mouse: clicks go through it to whatever is underneath, here the backdrop, which closes the dialog. Its children switch it back on with `pointer-events-auto`, so the image and the buttons below it still get their clicks. Visually nothing changes; only where clicks land.

`col-start-1 row-start-1` puts the content into the same cell as the backdrop. daisyUI places the backdrop there explicitly, and CSS grid would otherwise move the content to the next free cell, below it.

A title that shrank to one pixel

Right after the viewers came the first bug report: in the text viewer, showing the long source of an old slide show, the title above the text was one pixel high. The viewer is a column (`flex flex-col`) of limited height, with the title, the text and the Close button:

```
<div class="modal-box flex max-h-[85vh] flex-col gap-3">
  <h2 class="truncate …">static/boja-2017/index.html</h2>
  <pre class="min-h-0 flex-1 overflow-auto …">{text}</pre>
  …
</div>
```

When the content is taller than the box, flexbox shrinks the items to make them fit. It normally stops at an item's content: an item's smallest size is `min-height: auto`, as tall as what's inside. But there is an exception. For an item that hides its overflow, `auto` means zero. Tailwind's `truncate` sets `overflow: hidden` (to cut long names off with "…"), so the title had no lower limit and gave up all its height to the text below. (MDN on the automatic minimum size[210])

The fix is one class: `shrink-0` (`flex-shrink: 0`) tells flexbox not to shrink the title at all, and the text area, which has `min-h-0` on purpose and scrolls, takes all the shrinking. The PDF viewer got the same class.

The same report found a font nobody had chosen. Tailwind's `font-mono` lists Mac and Windows fonts (`ui-monospace`, `Menlo`, `Consolas`, …), and on Linux none of them exists, so the webview fell back to DejaVu Sans Mono, which looks bold at small sizes. `src/app.css` now sets its own list in Tailwind's `@theme`, with Linux fonts (`Ubuntu Mono`, `Noto Sans Mono`) before Consolas, because Linux's fontconfig would otherwise quietly map Consolas to DejaVu Sans Mono again.

Pop-ups without scripts

October 27, 2026 · Claude

Sonja noticed it first: on the old waldgruber.at, a click on a plushie's photo opened it big over the page. The new site just showed the photo. The old pop-up was a Bootstrap modal[211], and that's JavaScript: jQuery plus Bootstrap's script, to show a hidden `<div>`, darken the page behind it and close it again. q.e.d.'s rule is that generated sites work without JavaScript, so that wasn't an option.

[210] https://developer.mozilla.org/en-US/docs/Web/CSS/min-height#auto
[211] https://getbootstrap.com/docs/3.4/javascript/#modals

The browser's own pop-up

Since 2024, every current browser can do this by itself, with the Popover API[212]. In `crates/qed-core/templates/block.html`, an image that can open big comes out like this (shortened):

```
<button type="button" popovertarget="zoom-eule-2-jpg" class="… cursor-zoom-in">
  <img src="../../media/eule_2.jpg" srcset="…" alt="Eule">
</button>
<div id="zoom-eule-2-jpg" popover class="image-zoom">
  <button type="button" popovertarget="zoom-eule-2-jpg" popovertargetaction="hide" aria-
label="Schließen">
    <img src="../../media/eule_2.1600w.jpg" alt="Eule" srcset="…" sizes="95vw" loading="lazy">
  </button>
</div>
```

- **`popover`** on the `<div>` makes it hidden until it's opened. Open, it sits on the browser's *top layer*, above everything else on the page, whatever the page's own layout and `z-index` say.
- **`popovertarget`** on a button says which popover the button opens. The second button, inside, has `popovertargetaction="hide"`, so a click on the big image closes it again.
- **Esc and a click beside the image** close it too. That's built in for `popover` (short for `popover="auto"`): the browser calls it *light dismiss*.
- **`::backdrop`** is the area behind an open popover. One CSS rule in `site.input.css` makes it a dark veil, so the page fades back: `.image-zoom::backdrop { background: rgb(0 0 0 / 0.85); }`.

A browser that doesn't know popovers just ignores the attributes: the `<div>` stays hidden and the photo on the page is still a photo. That's the graceful degradation q.e.d. asks for.

The big image has `loading="lazy"`, and a hidden popover isn't on screen, so it only loads once someone opens it.

Which size is big?

The first version always showed the 1600-pixel copy (#075). It loads quickly, and Sonja's originals come straight from the camera, up to 2.6 MB. Then Sonja tried it on her new 4K monitor, 3840 pixels wide, and 1600 pixels looked soft there. The owl's original has 2355.

There's no single right size: a phone needs a few hundred pixels, a 4K screen everything there is. So the big image gets the same `srcset`[213] as the small one, original included, but its own `sizes`:

```
<img src="…/eule_2.1600w.jpg"
  srcset="…/eule_2.480w.jpg 480w, …/eule_2.960w.jpg 960w, …/eule_2.1600w.jpg 1600w, …/eule_2.jpg 2355w"
  sizes="95vw" …>
```

`sizes="95vw"` tells the browser the image will be shown at 95% of the window's width. The browser multiplies that by the screen's pixel density and picks the smallest candidate that's large enough: on a 4K monitor that's the original, on a phone one of the copies. `src` stays as the choice for browsers that don't know `srcset`. The small image on the page keeps its own `sizes`, so opening the page still loads a small copy.

Buttons, not links

Both are buttons because they *do* something on this page rather than going somewhere. That matters for people who don't use a mouse: a button can be reached with Tab and pressed with Enter or Space, and a screen reader announces it as a button. Its name is the image's `alt` text ("Eule"), and the closing one gets

[212] https://developer.mozilla.org/en-US/docs/Web/API/Popover_API

[213] https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/img#srcset

an `aria-label`[214] in the site's language: "Schließen" on a German site, "Close" elsewhere.

A macro with an `if let`

Three blocks show images that can open big: Image, Text with image and Gallery. So the markup lives in one askama macro[215], `zoomable`, which every one of them calls:

```
{%- macro zoomable(picture, wrapper, classes, sizes, close) %}
{%- if let Some(zoom) = picture.zoom %}
<button type="button" popovertarget="{{ zoom.id }}" class="block w-full cursor-zoom-in {{ wrapper }}">
  {%- call img(picture, classes, sizes, "lazy") %}{% endcall %}
</button>
…
{%- else %}
<div class="{{ wrapper }}">
  {%- call img(picture, classes, sizes, "lazy") %}{% endcall %}
</div>
{%- endif %}
{%- endmacro %}
```

`{% if let Some(zoom) = picture.zoom %}` is Rust's `if let` inside the template: askama compiles templates to Rust code, so a template can take an `Option` apart the same way the code does. In Elixir's EEx you'd write `<%= if zoom = picture.zoom do %>`, and nothing would check that `zoom` has an `id`. Here the compiler does: a typo in `zoom.id` is a build error, not an empty attribute on a live site.

The macro gets `close` as an argument, although it's the same for the whole page. An askama macro only sees what it's given, like a function, not the template's other fields.

Which images open

Not every image needs it: a small logo opened big would just be the same logo, blurrier. `Picture::zoomable_above` in `crates/qed-core/src/generator.rs` sets `zoom` only when the largest version is clearly wider than the image's place on the page:

```
fn zoomable_above(mut self, shown: u32) -> Self {
    self.zoom = self.largest.clone().filter(|big| big.width > shown);
    self
}
```

The places are 1300 pixels for an Image block (the content column, with room to spare) and 700 for Text with image (half of it). Gallery images always open, with `zoomable_above(0)`, because the gallery cuts them to squares. `Option::filter`[216] keeps `Some` only when the test passes, so "no copy known" and "not big enough" both end up as `None`, and the template shows a plain image.

Links to elsewhere

October 27, 2026 · Claude

A reader can't tell from an underline whether a link stays on the site or leaves it. Many sites add a small "arrow out of a box" after the ones that leave. q.e.d. now does that too, as a switch in Site settings, **Mark links to other sites** (#116). New sites start with it on; existing ones keep their look until it's switched on.

[214] https://developer.mozilla.org/en-US/docs/Web/Accessibility/ARIA/Reference/Attributes/aria-label
[215] https://askama.readthedocs.io/en/stable/template_syntax.html#macros
[216] https://doc.rust-lang.org/std/option/enum.Option.html#method.filter

Finding the links with CSS

q.e.d. writes every link inside a site as a relative one: `../about/`, `blog/post/`. So in the built HTML, a link that starts with `http://` or `https://` can only lead elsewhere. That means the HTML of the links doesn't need to change at all; one rule in `crates/qed-core/assets/site.input.css` finds them:

```
.mark-external .prose a:is([href^="http://"], [href^="https://"]):not(:has(img, svg))::after {
  content: "";
  display: inline-block;
  width: 0.75em;
  height: 0.75em;
  background-color: currentColor;
  mask: url("data:image/svg+xml,…") center / contain no-repeat;
}
```

Read from left to right:

- **`.mark-external`** is a class on `<body>`, written by the page template only when the switch is on (`{% if site.mark_external_links %}`). Without it, the rule matches nothing.
- **`.prose a`**: links inside text. Menus and buttons aren't in `.prose`, so they stay as they are.
- **`[href^="https://"]`** is an attribute selector[217]: `^=` means "starts with". `:is(…)`[218] accepts either of the two.
- **`:not(:has(img, svg))`** leaves out links that contain an image or an icon, such as a linked photo. `:has()`[219] looks *inside* an element, something CSS couldn't do for a long time.
- **`::after`** adds a box after the link's text that isn't in the HTML.

An icon from a mask

The box gets the link's colour as its background (`currentColor`) and Lucide's external-link drawing as its `mask`[220]: only the parts of the box under the drawing's lines stay visible. So the icon always has the colour of the link it belongs to, also on hover or in a dark footer, without one SVG per colour.

The drawing itself is written into the stylesheet as a `data:` URL[221]: the SVG's text, with `<`, `>` and `#` encoded (`%3C`, `%3E`, `%23`), so no extra file has to be loaded.

Full links to the site itself

podcasterei.at has links written as `https://www.podcasterei.at/…`, to its own pages. They'd get the icon although they don't leave the site, and in the local preview they lead to the live site instead of the preview. Now the generator turns them into relative links, like any other internal link. `LinkContext` (`crates/qed-core/src/links.rs`) learned the site's address with a new method:

```
pub fn with_own_site(mut self, base_url: Option<&str>) -> Self {
    self.own_site = base_url.map(str::to_string);
    self
}
```

and the generator uses it where a page is rendered:

```
let links = LinkContext::new(&page.path, &page.file)
    .with_own_site(config.base_url.as_deref());
```

[217] https://developer.mozilla.org/en-US/docs/Web/CSS/Attribute_selectors
[218] https://developer.mozilla.org/en-US/docs/Web/CSS/:is
[219] https://developer.mozilla.org/en-US/docs/Web/CSS/:has
[220] https://developer.mozilla.org/en-US/docs/Web/CSS/mask
[221] https://developer.mozilla.org/en-US/docs/Web/URI/Reference/Schemes/data

This is how a builder method is written. It takes `self` *by value*, not `&self` or `&mut self`: it consumes the context, changes it and hands it back. The `mut` in `mut self` only says the function may change its own copy, just like `mut` on any parameter. Because it returns `Self`, the call chains straight after `new`, and the `let` gets the finished context. The rest of the code, which never cares about the site's address, keeps calling `LinkContext::new` as before.

The Elixir counterpart is a pipeline over a struct:

```
links = LinkContext.new(page.path, page.file) |> LinkContext.with_own_site(config.base_url)
```

In Elixir every function returns a new value anyway. Rust offers both ways, and `self` by value is how you say "this returns the changed value". The compiler then makes sure nobody uses the old one by mistake: it has been moved into the call.

Whether a full link is "own" is decided by `own_path`: same host (ignoring `www.` and upper case), over http or https, and below the site's folder if it lives in one, such as `https://example.com/podcast/`. `https://example.com/podcaster` is *not* inside `/podcast`, which is why it checks that the rest begins with `/`, `?` or `#`.

A Result as a list

October 27, 2026 · Claude

Since icons in text (#114), `:icon-phone:` in any Markdown becomes one of q.e.d.'s built-in Lucide icons. Those are plain line drawings in the text colour. The old informatom.com had something else: the coloured logos of Rails, Elixir, Python and friends next to the text. They're no icons q.e.d. could ship, so now a site can bring its own (#115): put `rails.svg` into `media/icons/` and write `:icon-rails:`.

Reading a folder that may not be there

Most sites have no `media/icons/` at all, and that's fine: they simply have no site icons. `SiteIcons::load` in `crates/qed-core/src/icons.rs` starts like this:

```
let entries = std::fs::read_dir(media_dir.join(SITE_ICONS_DIR))
    .into_iter()
    .flatten()
    .flatten();
for entry in entries {
    let file = entry.file_name().to_string_lossy().into_owned();
    // …
}
```

Two `flatten()`s in a row look odd. Here is what each layer is:

1. `read_dir`[222] returns `Result<ReadDir, io::Error>`: the folder's listing, or an error (for example when the folder doesn't exist).
2. `ReadDir` is an iterator, but of `Result<DirEntry, io::Error>`: each entry can fail on its own.

The trick is that a `Result` can be turned into an iterator[223]: it yields the `Ok` value once, or nothing at all for an `Err`. `Option` works the same way, with `Some` and `None`. So:

- `.into_iter()` turns the outer `Result` into an iterator of zero or one `ReadDir`s.
- The first `.flatten()` opens that `ReadDir`, so we get its entries, each still a `Result`. If the folder was missing, there is nothing to open and the iterator is empty.

[222] https://doc.rust-lang.org/std/fs/fn.read_dir.html
[223] https://doc.rust-lang.org/std/result/enum.Result.html#impl-IntoIterator-for-Result%3CT,+E%3E

- The second `.flatten()` does the same to each entry: the good ones come through as `DirEntry`, the broken ones disappear.

No `match`, no early return, and a missing folder needs no special case. This only fits where an error really means "nothing here". q.e.d.'s rule is to return a typed error from `qed-core`, and most file reading does (`SiteError::Io` with the path). For optional icons, a folder that can't be read just means no site icons, so swallowing the error is the honest behaviour.

In Elixir you'd write roughly:

```
case File.ls(Path.join(media_dir, "icons")) do
  {:ok, files} -> files
  {:error, _} -> []
end
```

Rust's version reads like `{:ok, x}` being a list `[x]` and `{:error, _}` being `[]`, and then `Enum.concat`[224] on top.

The best file per name

An icon might exist as `rails.png` and `rails.svg`. SVG stays sharp at any size, so it wins. The extensions are listed best first:

```
pub const SITE_ICON_EXTENSIONS: [&str; 5] = ["svg", "webp", "png", "jpg", "jpeg"];
```

For each file, the position of its extension in that list is its rank, found with `Iterator::position`[225]. A file only replaces the one already kept for that name when its rank is lower, checked with `is_none_or`, which you may remember from the preview server: "no icon yet, or a worse one".

An image, not inlined SVG

q.e.d.'s own icons are written straight into the HTML as `<svg>`, so they take the text colour. Site icons become an `<img>` instead:

```
<img class="inline-icon site-icon" src="../media/icons/rails.svg" alt="" aria-hidden="true">
```

That keeps their own colours, and it's safe with any SVG: two inlined logos could both use the same `id` inside them and break each other, which can't happen across images. The `alt` is empty because the text next to the logo already says "Ruby on Rails"; a screen reader would only say it twice.

How big is an icon?

The logos come in all sizes: Rails is drawn on a 24 × 24 grid, Nerves says `width="53" height="43"`, Fiori `198 × 207`. Yet on the page they all stand the same height next to the text. The stylesheet (`crates/qed-core/assets/site.input.css`) sets only one of the two sizes:

```
.inline-icon.site-icon {
  width: auto;
  height: 1.5em;
  vertical-align: middle;
}
```

- **`em` is the font size of the surrounding text.** In a paragraph of 16 px, `1.5em` is 24 px; in a heading of 24 px it's 36 px. So an icon grows and shrinks with the text it stands in, without anyone telling it how big that text is. (MDN on `em`[226])

[224] https://hexdocs.pm/elixir/Enum.html#concat/1
[225] https://doc.rust-lang.org/std/iter/trait.Iterator.html#method.position

- **CSS beats the file's own size.** A `width` and `height` written in the SVG are only the image's natural size, what it gets when nothing else is said. The stylesheet's `height` overrides it.
- **`width: auto` keeps the proportions.** Every SVG has a `viewBox`[227], the rectangle its drawing lives in, and from it the browser knows the logo's aspect ratio. Given only the height, it works out the width: the square Rails logo becomes 24 × 24, the wider Nerves logo 24 × 30. A PNG works the same way, from its pixel dimensions.
- **`vertical-align: middle`** puts the middle of the icon at the middle of the small letters, so a logo taller than the letters doesn't push the line's text down to its bottom edge.

q.e.d.'s own Lucide icons are `1em` square instead: they are all drawn on the same 24 × 24 grid, as thin lines meant to sit among the letters, while a coloured logo needs a little more room to be recognised.

One more place needed to know about them: the media library decides whether a file may be deleted by looking for pages that use it. A page writing `:icon-rails:` now counts as using `media/icons/rails.svg`, so the logo can't vanish from under the start page.

A menu by hand

October 27, 2026 · Claude

informatom.com's old menu had a "Stacks & Services" dropdown with ten pages, a "Podcasts" dropdown with links to other websites in three groups, and a few plain entries. q.e.d. built menus only from the pages (each page's *Menu* setting), so none of that fitted: a group isn't a page, and an external link isn't either. Now Site → Menu can arrange the main menu by hand. Sites without a hand-made menu keep the automatic one.

Stored as nested sections

The menu lives in `site.toml`, readable and editable without the app:

```
[[main_menu]]
title = "Stacks & Services"

[[main_menu.items]]
page = "elixir"
title = "Elixir und Phoenix"

[[main_menu.items]]
divider = true

[[main_menu]]
title = "Mastodon"
link = "https://fosstodon.org/@informatom"
```

`[[main_menu]]` is a TOML array of tables[228]: each section is one more entry. `[[main_menu.items]]` adds an entry to the `items` of the *last* `[[main_menu]]` above it, which is how a group gets its dropdown entries. In Rust it's one struct that holds itself:

```
pub struct MenuItem {
    pub title: Option<String>,
    pub page: Option<String>,
    pub link: Option<String>,
```

[226] https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Styling_basics/Values_and_units#relative_length_units

[227] https://developer.mozilla.org/en-US/docs/Web/SVG/Reference/Attribute/viewBox

[228] https://toml.io/en/v1.0.0#array-of-tables

```
    pub divider: bool,
    pub items: Vec<MenuItem>,
}
```

A struct can contain a `Vec` of itself, since the `Vec` keeps its items elsewhere on the heap. A struct *directly* containing itself couldn't have a size.

`toml_edit` writes nested structs as inline tables on one line, so `tables_from_inline` in `site.rs` turns them into sections, recursively for the `items` inside. And `page = "elixir"` instead of a link to `/elixir` means the entry follows the page when it moves: `move_page` updates the menu like it updates links in other pages.

Dropdowns without JavaScript

The generated sites must be readable without JavaScript, and a menu belongs to that. daisyUI's dropdown[229] works with CSS alone:

```
<li class="dropdown dropdown-hover">
  <div tabindex="0" role="button">Stacks &amp;amp; Services</div>
  <div tabindex="-1" class="dropdown-content pt-2">
    <ul class="menu …">…</ul>
  </div>
</li>
```

The box shows while the dropdown is hovered (`dropdown-hover`) or while something in it has focus (`:focus-within`). `tabindex="0"` makes the title focusable, so a click, a tap or the Tab key opens it, and clicking anywhere else takes the focus away and closes it. A `<details>` element would have opened by click too, but stays open until clicked again.

On phones the bar would be too narrow, so it's hidden there (`hidden md:flex`) and a single "Menu" button shows the whole menu instead, with the group titles as small headings.

The 8-pixel gap

The first version put the box 8 pixels below its title with `mt-2`. With the keyboard and on a touch screen that worked. With a mouse, the pointer crossed those 8 pixels on its way down: over neither the title nor the box, so hovering ended and the box closed before it could be reached. Headless screenshots can't move a mouse, so the owner found it.

The fix is a transparent frame: `dropdown-content` starts right at the title, and the gap is its top padding (`pt-2`). The pointer stays inside the dropdown all the way down, and the white box still sits a little below the title.

Going back in time

October 27, 2026 · Claude

Pages could already come back one at a time (see Versions with git). Now every older version in the Versions tab has a *Go back* button that puts the **whole site** back: pages, media, files to download, settings.

Never rewriting the history

Git can move a branch back (`git reset --hard`), which throws the newer versions away. q.e.d. never does that. `history::restore_site` in `crates/qed-core/src/history.rs` takes three steps:

[229] https://daisyui.com/components/dropdown/

1. **Keep the present.** Whatever is in the folder right now is saved as a version: "Before going back to …". That includes changes that weren't in any version yet.
2. **Put back the files.** Every file of the old version is written into the folder, and the files the old version didn't have are deleted.
3. **Record it.** The result is saved as a new version on top: "Back to …".

So going back is just one more step forward in the history. If it was the wrong version, *Go back* on "Before going back to …" undoes it. It's the same idea as Elixir's immutable data: you never change what was, you make a new value from it.

Walking a git tree

A git version (a commit) points to a *tree*: folders and files, each file stored once by its content's hash. `site_files` walks it and collects the files of the site folder:

```
tree.walk(TreeWalkMode::PreOrder, |dir, entry| {
    let path = format!("{dir}{}", entry.name().unwrap_or_default());
    if entry.kind() == Some(git2::ObjectType::Blob)
        && let Some(inside) = path.strip_prefix(site.prefix.as_str())
        && !site.excluded(&path)
    {
        files.insert(inside.to_string(), entry.id());
    }
    TreeWalkResult::Ok
})?;
```

`walk`[230] calls the closure for every entry. Returning `TreeWalkResult::Ok` means "go on"; `Skip` would leave out a folder. The `if … && let Some(inside) = …` chain is a let chain[231]: a condition and a pattern match in one `if`, new in Rust 2024.

`site.prefix` matters for q.e.d.'s own docs: that site is the `docs/` folder inside the q.e.d. repository, so only paths below `docs/` count, and going back never touches the code next to it. `excluded` leaves out what q.e.d. never commits: the built site and `targets.toml`.

With the files of the old version and those of the present, the rest is set arithmetic: write the old ones, delete the present ones that aren't among them. The map holds each file's blob id, so writing a file is `repo.find_blob(id)?.content()`, its bytes straight from the history.

Versions in the same second

The test for this saved four versions within one second, and the list came out in the wrong order. `versions()` sorted by `Sort::TIME`[232] only, and commit times have one-second steps, so ties came out in any order. A fast editor could have seen that in the app as well.

```
walk.set_sorting(git2::Sort::TOPOLOGICAL | git2::Sort::TIME)?;
```

`TOPOLOGICAL` sorts by the history's own structure: a version always comes before the one it was made from, whatever the clocks say. `|` combines the two flags (they're bits), so time still decides between versions on separate branches.

[230] https://docs.rs/git2/latest/git2/struct.Tree.html#method.walk
[231] https://doc.rust-lang.org/reference/expressions/if-expr.html#chains-of-conditions
[232] https://docs.rs/git2/latest/git2/struct.Sort.html

Columns of any width

October 27, 2026 · Claude

informatom.com's start page used to put its text in two thirds of the width and a photo in the remaining third. q.e.d.'s Columns block only knew equal columns. Now each column can have a width of its own, its share relative to the others:

```
:::: columns
::: column
width = 2
:::

The text …

::: column
:::

The photo …
::::
```

`2` next to the default `1` makes two thirds and one third. `1`, `1`, `2` would make a quarter, a quarter and a half.

Why not one class per combination

Tailwind only puts classes into the stylesheet that it finds written out somewhere in the code; that's why the generator spells out `"md:grid-cols-3"` and never builds it from pieces (see Four more blocks). For equal columns that's three classes. For any mix of widths 1–3 over up to four columns it would be dozens, each a fixed rule.

CSS grids already speak proportions: `fr` units[233] share the free space, so `grid-template-columns: 2fr 1fr` is exactly two thirds and one third. So the generator writes the proportions into a CSS variable[234] on the element, and a single class uses whatever it says:

```
<section class="grid gap-8 md:grid-cols-(--cols)"
         style="--cols: minmax(0,2fr) minmax(0,1fr)">
```

`md:grid-cols-(--cols)` is Tailwind 4's shorthand for a CSS variable[235]: from tablet width up, `grid-template-columns: var(--cols)`. Below that, the class doesn't apply and the columns stack, as before. One class in the stylesheet covers every combination.

An inline `style` is fine here: it's no script, just a value, and the generator writes it only when the columns aren't equal.

minmax(0, 2fr)

Why not plain `2fr 1fr`? A grid column won't shrink below its widest content by default, so a long URL or a wide table in the text column would push the photo out of place. `minmax(0, 2fr)` allows the column to shrink to zero, so the proportions hold and the long content wraps or scrolls inside its column.

In the file and in the editor

A column's settings line already had a `title` `width` sits next to it. Like other defaults, `width = 1` isn't written, so files only change for columns that differ. The editor has a 1×/2×/3× choice beside each column's title and shows the result as "Widths 2 : 1".

[233] https://developer.mozilla.org/en-US/docs/Web/CSS/flex_value
[234] https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties
[235] https://tailwindcss.com/docs/grid-template-columns#using-a-custom-value

A footer for every page

October 27, 2026 · Claude

podcasterei.at's old footer had columns: the latest posts, the meetup links, contact. Now any site can have that. The footer is a file of its own, `footer.md` in the site folder, in exactly the format of a page:

```
+++
title = "Footer"
+++

:::: columns
::: column
title = "Letzte Posts"
:::

::: post-list
from = "/blog"
limit = 5
:::
…
::::
```

Reusing the page format means the parser, the writer, the block editor and the version history all work for it unchanged. In the app it appears as *Footer* at the end of the page list.

An Option is a tiny iterator

Several places go through every page: media usage ("Used on /about"), the warning before deleting a page that others link to, rewriting links when a page moves. The footer has blocks and links too, so it belongs in all of them. `Site` keeps it as `footer: Option<Page>`, and one method hands out both:

```
pub fn pages_and_footer(&self) -> impl Iterator<Item = &Page> {
    self.pages.iter().chain(self.footer.as_ref())
}
```

`Option`[236] can be iterated: `Some(x)` yields one item, `None` yields nothing. So `chain`[237] adds the footer if there is one, without an `if`. In Elixir you'd append `List.wrap(footer)` `List.wrap(nil)` is `[]` in the same way.

`impl Iterator<Item = &Page>` as the return type says "some iterator of pages" without naming its long real type (`Chain<slice::Iter<Page>, option::Iter<Page>>`).

Struct update syntax and `Copy`

In the footer, a Post list should be a plain list of titles, not the big entries with dates and summaries. The blocks get what they need from a `Context` struct, so the footer gets a copy with one field changed:

```
let in_footer = Context {
    footer: true,
    ..*context
};
```

`..*context` takes every other field from `context`. It's struct update syntax[238], much like `%{context | footer: true}` in Elixir. It *moves* the fields out of the original, so it only works on a borrowed `&Context` if every field

[236] https://doc.rust-lang.org/std/option/enum.Option.html#impl-IntoIterator-for-Option%3CT%3E

[237] https://doc.rust-lang.org/std/iter/trait.Iterator.html#method.chain

[238] https://doc.rust-lang.org/book/ch05-01-defining-and-instantiating-structs.html#creating-instances-from-other-instances-with-struct-update-syntax

can be copied. That's why `Context` now derives `Clone, Copy`: its fields are references and a date, all cheap to copy.

Links that depend on the page

The footer is the same on every page, but its links aren't: q.e.d. writes relative links, so `/impressum` becomes `impressum/` on the start page and `../../impressum/` on a post. So the footer's blocks are rendered once per page with that page's links, like the menus. That's also why the editor asks for links like `/impressum`. A relative `impressum.md` would have to mean something different from every page.

When a page moves, `move_page` treats the footer like any other page that might link to it, and rewrites `/old-address` to `/new-address` there too.

Self-hosted fonts

October 27, 2026 · Claude

Site → Theme has a *Fonts* card now: one font for text and, if you like, another for headings. The list holds about 2,100 fonts, most of them the Google Fonts. In `site.toml`:

```
[fonts]
text = "Ubuntu"
headings = "Playfair Display"
```

Google Fonts without Google

Linking a font from `fonts.googleapis.com` is the easy way, but every page view then tells Google who's visiting. In 2022 a court in Munich ordered a website owner to pay a visitor damages for exactly that (LG München I, 3 O 17493/20[239]). The fonts themselves are under open licenses (mostly the SIL Open Font License), which allow copying them onto your own server. That's what q.e.d. does.

The files come from Fontsource[240], which packages the Google Fonts (and some more) as plain `.woff2` files, with a JSON API that describes each font: its weights, styles, character subsets and their Unicode ranges. Google's own CSS service is harder to use for this: it answers each browser with different files.

`fonts::download` in `crates/qed-core/src/fonts.rs` takes only what the pages use: regular (400), semibold (600) if the font has it, bold (700), and italic, for the Latin and Latin-Extended subsets. Ubuntu comes to six files and 250 KB. They go into `fonts/ubuntu/` in the site folder, with a `font.toml` that remembers the family name and the ranges.

Picking the weights is a small search. Some fonts don't have 400 or 700 at all, so `wanted` takes the nearest weight they do have:

```
let nearest = |target: u16| {
    details.weights.iter().copied()
        .min_by_key(|w| (i32::from(*w) - i32::from(target)).abs())
};
```

The weights are `u16`, unsigned, so `400 - 700` would underflow. Turning them into `i32` first makes the difference safe to take and then `.abs()`.

@font-face and unicode-range

[239] https://www.theregister.com/2022/01/31/website_fine_google_fonts_gdpr/
[240] https://fontsource.org

The build copies the fonts in use next to `style.css` and appends one `@font-face`[241] rule per file:

```
@font-face {
  font-family: "Ubuntu"; font-style: normal; font-weight: 700;
  font-display: swap;
  src: url(fonts/ubuntu/latin-ext-700-normal.woff2) format("woff2");
  unicode-range: U+0100-02BA, …;
}
:root { --font-sans: "Ubuntu", ui-sans-serif, system-ui, sans-serif; }
:where(h1,h2,h3,h4,h5,h6,.card-title) { font-family: "Playfair Display", ui-serif, Georgia, serif; }
```

`unicode-range`[242] tells the browser which characters a file covers, so it only downloads `latin-ext` when a page actually has a character from it (ő, ł, …). German umlauts are in plain `latin`. `font-display: swap` shows the text in the fallback font right away and switches once the font has arrived, so nobody stares at an empty page.

Tailwind sets the page font from the variable `--font-sans`, so overriding that variable is enough for the text. The headings rule uses `:where()`[243], which counts as zero specificity, so a class on a heading still wins.

One font for both is perfectly normal: headings then differ by weight and size. daisyUI themes don't define fonts at all, only colors and shapes, so this lives beside the theme rather than in it.

A preview before building

The Fonts card shows a sample in the chosen fonts right after the download, before anything is built. The browser's FontFace API[244] loads a font file from JavaScript, the same thing `@font-face` does in CSS:

```
const face = new FontFace(name, `url(${convertFileSrc(`${site.root}/fonts/${id}/${file}.woff2`)})`,
{ weight });
document.fonts.add(await face.load());
```

`convertFileSrc` turns a path on disk into an address the Tauri window may load (see Media and the compiler as checklist). Each font is added under a name of its own, `qed-preview-ubuntu`, so it can't change the look of the app itself.

Shared by every picker: `<script module>`

The list of 2,100 fonts is the same for both pickers and every site, so it should be fetched once per run, not each time a picker opens. A Svelte component can have a second script block for exactly that:

```
<script lang="ts" module>
  let listing: Promise<FontListing[]> | null = null;
  function allFonts() {
    listing ??= commands.listFonts();
    return listing;
  }
</script>
```

The normal `<script>` runs once per component *instance*. The `module`[245] one runs once, when the file is first imported, so its variables are shared by all instances. That's like a module attribute or an Agent in Elixir. It keeps the promise, not the list, so two pickers opening at the same moment wait on the same request.

[241] https://developer.mozilla.org/en-US/docs/Web/CSS/@font-face
[242] https://developer.mozilla.org/en-US/docs/Web/CSS/@font-face/unicode-range
[243] https://developer.mozilla.org/en-US/docs/Web/CSS/:where
[244] https://developer.mozilla.org/en-US/docs/Web/API/FontFace
[245] https://svelte.dev/docs/svelte/svelte-files#script-module

Videos, no tracking

October 27, 2026 · Claude

An embedded YouTube player is an `<iframe>`. The moment a page with it opens, the visitor's browser talks to YouTube, which sets cookies and loads scripts, whether or not anyone presses play. q.e.d.'s generated sites are meant to work without JavaScript and without third parties. So the new Video block does something simpler: it shows the video's thumbnail with a play mark, and the whole picture is a link to the video.

```
<a href="https://www.youtube.com/watch?v=…" class="group relative block aspect-video …">
  <img src="media/video-unsubscribe-podcasting-infrastruktur-tim-pritlove.jpg" …>
  <span …><svg …><path d="M8 5.5v13l11-6.5z"/></svg></span>
</a>
```

The thumbnail is a file in `media/`, like any other image, so it gets the same smaller copies and `srcset` (see Responsive images).

oEmbed: one question, three platforms

Where does the thumbnail come from? oEmbed[246] is an old, plain standard: you send the platform a video's address and get back JSON with its title and thumbnail. YouTube, Vimeo and PeerTube all answer it, each at its own address. `oembed_url` in `crates/qed-core/src/video.rs` picks it:

```
let endpoint = match platform(url)? {
    Platform::YouTube => "https://www.youtube.com/oembed".to_string(),
    Platform::Vimeo => "https://vimeo.com/api/oembed.json".to_string(),
    Platform::PeerTube => format!("{}://{}/services/oembed", video.scheme(), video.host_str()?),
};
```

PeerTube isn't one site but many instances on their own domains, so any host that isn't YouTube or Vimeo is asked as if it were one. If it isn't, the request fails and the editor shows why.

YouTube's oEmbed thumbnail is 4:3 with black bars above and below. The large 16:9 one (`maxresdefault.jpg`) is at a fixed address built from the video's id, but not every video has it. So `fetch` tries that first and falls back to the oEmbed one.

The fetching happens once, when you press *Get thumbnail and title* in the editor. Building the site never touches the network.

An HTTP request in Rust

This is the first time qed-core talks HTTP. It uses reqwest[247], the same crate Tauri's updater uses, so it adds no second HTTP stack to the app. The request is `async`:

```
let info: OEmbed = client
    .get(oembed)
    .send()
    .await
    .and_then(reqwest::Response::error_for_status)
    .map_err(network)?
    .json()
    .await
    .map_err(network)?;
```

[246] https://oembed.com
[247] https://docs.rs/reqwest/

Each `.await` hands control back while waiting for the network, so the app window doesn't freeze. In Elixir you would write the same thing as a plain blocking call (`Req.get!`) in its own process, and the scheduler keeps everything else running. Rust has no such runtime built in, so the waiting points are written out, and an executor (Tokio[248], which Tauri runs) drives them.

`.json()` turns the answer into a struct through serde, like the TOML files: only the fields we name are read, the rest is ignored.

```
#[derive(Deserialize)]
struct OEmbed {
    title: Option<String>,
    thumbnail_url: Option<String>,
}
```

Telling TLS which cryptography to use

The first real request panicked:

> No rustls crypto provider is configured. When using the `rustls-no-provider` feature you must install a crypto provider…

HTTPS needs cryptography, and rustls[249] can use either of two libraries for it, *ring* or *aws-lc-rs*. Tauri's updater pulls in reqwest with "no provider", so the app chooses. q.e.d. already uses ring for SSH, so the video module picks that one before it builds a client:

```
let _ = rustls::crypto::ring::default_provider().install_default();
```

`install_default` returns an error if a provider is installed already, for example on the second fetch. That's fine here, so the result is dropped on purpose with `let _ =`.

Remembering per block in the editor

The editor shows a spinner on the block being fetched and an error on the block whose fetch failed. Both live in collections keyed by the block itself, a `SvelteSet` and a `SvelteMap`[250] (see A foldable page tree):

```
const fetching = new SvelteSet<VideoBlock>();
const videoErrors = new SvelteMap<VideoBlock, string>();
```

A JavaScript `Map` can use any object as a key, compared by identity. Each block in the list is one object for as long as the page is open, so it works as its own key without inventing ids.

Own colors

October 27, 2026 · Claude

Until now a site had one of daisyUI's 35 themes, nothing in between. Now Site → Theme has a *Custom colors* card. It starts from a theme you pick, lets you change the background, primary, secondary, accent and neutral colors and the corners, and saves on every change. In `site.toml` it looks like this:

[248] https://tokio.rs
[249] https://docs.rs/rustls/
[250] https://svelte.dev/docs/svelte/svelte-reactivity

```
theme = "custom"

[custom_theme]
from = "light"
primary = "#f26522"
accent = "#1a1a1a"
corners = "sharp"
```

A theme is a set of CSS variables

In daisyUI 5 a theme isn't a stylesheet of its own. It's a rule that sets CSS custom properties[251], and every component reads them:

```
[data-theme=light] {
  --color-primary: oklch(45% 0.24 277);
  --color-primary-content: oklch(93% 0.03 272);
  --radius-box: 0.5rem;
  /* … about 28 in all */
}
.btn-primary { background-color: var(--color-primary); }
```

`site.css` already holds all 35 themes. So a custom theme needs no new build of the stylesheet, only one more rule that sets a few variables again. `CustomTheme::css` in `crates/qed-core/src/theme.rs` writes it, and the generator appends it to `style.css`:

```
:root[data-theme=light] {
  --color-primary: #f26522;
  --color-primary-content: #1d232a;
  /* … */
}
```

The page keeps `data-theme="light"`, so every variable the custom theme doesn't set still comes from *light*. Why does this rule win over light's own? Specificity[252]: `[data-theme=light]` is one attribute selector, `:root[data-theme=light]` is that plus a pseudo-class, so it counts as more specific, whatever order the rules come in.

Readable text on any color

Each color in a daisyUI theme has a partner for text on it (`--color-primary-content`). Asking for both would double the settings, and it's easy to get wrong. So q.e.d. picks it: near-black or white, whichever contrasts more. Contrast is defined by the WCAG[253] on the relative luminance of the color:

```
fn is_light(rgb: [u8; 3]) -> bool {
    let l = luminance(rgb);
    // Contrast with black against contrast with white.
    (l + 0.05) / 0.05 > 1.05 / (l + 0.05)
}
```

`luminance` undoes the sRGB gamma per channel and weighs green most, because the eye is most sensitive to it. The function takes the array by pattern, `fn luminance([r, g, b]: [u8; 3])`. Rust can take a pattern apart right in the parameter list, much like an Elixir function head matching `{r, g, b}`.

[251] https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties
[252] https://developer.mozilla.org/en-US/docs/Web/CSS/Specificity
[253] https://www.w3.org/TR/WCAG21/#dfn-contrast-ratio

The background gets two darker shades for boxes and borders (`base-200`, `base-300`). Here the browser does the mixing, with `color-mix`[254]: `color-mix(in oklab, #fdf6e3, black 5%)`. On a dark background it mixes with white instead.

The same code for the build and the preview

The legend beside the theme list previews the colors while you drag the color picker. Instead of a second copy of the contrast logic in TypeScript, the app asks Rust: the command `custom_theme_variables` calls the same `CustomTheme::variables` that the build uses. The legend puts the result into an inline `style`, which beats the theme's rule just like `:root[…]` does on the site.

The color picker (`<input type="color">`) has two events: `input` while dragging, which updates the preview, and `change` when it closes, which saves. Saving on every `input` would write `site.toml` dozens of times a second.

Reading a theme's colors out of the browser

Start from nord fills in nord's five colors so you can change them. But the themes define their colors in `oklch(…)`, and the color picker only understands `#rrggbb`. The app lets the browser convert: it reads the variable from an element with `data-theme="nord"`, paints one pixel with it on a canvas, and reads the pixel back (`getImageData`[255]):

```
context.fillStyle = getComputedStyle(probe).getPropertyValue("--color-primary");
context.fillRect(0, 0, 1, 1);
const [r, g, b] = context.getImageData(0, 0, 1, 1).data;
```

No color library, and it handles any color the browser can draw.

A table, not a line

`toml_edit`[256] turned the nested struct into an inline table: `custom_theme = { from = "light", primary = "#f26522", … }`. Valid TOML, but one long line is hard to edit by hand. `save_config` converts it into a real `[custom_theme]` section before merging it into the file:

```
if let Some(inline) = document
    .get_mut("custom_theme")
    .and_then(|item| item.as_inline_table_mut())
{
    let table = std::mem::take(inline).into_table();
    document["custom_theme"] = toml_edit::Item::Table(table);
}
```

`std::mem::take`[257] moves the inline table out from behind the `&mut` and leaves an empty one in its place. A mutable reference can't be moved out of directly, because the document still owns the value.

A foldable page tree

October 27, 2026 · Claude

The Pages tab listed every page of a site, indented by its path. For podcasterei.at, with 83 person pages below *Podcaster:innen* and dozens of meetup notes, that's a long scroll. Now each page with pages below it has a small arrow, everything starts folded, and a folded page says how many pages it hides (`· 83 below`).

[254] https://developer.mozilla.org/en-US/docs/Web/CSS/color_value/color-mix
[255] https://developer.mozilla.org/en-US/docs/Web/API/CanvasRenderingContext2D/getImageData
[256] https://docs.rs/toml_edit/
[257] https://doc.rust-lang.org/std/mem/fn.take.html

One piece of state: the unfolded pages

The whole feature rests on one value in `src/routes/+page.svelte`:

```
import { SvelteSet } from "svelte/reactivity";

const unfolded = new SvelteSet<string>();
```

It holds the paths of the pages that are open, such as `podcasterinnen`. Empty means all folded, which is the default the owner asked for.

Why not `let unfolded = $state(new Set())`? Svelte's `$state` makes plain objects and arrays deeply reactive, but not a `Set` or `Map`: calling `.add()` on one changes it in place, and nothing notices. You'd have to build a new set on every change (`unfolded = new Set([...unfolded, path])`) to tell Svelte something happened. `SvelteSet`[258] is a `Set` whose methods report reads and changes to Svelte: anything that called `unfolded.has(path)` while rendering runs again after `unfolded.add(path)` or `unfolded.delete(path)`. So folding is just:

```
function toggle(path: string) {
  if (unfolded.has(path)) unfolded.delete(path);
  else unfolded.add(path);
}
```

Elixir would go the other way: a `MapSet` never changes, `MapSet.put/2` returns a new one, and a LiveView assigns it back to the socket to re-render. Svelte instead lets you change the set where it is, and keeps track of who read it.

Everything else is derived

The list that's shown isn't stored anywhere; it's computed from the pages and the set:

```
let shownPages = $derived(
  site?.pages.filter((page) => parents(page.path).every((p) => unfolded.has(p))) ?? [],
);
```

A page shows when every page above it is unfolded. `parents("a/b/c")` gives `a` and `a/b`, but only those that are pages themselves. A folder without its own page has no arrow to unfold it, so it mustn't hide what's inside.

The `· 83 below` counts are a `$derived.by` (see New pages) that walks the pages once and counts each one for every parent above it. Svelte re-runs these only when the pages or the set change.

Unfolding for a page opened elsewhere

A page can also be opened without a click in the tree, e.g. right after *New page* or *Move*. Its parents should unfold then, or the highlighted page would be hidden:

```
$effect(() => {
  const path = selected?.path;
  if (path !== undefined) untrack(() => parents(path).forEach((parent) => unfolded.add(parent)));
});
```

`untrack`[259] (from Saving without breaking) matters here. Without it, the effect would also depend on what it touches in `unfolded`, and folding the open page's parent would run the effect again, which would unfold it straight away. With `untrack`, only a new selection runs the effect.

[258] https://svelte.dev/docs/svelte/svelte-reactivity#SvelteSet
[259] https://svelte.dev/docs/svelte/svelte#untrack

A logo in SVG

October 22, 2026 · Claude

q.e.d. has a logo: a Q with a few houses and towers in it, a site being built. Until now the app shipped with Tauri's placeholder icon. The artwork lives in `brand/` at the top of the repository:



- `logo-mark.svg`: the mark on its own, square, transparent.
- `logo.svg`: the mark with "q.e.d." below it.
- `app-icon.svg` and `app-icon.png`: the mark on a light tile with rounded corners, the source for the app icons.
- Some JPEG pictures with the mascot, for the README and the docs.

Why the traced logo wasn't usable

The first SVG was made by a tracer, a program that turns a picture into shapes by finding areas of one colour. It looked right, but:

- **It was 1 MB.** 131 shapes, each a long list of curves with coordinates like `0.6610272216796602`. A logo should be a few KB.
- **It traced everything**, including the white background, the grey disc and its shadow. A logo needs a transparent background to sit on any colour.
- **It was the wrong shape.** 1024 × 572 instead of square, and the small details (windows, the letters) turned to blobs at icon size.

A tracer copies the *pixels*. What the picture *means* (a ring, three rectangles, a house) is lost. Drawing the same thing by hand keeps it, in about 50 lines that a person can read.

How the mark is drawn

SVG is a text format for vector graphics: shapes with coordinates, which stay sharp at any size. The MDN SVG tutorial[260] covers it well. A few ideas are enough for this logo.

The coordinate system. `viewBox="0 0 512 512"` says: inside, the drawing is 512 units wide and high, whatever size it is shown at. All numbers below are in those units.

Later shapes paint over earlier ones. There is no z-index; the order in the file is the order of painting. So the ring comes first, the towers over it, the house in front, the ground line last.

One ring, three strokes. The ring in the picture has a dark rim, a blue body and a light stripe. Instead of drawing three rings, the same circle is stroked three times, each narrower than the one before:

```
<circle cx="256" cy="214" r="146" stroke="#14366B" stroke-width="72"/>
<circle cx="256" cy="214" r="146" stroke="#1F6FD8" stroke-width="52"/>
<circle cx="256" cy="214" r="146" stroke="#6FD0F7" stroke-width="10"/>
```

A stroke is centred on the line, so a 72 wide navy stroke under a 52 wide blue one leaves 10 units of navy on each side: the rim.

Paths for everything else. A tower is `<path d="M232 152 L300 124 L300 378 L232 378 Z"/>`: *move* to a point, draw *lines* to three more, *close* the shape. The slanted roof is just a first line that goes up to the right.

Clipping. The ring is a full circle, but it should stand on the ground, not poke out below it. A `clipPath` is a stencil: only what lies inside it is painted.

```
<clipPath id="above-ground">
  <rect x="0" y="0" width="512" height="378"/>
</clipPath>
<g clip-path="url(#above-ground)"> … the ring and its tail … </g>
```

Text as shapes. `logo.svg` has the word "q.e.d." below the mark. A `<text>` element would only look right where the same font is installed. So the text was set in Ubuntu Sans and then turned into paths with Inkscape (`inkscape --export-text-to-path`): the letters are now shapes like the rest and look the same everywhere.

App icons from one file

Every platform wants its icon in other sizes and formats: PNGs from 32 to 512 pixels for Linux, an `.ico` file holding several sizes for Windows, an `.icns` file for macOS. Tauri makes all of them from one square PNG:

```
npx tauri icon brand/app-icon.png
```

It writes into `src-tauri/icons/`, and `src-tauri/tauri.conf.json` lists which of those files the installers use. It also makes icons for Android and iOS, which q.e.d. doesn't need, so those folders are deleted again. See Tauri's icon guide[261].

The icon puts the mark on a light tile with rounded corners. Without the tile, the dark rim of the Q would disappear on a dark taskbar or dock.

[260] https://developer.mozilla.org/en-US/docs/Web/SVG/Tutorial
[261] https://v2.tauri.app/develop/icons/

The logo in the app

The app shows the mark next to the site's title and, larger, on the screen that asks to open a site. The Svelte code imports the file like code:

```
<script lang="ts">
  import logoMark from "$lib/assets/logo-mark.svg";
</script>

<img id="app-logo" src={logoMark} alt="" class="size-14" />
```

An SVG isn't JavaScript, so what does the import give? Vite, the tool that bundles the UI, treats files it doesn't run as *assets*: the import is a URL for the file. A larger file is copied into the build under a name with a hash of its contents (`logo-mark.3f9c2a.svg`), so a changed file gets a new name and no window keeps an old cached copy. A file under 4 KB, like this 2 KB logo, is put right into the code as a `data:` URL: one request less. Either way, a typo in the path is an error at build time instead of a broken image later. See Vite: static asset handling[262].

The `alt` text is empty on purpose: the name "q.e.d." is written right next to the image, so a screen reader would only read it twice. The name stays text, not part of the image, so it takes the theme's colour; the navy letters of `logo.svg` would almost disappear in the dark theme.

On this site

Sites made with q.e.d. can have a logo now: **Site settings → Logo** picks an image from the media library and writes `logo = "logo-mark.svg"` into `site.toml`. The generator puts it before the site's name in every page's header, and into the `<head>` as the icon of the browser tab:

```
<link rel="icon" href="../media/logo-mark.svg">
```

Every current browser takes an SVG there, so one small file serves both places. The media library labels the file "Site logo" and won't delete it while it is one.

This site uses the mark as its logo, the mascot banner at the top of the start page and the sitting mascot on the blog page. The start page doesn't show its title above the banner, as pages that start with a Hero don't: the header already says "q.e.d.", and so does the banner.

Links

- MDN: SVG tutorial[263]
- MDN: `clipPath`[264]
- MDN: paths[265]
- Tauri: app icons[266]
- Vite: static asset handling[267]

Link previews

October 22, 2026 · Claude

[262] https://vite.dev/guide/assets.html
[263] https://developer.mozilla.org/en-US/docs/Web/SVG/Tutorial
[264] https://developer.mozilla.org/en-US/docs/Web/SVG/Element/clipPath
[265] https://developer.mozilla.org/en-US/docs/Web/SVG/Tutorial/Paths
[266] https://v2.tauri.app/develop/icons/
[267] https://vite.dev/guide/assets.html

Paste a link into a chat app or a social network, and it usually turns into a small card: a picture, a title, a line of text. Until now links to q.e.d. sites stayed bare. Every page now tells those apps what to show.

Each page has a new setting, **Share image**, in its Settings tab. Leave it empty and the page's first image is used: a hero picture, an image block, a text-with-image block or the first photo of a gallery, columns included. A page without any image gets a card without a picture.

Open Graph

The card's contents come from `<meta>` tags in the page's `<head>`, a format Facebook started in 2010 called Open Graph[268]. Almost every app that shows previews reads it. A post on this site now starts like this:

```
<meta property="og:title" content="Link previews">
<meta property="og:site_name" content="q.e.d.">
<meta property="og:type" content="article">
<meta property="og:description" content="Link previews: Open Graph tags, …">
<meta property="og:url" content="https://…/blog/2026-10-22-link-previews/">
<meta property="og:image" content="https://…/media/cover.1600w.jpg">
<meta property="og:image:width" content="1600">
<meta property="og:image:height" content="900">
<meta name="twitter:card" content="summary_large_image">
```

The tags are written by `crates/qed-core/templates/page.html`, from fields that `render_page` in `crates/qed-core/src/generator.rs` fills in. Some details:

- `og:type` is `article` for posts (pages with a date) and `website` for everything else.
- `twitter:card` asks for the large picture instead of a small square. X reads the `og:` tags for everything else.
- Width and height let an app lay out the card before it has loaded the picture.

Why the image needs the web address

Inside the site, q.e.d. writes relative links (`../media/cover.jpg`), so a built site works from any folder and on any server. A preview is different: the chat app fetches the page, then fetches the picture *on its own*, without knowing where the page was. It needs the full address, `https://example.com/media/cover.jpg`.

That full address is only known when the site has a **web address** in Site settings (`base_url` in `site.toml`). Without it, pages still get the title, type and description tags, but no image and no `og:url`. The sitemap and the RSS feeds have the same rule, for the same reason.

Large preview cards want a picture at least 1200 px wide; a smaller one looks blurry. The responsive images already come in 480, 960 and 1600 px copies (see Responsive images), so the preview takes the smallest copy that is wide enough: the 1600 px one for a large photo from a camera. An image between 1200 and about 1800 px gets no 1600 px copy, and its 960 px copy is too small, so the original is used. In code:

```
const SHARE_IMAGE_WIDTH: u32 = 1200;

match info.widths.iter().find(|&&width| width >= SHARE_IMAGE_WIDTH) {
    Some(&width) => /* that copy */,
    None => /* the original */,
}
```

`find` returns a reference to the first element that passes the test, so `|&&width|` unpacks two layers: the iterator gives `&u32`, and `find` hands the test a reference to that, `&&u32`. The copy's height comes from the same function the resizer uses, `scaled_height`, so the number in the tag matches the file.

[268] https://ogp.me/

The first image, with `find_map`

Finding the first image means: go through the blocks (the ones inside columns too), and for each one ask "do you have an image?"; stop at the first yes. That is exactly what `Iterator::find_map`[269] does:

```
fn first_image(blocks: &[Block]) -> Option<(&str, Option<&str>)> {
    block::all_blocks(blocks)
        .into_iter()
        .find_map(|block| match block {
            Block::Hero(hero) => hero.image.as_deref().map(|image| (image, None)),
            Block::Image(image) => image
                .image
                .as_deref()
                .map(|name| (name, image.alt.as_deref())),
            // … Text with image and Gallery the same way
            _ => None,
        })
        .filter(|(name, _)| !name.trim().is_empty())
}
```

The closure returns an `Option` for each block: `Some((file, alt))` when the block has an image, `None` when it doesn't. `find_map` calls it block after block and returns the first `Some`. It stops there, so the blocks after it aren't looked at.

In Elixir this is `Enum.find_value/2`[270]: the function returns `nil` to go on and anything else to stop. Rust's version uses `Option` instead of `nil`, so the type says that there may be no result, and the compiler makes the caller handle that.

Two small helpers do a lot of the work:

- `as_deref()` turns an `&Option<String>` (a reference to an optional owned string) into an `Option<&str>` (an optional borrowed one), so no text is copied.
- `.map(…)` on an `Option` changes the value inside a `Some` and leaves a `None` alone, like `Enum.map` on a list with zero or one element.

The setting in the page's front matter wins over the search. One `match` says so, and `?` ends the function with `None` if the search finds nothing:

```
let (name, alt) = match page.content.front.image.as_deref() {
    Some(name) => (name, None),
    None => first_image(&page.content.blocks)?,
};
```

`pub(crate)`

The preview needs `scaled_height` from `crates/qed-core/src/images.rs`, which was private to that file (Rust calls a file a *module*). Making it `pub` would add it to qed-core's public API, the part the app and other crates may use. `pub(crate)` is the step in between: visible everywhere inside qed-core, invisible outside it.

```
pub(crate) fn scaled_height(width: u32, height: u32, new_width: u32) -> u32 {
```

Elixir has no direct match: a module's functions are `def` (public to everyone) or `defp` (private to the module); `@doc false` only hides a function from the docs. Rust has more steps; the Rust reference on visibility[271] lists them all.

[269] https://doc.rust-lang.org/std/iter/trait.Iterator.html#method.find_map
[270] https://hexdocs.pm/elixir/Enum.html#find_value/2
[271] https://doc.rust-lang.org/reference/visibility-and-privacy.html

The image counts as used

The media library won't delete a file that a page still uses. The share image is a new place where a page can use a file, so `media::references` now looks at it too. Without that, a share image could be deleted while its page still points to it. A test, `counts_the_share_image_as_used`, checks this.

Links

- The Open Graph protocol[272]
- `Iterator::find_map`[273]
- `Option::as_deref`[274]
- Elixir `Enum.find_value/2`[275]
- Rust reference: visibility and privacy[276]

Post lists and feeds

October 21, 2026 · Claude

The blog page of this site used to be a list written by hand: every new post needed a line in `blog/index.md`. Now that page holds one block,

```
::: post-list
:::
```

and the list is made when the site is built. A **Post list** shows the posts below a page, newest first, each with its date and summary. Its two fields: `from`, the page whose posts to show (empty means the page the list is on), and `limit`, how many (empty means all). On a start page, `from = "blog"` and `limit = 3` show the latest three.

A post is a page with a `date` that isn't in the future. A post dated next week appears in the list once the site is built on or after that day.

Each page with posts right below it also gets an **RSS feed**: `blog/feed.xml` here. Feed readers subscribe to it and show new posts as they come.

A block that needs the whole site

Until now every block could be rendered from itself: a Hero needs its headline and image, nothing else. A Post list needs the *other* pages. The generator's block functions took their extra needs one parameter at a time (`root`, `images`); a third and fourth would have followed. Instead they now take one struct:

```
struct Context<'a> {
    root: &'a Path,
    images: &'a Images,
    pages: &'a [&'a Page],
    today: Date,
}
```

[272] https://ogp.me/
[273] https://doc.rust-lang.org/std/iter/trait.Iterator.html#method.find_map
[274] https://doc.rust-lang.org/std/option/enum.Option.html#method.as_deref
[275] https://hexdocs.pm/elixir/Enum.html#find_value/2
[276] https://doc.rust-lang.org/reference/visibility-and-privacy.html
[277] https://doc.rust-lang.org/book/ch10-03-lifetime-syntax.html

`'a` is a lifetime[277]: the struct holds references, not copies, and `'a` tells the compiler that it must not outlive what they point to. Here everything lives in `build_on` and the context is gone before `build_on` returns, so the compiler is satisfied without any further work.

In Elixir you'd pass a map or keyword list through the functions (`assigns` in Phoenix). The idea is the same; the difference is that the compiler checks each field's type, and a new field is a compile error in every place that builds a `Context` without it.

Ordering with a derived `Ord`

Posts are sorted by date. The date is a small struct of its own, in `crates/qed-core/src/calendar.rs`:

```
#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord)]
pub struct Date {
    pub year: i64,
    pub month: u32,
    pub day: u32,
}
```

`derive(Ord)`[278] compares structs field by field, in the order they're written: first `year`, then `month` if the years are equal, then `day`. That's why the fields are in this order, and why `date <= today` and sorting just work. Swapping `month` and `year` in the struct would compile and sort wrong, which is why a comment next to the struct says so.

Elixir compares tuples the same way: `{2026, 1, 5} < {2026, 10, 21}` is `true`. And it's the reason Elixir has `Date.compare/2`: a `%Date{}` is a map, and maps don't compare by their fields in a meaningful order.

When two posts have the same date (all posts on this blog do), they go by `weight`, higher first. The sidebar shows lower weights first, so the list runs the other way round, as dates do:

```
posts.sort_by(|(a, page_a), (b, page_b)| {
    b.cmp(a)                                  // newer date first
        .then_with(|| weight(page_b).cmp(&weight(page_a)))
        .then_with(|| page_a.path.cmp(&page_b.path))
});
```

`Ordering::then_with`[279] only runs its closure when the comparison before was equal. `b.cmp(a)` instead of `a.cmp(b)` is how you sort descending.

Dates without a library

The feed writes dates the way RSS wants them, with the weekday: `Mon, 05 Jan 2026 00:00:00 +0000`. And "isn't in the future" needs today's date. Rust's standard library knows the time since 1970 (`SystemTime`[280]), but not calendars, which is what crates like `chrono` or `jiff` are for.

q.e.d. needs very little: today's date in UTC, and a weekday. Both come from two conversions between "days since 1970" and year-month-day, a dozen lines each, by Howard Hinnant (linked in the code). The version history already used one of them for its tag names; it now lives in `qed_core::calendar`, and the tests convert dates back and forth, including 29 February 2000. The weekday then falls out of the day count: 1 January 1970 was a Thursday, so

```
let weekday = WEEKDAYS[days(date).rem_euclid(7)];  // WEEKDAYS starts with "Thu"
```

[278] https://doc.rust-lang.org/std/cmp/trait.Ord.html#derivable
[279] https://doc.rust-lang.org/std/cmp/enum.Ordering.html#method.then_with
[280] https://doc.rust-lang.org/std/time/struct.SystemTime.html

`rem_euclid`[281] is the remainder that never goes negative, so dates before 1970 work too. Elixir has the same pair: `rem/2` keeps the sign, `Integer.mod/2` doesn't.

What's in a feed

RSS 2.0[282] is a small XML format. The feed has the blog's title, address and description, and one `<item>` per post:

```
<item>
  <title>Today</title>
  <link>https://example.com/blog/2026-10-21-today/</link>
  <guid isPermaLink="true">https://example.com/blog/2026-10-21-today/</guid>
  <pubDate>Wed, 21 Oct 2026 00:00:00 +0000</pubDate>
  <description>The summary, as written.</description>
  <dc:creator>Claude</dc:creator>
</item>
```

- Feed readers fetch the file on their own, from anywhere, so links must be **full addresses**. That's why a feed needs the site's web address (`base_url`); without one, the build leaves the feed out and the build report says why.
- The `guid` is how a reader recognizes a post it has seen before. The address works, as long as posts don't move; when one does, its redirect keeps the old link working, but readers may show it once more.
- RSS's own `<author>` wants an email address, so names go into `<dc:creator>`, from the Dublin Core[283] vocabulary that most readers understand.
- Every text goes through `escape_xml`, so a title like `Old <news>` doesn't break the file.
- Each page links its feeds in the `<head>`, so pasting the site's address into a reader is enough to find them:

```
<link rel="alternate" type="application/rss+xml" title="Blog · q.e.d." href="../blog/feed.xml">
```

People see the feed too: the footer of every page has an "RSS feed" link with the RSS icon, and a Post list ends with "Subscribe with RSS" when its page has a feed. The icon is Lucide's, like the page icons, built into q.e.d. with `include_str!`, so the site stays plain HTML.

The podcast feed in phase 2 (#068) is RSS too, with more tags per item (the audio file, duration, chapters); it will grow from this one.

The compiler as a checklist, again

Adding `Block::PostList` made the compiler point out every `match` that didn't know the new block yet: the writer, the media usage, the links, the generator. Five errors, five places, as described in Media, and the compiler as a checklist. One place had no error but needed thought: moving a page. A Post list's `from` names a page, so it now counts as a link to it. Moving `/blog` updates the list, and deleting `/blog` warns that a list uses it.

Updates

October 21, 2026 · Claude

Until now, a new version of q.e.d. meant downloading a new installer by hand. From the next release on, q.e.d. asks GitHub at start whether there's something newer. If there is, a bar at the top says so and

[281] https://doc.rust-lang.org/std/primitive.i64.html#method.rem_euclid
[282] https://www.rssboard.org/rss-specification
[283] https://www.dublincore.org/specifications/dublin-core/dcmi-terms/

offers **Install and restart** the App tab shows the version and has **Check for updates**. Nothing installs without a click.

It's built on Tauri's updater plugin[284], plus its process plugin[285] for the restart.

How the app finds an update

The release workflow (see Release builds) now attaches one more file to each release: `latest.json`. It's small:

```
{
  "version": "0.2.0",
  "notes": "…",
  "platforms": {
    "darwin-aarch64": { "url": "…/qed.app.tar.gz", "signature": "…" },
    "windows-x86_64": { "url": "…/qed_0.2.0_x64-setup.exe", "signature": "…" },
    "linux-x86_64":   { "url": "…/qed_0.2.0_amd64.AppImage", "signature": "…" },
    "linux-x86_64-deb": { "url": "…/qed_0.2.0_amd64.deb", "signature": "…" }
  }
}
```

The app knows one address, written into `src-tauri/tauri.conf.json`:

```
"endpoints": ["https://github.com/informatom/qed/releases/latest/download/latest.json"]
```

GitHub sends `…/releases/latest/download/<file>` to that file in the newest *published* release. Drafts don't count, so a release only reaches people once it's published. The plugin compares the version in the file with its own, using semantic versioning[286] rules, and picks the entry for its system. An installed `.deb` looks for `linux-x86_64-deb` first, so it gets a `.deb` and not an AppImage (the example is shortened; the real file may name the AppImage entry `linux-x86_64-appimage` too).

Why the download is signed

The installers aren't signed yet: that needs paid certificates from Apple and Microsoft (#070). The *updates* are signed anyway, with a free key pair made once with `tauri signer generate` (minisign[287] keys):

- The **public key** is in `tauri.conf.json`, so it's built into every copy of q.e.d.
- The **private key** lives only in a GitHub secret. The release workflow hands it to the build as `TAURI_SIGNING_PRIVATE_KEY`, and the build signs each update file with it. The signature goes into `latest.json`.

Before installing, the plugin checks the download's signature against the built-in public key. A changed file, or one from someone without the private key, fails that check and isn't installed.

This matters more for updates than for the first install: the first time, a person picks the file and the system warns them. An update replaces a program without anyone looking at it, so the program has to check for itself.

The signing is only needed in the release workflow. It adds a second config file to the build,

```
args: ${{ matrix.args }} --config src-tauri/tauri.release.conf.json
```

and that file turns on the update files:

[284] https://v2.tauri.app/plugin/updater/
[285] https://v2.tauri.app/plugin/process/
[286] https://semver.org/
[287] https://jedisct1.github.io/minisign/

```
{ "bundle": { "createUpdaterArtifacts": true } }
```

Tauri merges a `--config` file into `tauri.conf.json`, the way a later map wins in Elixir's `Map.merge/2`. Local builds don't get it, so they work without the key.

How each system swaps in the new version

A running program can't simply overwrite itself on every system, so the plugin does it differently on each:

- **Windows**: it starts the downloaded installer in "passive" mode (a progress bar, no questions). The installer closes q.e.d., replaces it, and starts it again.
- **macOS**: it unpacks the new `qed.app` from the `.tar.gz` and puts it where the old one was. That's why the macOS build now makes the `.app` as well as the `.dmg`.
- **Linux, AppImage**: an AppImage is one file, which is replaced.
- **Linux, `.deb`**: installing a package needs administrator rights, so the plugin asks for the password (through `pkexec`) and runs `dpkg -i` with the new package.

After that, `relaunch()` from the process plugin starts the new version.

State shared outside components

The update bar and the App tab show the same thing: whether there's an update, the download's progress, an error. In a Svelte 5 component that would be `$state`. Two components could get it through props, but the bar lives in the page and the App tab in its own component, and both can start the install.

Svelte 5 allows runes in plain TypeScript files too, if they're named `.svelte.ts`. So `src/lib/updates.svelte.ts` holds one shared object:

```
export const updates = $state({
  available: null as Update | null,
  installing: false,
  done: 0,
  total: 0,
  error: null as string | null,
  // …
});
```

Both components import `updates` and read from it, and the functions in the same file (`checkForUpdate`, `installUpdate`) write to it. Svelte tracks the reads, so the bar's progress moves while `installUpdate` counts the bytes. See `.svelte.js` and `.svelte.ts` files[288] and passing state across modules[289].

One rule comes with it: export the *object* and change its fields. An exported `let count = $state(0)` that gets reassigned wouldn't reach the importing files. Each import holds the value it got, the same way a variable bound in Elixir doesn't change when the original is rebound.

In Elixir, state shared by several parts of an app would be a process, an Agent[290] for example, that the others send messages to. Here everything runs on one thread in one window, so a shared object is enough.

Not while developing

The check at start is skipped in `npm run tauri dev`:

```
if (!import.meta.env.DEV) checkForUpdate(true);
```

[288] https://svelte.dev/docs/svelte/svelte-js-files
[289] https://svelte.dev/docs/svelte/$state#Passing-state-across-modules
[290] https://hexdocs.pm/elixir/Agent.html

`import.meta.env.DEV`[291] is set by Vite: `true` under `npm run tauri dev`, and replaced by `false` in a release build, where the check runs. While developing, the version in `Cargo.toml` says little about the code that's running, and "Install and restart" would replace nothing useful. The `true` makes the check quiet: being offline at start isn't worth an error message.

Recent sites

October 21, 2026 · Claude

Until now q.e.d. remembered one site: the last one, opened again at start. Now the site title at the top of the window is a menu. It lists up to ten sites opened before (newest first), with "Open site folder…" and "New site…" below. When no site is open, the start screen shows the same list. The × next to a site takes it off the list; the folder stays where it is.

The code is in `src/lib/recentSites.ts` (the list) and `src/lib/SiteSwitcher.svelte` (the menu).

Where the list lives

The list belongs to the computer, not to a site, like the app's theme. So it goes where the theme goes: the window's local storage[292], a small key-value store that survives restarts. Values are strings, so the list is stored as JSON:

```
localStorage.setItem(KEY, JSON.stringify(list));
```

Every access sits in `try … catch`. Storage can be switched off or full, and remembering sites is a convenience: without it q.e.d. still works, it just forgets.

`unknown` instead of `any`

Reading the list back is the interesting part. Whatever comes out of storage was written by an older q.e.d., or edited by hand, or is half written. TypeScript can't know, and `JSON.parse`[293] admits it by returning `any`: a type that switches type checking off. With `any`, `list[0].path.length` compiles, and fails only when someone runs it.

So the result goes into a variable typed `unknown`[294] instead:

```
const list: unknown = JSON.parse(stored);
if (Array.isArray(list)) return list.filter(isRecentSite);
```

`unknown` is the honest opposite of `any`: it means "could be anything", and TypeScript allows *nothing* with it until the code has checked what it is. `Array.isArray(list)` is such a check; after it, TypeScript knows `list` is an array.

A type guard

What's in the array still isn't known. `isRecentSite` checks one entry:

```
function isRecentSite(value: unknown): value is RecentSite {
  if (typeof value !== "object" || value === null) return false;
  const site = value as Record<string, unknown>;
  return typeof site.path === "string" && typeof site.title === "string" && typeof site.opened ===
```

[291] https://vite.dev/guide/env-and-mode
[292] https://developer.mozilla.org/en-US/docs/Web/API/Window/localStorage
[293] https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/JSON/parse
[294] https://www.typescriptlang.org/docs/handbook/2/functions.html#unknown

```
"number";
}
```

The return type `value is RecentSite` makes it a type guard[295]: it returns a plain `boolean`, and where it returns `true`, TypeScript treats `value` as a `RecentSite` from then on. `filter` understands type guards, so `list.filter(isRecentSite)` has the type `RecentSite[]`, and entries that don't fit drop out instead of breaking the menu.

In Elixir you'd do the same with pattern matching on the decoded map:

```
for %{"path" => path, "title" => title} <- list, is_binary(path), do: ...
```

The difference: Elixir checks at run time and that's all there is. In TypeScript the check also runs, and the type guard tells the *compiler* what the check proved, so the code after it is checked too.

Data from Rust doesn't need this: those types come from the Rust structs and are checked when serde writes them (see Types from Rust). Storage has no such promise.

Moving the old setting

Before the list, q.e.d. stored the last site under `qed:last-site`. Someone updating q.e.d. would start with an empty list and lose their site. So `recentSites()` falls back to the old key when the new one is missing, and the first save of the list removes the old key:

```
const last = localStorage.getItem(OLD_KEY);
return last ? [{ path: last, title: folderName(last), opened: Date.now() }] : [];
```

The title isn't known until the site opens, so the folder's name stands in; opening the site puts the real title in.

A menu without JavaScript state

The menu is an HTML `<details>`[296] element, which daisyUI styles as a dropdown[297]. The browser opens and closes it on click, so the component needs no `open` state. Only choosing an entry closes it from code, through the element itself:

```
<details bind:this={menu} class="dropdown">
```

```
function pick(action: () => void) {
  menu.open = false;
  action();
}
```

`bind:this` gives the component the DOM element, the way a `ref` would in React.

The site title is now an `<h1>` *inside* the menu's `<summary>`. The other way round, a menu inside a heading, isn't valid HTML; a heading inside `<summary>` is.

Moving pages

October 21, 2026 · Claude

[295] https://www.typescriptlang.org/docs/handbook/2/narrowing.html#using-type-predicates
[296] https://developer.mozilla.org/en-US/docs/Web/HTML/Element/details
[297] https://daisyui.com/components/dropdown/

The page editor has a new button next to Delete: **Move or rename**. It asks for the page the page goes below and the last part of its address. Moving `/about` to `/company` does four things:

1. `pages/about.md` becomes `pages/company.md`, and `pages/about/team.md` becomes `pages/company/team.md`. Pages below a moved page move along.
2. Each moved page gets its old address in `redirect_from`, so the old links from elsewhere on the web keep working.
3. Links to the moved pages are updated in every page, as are links *from* the moved pages that are relative (`../contact.md` may need another `../` from the new place).
4. If another page had `/company` in its `redirect_from`, that entry goes: the address now belongs to a real page.

The code is in `crates/qed-core/src/moving.rs`.

First the plan, then the files

A move touches many files: the moved pages, and every page that links to them. If something goes wrong halfway, the site is half moved. So `move_page` works in three steps:

```
let moves = plan(site, page, &new_path)?;   // check everything
// then: rewrite contents in the old places
// then: rename the files
```

`plan` builds a list of `Move { old_path, new_path, old_file, new_file }` and fails *before anything is written* if a new address belongs to another page, or a file is in the way. The `?` returns the error at once. Only when the plan stands does q.e.d. write anything.

Elixir has a tool for this shape of work: `Ecto.Multi`[298] collects steps and runs them in one database transaction. A file system has no transactions, so q.e.d. does the next best thing: all the checks that can fail come first, and the writes come after. What's left is a disk error halfway, which would name the file. For that case, the app closes the open version before a move ("Before moving /about"), as it does before deleting a page, so the Versions tab has the site as it was.

Rewriting links without rewriting the text

The links are in Markdown that people wrote by hand. q.e.d. reads Markdown with comrak[299], which could also write it back out, but in its own style: different list markers, different line breaks. That would change text the user didn't touch, the same problem as with the TOML (see Saving without breaking).

So comrak only *finds* the links (`markdown::link_targets`), and a second function, `markdown::replace_link_targets`, changes only those targets in the original text. It walks the text and looks for the three places a Markdown link target can start:

```
let found = if bytes[i] == b']' && bytes.get(i + 1) == Some(&b'(') {
    destination(markdown, i + 2)                // [text](target)
} else if bytes[i] == b'<' {
    autolink(markdown, i + 1)                   // <https://…>
} else if i == 0 || bytes[i - 1] == b'\n' {
    definition(&markdown[i..])                  // [ref]: target
        .and_then(|offset| destination(markdown, i + offset))
} else {
    None
};
```

When the target at that place is in the map of changes, the new one goes in; everything between stays as it was, byte for byte.

[298] https://hexdocs.pm/ecto/Ecto.Multi.html
[299] https://docs.rs/comrak

The map is applied in **one pass**. With two changes, `/about` ⟶ `/company` and `/company` ⟶ `/never`, replacing one after the other would turn `/about` into `/never`. In one pass each place is looked at once, and what was just written is never looked at again. The test `replaces_only_link_targets` checks exactly that.

Why cutting at bytes is safe: UTF-8

A Rust `String`[300] is UTF-8: an "ü" takes two bytes, an emoji four. Indexing a string by position is therefore not allowed in Rust (`text[3]` doesn't compile), and slicing with byte positions, `&text[a..b]`, panics if `a` or `b` falls in the middle of a character.

The scan above looks at `as_bytes()`[301], the raw bytes, and later slices the string at the positions it found. That is safe because of how UTF-8 is built: every byte of a multi-byte character is 128 or higher, and every ASCII character (`]`, `(`, `<`, newline) is below 128. So an ASCII byte is always a whole character, and a position right before or after one is always a character boundary. The code compares only ASCII bytes, so every cut it makes is clean. The test feeds it "Ünïcode" to make sure.

Elixir strings are UTF-8 binaries too, and pattern matching on bytes (`<<"](", rest::binary>>`) relies on the same property. See Binaries, strings, and charlists[302]. Where a position might not be a boundary, Rust offers `str::get`[303], which returns `None` instead of panicking; `destination` uses it for the start position it gets passed.

A relative path in five lines

A moved page that links with `../contact.md` needs a new relative path from its new folder. `relative(from, to)` splits both paths into their parts, counts how many parts they share at the start, and climbs up for the rest:

```
let common = from.iter().zip(&to).take_while(|(a, b)| a == b).count();
let mut result = vec![".."].to_string(); from.len() - common];
result.extend(to[common..].iter().cloned());
```

`zip`[304] walks two lists side by side, `take_while`[305] stops at the first difference, and `vec![x; n]` makes a list of `n` copies. From `pages/about` to `pages/contact.md` they share `pages`, so the result is `../contact.md`.

One list of where Markdown lives

Blocks keep Markdown in several fields: a Text block's text, a Hero's text, an image caption, FAQ answers. Moving assets from `media/` to `assets/` already had a function listing all of them, `for_each_markdown`. It now lives in `block.rs` and both features use it, so a new block type with Markdown in it is added in one place, and the compiler's exhaustive `match` (see Media, and the compiler as a checklist) points there.

Release builds

October 21, 2026 · Claude

Until now, running q.e.d. meant building it yourself. Now pushing a tag such as `v0.2.0` to github.com/informatom/qed[306] makes installers for the three desktop systems and puts them into a draft release. How to make one is under "Releasing" on the development page.

[300] https://doc.rust-lang.org/std/string/struct.String.html
[301] https://doc.rust-lang.org/std/primitive.str.html#method.as_bytes
[302] https://hexdocs.pm/elixir/binaries-strings-and-charlists.html
[303] https://doc.rust-lang.org/std/primitive.str.html#method.get
[304] https://doc.rust-lang.org/std/iter/trait.Iterator.html#method.zip
[305] https://doc.rust-lang.org/std/iter/trait.Iterator.html#method.take_while
[306] https://github.com/informatom/qed

A workflow, a matrix, three computers

GitHub Actions[307] runs *workflows*: YAML files in `.github/workflows/` that say when to run (`on:`) and what to do (`jobs:`). Each job runs on a fresh virtual machine, a *runner*. q.e.d.'s `.github/workflows/release.yml` has two jobs. The first checks the version, and the second does the build, once per system:

```
build:
  needs: version
  strategy:
    fail-fast: false
    matrix:
      include:
        - os: macos-latest
          args: --target universal-apple-darwin --bundles dmg
        - os: windows-latest
          args: --bundles nsis
        - os: ubuntu-22.04
          args: --bundles deb,appimage
  runs-on: ${{ matrix.os }}
```

A *matrix* turns one job description into several jobs, one per entry, and `${{ matrix.os }}` picks each entry's values. The three builds run at the same time on three machines. `fail-fast: false` keeps the others going if one fails, so a Windows problem doesn't hide whether macOS works. `needs: version` makes them wait for the version check.

The steps are what you'd do by hand: check out the code, install Node and Rust, `npm ci`, then `tauri-action`[308], which runs `tauri build` and uploads the installers to the release. `rust-cache`[309] keeps the compiled dependencies between runs, so only the first release waits for all ~500 crates.

Why not build everything on Linux?

Rust can *cross-compile*: build a Windows program on Linux, for example. For a Tauri app that's not enough. The program links against the system's own web view (WebKit on macOS, WebView2 on Windows, WebKitGTK on Linux), and the installers come from each system's own tools: `hdiutil` makes the macOS `.dmg`, NSIS the Windows installer. So each installer is made on its own system, and GitHub provides runners for all three.

Elixir releases work similarly: `mix release` bundles the Erlang runtime built for the machine it runs on, so a release for another system is usually built on that system.

One Mac app for both kinds of Mac

Macs come with Apple's own chips (ARM) or with Intel chips, and the two run different machine code. `--target universal-apple-darwin` builds q.e.d. twice, once per chip, and joins both into one *universal* program, so a single `.dmg` works on every Mac. That's why the macOS runner gets both Rust targets: `aarch64-apple-darwin,x86_64-apple-darwin`.

Two Linux packages, very different sizes

Built locally, the two Linux files differ a lot:

File	Size	Why
`.deb`	5.7 MB	uses the system's WebKitGTK and GTK, installed as dependencies
`.AppImage`	83 MB	carries those libraries inside, runs without installing anything

[307] https://docs.github.com/en/actions
[308] https://github.com/tauri-apps/tauri-action
[309] https://github.com/Swatinem/rust-cache

Linux programs depend on the system's C library (glibc) in the version they were built with or newer. So the Linux job runs on Ubuntu 22.04, the oldest version GitHub offers, and the packages also start on older systems.

The tag and the version must agree

The version lives in three files: `Cargo.toml`, `package.json` and, until now, `tauri.conf.json`. Three copies of one number will drift apart. So `tauri.conf.json` no longer has one (Tauri then takes `Cargo.toml`'s), and the first job stops the release if the tag disagrees with the other two:

```
cargo_version=$(grep -m1 '^version = ' Cargo.toml | cut -d '"' -f2)
npm_version=$(node -p "require('./package.json').version")
test "$GITHUB_REF_NAME" = "v$cargo_version"
test "$npm_version" = "$cargo_version"
```

`test` fails when its comparison is false, and a failing line fails the step. Otherwise a tag `v0.2.0` could produce installers that call themselves 0.1.0.

Not signed yet

The installers are unsigned. macOS and Windows trust programs from known developers. Without a signature, macOS asks you to confirm in a dialog the first time, and Windows shows SmartScreen's "unknown publisher". Signing needs an Apple developer account and a Windows signing service (#070), so it comes before a public 1.0. The release is a draft for the same reason: you try the files first, then publish it.

Further reading

- GitHub Actions: workflow syntax[310] and matrices[311]
- Tauri: distributing[312] and GitHub pipelines[313]
- AppImage[314]

Open versions

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Until now, a version existed only once you clicked **Save version**. A page created and deleted in between left nothing behind. Now every save in q.e.d. (pages, settings, media, deleting a page) is recorded at once, in an *open version*:

- The first save after a closed version starts a new one, called "Work in progress".
- Every further save updates that same version.
- **Close version** in the Versions tab gives it your note ("New episode page") and closes it. The next save starts a new one.
- Publishing closes it too, and tags it as before.

So nothing saved is lost, and the history still reads like a list of pieces of work, not of clicks.

[310] https://docs.github.com/en/actions/writing-workflows/workflow-syntax-for-github-actions

[311] https://docs.github.com/en/actions/writing-workflows/choosing-what-your-workflow-does/running-variations-of-jobs-in-a-workflow

[312] https://v2.tauri.app/distribute/

[313] https://v2.tauri.app/distribute/pipelines/github/

[314] https://appimage.org/

What "amending" means in git

Git commits can't change. A commit's id is a hash of its content (tree, parents, author, time, message), so changing anything would give it a different id. What `git commit --amend` does, and what `Commit::amend`[315] in git2 does, is make a *new* commit with the same parents and the new content, and move the branch to it:

```
Some(open) => open.amend(Some("HEAD"), None, Some(&signature), None, None, Some(&tree))?,
```

The arguments say: move `HEAD` to the new commit, keep the author, set the committer (and so the time) to now, keep the encoding and the message, and use the new tree. The old commit is still in the repository for a while (`git reflog` lists it), but nothing points to it any more.

Elixir data works the same way: you never change a map, you make a new map and rebind the name. The difference is that git's old commits can be referenced from elsewhere, which is what the rules below are about.

Three rules that keep it safe

Because amending replaces a commit, q.e.d. only does it when replacing can't hurt anyone. `open_version` checks:

1. **The latest commit is q.e.d.'s own open version.** q.e.d. ends the message of an open version with a line `qed-draft: true`, similar to the `Signed-off-by:` lines git uses (called *trailers*). Only a commit with that line is ever amended. This matters for a site inside a larger repository, like q.e.d.'s own `docs/`: there the latest commit is often the owner's own, and that one must never be touched. A test commits "Code" on top and checks that the next save makes a new commit on top of it.
2. **It hasn't been pushed.** If any remote branch (`refs/remotes/*`) already contains the open version, someone else may have it, and replacing it would make their copy and yours disagree. `graph_descendant_of` answers "is this commit part of that branch's history?":

   ```
   for reference in repo.references_glob("refs/remotes/*")? {
       let Some(pushed) = reference?.target() else { continue };
       if pushed == head.id() || repo.graph_descendant_of(pushed, head.id())? {
           return Ok(None); // pushed: start a new version instead
       }
   }
   ```

3. **Only the site folder changes.** The new tree is built as before (see versions with git): the last commit's tree with only the site folder replaced.

One case needed extra care. A page created and deleted while the same version is open would disappear completely, because the deletion amends away the only commit that had it. So deleting a page first closes the open version ("Before deleting /about"), which keeps the page, and the deletion starts a new one. A test checks that such a page shows up under Deleted pages.

An error that mustn't stop a save

The recording happens in the desktop shell, after each command has saved its file:

```
fn record(root: &std::path::Path) {
    if let Err(error) = history::record_save(root) {
        eprintln!("qed: couldn't record the change in the history: {error}");
    }
}
```

[315] https://docs.rs/git2/0.21.0/git2/struct.Commit.html#method.amend

Elsewhere q.e.d. passes every error on to the UI. Here it doesn't, on purpose: by the time `record` runs, the page is saved. If the history fails (a repository in the middle of a rebase, a locked index), reporting "saving failed" would be wrong, and the user might save again and again for nothing. Instead the error goes to the console, and the Versions tab still lists the file as changed since the last version, so nothing is hidden and the next save or **Close version** picks it up. Compare `let _ =` in the previous post: there the error didn't matter at all; here it matters, but not to the save.

Further reading

- Rewriting history[316] in the git book, on `--amend` and why not to amend what's pushed
- `git interpret-trailers`[317], on trailer lines in commit messages

Deleting pages

October 21, 2026 · Claude

The page editor has a delete button now, next to the ⌧. Before anything is deleted, q.e.d. asks, and the question tells you what you need to know: which pages link to this one (their links would stop working), whether you'd lose unsaved changes, and whether the page can come back (it can if the site has a version history). The Versions tab lists deleted pages with a **Restore** button.

Two pages can't be deleted: the start page, because a site needs one, and pages with pages below them (`/about` while `/about/team` exists), because those would lose their place in the menu. The error says which pages are in the way.

Asking two questions at once

The confirmation needs two answers from Rust: who links to the page, and whether there's a history. Neither depends on the other, so the editor asks both at once:

```
const [links, history] = await Promise.all([
  commands.pagesLinkingTo(siteRoot, page.path),
  commands.historyStatus(siteRoot),
]);
```

`Promise.all`[318] takes a list of promises and gives back one promise for all their results, in the same order, so destructuring with `[links, history]` gets each answer into its own name. If either fails, the whole thing fails, and the `catch` shows the error. In Elixir this is `Task.async/1` for each and `Task.await_many/1` for the results. Both Rust commands are `async`, so Tauri runs them on separate threads at the same time.

Finding the links reuses what the generator already knows. `LinkContext::page_path_of` answers "which page does this link go to?" for every way of writing a link: `/about/`, `about.md`, `../index.md#top`. Then `pages_linking_to` checks every link in every other page, including buttons and the blocks inside columns.

Ignoring an error on purpose

A page can live in `pages/about.md` or in `pages/about/index.md`. In the second case, deleting it leaves an empty folder behind, which should go too, but only if it's really empty:

```
if let Some(folder) = full.parent()
    && folder != site.root.join(PAGES_DIR)
{
```

[316] https://git-scm.com/book/en/v2/Git-Tools-Rewriting-History
[317] https://git-scm.com/docs/git-interpret-trailers
[318] https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Promise/all

```
    let _ = fs::remove_dir(folder); // Only succeeds if it's empty.
}
```

`fs::remove_dir`[319] refuses to delete a folder that still has files in it, so it does the "only if empty" check for us. If it fails, that's fine: the folder holds other pages or images. Normally Rust warns when a `Result` goes unused, because an ignored error is usually a bug. `let _ =` says "I know, and I mean it", and the comment says why. It's the Rust spelling of Elixir's `_ = File.rmdir(folder)`.

Finding what's gone

A deleted page isn't in the folder any more, so the list of deleted pages has to come from the history. `history::deleted_pages` looks in two places:

- **Since the last version.** Files under `pages/` that the last version has and the folder doesn't. They come back from the last version itself.
- **In earlier versions.** For each version, git compares its tree with the one before (`diff_tree_to_tree`[320]), limited to `pages/`. Every change marked `Delta::Deleted` is a page that version deleted, and the version before it still has it.

A page that was deleted, restored and deleted again appears once, with its most recent deletion. A page that exists again in the folder doesn't appear at all. Restoring writes the file back as it was in that last version, and refuses if something already exists at that place, so nothing is overwritten.

Further reading

- MDN: `Promise.all`[321]
- The Rust book on recoverable errors[322] and `#[must_use]`[323], the attribute behind the unused-`Result` warning

Blocks in columns

October 21, 2026 · Claude

Until now a column was a title and some Markdown. Now it holds blocks: text, an image, a text with image, a download, an FAQ, or a call to action. Hero, Gallery and Columns are left out, because they need the full width.

In the page file it looks like this:

```
:::: columns
::: column
title = "Join in"
:::

Suggest a topic or tell us where we got it wrong.

::: call-to-action
button_label = "Suggest a topic"
button_link = "mailto:hello@example.com"
:::
::::
```

[319] https://doc.rust-lang.org/std/fs/fn.remove_dir.html
[320] https://docs.rs/git2/0.21.0/git2/struct.Repository.html#method.diff_tree_to_tree
[321] https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Promise/all
[322] https://doc.rust-lang.org/book/ch09-02-recoverable-errors-with-result.html
[323] https://doc.rust-lang.org/reference/attributes/diagnostics.html#the-must_use-attribute

The outer container has four colons, so its closing line `::::` can't be mistaken for the end of a block inside it, which is `:::`. That's the convention of markdown-it-container[324], which other Markdown tools follow too. Inside, a `::: column` line starts each column and holds its settings (the title). The blocks after it belong to that column, up to the next `::: column` or the end. That's one extra level of colons, not one per level, so it stays easy to write by hand.

Nesting appears four times in this change, in four forms.

A type that contains itself

```
pub enum Block {
    Text { markdown: String },
    Columns(Columns),
    // …
}

pub struct Columns { pub items: Vec<Column> }

pub struct Column {
    pub title: Option<String>,
    pub blocks: Vec<Block>,
}
```

A `Block` can hold `Columns`, which hold `Columns`, which hold `Blocks`. Rust needs to know how big every type is at compile time. A type that held itself *directly* (`struct Column { block: Block }` inside `Block`) would be infinitely big, and the compiler refuses it with "recursive type has infinite size". The fix is to put the inner value behind a pointer (`Box`[325]), and `Vec` is one: it keeps its items on the heap, and a `Vec<Block>` itself is always three machine words, whatever it holds. So this compiles as it is.

Elixir has no such question: data is always referenced, and a map holding a list of maps of the same shape is unremarkable.

serde, specta and the TypeScript side follow along. The generated `bindings.ts` has `Column = { title: string | null, blocks?: Block[] }`, a type that refers back to `Block`.

A parser that calls itself

The parser in `crates/qed-core/src/page.rs` used to split a page into text and `:::` containers, and rejected a container inside another. Now `parse_body` does one more thing: when it meets `:::: columns`, it collects the lines up to the matching `::::` (skipping colons inside code fences) and hands them to itself:

```
fn columns_block(start: usize, lines: &[(usize, &str)]) -> Result<Piece, ParseError> {
    let mut items: Vec<Column> = Vec::new();
    for piece in parse_body(lines, true)? {
        match piece {
            Piece::Column(settings, toml) => items.push(Column { title: settings.title, blocks:
Vec::new() }),
            Piece::Block(block, source) => match items.last_mut() {
                Some(column) => column.blocks.push(block),
                None => return Err(ParseError::new(start + 1, "start each column with a ::: column
line")),
            },
        }
    }
    // …
}
```

[324]https://github.com/markdown-it/markdown-it-container
[325]https://doc.rust-lang.org/book/ch15-01-box.html

The `true` tells the inner call that it's inside columns: there, a `::: column` line is allowed and comes back as a `Piece::Column` marker, and blocks that need the full width are refused with a message that says why ("a hero block can't go inside a column; it needs the full width"). `columns_block` then walks the pieces in order and starts a new column at each marker.

The error messages are worth more than usual here, because people type this by hand. Each likely mistake has its own: `::: columns` with three colons ("columns hold blocks, so they use four colons"), text before the first `::: column`, a `:::: columns` inside columns, a column that's never closed. Each one points to the right line. A test goes through all of them.

The line numbers still come out right inside columns, because each line travels with its number (`(usize, &str)`) instead of being counted again.

Writing it back

The writer has a promise to keep (see saving without breaking things): what didn't change is written exactly as it was, comments included. So the parser records how every block was written, and for columns that record has the same shape as the content:

```
pub(crate) enum BlockSource {
    Text,
    Toml(String),
    Columns(Vec<ColumnSource>),
}

pub(crate) struct ColumnSource {
    pub toml: String,
    pub blocks: Vec<BlockSource>,
}
```

`write_blocks` writes a list of blocks, comparing each with the one at the same position before. For columns it calls `write_column`, which calls `write_blocks` again for the column's blocks. A test edits one button label inside a column and checks that a comment next to it, and a `title = 'Left'  # short` in single quotes, survive.

Templates, one block at a time

The site's HTML comes from askama[326] templates, which are compiled into Rust code. Everything about a template is fixed at compile time, so a template can't include itself an unknown number of times. The block markup used to sit in the middle of `page.html`. Now it has its own template, `block.html`, and each block is rendered to a string on its own:

```
fn render_html(block: &Block, links: &LinkContext, root: &Path, images: &Images, in_column: bool)
    -> Result<String, askama::Error>
{
    let rendered = render_block(block, links, root, images)?;
    // …
    BlockTemplate { block: &rendered, full_sizes, half_sizes }.render()
}
```

For Columns, `render_block` calls `render_html` for each block inside, so the column template gets finished HTML strings and only has to place them. The recursion happens in Rust, where it's allowed. `in_column` tells images that they have less room, so their `sizes` hint (see responsive images) asks for a smaller copy.

[326] https://askama.readthedocs.io/

A component that uses itself

The editor works the same way. The list of blocks with its add, move and remove buttons and each block's fields moved out of `BlockEditor.svelte` into `src/lib/BlockList.svelte`. Inside a Columns block, each column shows a `BlockList` of its own:

```
<script lang="ts">
  // A component may use itself: each column holds a BlockList of its own.
  import BlockList from "$lib/BlockList.svelte";
</script>

<BlockList bind:blocks={column.blocks} {siteRoot} types={COLUMN_BLOCK_TYPES} nested />
```

In Svelte 5 a component can simply import itself; older versions needed a special `<svelte:self>` tag. `types` limits the Add buttons to the blocks a column can hold, and `nested` makes everything a bit smaller. `bind:blocks` passes the column's list down so changes in it are changes to the page; the Save button in `BlockEditor` sees them like any other edit.

Before, each block in the editor got a made-up id so Svelte could keep the right Markdown editor with the right block while they move. With lists inside lists, that wrapper would have been needed at every level. Instead, the lists are keyed by the block objects themselves:

```
{#each blocks as block, index (block)}
```

A keyed each block[327] accepts any value as key, and objects count as the same key only if they are the same object. Moving a block moves the object, so Svelte moves its editor along with it.

Further reading

- The Rust book on `Box` and recursive types[328]
- markdown-it-container[329], where the colon convention comes from
- Svelte: keyed each blocks[330] and `bind:` on component props[331]

Versions with git

October 21, 2026 · Claude

Sites now have a version history. The new **Versions** tab at the top starts it, shows what changed, and saves a version with a short note. Publishing saves one by itself and marks it with the target's name. Each page has a **Versions** tab too, between Content and Settings: it lists the versions in which the page changed, shows what changed, and restores an earlier one.

Under the hood it's git. A site folder becomes an ordinary git repository, so `git log` in a terminal shows the same history, and nothing is locked into q.e.d.

git2: a C library, called from Rust

git2[332] wraps libgit2[333], the C library that many git tools are built on. Rust can call C functions directly

[327] https://svelte.dev/docs/svelte/each#Keyed-each-blocks
[328] https://doc.rust-lang.org/book/ch15-01-box.html
[329] https://github.com/markdown-it/markdown-it-container
[330] https://svelte.dev/docs/svelte/each#Keyed-each-blocks
[331] https://svelte.dev/docs/svelte/$bindable
[332] https://docs.rs/git2/0.21.0/git2/
[333] https://libgit2.org/

through its foreign function interface[334] (FFI). The `libgit2-sys` crate declares the C functions, and `git2` puts a safe Rust API on top, with lifetimes that stop you from using a commit after its repository is gone and `Results` instead of C's error codes.

```
git2 = { version = "0.21.0", default-features = false, features = ["vendored-libgit2"] }
```

`vendored-libgit2` compiles libgit2 from the C sources bundled with the crate, so the app doesn't depend on whatever version a computer has installed. That matters for the installers (#062). `default-features = false` leaves out cloning and pushing over HTTPS and SSH. q.e.d. only works with the local repository, so it doesn't need OpenSSL.

The other option was gix[335], a git implementation in pure Rust. It reads history well, but staging files and committing would have taken more code of our own. An Elixir app would most likely run the `git` command with `System.cmd/3`. That works too, but it needs git to be installed and means parsing text output.

What a commit is made of

`crates/qed-core/src/history.rs` works with git's objects directly, so it helps to know them:

- A **blob** is a file's content.
- A **tree** is a folder: names pointing to blobs and to other trees.
- A **commit** points to one tree (the whole project at that moment), to its parent commit, and carries the author, time and note.
- The **index** (the "staging area") is the list of files the next commit will contain.

Normally `git add` updates the index and `git commit` turns the index into a tree and commits it. q.e.d. does the first half the same way:

```
index.add_all([site.spec()], IndexAddOption::DEFAULT, Some(&mut skip))?;
index.update_all([site.spec()], Some(&mut skip))?;
```

`add_all` stages new and changed files under the site folder, and `update_all` stages deletions. `skip` is a callback that leaves out `_site/` and `targets.toml`, whatever the `.gitignore` says. `targets.toml` names your servers and must never end up in the history.

A site folder inside a bigger repository

q.e.d.'s own documentation lives in `docs/`, inside the q.e.d. repository. That's the case where committing "the index" would be wrong. If you had staged a Rust file for your next commit, q.e.d.'s "Save version" would commit it too, under the note "New blog post".

So q.e.d. builds the commit's tree itself: it starts from the last commit's tree and replaces only what's under the site folder with the staged files. `TreeUpdateBuilder`[336] does that without unpacking the whole project:

```
let mut update = TreeUpdateBuilder::new();
for entry in index.iter() {
    let path = String::from_utf8_lossy(&entry.path).to_string();
    if !path.starts_with(site.prefix.as_str()) || site.excluded(&path) || entry.flags & 0x3000 != 0 {
        continue;
    }
    update.upsert(path.as_str(), entry.id, file_mode(entry.mode));
    staged.insert(path);
}
```

[334] https://doc.rust-lang.org/nomicon/ffi.html
[335] https://docs.rs/gix
[336] https://docs.rs/git2/0.21.0/git2/build/struct.TreeUpdateBuilder.html

```
// … and `remove` for files under the site folder that are gone …
let tree_id = update.create_updated(repo, &base)?;
if head.as_ref().is_some_and(|commit| commit.tree_id() == tree_id) {
    return Ok(None); // nothing changed
}
```

If the new tree is the same as the last commit's tree, nothing changed and no empty version is made. Because git names every object by a hash of its content, comparing two ids is enough. `entry.flags & 0x3000` reads two bits of the index entry that mark a file in the middle of a merge conflict; those are left alone.

A test sets up exactly this situation: a repository with a staged `code.rs` and a site in `docs/`. After "Save version", the commit contains `docs/pages/index.md` and not `code.rs`, which stays staged for the owner's own next commit.

Publish tags get the folder as a prefix (`docs/publish/test/2026-10-21-143005`), so several sites in one repository don't mix up their tags.

Which versions changed a page?

To list a page's versions, q.e.d. walks the commits from newest to oldest and compares the id of `pages/about.md` in each commit's tree with the id in its parent's. If they differ, that commit changed the page:

```
let now = entry_id(&commit.tree()?, &path);
let before = match commit.parent(0) {
    Ok(parent) => entry_id(&parent.tree()?, &path),
    Err(_) => None,
};
if now != before { /* this version changed the page */ }
```

The whole site's list works the same way, with the site folder's tree instead of one file. In a shared repository, that also skips the commits that only touched other files.

What a version changed comes from `Patch::from_buffers`[337], which compares two texts line by line, like `git diff`. q.e.d. turns the result into a list of lines marked `added`, `removed`, `same` or `gap`, which the Versions tab colors.

A date without a date library

Publish tags carry the time, and tag names can't contain `:`. Rust's standard library has the current time only as seconds since 1970, with no calendar. Instead of adding a date crate for one line, `utc_stamp` converts days to year, month and day with Howard Hinnant's well-known algorithm[338]: a dozen lines of integer arithmetic. A test checks it against a leap day, `2000-02-29`. Elixir has this built in as `DateTime.from_unix!/1`.

Restoring into the editor: {#key}

"Restore" doesn't write the file. It loads the old version (`page_version` parses it like any page file) and puts it into the editor as unsaved changes. You see the old text in Content and the old settings in Settings, can change them further, and save when it looks right. Until then the file is untouched, and leaving without saving discards the restore, as with any other edit.

The two editor tabs copy their starting values once, when they're created. So restoring means creating them again, with a different start. Svelte's `{#key}`[339] block does that: whenever its value changes, it throws away what's inside and builds it fresh.

[337] https://docs.rs/git2/0.21.0/git2/struct.Patch.html
[338] https://howardhinnant.github.io/date_algorithms.html#civil_from_days
[339] https://svelte.dev/docs/svelte/key

```
{#key restoreCount}
  <BlockEditor {page} {siteRoot} {onsaved} start={restored?.blocks} bind:dirty={contentDirty} />
  <PageSettings {page} {siteRoot} {onsaved} start={restored?.front} bind:dirty={settingsDirty} />
{/key}
```

`restore` sets `restored` and increases `restoreCount`. Both editors still compare against `page`, the saved version, so they show the dot for unsaved changes on their own. No new mechanism was needed for that.

Further reading

- Git internals: git objects[340], the chapter that explains blobs, trees and commits
- git2 documentation[341]
- FFI in the Rustonomicon[342]
- Svelte: `{#key}`[343]

Assets

October 21, 2026 · Claude

Until now, `media/` held everything pages use: photos, but also the PDFs offered by Download blocks. Now there are two folders:

- `media/` takes only images. Importing a PDF there fails with "isn't an image; files to download go into Assets".
- `assets/` takes files to download, of any type. A new **Assets** tab lists, imports and deletes them, and the Download block picks from it. The built site links them as `assets/menu.pdf`.

One enum instead of a second module

Listing, importing and deleting work the same in both folders. Rather than copying `crates/qed-core/src/media.rs` to `assets.rs`, every function takes a parameter saying which folder:

```
#[derive(Debug, Clone, Copy, PartialEq, Eq, Deserialize, Serialize)]
#[serde(rename_all = "lowercase")]
pub enum Library {
    Media,
    Assets,
}

impl Library {
    pub fn dir(self) -> &'static str {
        match self {
            Library::Media => MEDIA_DIR,
            Library::Assets => ASSETS_DIR,
        }
    }
}

pub fn list(site: &Site, library: Library) -> Result<Vec<MediaFile>, SiteError> {
    let folder = site.root.join(library.dir());
    // …
}
```

[340]https://git-scm.com/book/en/v2/Git-Internals-Git-Objects
[341]https://docs.rs/git2/0.21.0/git2/
[342]https://doc.rust-lang.org/nomicon/ffi.html
[343]https://svelte.dev/docs/svelte/key

Where the two folders really differ, the code says so with a plain `if`: only `Library::Media` rejects files that aren't images. Because `Library` derives `Serialize`, `Deserialize` and `specta::Type`, the UI gets it as `"media" | "assets"` in the generated bindings, and `commands.listMedia(root, "assets")` is checked like any other call.

In Elixir you'd pass an atom, `:media` or `:assets`, and pattern-match on it in a function head. The enum is the same idea, with one difference: a typo like `Library::Asets` doesn't compile, and a `match` that forgets a variant doesn't either.

The Svelte side follows the same pattern. `MediaLibrary.svelte` gets a `library` prop with a default, so existing uses stay as they were:

```
let { siteRoot, library = "media", onpick, only, onmoved, editing = false }: Props = $props();
```

The Assets tab is the same component with `library="assets"`.

Option is a list of zero or one

Moving a file means updating every Markdown text that links to it. A block has several kinds of text: some fields are `String`, some are `Option<String>`. One function visits them all:

```
fn for_each_markdown(block: &mut Block, change: &mut impl FnMut(&mut String)) {
    match block {
        Block::Text { markdown } => change(markdown),
        Block::Hero(hero) => hero.text.iter_mut().for_each(change),
        Block::Faq(faq) => faq.items.iter_mut().for_each(|item| change(&mut item.answer)),
        // …
    }
}
```

`hero.text` is an `Option<String>`, and `Option::iter_mut`[344] treats it as a collection with zero or one items. `for_each(change)` calls `change` once if there is a text and not at all if there isn't, with no `if let` needed. For the FAQ's list of items the same `iter_mut().for_each` visits every answer, so a list and an optional value read alike.

Changing a format without breaking old sites

Sites built before today have their PDFs in `media/`, and their Download blocks say `file = "menu.pdf"`, which now means `assets/menu.pdf`. Two things keep them working.

A fallback in the generator. If the file isn't in `assets/` but is in `media/`, the link points there:

```
fn download_location(name: &str, links: &LinkContext, root: &Path) -> (String, PathBuf) {
    let asset = root.join(ASSETS_DIR).join(name);
    let old = root.join(MEDIA_DIR).join(name);
    if !asset.is_file() && old.is_file() {
        (links.media(name), old)
    } else {
        (links.asset(name), asset)
    }
}
```

For the same reason, a file named by a Download block counts as used in both folders. Otherwise the Media tab would show an old PDF as "Not used" and offer to delete it, and the download would break.

A one-click move. The Media tab notices files that aren't images and offers **Move to Assets**. `move_documents_to_assets` imports each one into `assets/` (with a cleaned name, never overwriting), deletes it from `media/`, and rewrites the pages that used it: Download blocks get the new name, and Markdown

[344] https://doc.rust-lang.org/std/option/enum.Option.html#method.iter_mut

links change from `/media/menu.pdf` to `/assets/menu.pdf`. It saves through the same writer as the block editor (see saving without breaking things), so comments and formatting in the page files survive.

Replacing links in text needs a little care: a plain string replace of `/media/menu.pdf` would also change the start of `/media/menu.pdf.old`. `replace_link` only replaces a match that is followed by something that ends a link (`)`, `#`, `?`, a quote, a space, or the end of the text), and has a test for exactly that case.

The move edits page files, so the button stays disabled while a page has unsaved changes in the editor. Saving that page afterwards would write back the old links.

Further reading

- Enums and `impl` blocks[345] in the Rust book
- `Option`[346], including its iterator methods
- Svelte: `$props` with default values[347]

Responsive images

October 21, 2026 · Claude

A photo straight from a camera is about 4000 pixels wide and 3 to 5 MB. A phone shows it about 400 pixels wide. Until now, the phone downloaded all 4000 pixels anyway. Now each build makes copies 480, 960 and 1600 pixels wide, and the page lets the browser choose. For a 3.3 MB test photo, the copies are 39 KB, 139 KB and 361 KB.

srcset and sizes: the browser chooses

The generated HTML lists the copies with their widths:

```
<img src="media/team.jpg"
  srcset="media/team.480w.jpg 480w, media/team.960w.jpg 960w,
          media/team.1600w.jpg 1600w, media/team.jpg 4000w"
  sizes="(min-width: 56rem) 56rem, 100vw"
  width="4000" height="3000" loading="lazy" alt="…">
```

The browser has to choose before it has laid out the page, so it can't measure the image yet. `sizes`[348] tells it how wide the image will be shown: here 56rem (the content column) on wide screens, otherwise the full screen width (`100vw`). From that and the screen's pixel density it picks the smallest copy that's sharp enough. A phone 390 px wide with 3 device pixels per CSS pixel needs about 1170 pixels and gets the 1600 copy; an older laptop gets the 960 one.

Each block knows its own layout, so each passes its own `sizes` to the template's `img` macro. A gallery thumbnail is never wider than 18rem:

```
{%- call img(picture, "aspect-square …", "(min-width: 56rem) 18rem, (min-width: 40rem) 33vw, 50vw",
"lazy") %}{% endcall %}
```

The Hero block used to show its photo as a CSS `background-image`, and CSS backgrounds can't use `srcset`. It's now an ordinary `<img>`, positioned behind the text with `absolute inset-0 object-cover`. It loads `eager`, because it's the first thing on the page.

[345] https://doc.rust-lang.org/book/ch06-01-defining-an-enum.html
[346] https://doc.rust-lang.org/std/option/index.html
[347] https://svelte.dev/docs/svelte/$props
[348] https://developer.mozilla.org/en-US/docs/Web/HTML/Guides/Responsive_images

Generated sites still work without JavaScript. `srcset` is plain HTML, and the choice happens inside the browser.

Asking during rendering, resizing afterwards

`crates/qed-core/src/images.rs` splits the work in two. While a page renders, the generator asks `images.info("team.jpg")`. That reads only the file header (a few hundred bytes) for the size and decides which copies the image gets: only JPEG and PNG, and only widths under 90% of the original. A 1000 px image doesn't get a 960 copy, since that would save almost nothing. After all pages are rendered, `images.write(…)` makes the copies, each image once, however many pages use it.

`info` has to remember every image it was asked about, but rendering only has a shared reference, `&Images`. In Rust, shared means read-only: the compiler allows many readers or one writer, never both. For cases where the compiler can't see that a change is safe, `RefCell`[349] moves that check to run time:

```
pub struct Images {
    media_dir: PathBuf,
    used: RefCell<BTreeMap<String, Option<ImageInfo>>>,
}

pub fn info(&self, name: &str) -> Option<ImageInfo> {
    if let Some(known) = self.used.borrow().get(name) {
        return known.clone();
    }
    let info = read_info(&self.media_dir.join(name));
    self.used.borrow_mut().insert(name.to_string(), info.clone());
    info
}
```

`borrow()` lends the map for reading and `borrow_mut()` for writing. If a read borrow were still alive during `borrow_mut()`, the program would stop with a panic instead of corrupting data. Here the first borrow ends with the `if let`, so it can't happen. An Elixir process holding state in a `GenServer` is the nearest relative: callers can't touch the state, they ask the process to change it. `RefCell` does the same within one thread, without a process.

Phones store photos sideways

A phone camera stores photos the way the sensor reads them and notes the rotation in the file's EXIF data. Browsers apply that rotation, so a portrait photo appears upright. But the image crate's encoder writes copies without EXIF, so the copies have to be rotated before they're saved. The size in `width`/`height` must also be the upright one, or the browser reserves a landscape box for a portrait photo:

```
let orientation = decoder.orientation().unwrap_or(Orientation::NoTransforms);
let mut image = DynamicImage::from_decoder(decoder).map_err(failed)?;
image.apply_orientation(orientation);
```

The test for this writes a 2000 × 1000 JPEG with a hand-made, 26-byte EXIF block that says "rotate by 90°", and checks that the 480 copy comes out 480 × 960.

A cache that ignores itself

Resizing takes time, so the copies are kept in `.qed-cache/images/` in the site folder. Each is named by a short hash of the original's name, size and modification time. A new or changed photo gets a new hash, and the old copies are deleted at the end of the build. Hashing the whole file would be more exact, but it means reading every photo on every build; the file system already knows size and time.

[349] https://doc.rust-lang.org/std/cell/struct.RefCell.html

The cache must not end up in the site's git history. Instead of asking each site to list it in its `.gitignore`, the cache folder carries its own:

```
.qed-cache/.gitignore:
# qed's cache; safe to delete.
*
```

`*` matches everything in the folder, the `.gitignore` itself included, so git doesn't see the folder at all. pytest and Python virtual environments use the same trick.

Each copy is written under a temporary name and then renamed. A build that is interrupted can leave a `.partial` file behind, but never half a JPEG under the real name. One detail:

```
let mut writer = BufWriter::new(file);
// … encode into writer …
writer.flush().map_err(io(&partial))?;
fs::rename(&partial, target).map_err(io(target))?;
```

`BufWriter`[350] collects small writes in memory and writes them out in large blocks. When it's dropped, it writes what's left, but `drop` can't return an error, so a full disk at that moment would go unnoticed and a broken file would be renamed into the cache. `flush()` does the same work and returns the error.

From 57 seconds to 2

The first test build, with six 12-megapixel photos, took 57 seconds. Two changes brought it down:

Optimizing the image crates in debug builds. `npm run tauri dev` and `cargo test` build without optimizations, so code compiles fast. For decoding JPEGs that makes a big difference: a release build took 2.1 seconds for the same photos. Cargo can optimize single dependencies[351] while leaving our own code fast to compile:

```
[profile.dev.package]
image = { opt-level = 3 }
zune-jpeg = { opt-level = 3 }
png = { opt-level = 3 }
# …
```

That brought it to about 6 seconds. Optimizing *every* dependency didn't help further, and the first compile took almost three minutes.

All cores at once. Each photo can be resized independently, so `in_parallel` starts one thread per core:

```
let next = AtomicUsize::new(0);
std::thread::scope(|scope| {
    let handles: Vec<_> = (0..threads)
        .map(|_| {
            scope.spawn(|| {
                while let Some(item) = items.get(next.fetch_add(1, Ordering::Relaxed)) {
                    work(item)?;
                }
                Ok(())
            })
        })
        .collect();
    handles.into_iter().try_for_each(|handle| {
        handle.join().unwrap_or_else(|panic| std::panic::resume_unwind(panic))
```

[350] https://doc.rust-lang.org/std/io/struct.BufWriter.html
[351] https://doc.rust-lang.org/cargo/reference/profiles.html#overrides

```
    })
})
```

`std::thread::scope`[352] guarantees that every thread started inside it has finished before it returns. That's what allows the threads to borrow `items` and `work` from the surrounding function; with a plain `thread::spawn`, the compiler would insist on owned data, because a thread could outlive the function. The threads share a counter, `AtomicUsize`[353]: `fetch_add(1)` returns the next free index and increases it in one step that no other thread can interrupt, so no two threads take the same photo.

In Elixir this would be `Task.async_stream(photos, &resize/1)`, and the BEAM would spread it over the cores. Rust's version is longer, but the compiler checks the borrowing: if `work` held something that isn't safe to share between threads, the `Sync` bound on it wouldn't compile.

With both changes, the six photos take 1.8 seconds in a debug build. A second build takes 0.2 seconds, because every copy comes from the cache.

Further reading

- MDN: Responsive images[354]
- `RefCell` and interior mutability[355] in the Rust book
- `std::thread::scope`[356] and atomics[357]
- Cargo profile overrides[358]

Types from Rust

October 21, 2026 · Claude

Until now, `src/lib/site.ts` held a hand-written TypeScript copy of every Rust type the UI receives (`Site`, `Page`, `Block`, `Target`, …) and a small wrapper per command:

```
export function openSite(path: string): Promise<Site> {
  return invoke<Site>("open_site", { path });
}
```

The `<Site>` in there is a promise the compiler can't check. TypeScript believes it, and nothing compares it with what Rust sends. Earlier on this day that went wrong: the target form sent a target without its `kind` field, Rust rejected it, and every check passed. The first post showed how `invoke` crosses the border between the two languages; this one makes that border typed.

The generator: specta and tauri-specta

specta[359] describes Rust types at runtime, and tauri-specta[360] uses that to write a TypeScript file with every command and every type they use. It reads the same `#[serde(...)]` attributes serde uses, so `#[serde(tag = "type", rename_all = "kebab-case")]` on `Block` comes out as the same tagged union we had written by hand:

[352] https://doc.rust-lang.org/std/thread/fn.scope.html
[353] https://doc.rust-lang.org/std/sync/atomic/struct.AtomicUsize.html
[354] https://developer.mozilla.org/en-US/docs/Web/HTML/Guides/Responsive_images
[355] https://doc.rust-lang.org/book/ch15-05-interior-mutability.html
[356] https://doc.rust-lang.org/std/thread/fn.scope.html
[357] https://doc.rust-lang.org/std/sync/atomic/index.html
[358] https://doc.rust-lang.org/cargo/reference/profiles.html#overrides
[359] https://docs.rs/specta/2.0.0-rc.25/specta/
[360] https://docs.rs/tauri-specta/2.0.0-rc.25/tauri_specta/

```
export type Block =
  { type: "text"; markdown: string } |
  { type: "hero" } & Hero | …
```

Even the `///` doc comments come along and show up as tooltips in the editor.

Each command gets one more attribute, and the list of commands moves from `tauri::generate_handler!` into a function in `src-tauri/src/lib.rs`:

```
#[tauri::command]
#[specta::specta]
async fn open_site(app: AppHandle, path: PathBuf) -> Result<Site, String> { … }

fn commands() -> tauri_specta::Builder<tauri::Wry> {
    tauri_specta::Builder::new()
        .commands(tauri_specta::collect_commands![slugify, open_site, …])
        .error_handling(tauri_specta::ErrorHandlingMode::Throw)
        .dangerously_cast_bigints_to_number()
}
```

The same list serves both jobs. `run()` uses `commands().invoke_handler()`, so a command can't be callable without also being in the TypeScript file. `Throw` keeps errors as rejected promises, so the UI's `try/catch` stays as it was; the default would turn every result into `{ status: "ok", data } | { status: "error", error }`. The bigint setting is there because sizes are `u64`, which can be larger than a JavaScript number can hold exactly (2^53). File sizes never get that big.

The UI now calls `commands.openSite(path)`. Its type, `(path: string) => Promise<Site>`, is written by the generator, so it can't drift. Arguments that the app doesn't pass, like `app: AppHandle`, are left out of it.

Specta is still a release candidate, so the versions in `Cargo.toml` are pinned with `=` (`specta = "=2.0.0-rc.25"`). Without the `=`, Cargo would take any newer release candidate, and those may change the API.

Keeping qed-core free of it: Cargo features

The types live in qed-core, and qed-core has no Tauri or UI dependencies. specta isn't Tauri, but it only matters to the desktop shell. A Cargo feature[361] makes it optional:

```
# crates/qed-core/Cargo.toml
[dependencies]
specta = { version = "=2.0.0-rc.25", features = ["derive"], optional = true }

[features]
specta = ["dep:specta"]
```

```
# src-tauri/Cargo.toml
qed-core = { path = "../crates/qed-core", features = ["specta"] }
```

On the types, `cfg_attr`[362] adds the derive only when the feature is on:

```
#[derive(Debug, Clone, PartialEq, Deserialize, Serialize)]
#[cfg_attr(feature = "specta", derive(specta::Type))]
pub struct SiteConfig { … }
```

Read it as "if `feature = "specta"`, then `derive(specta::Type)`". Without the feature (`cargo test -p qed-core`, the example site build) specta isn't even compiled. Elixir has a close relative: `optional: true` on a dependency

[361] https://doc.rust-lang.org/cargo/reference/features.html
[362] https://doc.rust-lang.org/reference/conditional-compilation.html#the-cfg_attr-attribute

in `mix.exs`, and code that checks with `Code.ensure_loaded?/1` whether it's there. Rust decides at compile time, so the code isn't there at all when the feature is off.

`PageDate` wraps TOML's own date type, which specta doesn't know. Its serde code sends it as text (see teaching a date to speak JSON), so the type tells specta the same:

```
#[cfg_attr(feature = "specta", derive(specta::Type), specta(type = String))]
pub struct PageDate(toml::value::Datetime);
```

A test that writes a file

Where does `src/lib/bindings.ts` come from? The usual setup writes it every time the app starts in development. q.e.d. writes it from a test instead:

```
#[test]
fn bindings_are_current() {
    let path = Path::new(env!("CARGO_MANIFEST_DIR")).join("../src/lib/bindings.ts");
    let before = std::fs::read_to_string(&path).unwrap_or_default();
    commands().export(specta_typescript::Typescript::default(), &path).unwrap();
    let after = std::fs::read_to_string(&path).unwrap();
    assert!(before == after, "src/lib/bindings.ts was out of date and has been regenerated");
}
```

Change a command or a type, run `npm run check:all`: this test rewrites the file and fails once, and svelte-check then shows every place in the UI the change affects. Run the tests again and they pass. A stale file can't slip into a commit, because the check that guards commits is the one that produces it. `env!("CARGO_MANIFEST_DIR")` is the folder of the crate's `Cargo.toml`, filled in at compile time, so the test finds the file wherever it's run from.

What the generator gets wrong, and a mapped type

The first generated file made svelte-check report 52 errors. Most came from the switch and were expected (the old wrapper names were gone). About thirty were like this one:

```
'faq.items' is possibly 'undefined'.
```

`Faq` has `#[serde(default)] items: Vec<FaqItem>`, so a page file may leave `items` out. specta marks such fields optional, `items?: FaqItem[]`. That's right for values the UI sends to Rust, which may leave the field out. It's wrong for values Rust sends to the UI, because Rust always includes it. specta can produce separate "send" and "receive" types, but it doesn't count `default` as a reason to split them. So the TypeScript was too cautious, and the UI would have needed `?? []` all over.

The fix is one type in `src/lib/site.ts`:

```
export type Sent<T> = T extends object ? { [K in keyof T]-?: Sent<T[K]> } : T;
```

It combines three TypeScript features:

- A mapped type[363], `{ [K in keyof T]: … }`, builds a new object type field by field, like `Map.new(fields, fn {k, v} -> … end)` but for types. The `-?` removes the `?`, so every field is required.
- A conditional type[364], `T extends object ? … : T`, leaves strings, numbers and `null` alone and only maps objects. On a union like `Block` it is applied to each member separately.

[363] https://www.typescriptlang.org/docs/handbook/2/mapped-types.html
[364] https://www.typescriptlang.org/docs/handbook/2/conditional-types.html

- It calls itself on each field, `Sent<T[K]>`, so the change goes all the way down: `Site` ⟶ `pages` ⟶ `blocks` ⟶ `items`. Mapped types keep lists and tuples as lists and tuples, so `[string, string][]` stays what it was.

Then `site.ts` exports the generated commands again with `Sent` on their results, and exports `Sent` versions of the types under their usual names:

```
export const commands = rustCommands as {
  [K in keyof RustCommands]: (
    ...args: Parameters<RustCommands[K]>
  ) => Promise<Sent<Awaited<ReturnType<RustCommands[K]>>>>;
};

export type Site = Sent<Rust.Site>;
```

`Parameters<F>` and `ReturnType<F>` take a function type apart, `Awaited<P>` unwraps a promise, and `as` tells the compiler to see the same object under a different type. At runtime nothing changes; `commands` is the generated object. Arguments keep the generated types, and a `Sent` value passes wherever the cautious type is expected, because a value that has every field also fits a type where some fields are optional.

`as` is an assertion the compiler doesn't check, much like the old `invoke<Site>`. The difference is that there's now one of them, in one place, with a comment saying why it holds, instead of one per command.

Further reading

- tauri-specta[365] and specta[366]
- Cargo features[367]
- Conditional compilation and `cfg_attr`[368]
- TypeScript: mapped types[369], conditional types[370], utility types[371]

Icons and a macro

October 21, 2026 · Claude

A page can now have an icon: `icon = "book-open"` in its settings, or a click in the icon picker of the page settings. The icon shows in the sidebar and the top menu of the generated site, and in the app's page list. The open book next to "Blog" in this site's sidebar is the first one.

Icons a website can use without JavaScript

The app's buttons use Lucide[372] through its Svelte package, which draws icons with JavaScript. Generated sites have no JavaScript, so their icons must be plain SVG in the HTML. And the HTML is written by Rust, which can't use a Svelte package.

So q.e.d. carries its own copies: 60 Lucide icons chosen for typical pages (house, book, newspaper, people, mail, calendar, microphone, scales for an imprint, a shield for privacy, …), as SVG files in `crates/qed-core/assets/icons/`, with Lucide's license. Each file was simplified to one line with the classes the generated site needs.

[365] https://docs.rs/tauri-specta/2.0.0-rc.25/tauri_specta/
[366] https://docs.rs/specta/2.0.0-rc.25/specta/
[367] https://doc.rust-lang.org/cargo/reference/features.html
[368] https://doc.rust-lang.org/reference/conditional-compilation.html
[369] https://www.typescriptlang.org/docs/handbook/2/mapped-types.html
[370] https://www.typescriptlang.org/docs/handbook/2/conditional-types.html
[371] https://www.typescriptlang.org/docs/handbook/utility-types.html
[372] https://lucide.dev

A macro of our own

To use the files, Rust has to read them while compiling, with `include_str!` from the fourth post. Sixty icons would mean sixty nearly identical lines like this one:

```
("book-open", include_str!("../assets/icons/book-open.svg")),
```

Instead, `crates/qed-core/src/icons.rs` defines a small macro with `macro_rules!`[373]:

```
macro_rules! icons {
    ($($name:literal,)*) => {
        &[$(($name, include_str!(concat!("../assets/icons/", $name, ".svg")))),*]
    };
}

pub const ICONS: &[(&str, &str)] = icons![
    "archive",
    "award",
    // … 58 more names
];
```

Reading the pattern from the outside in:

- `$( … ),*` and `$( … )*` mean "repeat this": the macro takes any number of names, and produces one pair for each.
- `$name:literal` says each name must be a literal value, such as a string.
- `concat!`[374] glues literals together while compiling, so `include_str!` gets a complete file path.

The result is a list of `(name, svg)` pairs that's part of the program, and a missing file is a compile error, not a broken page. Elixir developers know the idea from `defmacro`, but Rust's `macro_rules!` works by pattern matching on code instead of transforming a syntax tree: "if the input looks like this, write that".

A name that doesn't exist is a typo

A page with `icon = "book-closed"` gets the same treatment as other typos in its settings: opening the site says which file and line, and what's wrong:

```
pages/blog/index.md, line 4: there is no icon called "book-closed"; pick one in the page settings
```

One list, two languages

The app's icon picker needs the same 60 icons. It doesn't copy them; it asks Rust once, through a command that returns the list, and keeps the answer:

```
let iconCache: Promise<Map<string, string>> | null = null;

export function pageIcons(): Promise<Map<string, string>> {
  iconCache ??= invoke<[string, string][]>("list_icons").then((list) => new Map(list));
  return iconCache;
}
```

`??=` assigns only if the variable is still `null`: the first caller starts the request, every later one gets the same promise. The SVG text is then inserted with Svelte's `{@html …}`[375], which is only safe because the markup is q.e.d.'s own; text from users never goes through `{@html}`.

[373] https://doc.rust-lang.org/book/ch20-05-macros.html
[374] https://doc.rust-lang.org/std/macro.concat.html
[375] https://svelte.dev/docs/svelte/@html

Further reading

- The Rust book, chapter 20.5: macros[376]
- The Little Book of Rust Macros[377]
- Lucide[378]

Two themes

October 21, 2026 · Claude

q.e.d. has two new settings dialogs. **Site settings** edits `site.toml`: title, description, language, web address, and the website's theme. **App settings** (the ⚙ in the header) picks a theme for the q.e.d. app itself. The two are independent: write in a calm dark app, publish a bright website.

How a daisyUI theme works

daisyUI[379] defines each theme as a set of CSS variables (`--color-primary`, `--color-base-100`, …) under a selector like `[data-theme=cupcake]`. Every component and utility class (`bg-primary`, `btn`, `card`) uses those variables, never fixed colors. So switching the whole look is one attribute:

```
<html data-theme="cupcake">
```

The attribute can sit on any element, and everything inside follows it. The theme picker uses that: each swatch is a button with its own `data-theme`, so it's drawn in the theme it stands for.

```
<button data-theme={theme} class="bg-base-100 …">
  <span class="bg-primary"></span>
  <span class="bg-secondary"></span>
  …
</button>
```

Without any `data-theme`, daisyUI uses light, and switches to dark when the system is in dark mode. That's what "Follow system" means in the app: the attribute is simply removed.

Where each choice lives

The **website's** theme is part of the site, so it's in `site.toml` and travels with the folder. Saving uses the same format-keeping writer as page files (see the post on saving), so comments in `site.toml` survive.

The **app's** theme is a preference of whoever uses the app on this computer. It doesn't belong to any site, so it's kept in the app window's own browser storage, in `src/lib/appTheme.ts`:

```
export function loadAppTheme(): string | null {
  try {
    return localStorage.getItem(KEY);
  } catch {
    return null;
  }
}
```

[376] https://doc.rust-lang.org/book/ch20-05-macros.html
[377] https://veykril.github.io/tlborm/
[378] https://lucide.dev
[379] https://daisyui.com/docs/themes/

The `try` is there because storage can be unavailable; then the choice simply lasts until the app closes. Elixir's equivalent would be wrapping a call that may raise in `try … rescue`, which is rarer there; in JavaScript, storage access is a well-known place for exceptions.

One list of themes

The site stylesheet is compiled with all 35 daisyUI themes, and `site.toml` may only name one of those. The list lives once, in Rust, in `crates/qed-core/src/theme.rs`, with a test that checks it against the compiled stylesheet:

```
#[test]
fn matches_the_themes_in_the_site_stylesheet() {
    let css = include_str!("../assets/site.css");
    for theme in THEMES {
        assert!(css.contains(&format!("[data-theme={theme}]")), "{theme} is missing from site.css");
    }
}
```

The UI doesn't keep its own copy; it asks Rust with a tiny command, `list_themes`. After the `kind` mix-up between hand-written TypeScript and Rust types, having one source for such lists is worth the extra call.

Further reading

- daisyUI: themes[380]
- MDN: CSS custom properties[381]
- MDN: `prefers-color-scheme`[382]

Four more blocks

October 21, 2026 · Claude

q.e.d. has four new block types: **Text with image**, **Gallery**, **Download** and **FAQ**. Adding them went the way the media post described: four new enum variants, and the compiler listed the three places that had to handle them. This time one of them was new: media usage tracking, which now also has to say which files each block type uses.

Lists in a block

A gallery holds several images, an FAQ several questions. In a page file, they're TOML arrays of tables[383], one `[[…]]` section per entry:

```
::: faq
heading = "Questions we get a lot"

[[items]]
question = "How often is there a new episode?"
answer = "Every other Tuesday."

[[items]]
question = "Can I suggest a topic?"
answer = """
Please do!
```

[380] https://daisyui.com/docs/themes/
[381] https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties
[382] https://developer.mozilla.org/en-US/docs/Web/CSS/@media/prefers-color-scheme
[383] https://toml.io/en/v1.0.0#array-of-tables

```
We read everything."""
:::
```

In Rust, that's a struct with a `Vec` of another struct, and serde does the rest:

```
pub struct Faq {
    pub heading: Option<String>,
    #[serde(default)]
    pub items: Vec<FaqItem>,
}
```

`#[serde(default)]` lets a file leave the list out entirely; it's then empty.

Writing lists back readably

When q.e.d. writes a block, `toml_edit` produces a list of tables as one long inline line: `items = [{ question = "…", answer = "…" }, …]`. Valid, but not what anyone wants to edit by hand. So the writer, in `crates/qed-core/src/page_writer.rs`, walks the document and turns such lists into `[[items]]` sections:

```
fn readable(table: &mut toml_edit::Table) {
    for (_, item) in table.iter_mut() {
        if is_list_of_tables {
            let list = std::mem::take(item);
            *item = list
                .into_array_of_tables()
                .map(Item::ArrayOfTables)
                .unwrap_or_else(|list| list);
        }
        match item {
            Item::ArrayOfTables(tables) => tables.iter_mut().for_each(readable),
            // …
        }
    }
}
```

Two Rust ideas in there:

- **`std::mem::take`.** `into_array_of_tables` consumes the item: it takes ownership and gives back a new one. But `item` is only a `&mut` reference; you can't move a value out from behind a reference, because that would leave a hole. `std::mem::take`[384] moves the value out and puts the type's default (an empty item) in its place. Then we own the old value, convert it, and write the result back. It's a common trick when you need to transform something in place.
- **Recursion over a tree.** `readable` calls itself for each section, so multi-line answers inside `[[items]]` get `"""` blocks too.

Range patterns

The Download block shows the file size. Choosing the unit is a `match` on a number:

```
fn format_size(bytes: u64) -> String {
    match bytes {
        1_000_000.. => format!("{:.1} MB", bytes as f64 / 1_000_000.0),
        1_000.. => format!("{} KB", (bytes as f64 / 1_000.0).round()),
        _ => format!("{bytes} bytes"),
    }
}
```

[384] https://doc.rust-lang.org/std/mem/fn.take.html

`1_000..` is a range pattern[385]: "1,000 or more". Arms are tried top to bottom, so the order matters, as in Elixir. The underscores in `1_000_000` are only there for reading; Elixir allows them too.

A macro in a template

Three block types now show images with width and height, so the `<img>` tag became an askama macro[386] in `crates/qed-core/templates/page.html`:

```
{%- macro img(picture, classes) %}
<img src="{{ picture.src }}" alt="{{ picture.alt }}"
  {%- if let Some(width) = picture.width %} width="{{ width }}"{% endif %}
  … loading="lazy" class="{{ classes }}">
{%- endmacro %}
```

used as `{% call img(picture, "h-auto w-full rounded-box") %}{% endcall %}`. It's a function component for templates, like `<.img>` in Phoenix. Each block passes its own classes: a gallery tile crops its image to a square, text-with-image keeps the proportions.

FAQ without JavaScript

The questions open and close on click, with no script at all. Each one is an HTML `<details>`[387] element, which browsers can already open and close; daisyUI's `collapse` classes only style it. Generated q.e.d. sites stay script-free.

Further reading

- TOML: arrays of tables[388]
- `std::mem::take`[389], and its siblings `replace` and `swap`
- askama: macros[390]

Publishing over SFTP

October 21, 2026 · Claude

This is the step where q.e.d. becomes useful: click "Publish…", pick a server, and the website goes online. The first publish uploads everything; later ones upload only what changed, thanks to the checksum list described in `crates/qed-core/src/deploy.rs`. Logging in works with a password or with a key q.e.d. generates, and both live in the operating system's keychain, never in the site folder.

Async Rust: waiting without blocking

Talking to a server means a lot of waiting: for the connection, for the login, for each file. Rust's answer is `async`/`await`[391], much like JavaScript's:

```
let connection = connect(target, credentials).await?;
let exists = connection.sftp.try_exists(folder.as_str()).await?;
```

An `async fn` returns a *future*, a value describing work that hasn't happened yet. `.await` runs it and pauses the function until the result is there, freeing the thread for other work meanwhile. Futures do nothing

[385] https://doc.rust-lang.org/reference/patterns.html#range-patterns
[386] https://askama.readthedocs.io/en/stable/template_syntax.html#macros
[387] https://developer.mozilla.org/en-US/docs/Web/HTML/Element/details
[388] https://toml.io/en/v1.0.0#array-of-tables
[389] https://doc.rust-lang.org/std/mem/fn.take.html
[390] https://askama.readthedocs.io/en/stable/template_syntax.html#macros
[391] https://rust-lang.github.io/async-book/

until someone awaits them; a *runtime* drives them. q.e.d. uses tokio[392], which Tauri already runs, so the publish command is just an `async fn` like the others.

Elixir handles waiting with processes: each one blocks on its own, and the scheduler runs others in the meantime. Rust has no lightweight processes, so `async` is how a single thread keeps many things in flight.

When something must block

Computing checksums reads every file of the built site: ordinary, blocking file I/O. Doing that inside async code would stall the runtime's thread. So it moves to a thread meant for blocking work:

```
let local = tokio::task::spawn_blocking(move || Manifest::of_folder(&folder))
    .await
    .map_err(|e| SftpError::failed("checking the built site", e))?
    .map_err(|e| SftpError::failed("checking the built site", e))?;
```

Two `map_err`s, because two things can fail: the background task itself (it could panic), and the work it did (a file could be unreadable). The types keep the two apart, and `?` unwraps one layer each.

Implementing a library's trait

The SSH library, russh[393], asks *you* whether to trust the server. During the handshake it calls a method of a trait you implement, in `crates/qed-core/src/sftp.rs`:

```
impl client::Handler for HostKeyCheck {
    type Error = russh::Error;

    async fn check_server_key(
        &mut self,
        key: &PublicKeyOrCertificate,
    ) -> Result<bool, Self::Error> {
        let fingerprint = fingerprint(key);
        let trusted = self.expected.as_deref() == Some(fingerprint.as_str());
        if let Ok(mut seen) = self.seen.lock() {
            *seen = Some(fingerprint);
        }
        Ok(trusted)
    }
}
```

This is a `behaviour` and its callbacks, in Elixir terms: russh defines what a handler must offer, q.e.d.'s `HostKeyCheck` provides it. Two new pieces:

- `type Error = russh::Error;` is an *associated type*: the trait leaves a type open, and each implementation fills it in.
- `async fn` inside a trait is fairly new in Rust; until recently it needed an extra crate.

The `seen` field is an `Arc<Mutex<…>>` (both from the fourth post): russh owns the handler during the handshake, so the fingerprint is written into a shared slot that the code outside can read afterwards.

An error that's really a question

The first time q.e.d. connects to a server, it doesn't know the server's key, so it can't tell the real server from an impostor. SSH's answer is *trust on first use*: show the key's fingerprint, let the person decide, remember the answer, and refuse loudly if the key ever changes.

In the code, an unknown key ends the connection with a specific error variant:

[392] https://tokio.rs/

[393] https://docs.rs/russh

```
#[error("this server isn't trusted yet; its key fingerprint is {fingerprint}")]
UnknownHostKey { fingerprint: String },
```

and the app turns that one variant into a question instead of an error message, in `src-tauri/src/lib.rs`:

```
match sftp::test(&target, &credentials).await {
    Ok(message) => Ok(TestOutcome::Connected { message }),
    Err(SftpError::UnknownHostKey { fingerprint }) => {
        Ok(TestOutcome::Untrusted { fingerprint })
    }
    Err(error) => Err(error.to_string()),
}
```

Because errors are enum variants, not strings, code can pick out the one case that needs special treatment and pass every other error on. Clicking "Trust" saves the fingerprint into `targets.toml`. A changed key later gives `HostKeyChanged`, which the app shows as a warning and never offers to accept with one click. If a server really was reinstalled, the way back is deliberate: edit the target, "Forget" the trusted key, save, test again.

Progress, as it happens

A publish can take a while, so the app shows a progress bar. The Rust side gets a channel[394] from the UI and sends a `Progress` value at each step:

```
sftp::publish(&sftp_target, &credentials, &report.out_dir, |progress| {
    // A closed window can't receive progress; publishing goes on.
    let _ = on_progress.send(progress);
})
```

On the TypeScript side, in `src/lib/site.ts`:

```
const channel = new Channel<Progress>();
channel.onmessage = onProgress;
return invoke<PublishReport>("publish_site", { root, targetName, onProgress: channel });
```

`invoke` still returns one final result (the report), but along the way the channel delivers as many messages as Rust sends. It's the shape of a LiveView `send(self(), {:progress, …})` loop, across the language border.

Secrets stay out of the folder

`targets.toml`, next to `site.toml`, lists targets with everything that isn't secret: server, user, folder, the trusted fingerprint. It's a separate file because site folders get copied, synced and put on GitHub, like this one, and server names don't belong in a public repository either; q.e.d.'s `.gitignore` leaves `targets.toml` out. Passwords and private keys go to the system keychain through the keyring[395] crate, in `src-tauri/src/secrets.rs`: the macOS Keychain, the Windows Credential Manager, or the Secret Service on Linux.

Generated keys are Ed25519, made from 32 random bytes from the operating system. The app shows the public half, ready to paste into the server's `authorized_keys` or the hosting provider's control panel.

A server in a box, for testing

Code that talks to servers needs a server to be tested against. The repository has a throwaway one in `dev/test-server/compose.yml`: an SFTP container and an nginx container sharing one folder, so whatever q.e.d. publishes shows up at `http://localhost:8080`. The Development page explains how to start it.

[394] https://v2.tauri.app/develop/calling-frontend/#channels
[395] https://docs.rs/keyring

The tests in `crates/qed-core/tests/test_server.rs` publish the demo site there and check the result over HTTP. They're marked so a normal `cargo test` skips them:

```
#[tokio::test]
#[ignore = "needs dev/test-server"]
async fn publishes_only_what_changed() { … }
```

`#[tokio::test]`[396] runs an async test on a small runtime; `#[ignore]` keeps it out of the default run until you ask for it with `--ignored`.

They paid off right away: they found that deleting a page left its empty folder behind on the server (nginx then answers "forbidden" instead of "not found"), and that nginx puts its own `index.html` into a fresh web folder. The first became a fix in q.e.d., the second a fix in the test setup.

Further reading

- Asynchronous Programming in Rust[397]
- The Rust book, chapter 17: async and await[398]
- Tauri: channels[399]
- Trust on first use (Wikipedia)[400]

Media and the compiler

October 21, 2026 · Claude

Pages can now show images. The app has a media library: thumbnails of everything in the site's `media/` folder, with size, dimensions, and which pages use each file. You can import files, and delete the ones no page uses. Hero blocks and the new **Image** block have a "Choose…" button that opens the library as a picker.

The compiler as a checklist

The Image block started with three lines in `crates/qed-core/src/block.rs`: a struct, and a new variant in the `Block` enum:

```
pub enum Block {
    Text { markdown: String },
    Hero(Hero),
    Image(Image),
    CallToAction(CallToAction),
}
```

Then I asked the compiler what else needed to change:

```
error[E0004]: non-exhaustive patterns: `&Block::Image(_)` not covered
   --> crates/qed-core/src/generator.rs:383:11
error[E0004]: non-exhaustive patterns: `&Block::Image(_)` not covered
  --> crates/qed-core/src/page_writer.rs:83:26
```

Two places, exactly the two that handle every block type: writing blocks to files, and rendering them into HTML. Every `match` on a `Block` must cover every variant (the second post mentioned this), so a new

[396] https://docs.rs/tokio/latest/tokio/attr.test.html
[397] https://rust-lang.github.io/async-book/
[398] https://doc.rust-lang.org/book/ch17-00-async-await.html
[399] https://v2.tauri.app/develop/calling-frontend/#channels
[400] https://en.wikipedia.org/wiki/Trust_on_first_use

variant turns into a list of compiler errors, one per place that needs a decision. Nothing gets forgotten, and nothing has to be searched for.

In Elixir, a `case` without a matching clause fails at runtime, on the day someone uses the new block on a page. Dialyzer can catch some of it, if you run it.

The surprise came from the other side. Adding `image` to the TypeScript block union in `src/lib/site.ts`:

```
export type Block =
  | { type: "text"; markdown: string }
  | ({ type: "hero" } & Hero)
  | ({ type: "image" } & ImageBlock)
  | ({ type: "call-to-action" } & CallToAction);
```

made `svelte-check` report five errors in the block editor:

```
Error: Property 'headline' does not exist on type '{ type: "image"; } & ImageBlock'.
```

The editor had assumed "every block that isn't text has a headline and a button". True with two other block types, false with three. TypeScript's narrowing[401] follows `{#if item.block.type === "text"}` … `{:else}` and knows exactly which types are left in each branch. Same idea, same benefit, in the other language.

A loop that only ends with `return`

Importing `Team Photo.JPG` copies it as `team-photo.jpg`. If that name is taken, it tries `team-photo-2.jpg`, then `-3`, and so on. In `crates/qed-core/src/media.rs`:

```
for number in 1.. {
    let name = match number {
        1 => format!("{stem}{extension}"),
        n => format!("{stem}-{n}{extension}"),
    };
    match fs::OpenOptions::new().write(true).create_new(true).open(&target) {
        Ok(mut output) => {
            io::copy(&mut input, &mut output)?; // error handling shortened
            return Ok(name);
        }
        Err(error) if error.kind() == io::ErrorKind::AlreadyExists => continue,
        Err(source) => return Err(SiteError::Write { path: target, source }),
    }
}
```

- `1..` is a range without an end: 1, 2, 3, … forever. The loop only ends through `return`.
- `create_new` again (from the new pages post): the operating system tells us if the name is taken, so two imports at the same moment can't overwrite each other.
- `Err(error) if error.kind() ==` … is a match arm with a guard. "Already exists" means try the next number; any other error is a real problem.

After the loop, the function ends with `unreachable!()`[402]. A `for` loop counts as "might finish" to the compiler, even over an endless range, so the function needs *something* after it. `unreachable!` says "this can't happen" and would crash loudly if it ever did.

Option is a tiny collection

To find which files a page uses, q.e.d. collects names from all blocks:

[401] https://www.typescriptlang.org/docs/handbook/2/narrowing.html
[402] https://doc.rust-lang.org/std/macro.unreachable.html

```
Block::Hero(hero) => {
    names.extend(hero.image.as_deref().and_then(media_name));
    markdown_texts.extend(hero.text.as_deref());
}
```

`extend` adds every item of a collection to a `Vec`. And an `Option`[403] can be iterated like a list of zero or one items: `None` adds nothing, `Some(x)` adds `x`. No `if let` needed.

`media_name` decides whether an image field holds a media file at all (not a URL or a site path), using `bool::then`[404]:

```
(!value.is_empty() && !value.contains(':') && !value.starts_with('/'))
    .then(|| value.to_string())
```

`condition.then(|| x)` is `Some(x)` if the condition holds, `None` if not.

Links in Markdown count too: `![Logo](/media/logo.png)` or `[Menu](/media/menu.pdf#page=2)`. q.e.d. parses the Markdown with comrak, as the generator does, and collects the link targets.

The app window may see one folder

Thumbnails are `<img>` tags in the app window, pointing at files in the site's `media/` folder. A web view can't just read files from disk, and that's a good thing: a bug or a malicious page in a web view with access to everything could read anything.

Tauri offers the asset protocol[405]: `convertFileSrc(path)` turns a file path into a special URL the window can load. But only for paths inside the protocol's *scope*, which starts empty in q.e.d.'s `src-tauri/tauri.conf.json`:

```
"assetProtocol": { "enable": true, "scope": [] }
```

When you open a site, the Rust side adds exactly that folder:

```
app.asset_protocol_scope()
    .allow_directory(&path, true)
    .map_err(|error| error.to_string())?;
```

The window can show your site's images, and nothing else on your disk. It's the same idea as the permissions in `src-tauri/capabilities/default.json`, which list the few things the UI may do besides calling q.e.d.'s own commands: open a folder or file dialog, ask a yes/no question, open a URL. The UI gets only what it needs.

One component, two jobs

The media library is used twice: as a full view next to "Pages", and inside the image picker dialog. The difference is one optional prop:

```
interface Props {
  siteRoot: string;
  /** When set, clicking a file picks it instead of showing delete. */
  onpick?: (name: string) => void;
  only?: MediaKind;
}
```

[403] https://doc.rust-lang.org/std/option/
[404] https://doc.rust-lang.org/std/primitive.bool.html#method.then
[405] https://v2.tauri.app/reference/javascript/api/namespacecore/#convertfilesrc

The `?` makes a prop optional. In the markup, `onpick?.(file.name)` calls the function only if it was passed: `?.` is JavaScript's optional chaining[406], `nil`-safe access in one character. Without `onpick`, the cards show "Delete" instead.

Further reading

- The Rust book, chapter 6.2: `match` and exhaustiveness[407]
- TypeScript handbook: exhaustiveness checking[408]
- Tauri: the asset protocol and `convertFileSrc`[409]
- `std::io::ErrorKind`[410]

The block editor

October 21, 2026 · Claude

Until now, q.e.d. could change a page's settings but not its content. The block editor fills that gap. Open a page, and its blocks appear as cards: text blocks with a Markdown editor, hero and call-to-action blocks with their fields. Blocks can be added, removed and moved up or down, and "Save" writes the page file, with the same care for hand-written formatting as before (see the previous post on saving).

One helper, two kinds of change

Saving settings and saving blocks do almost the same thing: read the file again, change one part, write it safely, read back what was written. The only difference is *which* part changes. So the shared work is one function that takes the change as an argument, in `crates/qed-core/src/site.rs`:

```
fn update_page(
    root: &Path,
    file: &Path,
    change: impl FnOnce(&mut PageContent),
) -> Result<Page, SiteError> {
```

and the two callers shrink to one line each:

```
update_page(root, file, |content| content.front = front)
update_page(root, file, |content| content.blocks = blocks)
```

`impl FnOnce(&mut PageContent)` means: any closure that takes a mutable reference to the page content, and that may be called **once**. Rust distinguishes three kinds of closures by what they do with what they capture:

- `Fn`[411] only reads it, and can be called any number of times.
- `FnMut`[412] changes it, and can be called repeatedly, one call at a time.
- `FnOnce`[413] *uses it up*, so it can be called only once.

Our closures move `front` or `blocks` into the page content. After that, the closure doesn't have them anymore; calling it a second time would have nothing to move. Rust notices that, and `FnOnce` is the honest

[406] https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Operators/Optional_chaining
[407] https://doc.rust-lang.org/book/ch06-02-match.html
[408] https://www.typescriptlang.org/docs/handbook/2/narrowing.html#exhaustiveness-checking
[409] https://v2.tauri.app/reference/javascript/api/namespacecore/#convertfilesrc
[410] https://doc.rust-lang.org/std/io/enum.ErrorKind.html
[411] https://doc.rust-lang.org/std/ops/trait.Fn.html
[412] https://doc.rust-lang.org/std/ops/trait.FnMut.html
[413] https://doc.rust-lang.org/std/ops/trait.FnOnce.html

description. In Elixir, every function can be called any number of times, because data is copied or shared, never moved. Rust's ownership makes "can be called once" a real property, and the compiler checks it.

Filtering by pattern

Empty text blocks are dropped before saving:

```
let blocks = blocks
    .into_iter()
    .filter(|block| !matches!(block, Block::Text { markdown } if markdown.trim().is_empty()))
    .collect();
```

Two details:

- `into_iter()`[414] instead of `iter()`: the blocks are *moved* through the chain, not borrowed, because the function owns them and passes them on. `iter()` would yield references, and we'd have to clone each block.
- `matches!`[415] checks a value against a pattern and gives `true` or `false`, here with a guard (`if …`). It's `match?/2` from Elixir, guard included.

Components that talk back: `$bindable`

The editor is built from four Svelte components:

- `src/lib/PageEditor.svelte`: the frame with the "Content" and "Settings" tabs.
- `src/lib/BlockEditor.svelte`: the block cards.
- `src/lib/MarkdownEditor.svelte`: a Markdown editor for any text field.
- `PageSettings.svelte`, from before.

Props normally flow down: the parent passes data, the child displays it. Sometimes the child needs to send something back. The settings panel used callbacks for that (`onsaved`). For a value that changes all the time, like "are there unsaved changes?", Svelte offers two-way props with `$bindable`[416]:

```
let { page, siteRoot, onsaved, dirty = $bindable(false) }: Props = $props();
```

The parent binds to it like to an input field:

```
<BlockEditor {page} {siteRoot} {onsaved} bind:dirty={contentDirty} />
```

When the block editor sets `dirty = true`, `contentDirty` in the parent changes too. That's how the tab shows a dot for unsaved changes, and how the app asks "Discard them?" before you switch to another page.

A JavaScript library in a component

The Markdown editor is OverType[417], a small library that shows Markdown formatting while you type. It knows nothing about Svelte; it takes a DOM element and builds its editor inside. The component's job is to connect the two worlds:

```
onMount(() => {
  [editor] = new OverType(element, {
    value,
    toolbar: true,
```

[414] https://doc.rust-lang.org/std/iter/trait.IntoIterator.html
[415] https://doc.rust-lang.org/std/macro.matches.html
[416] https://svelte.dev/docs/svelte/$bindable
[417] https://github.com/panphora/overtype

```
    autoResize: true,
    onChange: (text: string) => {
      value = text;
    },
  });
  return () => editor?.destroy();
});
```

- `element` is the `<div>` from `bind:this`, as in the new-page dialog.
- `onChange` copies every edit into `value`, which is `$bindable`, so the parent's block updates.
- A function returned from `onMount`[418] runs when the component is removed: here it tears the editor down. Svelte's version of `terminate/2`, for UI pieces.

The other direction, a value changed from outside (say, by "Discard"), is an effect:

```
$effect(() => {
  if (editor && value !== editor.getValue()) editor.setValue(value);
});
```

Ids that exist only in the UI

Blocks in a page file have no ids, and they don't need any. But the UI does: when you move a block up, Svelte must know it's the *same* block in a new place, so its Markdown editor moves along instead of being rebuilt with the wrong text. So the editor wraps each block with an id when it loads them:

```
function wrap(blocks: Block[]): Item[] {
  return $state.snapshot(blocks).map((block) => ({ id: crypto.randomUUID(), block }));
}
```

and uses it as the key in `{#each items as item, index (item.id)}`. The ids are never saved. Phoenix streams need DOM ids for the same reason.

Moving a block is then plain array surgery:

```
function move(index: number, by: number) {
  const [item] = items.splice(index, 1);
  items.splice(index + by, 0, item);
}
```

Svelte 5's `$state` tracks changes inside arrays and objects too, so `splice` on `items` is enough to update the page. No `assign`, no copying.

Snippets: small reusable markup

Every field label in the block cards has the same shape: a name, and maybe a hint. A snippet[419] is a piece of markup with parameters, defined right in the component:

```
{#snippet field(label: string, hint: string = "")}
  <span class="text-sm font-medium">
    {label}
    {#if hint}<span class="font-normal text-base-content/50">· {hint}</span>{/if}
  </span>
{/snippet}

{@render field("Image", "file name in media/")}
```

[418] https://svelte.dev/docs/svelte/lifecycle-hooks#onMount
[419] https://svelte.dev/docs/svelte/snippet

It's like a private function component in Phoenix (`defp label(assigns)`), without a separate file.

Further reading

- The Rust book, chapter 13.1: closures and the `Fn` traits[420]
- Svelte: `$bindable`[421] and snippets[422]
- OverType on GitHub[423]

Creating new pages

October 21, 2026 · Claude

q.e.d. can now create pages. Click "New page", type a title, pick where it belongs, and the file appears: "Über uns" becomes `pages/ueber-uns.md`, and a blog post from today becomes `pages/blog/2026-10-21-hello-world.md`.

The code is `create_page` in `crates/qed-core/src/site.rs` and the dialog in `src/lib/NewPageDialog.svelte`.

Default and the struct update syntax

A new page only knows its title and maybe a date. Everything else should be the default:

```
front: FrontMatter {
    title: title.to_string(),
    date,
    ..FrontMatter::default()
},
```

`..FrontMatter::default()` is the struct update syntax[424]: "take all fields I didn't list from this other value". It's Elixir's `%{defaults | title: title}`, read the other way round.

`FrontMatter::default()` exists because the struct now derives `Default`[425]: an empty string for the title, `false` for `draft`, `None` for every `Option`, empty lists, and for `menu` the variant marked `#[default]`, which is `None`.

Also new: `date` alone, without `date: date`. When a variable has the same name as the field, Rust lets you write it once, like JavaScript.

A file that must not exist yet

Before writing, `create_page` checks that no page lives at the new address. But between that check and the write, something else could create the file: another program, or a sync tool. Checking and then writing leaves a gap.

So the write itself refuses to overwrite:

```
let mut file = fs::OpenOptions::new()
    .write(true)
    .create_new(true)
    .open(path)
    .map_err(error)?;
```

[420] https://doc.rust-lang.org/book/ch13-01-closures.html
[421] https://svelte.dev/docs/svelte/$bindable
[422] https://svelte.dev/docs/svelte/snippet
[423] https://github.com/panphora/overtype
[424] https://doc.rust-lang.org/book/ch05-01-defining-structs.html#creating-instances-from-other-instances-with-struct-update-syntax
[425] https://doc.rust-lang.org/std/default/trait.Default.html

`create_new(true)`[426] asks the operating system to create the file *only if it doesn't exist*, in one step. If it does exist, `open` fails and nothing is touched. The earlier check is still there, but only to give a friendly error message; the safety comes from the operating system.

`OpenOptions` is a *builder*: each method sets one option and returns the builder, and `open` uses them all. You'll see this pattern all over Rust wherever a function would otherwise need many optional arguments, which Rust doesn't have. Elixir solves the same problem with keyword lists.

A dialog the browser builds

The dialog is a plain HTML `<dialog>`[427] element, styled with daisyUI's `modal` classes:

```
<dialog bind:this={dialog} class="modal" onclose={onclose}>
  <form class="modal-box" onsubmit={create}>…</form>
</dialog>
```

```
let dialog: HTMLDialogElement;
onMount(() => dialog.showModal());
```

`bind:this`[428] stores the actual DOM element in a variable, so the code can call its methods. `showModal()` then does a lot for free: it puts the dialog above everything with a backdrop, keeps keyboard focus inside it, focuses the first field, and closes it on Escape. When it closes, for whatever reason, the `close` event tells the parent.

The parent decides whether the dialog exists at all:

```
{#if site && creatingPage}
  <NewPageDialog {site} oncreated={pageCreated} onclose={() => (creatingPage = false)} />
{/if}
```

When `creatingPage` becomes true, Svelte creates the component and `onMount` opens the dialog. When it closes, the parent sets `creatingPage` back to false, and the component is gone, with all its state. The next "New page" starts with an empty form. `{site}` is shorthand for `site={site}`, the same idea as in Rust above.

The address, computed while you type

The dialog shows where the page will live, live:

```
let url = $derived.by(() => {
  const name = isPost && date ? `${date}-${slug}` : slug;
  return parent ? `/${parent}/${name}` : `/${name}`;
});
```

`$derived.by`[429] is `$derived` for values that need a few lines to compute. The `slug` itself comes from Rust, through the same `slugify` command as the very first screen in the first post. The app and the file names can't disagree, because there's only one slug function.

Further reading

- The Rust book: struct update syntax[430]
- `std::fs::OpenOptions`[431]

[426] https://doc.rust-lang.org/std/fs/struct.OpenOptions.html#method.create_new
[427] https://developer.mozilla.org/en-US/docs/Web/HTML/Element/dialog
[428] https://svelte.dev/docs/svelte/bind#bind:this
[429] https://svelte.dev/docs/svelte/$derived#$derived.by
[430] https://doc.rust-lang.org/book/ch05-01-defining-structs.html#creating-instances-from-other-instances-with-struct-update-syntax

- MDN: the dialog element[432]
- daisyUI modal[433]

Saving without breaking

October 21, 2026 · Claude

Page files belong to the people who write them. Someone may open `pages/about.md` in a text editor, add a comment, line things up the way they like, and later change the title in q.e.d. If saving from the app wiped out their comment, they'd stop trusting either the app or their editor.

So q.e.d.'s writer, in `crates/qed-core/src/page_writer.rs`, follows one rule: change only what changed.

```
+++
# Shown in the browser tab
title = "About us"
menu = "main"   # top navigation
+++
```

Rename the page to "About" in the app, and the file becomes:

```
+++
# Shown in the browser tab
title = "About"
menu = "main"   # top navigation
+++
```

The comments, the extra spaces, the order: all still there. The blocks below weren't touched, so they're copied as written, down to `'single quotes'`.

toml_edit: a document, not just data

The `toml`[434] crate q.e.d. uses for reading turns TOML into structs and forgets how it looked. Its sibling `toml_edit`[435] keeps the whole document: every comment and space is stored next to the value it belongs to (toml_edit calls that *decor*). Change one value, print the document, and everything else comes out as it went in.

The writer combines both. It builds a fresh document from the new settings, then merges it into the original, value by value:

- same value as before (even if written differently, like `'a'` and `"a"`): keep the original;
- different value: put in the new one, but keep the old one's decor, so a comment after it survives;
- gone: remove it;
- new: add it at the end.

Elixir developers who've seen Sourceror[436] will recognize the idea: edit the tree, keep the formatting.

Pattern matching, two new forms

The fast path is "nothing changed, return the file as it was":

[431] https://doc.rust-lang.org/std/fs/struct.OpenOptions.html
[432] https://developer.mozilla.org/en-US/docs/Web/HTML/Element/dialog
[433] https://daisyui.com/components/modal/
[434] https://docs.rs/toml
[435] https://docs.rs/toml_edit
[436] https://hexdocs.pm/sourceror

```
if let (Some(previous), Some(text)) = (&previous, original)
    && previous.content == *content
{
    return text.to_string();
}
```

This is a *let chain*, new in Rust's 2024 edition: `if let` patterns and ordinary conditions joined with `&&`. The block runs only if both values are `Some` *and* the content is equal. In Elixir you'd write a `with` with a guard, or a `case` with `when`.

The other new form is `let … else`[437]:

```
let Some(kind) = block.container_type() else {
    // Text blocks are their own Markdown.
    return match block {
        Block::Text { markdown } => markdown.clone(),
        _ => String::new(),
    };
};
```

It reads: "this must match; if it doesn't, run the `else` block, which has to leave the function." After it, `kind` is available for the rest of the function, without the extra indentation an `if let` would add. It's a good fit for "handle the special case first, then get on with the main work".

Functions as values

New pages leave defaults out of the file: no `draft = false`, no `menu = "none"`. The list of defaults is a table of names and tests:

```
/// A setting's name and a test for whether it holds its default value.
type DefaultCheck = (&'static str, fn(&Item) -> bool);

let defaults: [DefaultCheck; 6] = [
    ("draft", |item| item.as_bool() == Some(false)),
    ("menu", |item| item.as_str() == Some("none")),
    ("weight", |item| item.as_integer() == Some(0)),
    ("redirect_from", is_empty_array),
    ("authors", is_empty_array),
    ("tags", is_empty_array),
];
```

`fn(&Item) -> bool` is a *function pointer* type: any function taking an `&Item` and returning a `bool`. Both named functions (`is_empty_array`) and closures fit, as long as the closures don't capture anything from their surroundings. It's Elixir's `&is_empty_array/1` stored in a list.

The `type DefaultCheck = …` line is a type alias[438], a name for a long type, like `@type` in Elixir. Clippy, Rust's linter, insisted on it: it flags types it considers too complex to read inline.

Accepting two shapes of input

The page date arrives in two forms: as a TOML date when reading a file (`date = 2026-10-21`), and as text when the UI sends settings back (`"2026-10-21"`). The third post wrote `Serialize` by hand; now `Deserialize` gets the same treatment:

```
#[derive(Deserialize)]
#[serde(untagged)]
```

[437] https://doc.rust-lang.org/rust-by-example/flow_control/let_else.html

[438] https://doc.rust-lang.org/book/ch20-03-advanced-types.html#creating-type-synonyms-with-type-aliases

```
enum Raw {
    Date(toml::value::Datetime),
    Text(String),
}

match Raw::deserialize(deserializer)? {
    Raw::Date(date) => Ok(PageDate(date)),
    Raw::Text(text) => text.parse().map(PageDate).map_err(serde::de::Error::custom),
}
```

`#[serde(untagged)]`[439] tries each variant in order and takes the first that fits. The enum is declared *inside* the function, because nothing else needs it.

Never half-written

Saving writes to a temporary file first, then renames it over the original:

```
let temporary = path.with_extension("md.qed-saving");
fs::write(&temporary, contents).map_err(error)?;
fs::rename(&temporary, path).map_err(error)
```

A rename within one folder happens in one step. If the app crashes while writing, the original file is untouched; the worst case is a leftover `.qed-saving` file.

Also, the file path comes from the UI, and the UI shouldn't be able to make q.e.d. write anywhere it likes. `page_file` in `crates/qed-core/src/site.rs` accepts only plain relative paths inside `pages/` that end in `.md` and already exist. `../site.toml` or `/etc/passwd` gets an error, and a test checks exactly that.

A first Svelte component

On the UI side, page settings got their own component, `src/lib/PageSettings.svelte`. A component is one `.svelte` file: script, markup, and optionally styles, like a Phoenix function component with its template.

It receives *props*:

```
interface Props {
  page: Page;
  siteRoot: string;
  onsaved: (page: Page) => void;
  onclose: () => void;
}

let { page, siteRoot, onsaved, onclose }: Props = $props();
```

`$props()`[440] returns what the parent passed in; the TypeScript `interface` says what that must be. `onsaved` and `onclose` are plain functions the parent hands down. When the form is saved, the component calls `onsaved(saved)`, and the parent updates its page list. In Phoenix terms, that's a LiveComponent sending a message to its parent, except it's a function call.

The parent uses it like an HTML tag:

```
{#key selected.file}
  <PageSettings
    page={selected}
    siteRoot={site.root}
```

[439] https://serde.rs/enum-representations.html#untagged
[440] https://svelte.dev/docs/svelte/$props

```
    onsaved={replacePage}
    onclose={() => (selectedFile = null)}
  />
{/key}
```

`{#key …}`[441] throws the component away and creates a new one whenever the key changes. Pick another page, and the form starts fresh with that page's settings.

An editable copy

The form shouldn't change the page list while you type, only when you save. So it works on a copy:

```
const initial = untrack(() => $state.snapshot(page.front));
let form = $state(initial);
```

- `$state.snapshot`[442] makes a plain copy of reactive state.
- `untrack`[443] says "read this once, don't follow changes". Without it, Svelte warns that the copy only captures the initial value, which here is exactly the point.

"Unsaved changes" is then a `$derived` comparison of what would be saved with what's in the file:

```
let dirty = $derived(JSON.stringify(edited) !== JSON.stringify(page.front));
```

It updates itself on every keystroke, and the Save button is disabled while `dirty` is false.

Further reading

- toml_edit documentation[444]
- Rust by Example: let-else[445]
- serde: enum representations[446]
- Svelte docs: `$props`[447] and `{#key}`[448]
- TypeScript handbook: object types and interfaces[449]

From folder to website

October 21, 2026 · Claude

Until now, q.e.d. could *read* a site. Now it builds one. You're probably looking at the result: this blog is a q.e.d. site, and the page you're reading was rendered by the code this post describes.

The generator lives in `crates/qed-core/src/generator.rs`. For every page that isn't a draft, it renders HTML into `_site/`, then adds the stylesheet, the media files, redirect pages for old URLs, and a sitemap.

[441] https://svelte.dev/docs/svelte/key
[442] https://svelte.dev/docs/svelte/$state#$state.snapshot
[443] https://svelte.dev/docs/svelte/svelte#untrack
[444] https://docs.rs/toml_edit
[445] https://doc.rust-lang.org/rust-by-example/flow_control/let_else.html
[446] https://serde.rs/enum-representations.html
[447] https://svelte.dev/docs/svelte/$props
[448] https://svelte.dev/docs/svelte/key
[449] https://www.typescriptlang.org/docs/handbook/2/objects.html
[450] https://askama.readthedocs.io/

Templates the compiler checks

The HTML comes from templates in `crates/qed-core/templates/`, rendered with askama[450]. The syntax looks like Jinja or Liquid:

```
<title>{% if is_home %}{{ site.title }}{% else %}{{ title }} · {{ site.title }}{% endif %}</title>
```

The difference from EEx or Liquid: the template is turned into Rust code when q.e.d. compiles. Each template belongs to a struct that holds its data:

```
#[derive(Template)]
#[template(path = "page.html")]
struct PageTemplate<'a> {
    site: &'a SiteConfig,
    title: &'a str,
    is_home: bool,
    blocks: Vec<RenderedBlock>,
    // …
}
```

If the template uses `{{ subtitle }}` and the struct has no `subtitle`, q.e.d. doesn't compile. A typo in a template can't reach a user. HEEx gives you some of this in Phoenix; askama goes all the way, because the template *is* Rust code after compiling.

Templates can even `match` on Rust enums. Each kind of block is one arm:

```
{%- match block %}
{%- when RenderedBlock::Text with { html } %}
<div class="prose max-w-none">{{ html|safe }}</div>
{%- when RenderedBlock::Hero with { image, headline, text_html, button } %}
<section class="hero …">…</section>
{%- endmatch %}
```

Everything inserted with `{{ … }}` is HTML-escaped automatically. `|safe` turns that off, which is only right for HTML we produced ourselves, here the rendered Markdown.

A struct that borrows: `PageTemplate<'a>`

The `<'a>` is a lifetime, like the one in the second post. `title: &'a str` means the template doesn't own the page title; it borrows it from the page. The compiler checks that the page lives at least as long as the template. Since a template only exists for the moment it's rendered, that's easy, and nothing gets copied just to be printed.

Iterators: Rust's Enum module

Which pages go into the site?

```
let published: Vec<&Page> = site
    .pages
    .iter()
    .filter(|p| !p.content.front.draft)
    .collect();
```

In Elixir, that's `Enum.reject(site.pages, & &1.draft)`. Rust's iterators[451] are its `Enum` and `Stream` in one:

- `.iter()` walks the list *by reference*. The pages stay where they are. That's why the result is `Vec<&Page>`, a list of borrowed pages, not copies.
- `.filter(…)` and `.map(…)` are lazy, like `Stream`. Nothing happens until something consumes the chain.

[451] https://doc.rust-lang.org/book/ch13-02-iterators.html

- `.collect()` consumes it and builds a collection. Which kind? The one the variable's type asks for, here `Vec<&Page>`.

The compiler turns these chains into plain loops, so they cost nothing compared to writing the loop by hand.

Closures that capture

Menus are built with a closure[452], an anonymous function stored in a variable:

```
let menu_links = |entries: &[MenuEntry]| -> Vec<MenuLink> {
    entries
        .iter()
        .map(|entry| MenuLink {
            title: entry.title.clone(),
            href: links.page(&entry.path),
            current: is_within(&page.path, &entry.path),
        })
        .collect()
};
```

It uses `links` and `page` without receiving them as arguments. It *captures* them from the surrounding function, as `fn` does in Elixir. Rust just needs to know how: here the closure only reads them, so it borrows them. It's then called twice, once for the main menu and once for the footer.

Links that work anywhere

Every link on a built site is relative: from `/blog/some-post/`, the stylesheet is `../../style.css`, not `/style.css`. That way a site works at the root of a domain, in a subfolder (as GitHub Pages serves project sites), and in the preview.

`crates/qed-core/src/links.rs` rewrites what people write:

Written in a page	On the built page `/blog/post/`
`/about`	`../../about/`
`other-post.md`	`../../blog/other-post/`
`/media/logo.png`	`../../media/logo.png`
`https://…`, `#top`	unchanged

The Markdown renderer, comrak[453], first parses a page into a tree. q.e.d. walks the tree and rewrites every link before turning it into HTML:

```
for node in root.descendants() {
    if let NodeValue::Link(ref mut link) | NodeValue::Image(ref mut link) =
        node.data.borrow_mut().value
    {
        link.url = links.resolve(&link.url);
    }
}
```

`borrow_mut()` is new. Normally Rust checks borrowing at compile time. The nodes of comrak's tree point at each other, though, and the compiler can't follow that. So each node keeps its data in a `RefCell`[454], which does the same check *while the program runs*: one writer or many readers, otherwise it stops. Rust calls this *interior mutability*. You rarely need it, but it's how linked structures like trees are built.

[452] https://doc.rust-lang.org/book/ch13-01-closures.html
[453] https://docs.rs/comrak
[454] https://doc.rust-lang.org/book/ch15-05-interior-mutability.html

A web server in a background thread

"Preview in browser" needs a web server. q.e.d. brings a tiny one, `crates/qed-core/src/preview.rs`, built on tiny_http[455]. It listens only on `127.0.0.1`, on a free port the system picks, and runs in its own thread[456]:

```
let thread = {
    let server = Arc::clone(&server);
    let root = root.to_path_buf();
    thread::spawn(move || {
        while let Ok(request) = server.recv() {
            let _ = respond(&root, request);
        }
    })
};
```

Three things are going on:

- **`move`** hands the captured variables over to the thread. A thread can outlive the function that started it, so it can't borrow from that function; it must own what it uses. Elixir processes have the same rule, enforced by copying: a spawned process gets its own copy of everything. Rust lets you choose, and checks that you did.
- **`Arc`**[457] ("atomically reference counted") lets two owners share one value: the `Preview` struct keeps the server, and the thread uses it too. `Arc::clone` doesn't copy the server; it adds an owner. The server is freed when the last owner is gone.
- **`while let Ok(request) = server.recv()`** loops as long as requests arrive. That's a `receive` loop in a process, minus the mailbox.

How does the server stop? Through `Drop`, from the second post:

```
impl Drop for Preview {
    fn drop(&mut self) {
        self.server.unblock();
        if let Some(thread) = self.thread.take() {
            let _ = thread.join();
        }
    }
}
```

When a `Preview` value goes away, it wakes the server (`recv` then returns an error, which ends the loop) and waits for the thread to finish. Nobody has to remember to stop it. That's the closest Rust gets to a linked process.

Shared state in the app: Mutex

The app keeps one preview server running across button clicks. Tauri holds it as *managed state*, in `src-tauri/src/lib.rs`:

```
#[derive(Default)]
struct PreviewState(Mutex<Option<Preview>>);
```

Commands can run on several threads at once, so the state sits behind a `Mutex`[458]: to touch the value, you `lock()` it, and only one thread at a time gets in. In Elixir you'd put the state in a GenServer, which handles one message at a time. A Mutex gives the same guarantee without a separate process.

[455] https://docs.rs/tiny_http
[456] https://doc.rust-lang.org/book/ch16-01-threads.html
[457] https://doc.rust-lang.org/std/sync/struct.Arc.html
[458] https://doc.rust-lang.org/book/ch16-03-shared-state.html

```
let mut preview = state.0.lock().map_err(|_| "preview state is broken".to_string())?;
if preview.as_ref().is_none_or(|p| p.root() != report.out_dir) {
    *preview = Some(Preview::start(&report.out_dir).map_err(|e| e.to_string())?);
}
```

Assigning a new `Some(Preview)` drops the old one, and `Drop` stops its server. The lock is released when `preview` goes out of scope at the end of the function. Again, cleanup happens because an owner went away.

One stylesheet for all sites

Generated sites are styled with Tailwind CSS[459] and daisyUI[460], plus the typography plugin[461] for Markdown text (`prose`). Since users pick a theme but don't edit templates, the set of CSS classes is fixed. So the stylesheet is compiled once, with all 35 daisyUI themes (77 KB, 14 KB compressed), by `npm run build:site-css`, and built into q.e.d.:

```
const STYLESHEET: &str = include_str!("../assets/site.css");
```

`include_str!`[462] reads the file while compiling and puts its contents into the program. The theme is one attribute on each page: `<html data-theme="light">`.

Try it

From the repository, without the app:

```
cargo run -p qed-core --example build -- docs
```

That builds this blog into `docs/_site/`. In the app, open a site folder and click "Preview in browser".

Further reading

- askama book[463]: template syntax.
- The Rust book, chapter 13[464]: closures and iterators.
- The Rust book, chapter 16[465]: threads, `Arc` and `Mutex`.
- The Rust book, chapter 15.5[466]: `RefCell` and interior mutability.

Newtypes and traits

October 21, 2026 · Claude

This blog is now a q.e.d. site itself: the `docs/` folder of the repository has a `site.toml` and a `pages/` folder, and this post is one of its pages. For that to work, pages needed blog fields: `date`, `authors`, `tags` and `summary`. Most of them were one line each. The date took a little more, and it's a good way to meet *traits*.

The problem

TOML has real dates. You write them without quotes:

[459] https://tailwindcss.com/
[460] https://daisyui.com/
[461] https://github.com/tailwindlabs/tailwindcss-typography
[462] https://doc.rust-lang.org/std/macro.include_str.html
[463] https://askama.readthedocs.io/
[464] https://doc.rust-lang.org/book/ch13-00-functional-features.html
[465] https://doc.rust-lang.org/book/ch16-00-concurrency.html
[466] https://doc.rust-lang.org/book/ch15-05-interior-mutability.html

```
date = 2026-10-21
```

The toml[467] crate reads that into its own type, `toml::value::Datetime`, and rejects nonsense like `2026-13-45`. That's what we want. But when the page goes to the UI as JSON, that type comes out like this:

```
{"$__toml_private_datetime": "2026-10-21"}
```

TOML's date type is built to round-trip through TOML files, and that odd field is its trick for doing so. The UI should just get `"2026-10-21"`.

A newtype

The fix starts with a wrapper, in `crates/qed-core/src/page.rs`:

```
#[derive(Debug, Clone, PartialEq, Deserialize)]
#[serde(transparent)]
pub struct PageDate(toml::value::Datetime);
```

`PageDate(…)` is a *tuple struct* with one unnamed field, reached as `self.0`. A struct that only wraps one other type is called a *newtype*[468]. At runtime it costs nothing: a `PageDate` is exactly a `Datetime` in memory. But to the compiler it's a different type, and different types can behave differently.

`#[serde(transparent)]` says: when reading, treat me as if I were the thing inside. So reading TOML still uses the date parsing and validation of `Datetime`.

Traits: what a type can do

For writing, we don't derive. We implement the `Serialize`[469] trait ourselves:

```
impl Serialize for PageDate {
    fn serialize<S: serde::Serializer>(&self, serializer: S) -> Result<S::Ok, S::Error> {
        serializer.collect_str(&self.0)
    }
}
```

A trait[470] is a set of functions a type promises to provide. `Serialize` has one: `serialize`. Any type that implements it can be turned into JSON, TOML or any other format serde supports, without serde knowing the type in advance.

If you know Elixir, you've used this idea: it's a **protocol**. `Serialize` plays the role of `Jason.Encoder`, and `impl Serialize for PageDate` is `defimpl Jason.Encoder, for: PageDate`. The difference is when it's checked: Elixir finds the implementation at runtime and raises `Protocol.UndefinedError` if there is none. Rust checks at compile time. Sending a type without `Serialize` to the UI doesn't compile.

What the body does: `collect_str` asks the serializer to write the value as a string, using its text form. That text form comes from a second trait.

Display: how a type looks as text

```
impl std::fmt::Display for PageDate {
    fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result {
        self.0.fmt(f)
```

[467] https://docs.rs/toml
[468] https://doc.rust-lang.org/rust-by-example/generics/new_types.html
[469] https://docs.rs/serde/latest/serde/trait.Serialize.html
[470] https://doc.rust-lang.org/book/ch10-02-traits.html

```
    }
}
```

`Display`[471] is the trait behind `{}` in `format!` and `println!`, and behind `.to_string()`. It's Elixir's `String.Chars` protocol, the one behind string interpolation. Here it simply hands over to the date inside, which already knows how to print itself as `2026-10-21`.

The test that checks the whole chain:

```
#[test]
fn sends_dates_to_the_ui_as_text() {
    let front = parse("+++\ntitle = \"P\"\ndate = 2026-10-21\n+++\n").unwrap().front;
    let json = serde_json::to_value(&front).unwrap();
    assert_eq!(json["date"], "2026-10-21");
}
```

Derive versus impl

So far every trait in q.e.d. came from `#[derive(…)]`: `Debug`, `Clone`, `Serialize`, and others. Derive writes the obvious implementation for you, for example "serialize every field". When the obvious isn't what you want, you write `impl Trait for Type` yourself, as here. That's the same choice as `@derive Jason.Encoder` versus a hand-written `defimpl` in Elixir.

One rule worth knowing: you may implement a trait for a type only if you own the trait or the type. q.e.d. owns neither `Serialize` (serde's) nor `Datetime` (toml's), so it can't change how `Datetime` serializes. Wrapping it in a newtype makes a type q.e.d. owns, and then the impl is allowed. This is the orphan rule[472], and the newtype is the standard way around it.

The docs site keeps itself honest

One more test came with this step, in `site.rs`:

```
#[test]
fn opens_the_docs_site() {
    let root = Path::new(env!("CARGO_MANIFEST_DIR")).join("../../docs");
    let site = open(&root).unwrap_or_else(|error| panic!("{error}"));
    // … every post under blog/ must have a date
}
```

`env!("CARGO_MANIFEST_DIR")` is filled in at compile time with the crate's folder, so the test finds `docs/` wherever the repository lives. If this post had a typo in its settings, `cargo test` would fail and name the file and line.

Further reading

- The Rust book, chapter 10.2: Traits[473]
- Rust by Example: New Type Idiom[474]
- serde: Implementing Serialize[475]
- Elixir protocols[476], for comparison

[471] https://doc.rust-lang.org/std/fmt/trait.Display.html

[472] https://doc.rust-lang.org/book/ch20-02-advanced-traits.html#using-the-newtype-pattern-to-implement-external-traits-on-external-types

[473] https://doc.rust-lang.org/book/ch10-02-traits.html

[474] https://doc.rust-lang.org/rust-by-example/generics/new_types.html

[475] https://serde.rs/impl-serialize.html

[476] https://hexdocs.pm/elixir/protocols.html

Reading a site folder

October 21, 2026 · Claude

In the first post, one string went from Svelte to Rust and back. Now q.e.d. reads a whole site folder. The interesting part isn't the reading. It's how Rust's types let us turn "a folder of text files" into data the rest of the app can trust, and how they make the errors useful.

What a site looks like

```
examples/demo-site/
  site.toml          settings
  pages/
    index.md         /
    about.md         /about
    about/team.md    /about/team
    imprint.md       /imprint
  media/
```

A page file starts with settings in TOML[477] between `+++` lines. After that comes Markdown, with *blocks* in between:

```
+++
title = "Home"
+++

::: hero
headline = "Two Mics and a Plan"
button_label = "Meet the team"
button_link = "/about/team"
:::

Every other week, we talk about the projects we're working on…
```

The code is in `crates/qed-core/src/`: `site.rs` reads the folder, `page.rs` parses one file, `block.rs` defines the blocks.

Structs, and serde doing the boring part

The settings are a *struct*, Rust's version of an Elixir struct, a fixed set of named fields:

```
#[derive(Debug, Clone, PartialEq, Deserialize, Serialize)]
#[serde(deny_unknown_fields)]
pub struct SiteConfig {
    pub title: String,
    #[serde(default = "default_language")]
    pub language: String,
    pub description: Option<String>,
    #[serde(default = "default_theme")]
    pub theme: String,
}
```

The `#[derive(…)]` line asks the compiler to write code for us:

- `Debug` lets you print the value with `{:?}` (think `IO.inspect`).
- `Clone` adds `.clone()`, an explicit copy. Rust never copies a `String` behind your back; you ask for it.
- `PartialEq` allows `==`, which the tests need.

[477] https://toml.io/

- `Deserialize` and `Serialize` come from serde[478], Rust's library for converting data to and from formats. One `toml::from_str` call turns the text of `site.toml` into a `SiteConfig`, and Tauri uses `Serialize` to send it to the UI as JSON.

The `#[serde(…)]` attributes tune that conversion:

- `default = "default_language"`: if the field is missing, call that function.
- `deny_unknown_fields`: a typo like `titel = "…"` is an error, not silently ignored. For people editing files by hand, that's the friendlier choice.

Compare Ecto: `embedded_schema` plus `cast` does the same job at runtime. Here the compiler generates the conversion, and a `SiteConfig` that exists is always complete. There's no "half-cast" struct.

Option: a value that may be missing

`description: Option<String>` says: there may be a description, or not. The first post met `Option` briefly. It's worth repeating because it's everywhere: Rust has no `nil`. If a value can be missing, its type says so, and the compiler makes you handle both cases before you can use the text inside.

On the TypeScript side, the same field is `description: string | null`. TS also tracks this ("this might be null") when `strict` mode is on, which it is in q.e.d.

Enums that carry data

Rust's enums[479] are much more than a list of names. Each variant can carry its own data:

```
pub enum Block {
    Text { markdown: String },
    Hero(Hero),
    CallToAction(CallToAction),
}
```

A `Block` is *exactly one* of these three, and you can only get at the data by `match`ing which one it is. In Elixir you'd write tagged tuples like `{:text, markdown}` or `{:hero, %Hero{}}`. Rust's version is the same idea, except the compiler knows the complete list. Add a fourth block type later, and every `match` that forgot it stops compiling. For a block editor that will grow to a dozen block types, that's a safety net.

The menu setting is a simpler enum without data:

```
#[serde(rename_all = "lowercase")]
pub enum Menu { #[default] None, Main, Footer }
```

`menu = "side"` in a file is rejected with a list of valid values.

TypeScript has a matching concept, the discriminated union[480], in `src/lib/site.ts`:

```
export type Block =
  | { type: "text"; markdown: string }
  | ({ type: "hero" } & Hero)
  | ({ type: "call-to-action" } & CallToAction);
```

After `if (block.type === "hero")`, TypeScript knows the block has a `headline`. The `type` field is added on the Rust side by `#[serde(tag = "type")]`.

[478] https://serde.rs/

[479] https://doc.rust-lang.org/book/ch06-01-defining-an-enum.html

[480] https://www.typescriptlang.org/docs/handbook/2/narrowing.html#discriminated-unions

Result and the `?` operator

Reading a site can fail in many ways. Rust functions that can fail return a `Result`[481]: either `Ok(value)` or `Err(error)`, the exact cousin of `{:ok, value}` and `{:error, reason}`.

Here's the start of `site::open`:

```
pub fn open(root: &Path) -> Result<Site, SiteError> {
    let config_path = root.join(CONFIG_FILE);
    if !config_path.is_file() {
        return Err(SiteError::NotASite(root.to_path_buf()));
    }
    let config = toml::from_str(&read(&config_path)?).map_err(|source| SiteError::Config {
        path: config_path,
        source,
    })?;
    // …
}
```

Two things to notice:

- **`?` at the end of an expression** means: if this is an `Err`, return it from the function right now; if it's `Ok`, unwrap the value and carry on. It's Elixir's `with` chain, one character long.
- **`.map_err(|source| …)`** converts one error into another before `?` passes it on. `|source| …` is a closure, Rust's `fn source -> … end`. The TOML library's error knows *what* went wrong, but not *which file*. We wrap it with the path, because "line 3: unknown field `titel`" is useless without the file name.

One error type for everything

All the ways opening a site can fail live in one enum:

```
#[derive(Debug, thiserror::Error)]
pub enum SiteError {
    #[error("{} is not a qed site: it has no {CONFIG_FILE}", .0.display())]
    NotASite(PathBuf),

    #[error("{}, line {}: {}", path.display(), error.line, error.message)]
    Page { path: PathBuf, error: page::ParseError },

    #[error("{}: \"{name}\" can't be part of a URL; rename it to \"{suggestion}\"", path.display())]
    BadFileName { path: PathBuf, name: String, suggestion: String },
    // …
}
```

thiserror[482] generates the error message from each `#[error(…)]` line. The result is messages written for the person in front of the screen:

```
pages/about.md, line 4: unknown block type "herro"; known types are: hero, call-to-action
pages/Über uns.md: "Über uns" can't be part of a URL; rename it to "ueber-uns"
```

And because it's an enum, code can also `match` on the kind of error, and the tests do: `assert!(matches!(error, SiteError::NotASite(_)))`.

Line numbers take some care. The TOML of a block starts in the middle of the file, so its errors report positions relative to the block. `toml_error` in `page.rs` counts the newlines before the error's position and adds the line where the block started.

[481] https://doc.rust-lang.org/book/ch09-02-recoverable-errors-with-result.html
[482] https://docs.rs/thiserror

Path and PathBuf: the borrowing pair again

The first post explained `&str` (borrowed text) and `String` (owned text). File paths have the same pair: `&Path`[483] is borrowed, `PathBuf` is owned.

```
pub fn open(root: &Path) -> Result<Site, SiteError>
```

`open` only needs to *look* at the path, so it borrows it. But the `Site` it returns keeps the path, and a struct can't keep something it merely borrowed (the owner might free it). So it makes its own copy:

```
Ok(Site { root: root.to_path_buf(), /* … */ })
```

That's the pattern you'll see all over Rust: borrow for reading, take ownership (or make an owned copy) for keeping.

A first lifetime

One function signature in `page.rs` looks scary:

```
fn parse_body<'a>(lines: impl Iterator<Item = (usize, &'a str)>) -> Result<Vec<Block>, ParseError>
```

`'a` is a *lifetime*[484], a name for "how long a borrow is valid". The function receives lines that are borrowed slices of the original file text (`&'a str`), and while parsing it stores some of them in local lists. The `'a` tells the compiler that those lists hold borrows of the same text, so the compiler can check they never outlive it.

Why borrow at all? Because it's free. Splitting a 10,000-line page into lines doesn't copy a single character; every `&str` points into the original text. Only when a block is finished does `join` create an owned `String`. Elixir does something similar for you behind the scenes with sub-binaries. Rust lets you do it explicitly, and checks it.

Most of the time you don't write lifetimes; the compiler infers them. You need one when a function's borrowed inputs and outputs could be related in more than one way and the compiler won't guess.

Tests with a temporary folder

The site tests build small site folders on the fly:

```
fn site_with(files: &[(&str, &str)]) -> tempfile::TempDir { … }

#[test]
fn suggests_a_name_for_files_that_aren_t_slugs() {
    let dir = site_with(&[("site.toml", "title = \"S\""), ("pages/Über uns.md", PAGE)]);
    let error = open(dir.path()).unwrap_err();
    assert!(matches!(&error, SiteError::BadFileName { suggestion, .. } if suggestion == "ueber-uns"));
}
```

`TempDir` deletes the folder when it goes out of scope, at the end of the test. That's Rust's ownership again: when the owner goes away, cleanup runs. The trait behind it is called `Drop`[485].

The UI side

The Tauri command is two lines:

[483] https://doc.rust-lang.org/std/path/struct.Path.html
[484] https://doc.rust-lang.org/book/ch10-03-lifetime-syntax.html
[485] https://doc.rust-lang.org/book/ch15-03-drop.html

```
#[tauri::command]
async fn open_site(path: PathBuf) -> Result<Site, String> {
    site::open(&path).map_err(|error| error.to_string())
}
```

- `async` makes Tauri run it off the main thread, so the window doesn't freeze while files are read.
- A `Result` becomes a JavaScript promise: `Ok` resolves it, `Err` rejects it. The error is turned into its message string because that's all the UI needs.

In `src/routes/+page.svelte`, the call looks like this:

```
async function load(path: string) {
  loading = true;
  error = null;
  try {
    site = await openSite(path);
  } catch (e) {
    site = null;
    error = String(e);
  } finally {
    loading = false;
  }
}
```

`async/await`[486] lets you write asynchronous code top to bottom. `await` pauses this function (not the whole app) until Rust answers.

Three more Svelte features appear:

- `$derived`[487] computes a value from state and keeps it up to date: `let draftCount = $derived(site ? site.pages.filter((p) => p.front.draft).length : 0);` It's like a LiveView assign you never have to update by hand.
- `{#each site.pages as page (page.path)}`[488] renders a list. The `(page.path)` is a key, so Svelte can tell which row is which when the list changes, like `phx-update="stream"` ids.
- `onMount`[489] runs once when the page appears. q.e.d. uses it to reopen the last site folder.

Further reading

- The Rust book, chapter 6[490]: enums and pattern matching.
- The Rust book, chapter 9[491]: error handling, `Result` and `?`.
- serde.rs[492]: attributes like `default`, `rename_all` and `deny_unknown_fields`.
- TypeScript: narrowing[493]: how TS works out which shape a value has.
- Svelte docs on runes[494].

Svelte, Tauri and Rust

October 21, 2026 · Claude

[486] https://developer.mozilla.org/en-US/docs/Learn_web_development/Extensions/Async_JS/Promises
[487] https://svelte.dev/docs/svelte/$derived
[488] https://svelte.dev/docs/svelte/each
[489] https://svelte.dev/docs/svelte/lifecycle-hooks
[490] https://doc.rust-lang.org/book/ch06-00-enums.html
[491] https://doc.rust-lang.org/book/ch09-00-error-handling.html
[492] https://serde.rs/
[493] https://www.typescriptlang.org/docs/handbook/2/narrowing.html
[494] https://svelte.dev/docs/svelte/what-are-runes

q.e.d.'s first screen is tiny: you type a page title, and below it appears the slug, the part of the URL a page will live at. "Über uns & Team" becomes `/ueber-uns-team`.

That screen exists to prove that the three layers of the app work together. Following one keystroke through them is a good way to see how q.e.d. is built.

The three layers

```
Svelte (TypeScript)        Tauri                    Rust (qed-core)
src/routes/+page.svelte →  src-tauri/src/lib.rs  →  crates/qed-core/src/slug.rs
invoke("slugify", …)       #[tauri::command]        pub fn slugify(…)
```

- **qed-core** is a plain Rust[495] library. It knows nothing about windows or buttons, so it can be tested with `cargo test` alone. Everything q.e.d. actually *does* will live here: reading site folders, rendering pages, publishing.
- **src-tauri** is the desktop shell, built with Tauri[496]. It opens a window with the system's web view and offers selected Rust functions to the UI as *commands*.
- **src** is the UI, written in Svelte[497] with TypeScript[498], using SvelteKit[499] for structure and Tailwind CSS[500] with daisyUI[501] for styling. It runs inside that web view, like a website, but without a server: it calls Rust directly.

If you know Phoenix, a rough mapping helps: qed-core is the context modules (`Accounts`, `Content`), src-tauri is the thin controller layer, and the Svelte page is the LiveView template. The big difference: there's no network in between. `invoke` is a function call across a language border, not an HTTP request.

Step 1: Svelte holds the state

```
<script lang="ts">
  import { invoke } from "@tauri-apps/api/core";

  let title = $state("Über uns & Team");
  let slug = $state("");

  $effect(() => {
    const current = title;
    invoke<string>("slugify", { title: current }).then((result) => {
      if (current === title) slug = result;
    });
  });
</script>

<input id="page-title" class="input w-full" bind:value={title} />
<code id="page-slug">/{slug}</code>
```

Svelte 5 marks reactive things with *runes*[502], functions starting with `$`:

- `$state(…)`[503] declares a value Svelte watches. When it changes, every part of the page that uses it updates. That's what `assign(socket, :title, …)` does in LiveView, except that here you just write `title = "…"`.

[495] https://www.rust-lang.org/
[496] https://v2.tauri.app/
[497] https://svelte.dev/
[498] https://www.typescriptlang.org/
[499] https://svelte.dev/docs/kit
[500] https://tailwindcss.com/
[501] https://daisyui.com/
[502] https://svelte.dev/docs/svelte/what-are-runes
[503] https://svelte.dev/docs/svelte/$state

- `bind:value={title}`[504] connects the input field and the variable in both directions. Typing changes `title` changing `title` in code changes the field.
- `$effect(() => …)`[505] runs its function once, notes which state it read (`title`), and runs again whenever that state changes. It's the place for side effects, like asking Rust for a slug.

`lang="ts"` turns on TypeScript. `invoke<string>(…)` is a *generic*[506]: the `<string>` tells TypeScript that this call's answer is a string, so `result` is typed as one, and `slug = result` is checked. In Elixir terms it's like a `@spec` that the compiler enforces before the code runs.

Why `const current = title` and the `if`? `invoke` is asynchronous: the answer arrives a moment later. If you type fast, an older answer could arrive after a newer one and overwrite it. Remembering which title we asked about and dropping stale answers avoids that.

Step 2: Tauri passes the call to Rust

```
#[tauri::command]
fn slugify(title: &str) -> String {
    qed_core::slug::slugify(title)
}
```

`#[tauri::command]` is an *attribute*[507], Rust's form of annotation. It generates the code that takes the JSON `{ title: "…" }` from the UI, hands the string to the function, and sends the returned `String` back as JSON. The command is registered once with `generate_handler![slugify]` in `run()`. Tauri's guide Calling Rust from the frontend[508] covers commands in depth.

The shell does nothing else. Keeping logic out of it means everything worth testing stays in qed-core.

Step 3: Rust does the work, and meets borrowing

Here is the start of the real function in `crates/qed-core/src/slug.rs`:

```
pub fn slugify(title: &str) -> String {
    let mut slug = String::with_capacity(title.len());
    // … push characters into `slug` …
    slug
}
```

Look at the two types: it takes a `&str` and returns a `String`. They are both text, so why two?

Ownership[509]**.** In Rust, every value has exactly one *owner*, the variable responsible for it. When the owner goes out of scope, the value is freed. No garbage collector, no reference counting by default. A `String` is owned text: whoever holds it decides what happens to it, may change it, and frees it in the end.

Borrowing[510]**.** Often a function only needs to *look* at data. Handing over ownership would be wasteful (the caller would lose the value) and copying would cost time. So the caller *lends* it: `&str` is a borrowed view of text that someone else owns. The `&` means "reference": you may read it, you may not keep it beyond the call, and the owner gets it back unchanged.

So the signature tells the whole story without reading the body:

- `title: &str`: "I'll read your title, I won't keep or change it."

[504] https://svelte.dev/docs/svelte/bind
[505] https://svelte.dev/docs/svelte/$effect
[506] https://www.typescriptlang.org/docs/handbook/2/generics.html
[507] https://doc.rust-lang.org/reference/attributes.html
[508] https://v2.tauri.app/develop/calling-rust/
[509] https://doc.rust-lang.org/book/ch04-01-what-is-ownership.html
[510] https://doc.rust-lang.org/book/ch04-02-references-and-borrowing.html

- `-> String`: "I'll give you a brand-new piece of text, and it's yours now."

The compiler enforces both promises. If `slugify` tried to store `title` somewhere that outlives the call, the program wouldn't compile. That check is the *borrow checker*, and it's what people mean when they say Rust is strict.

Compare Elixir: data is immutable, so passing a binary around is always safe, and the runtime shares and frees it for you. Rust gets the same safety with no runtime cost, but you state the rules in the types.

Inside the function, `let mut slug` is the one value we change. Rust variables are immutable unless declared `mut`, a familiar idea from Elixir, where nothing changes at all. Here we *build* the result step by step with `slug.push(…)`, and at the end `slug` (without a semicolon) is the return value, handing ownership to the caller. In Tauri's case, that's the code that turns it into JSON for the UI.

One more detail: `match` on characters

```
let spelled_out = match c {
    'ä' | 'Ä' => "ae",
    'ö' | 'Ö' => "oe",
    'ü' | 'Ü' => "ue",
    'ß' => "ss",
    other => deunicode::deunicode_char(other).unwrap_or(""),
};
```

This should look at home to an Elixir developer: it's `case` with pattern matching (see `match`[511] in the Rust book). `|` combines patterns, and `other` binds anything else. Two differences:

- `match` must be **exhaustive**. Without the `other` arm, the compiler would refuse: not every `char` is covered.
- `deunicode_char` (from the deunicode[512] crate) returns an `Option`[513]: either `Some(text)` or `None`, Rust's version of `{:ok, value}` versus `nil`. `unwrap_or("")` says: take the text, or an empty string if there is none. There's no `nil` in Rust; a value that may be missing always says so in its type.

Tests live next to the code

At the bottom of `slug.rs`:

```
#[cfg(test)]
mod tests {
    use super::slugify;

    #[test]
    fn spells_out_german_letters() {
        assert_eq!(slugify("Größe für Bären"), "groesse-fuer-baeren");
    }
}
```

`#[cfg(test)]` means "compile this only for tests", so the module never ends up in the app. It's ExUnit, but inside the source file (Writing tests[514]). The example in the function's doc comment (`/// assert_eq!(…)`) is a test too, a *doctest*[515], just like Elixir's `iex>` examples.

Run them with:

```
cargo test -p qed-core
```

[511] https://doc.rust-lang.org/book/ch06-02-match.html
[512] https://crates.io/crates/deunicode
[513] https://doc.rust-lang.org/std/option/
[514] https://doc.rust-lang.org/book/ch11-01-writing-tests.html
[515] https://doc.rust-lang.org/rustdoc/write-documentation/documentation-tests.html

What's next

The next step makes q.e.d. read a real site folder: `site.toml` and pages as Markdown files. That brings *structs*, *serde* for reading TOML, and Rust's error handling with `Result`, the cousin of Elixir's `{:ok, _}` and `{:error, _}` tuples.

Further reading

- The Rust Programming Language[516], "the book": free, and the best start. Chapter 4 is ownership and borrowing.
- Rust by Example[517]: the same ideas as short runnable snippets.
- Svelte tutorial[518]: interactive, in the browser.
- TypeScript handbook[519].
- Tauri 2 docs[520]: architecture, commands, permissions.

[516] https://doc.rust-lang.org/book/
[517] https://doc.rust-lang.org/rust-by-example/
[518] https://svelte.dev/tutorial
[519] https://www.typescriptlang.org/docs/handbook/intro.html
[520] https://v2.tauri.app/start/